

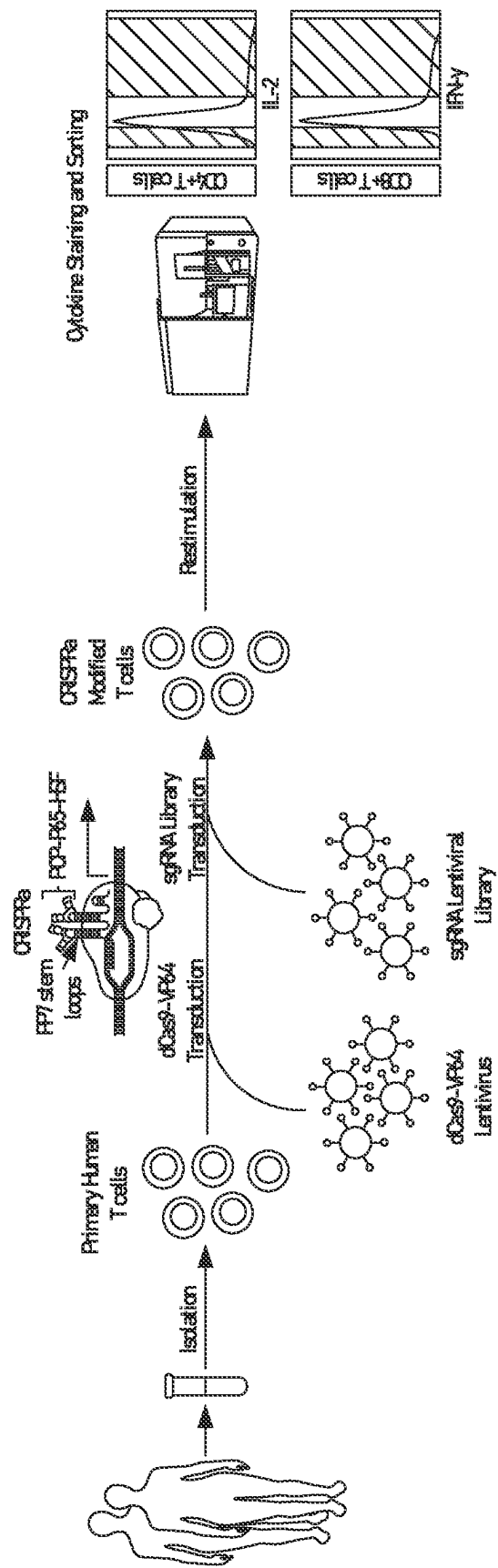


FIG.1A

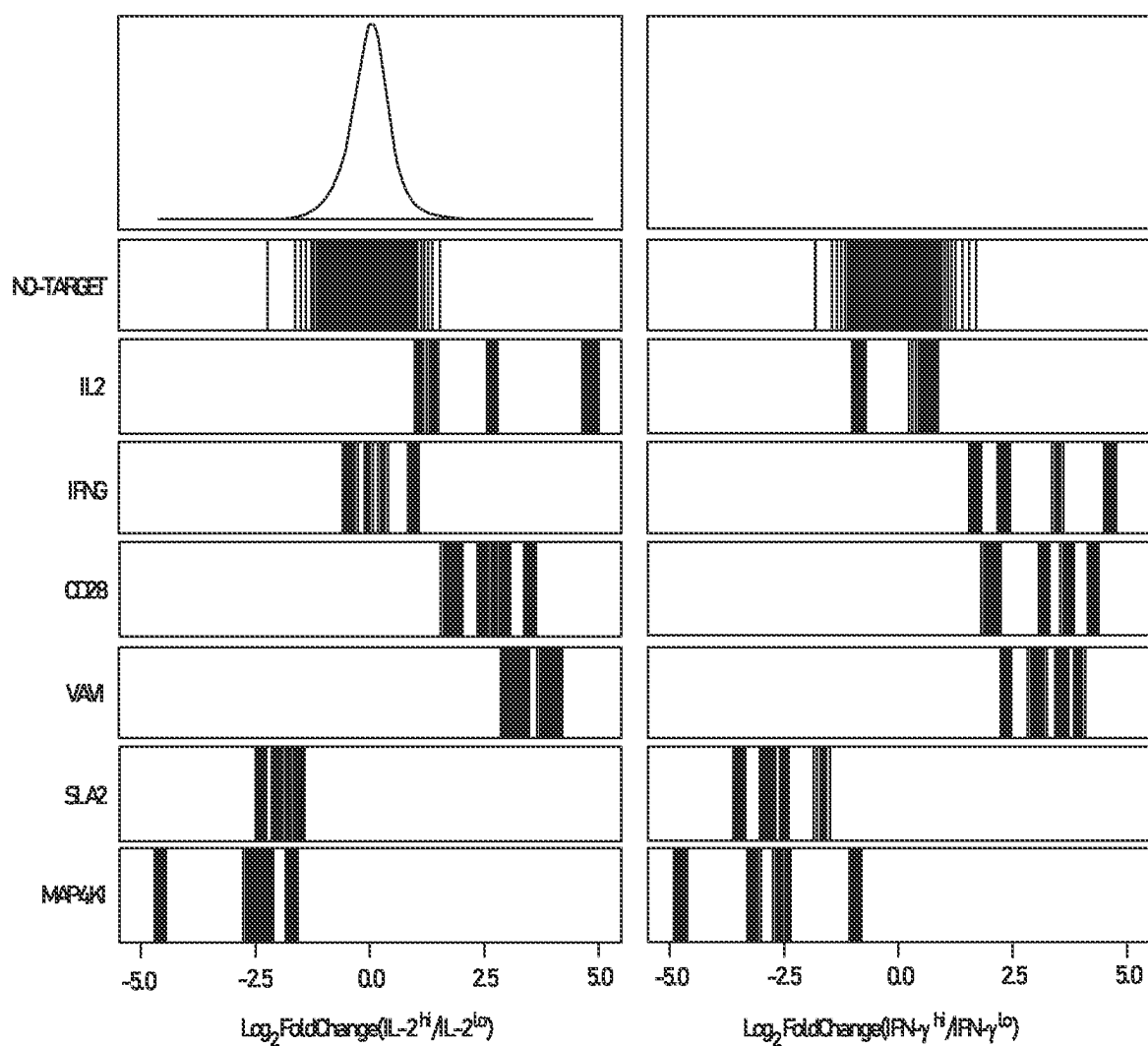


FIG.1B

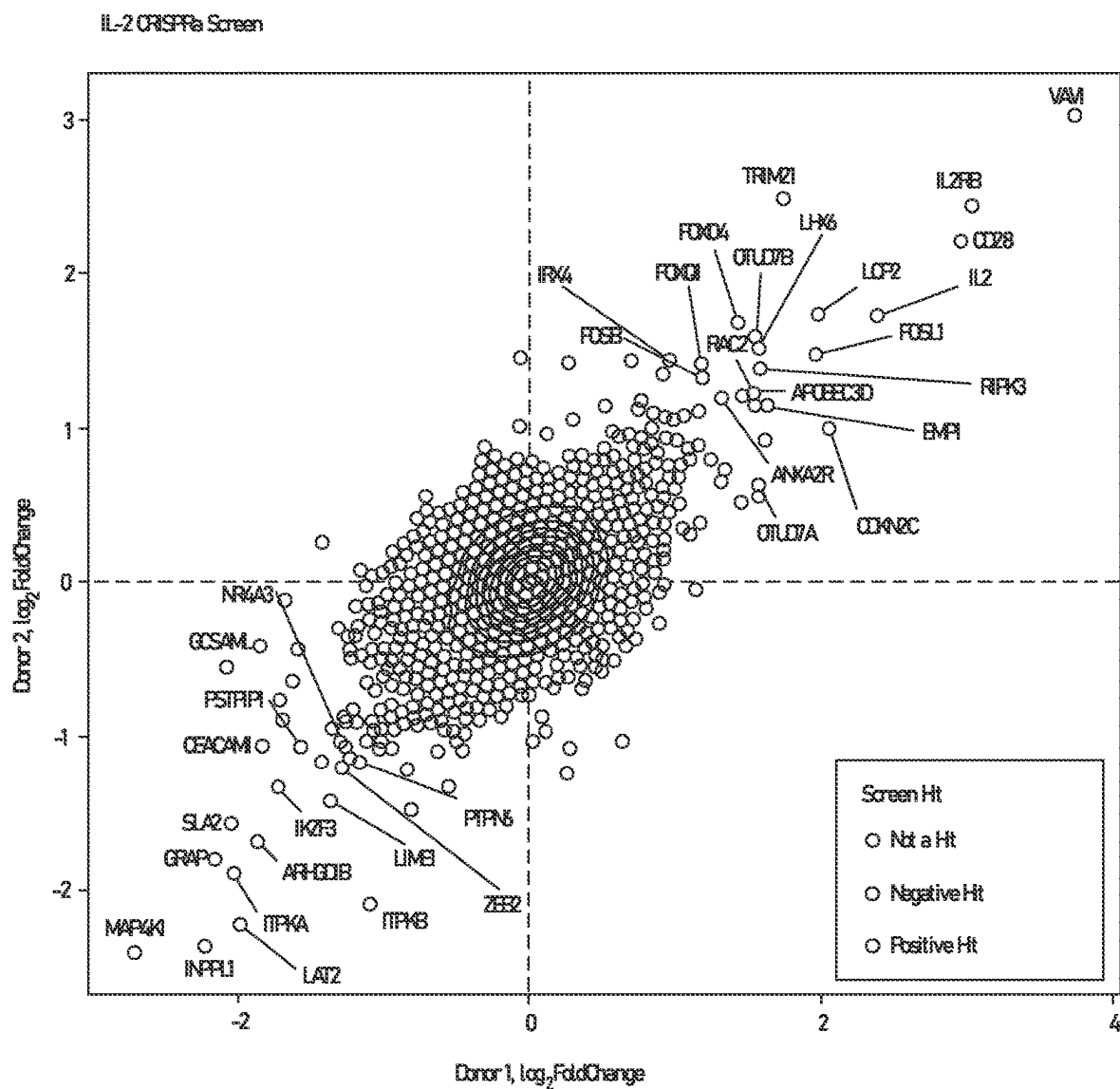


FIG. 1D

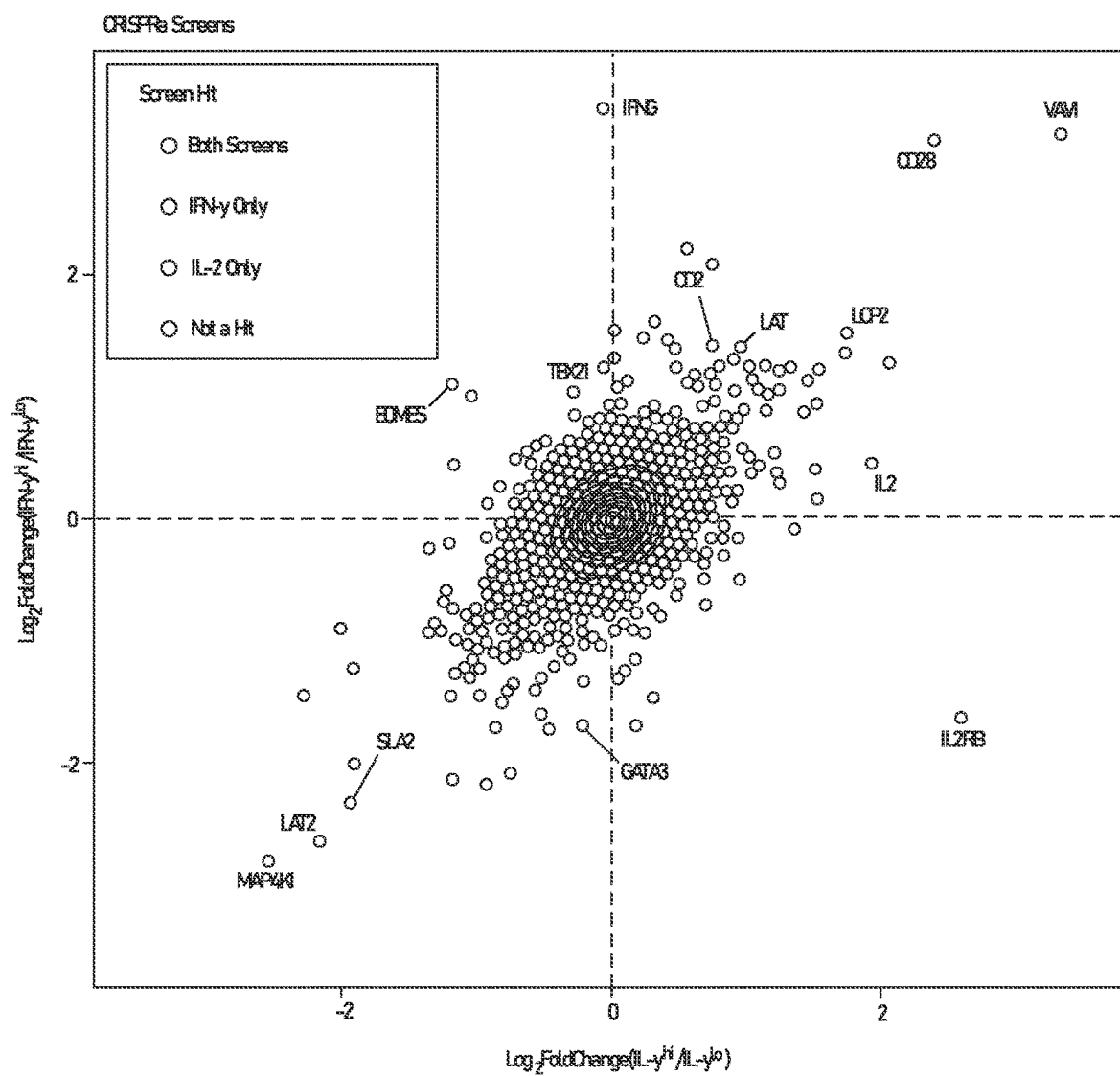


FIG.1E

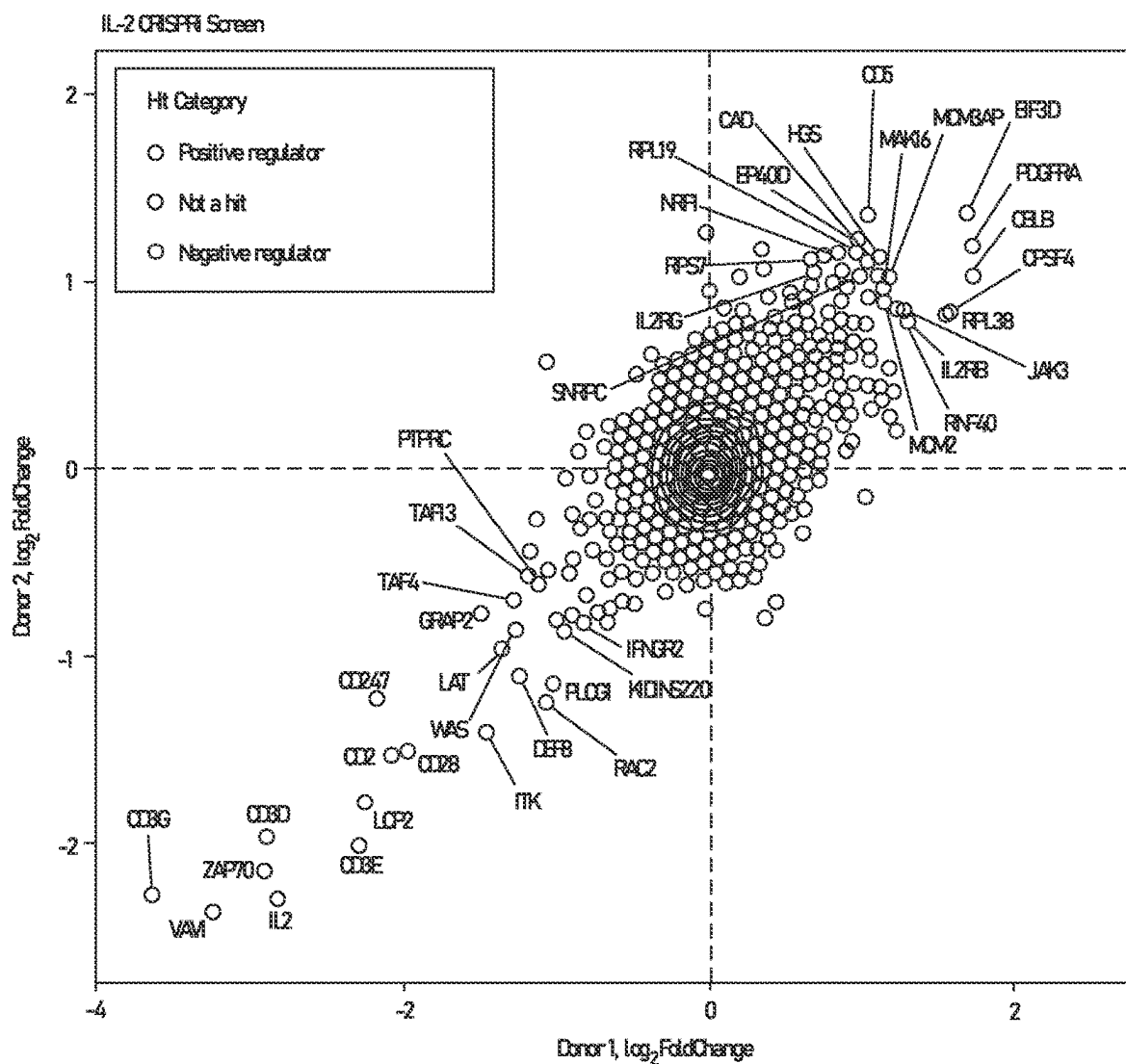


FIG.2A

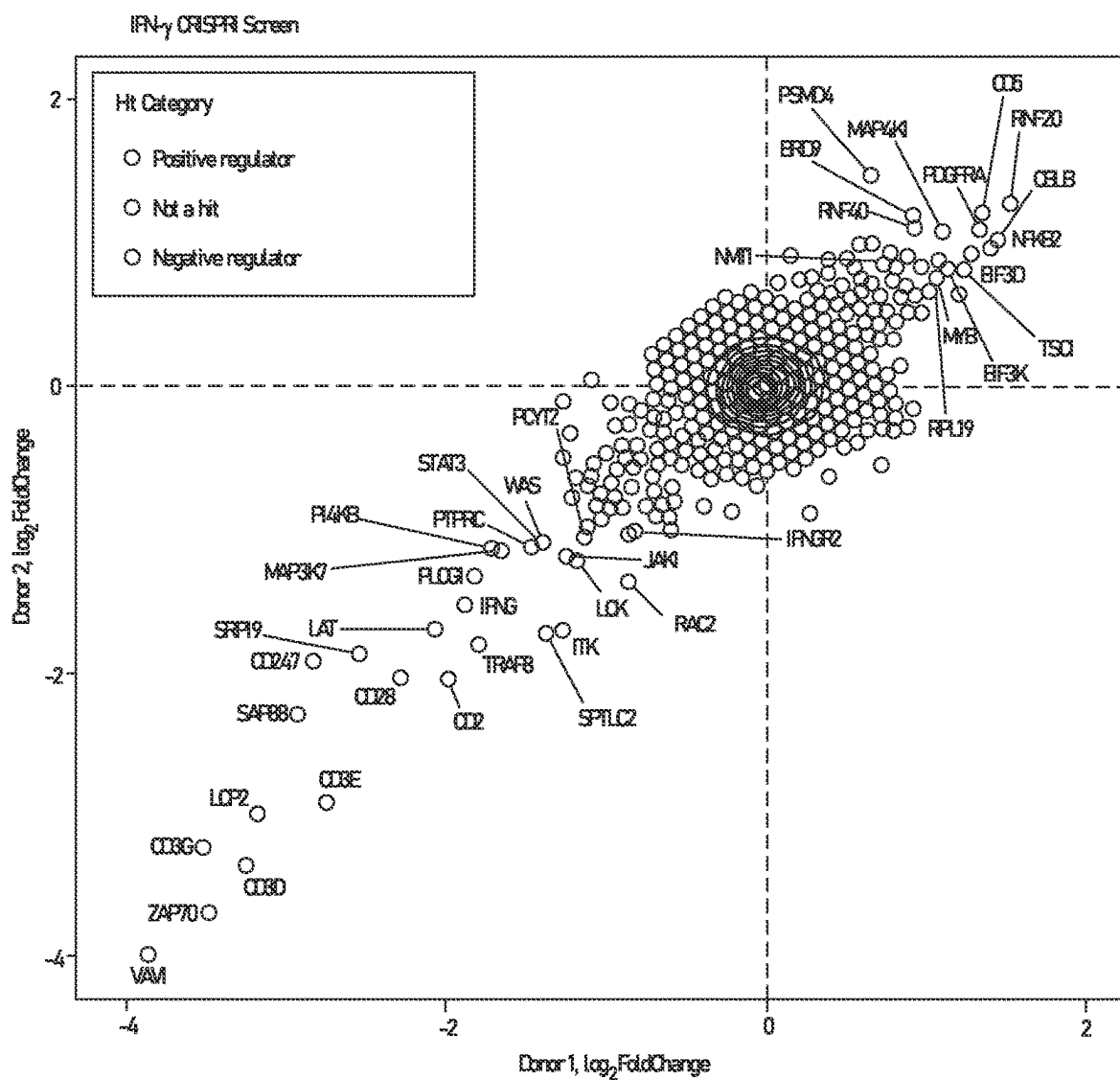


FIG.2B

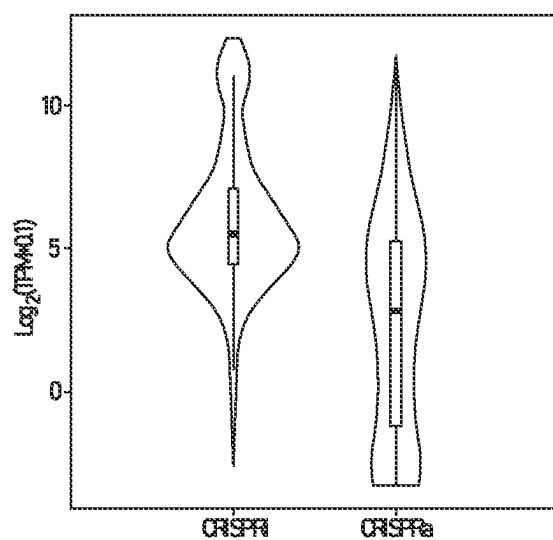


FIG.2C

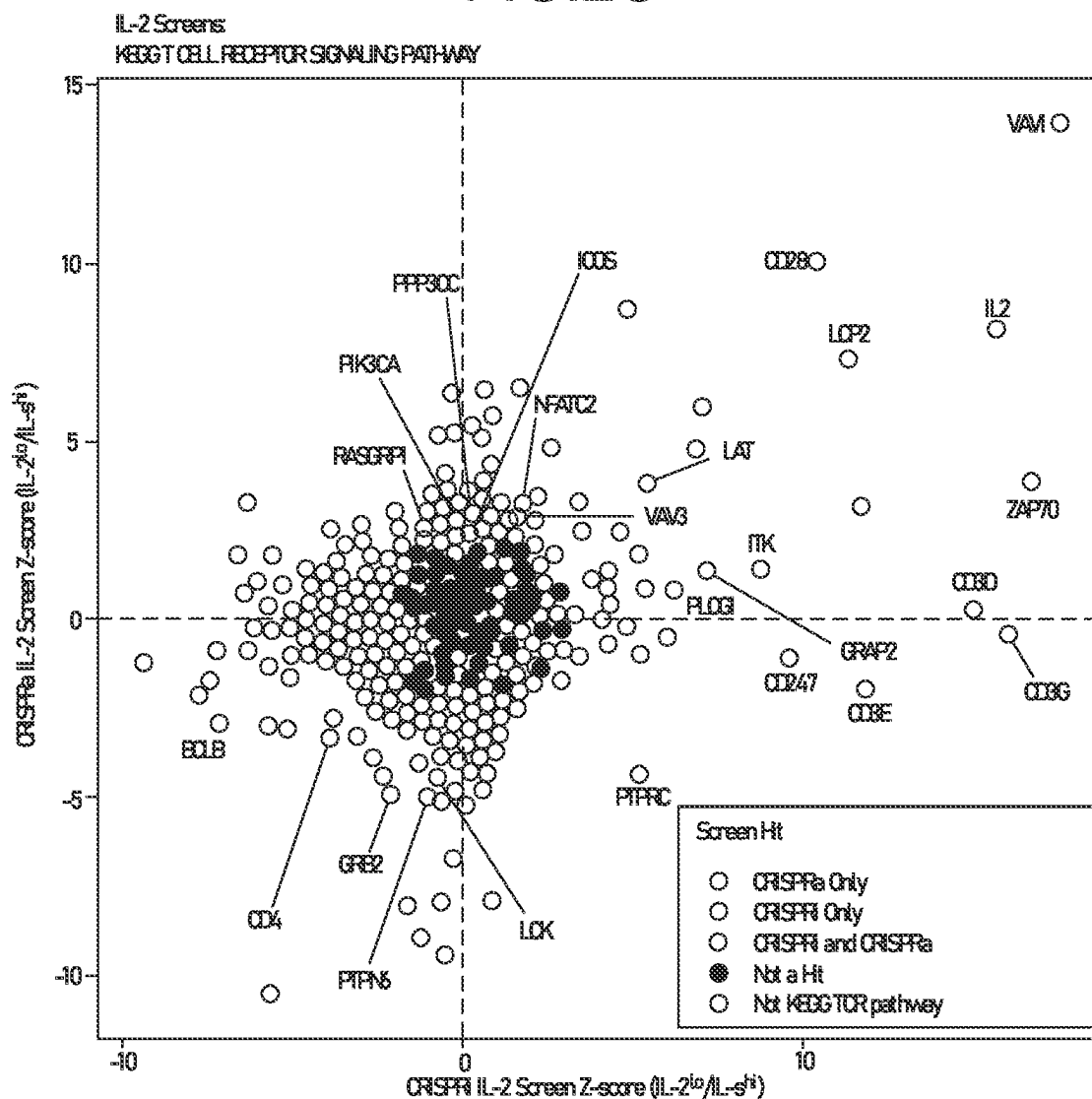


FIG.2D

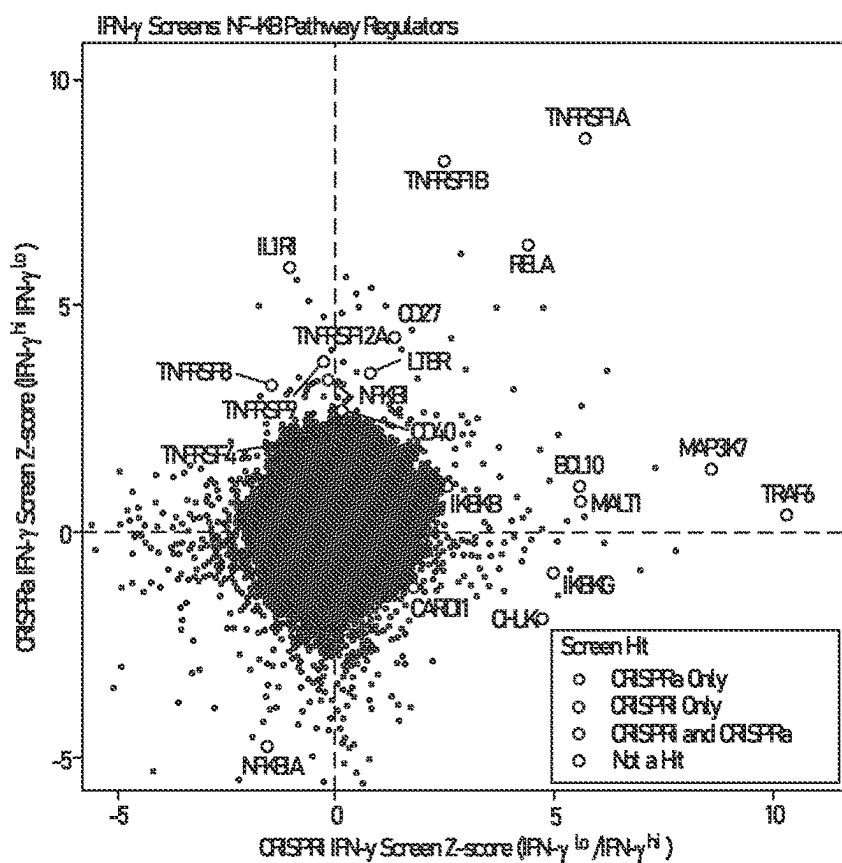


FIG.2E

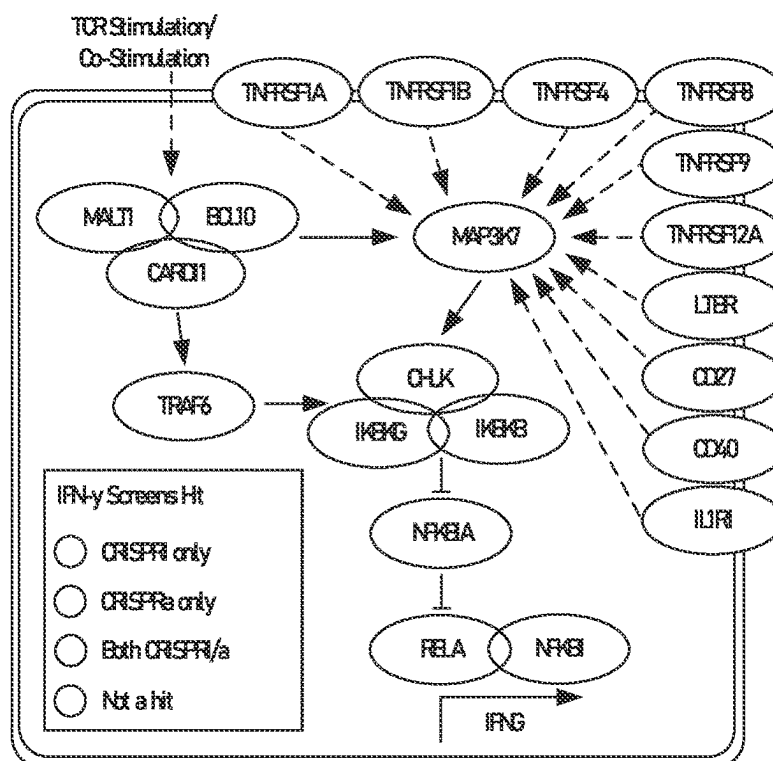
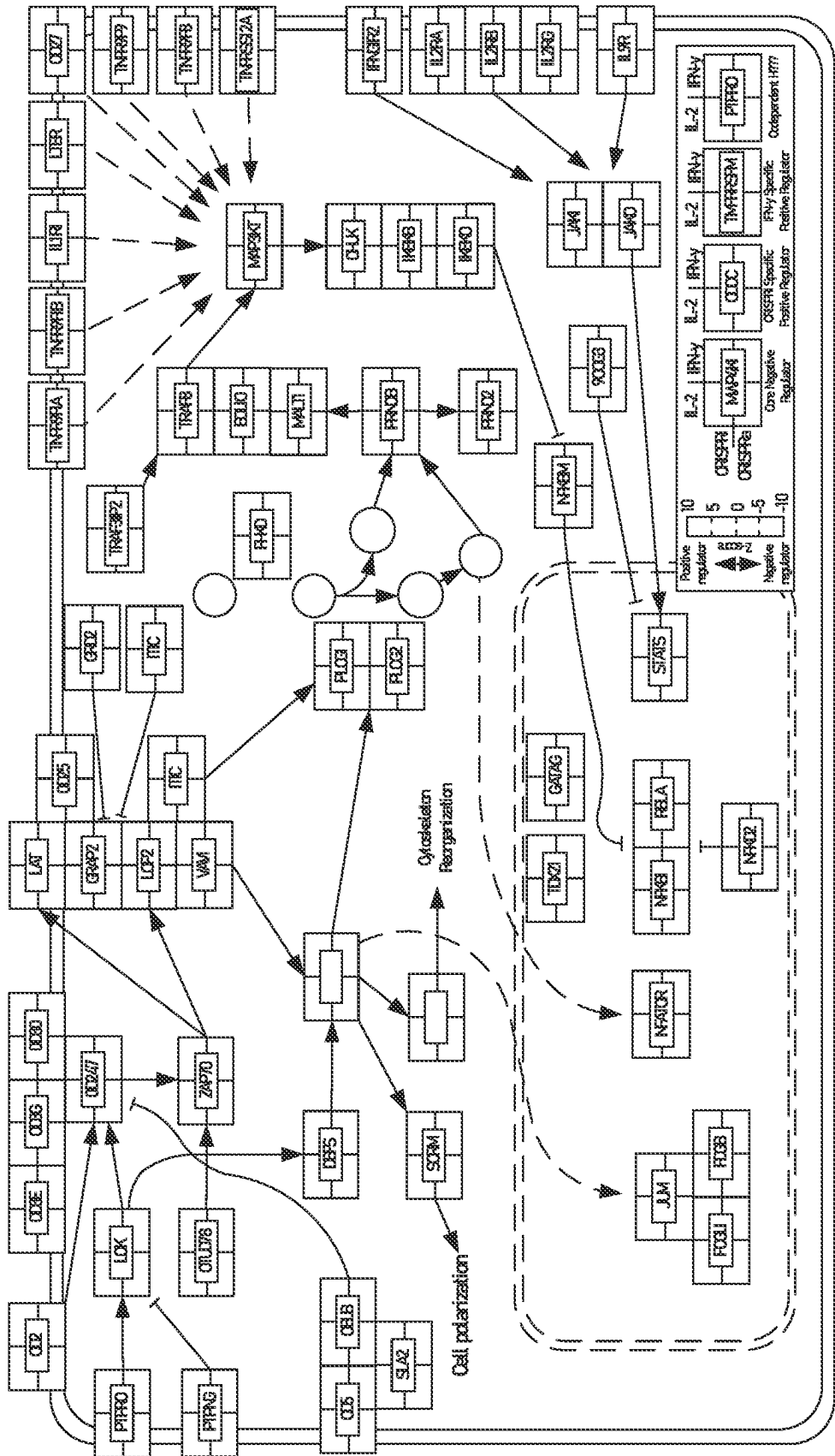


FIG.2F



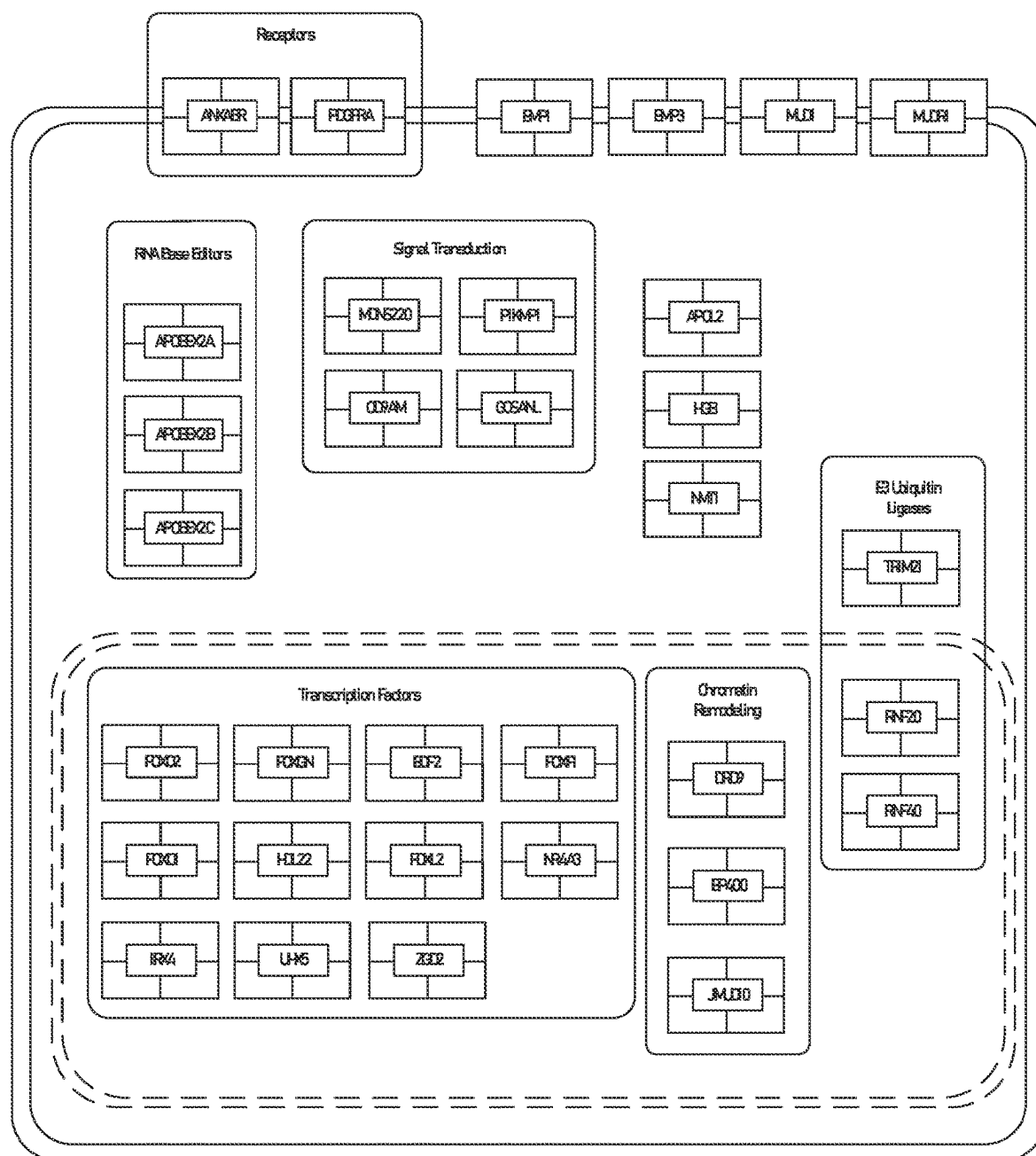


FIG.2H

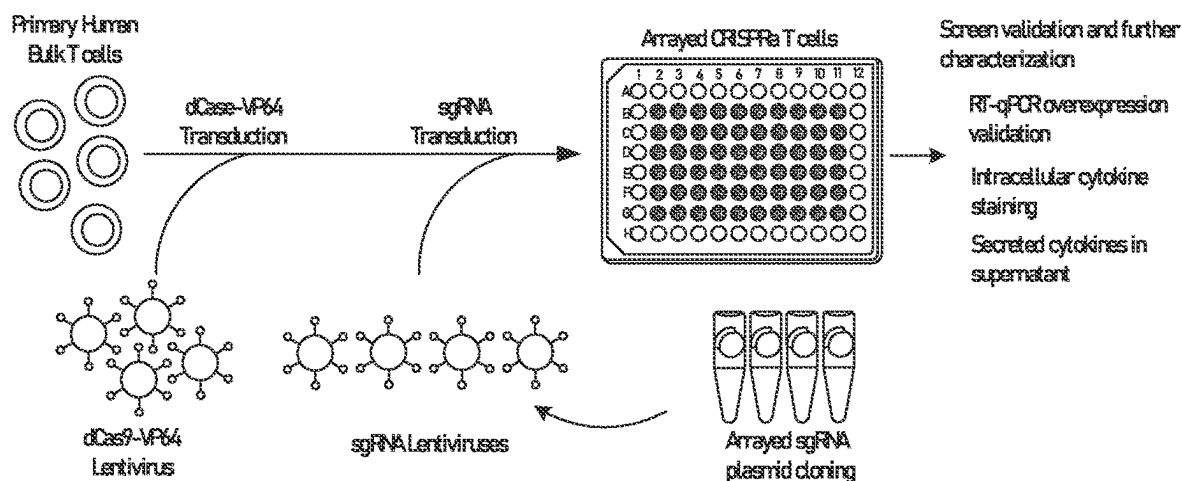


FIG.3A

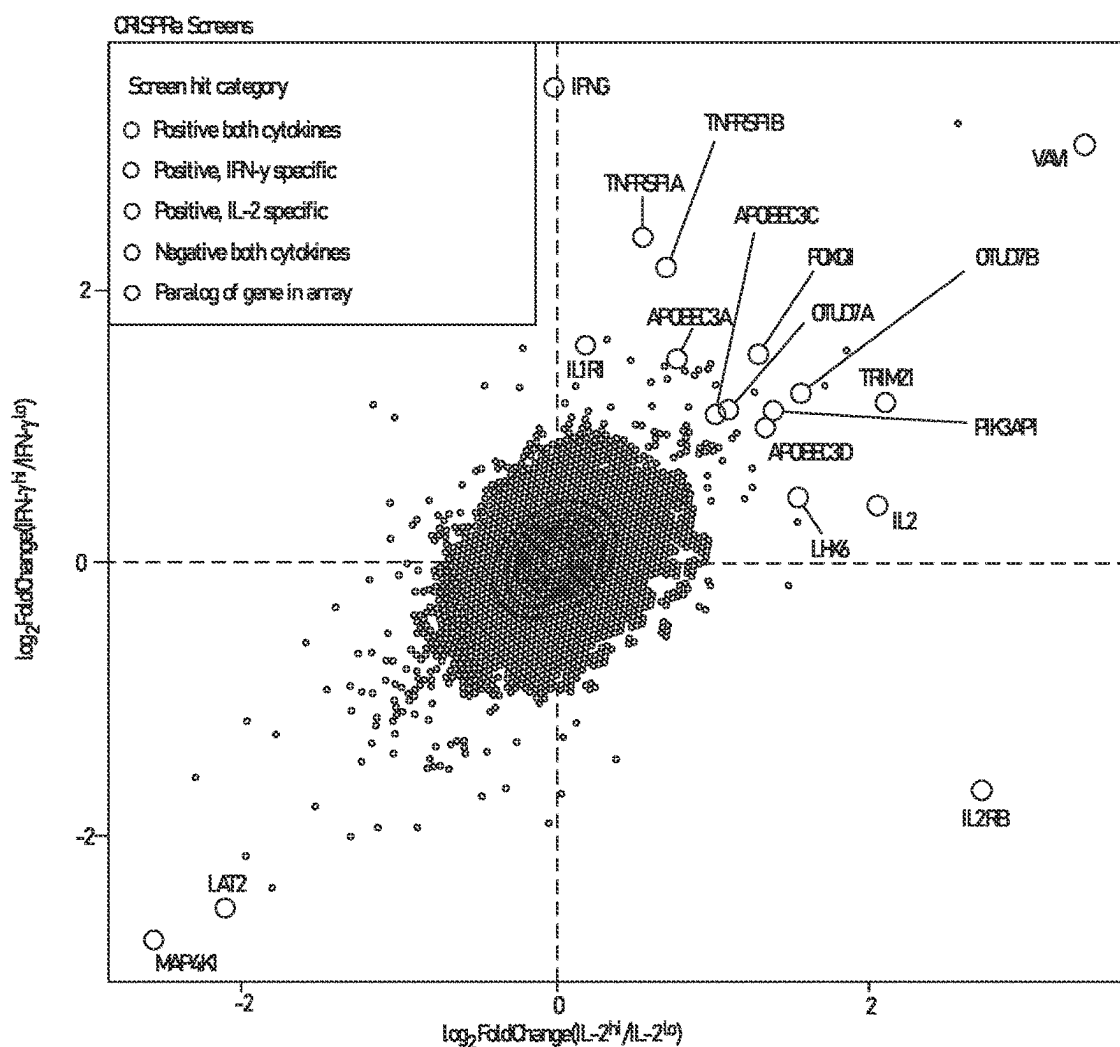


FIG.3B

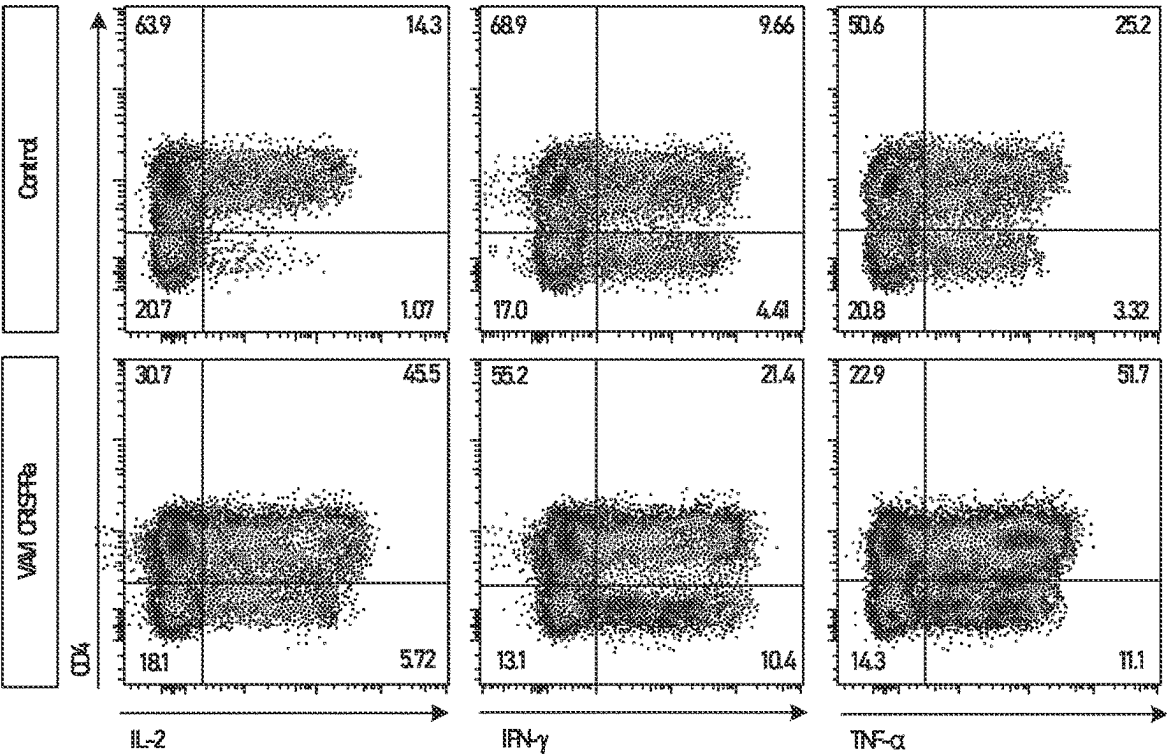


FIG.3C

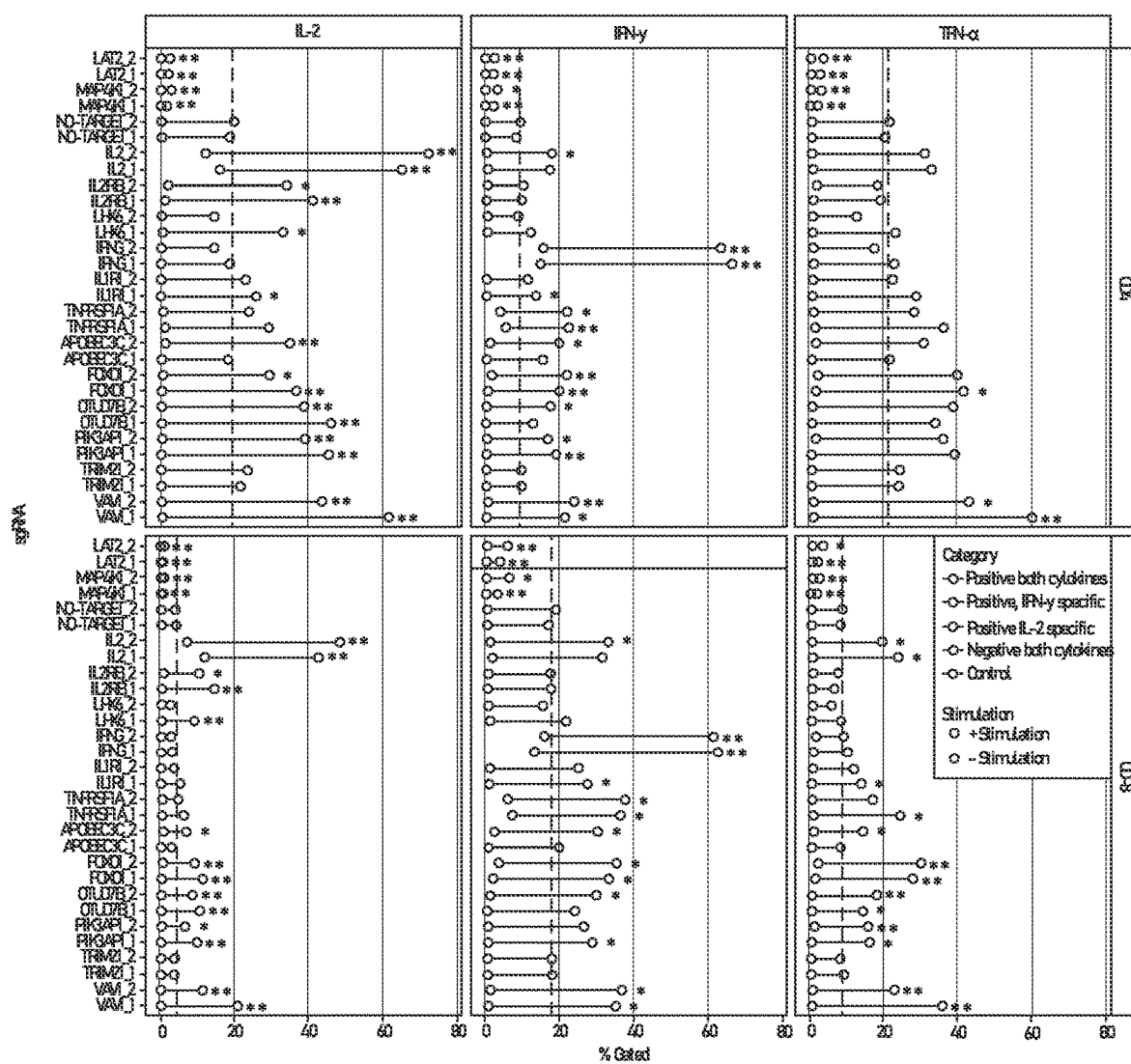


FIG.3D

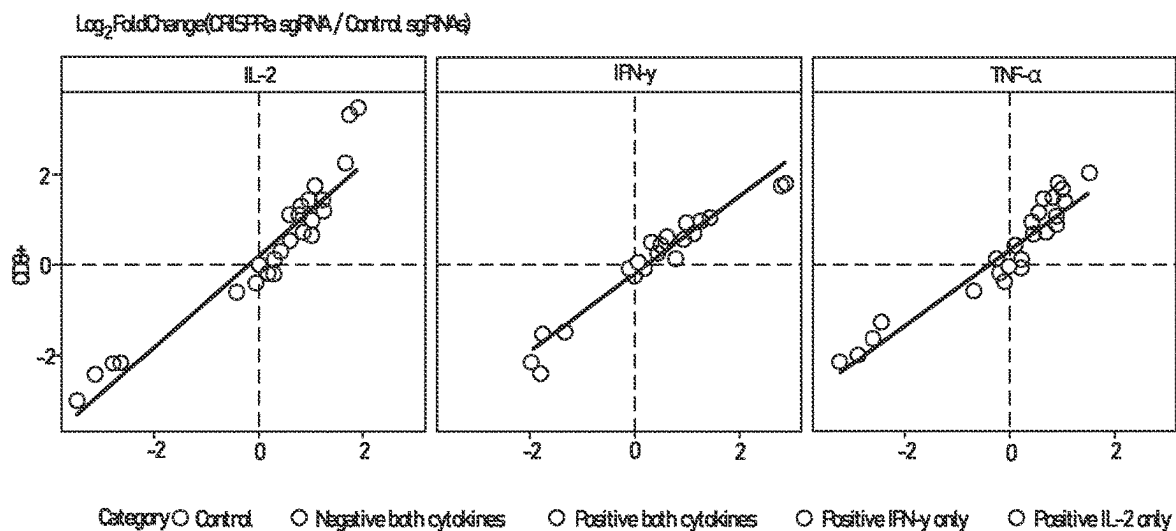


FIG.3E

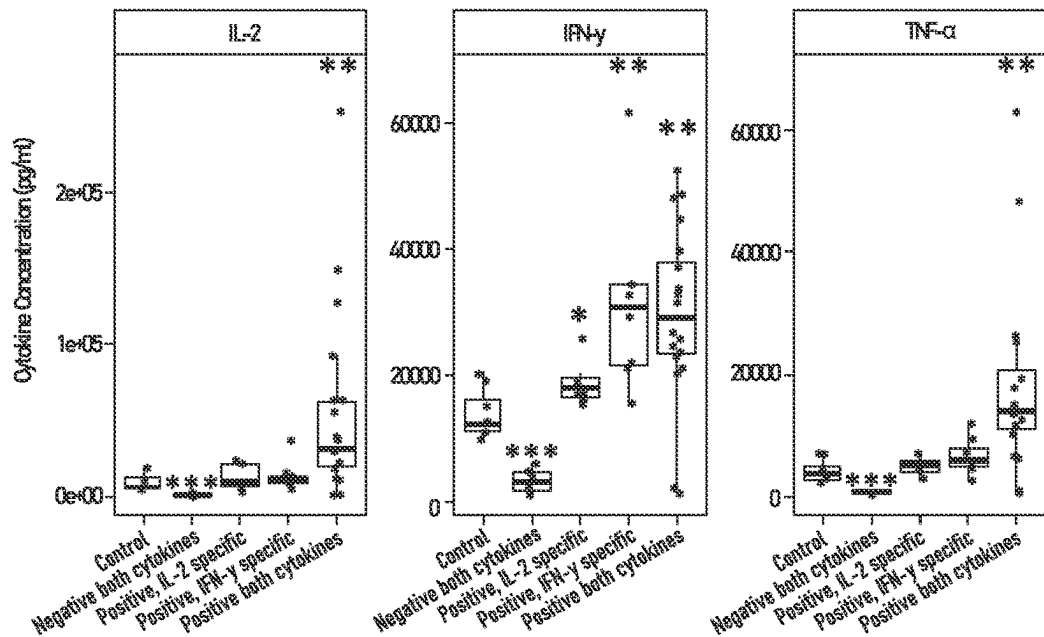


FIG.3F

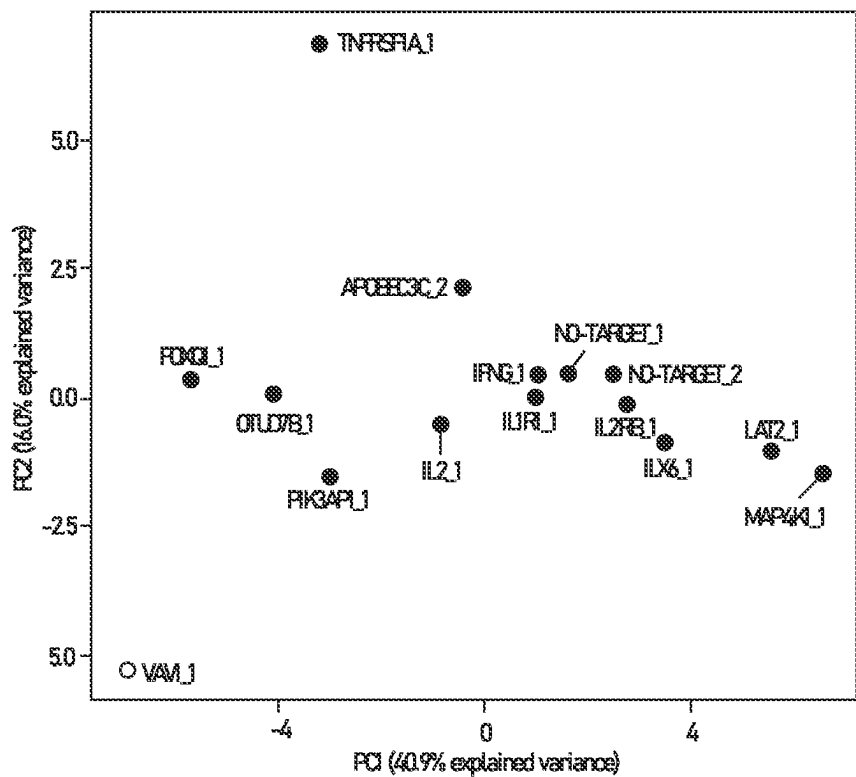


FIG.3G

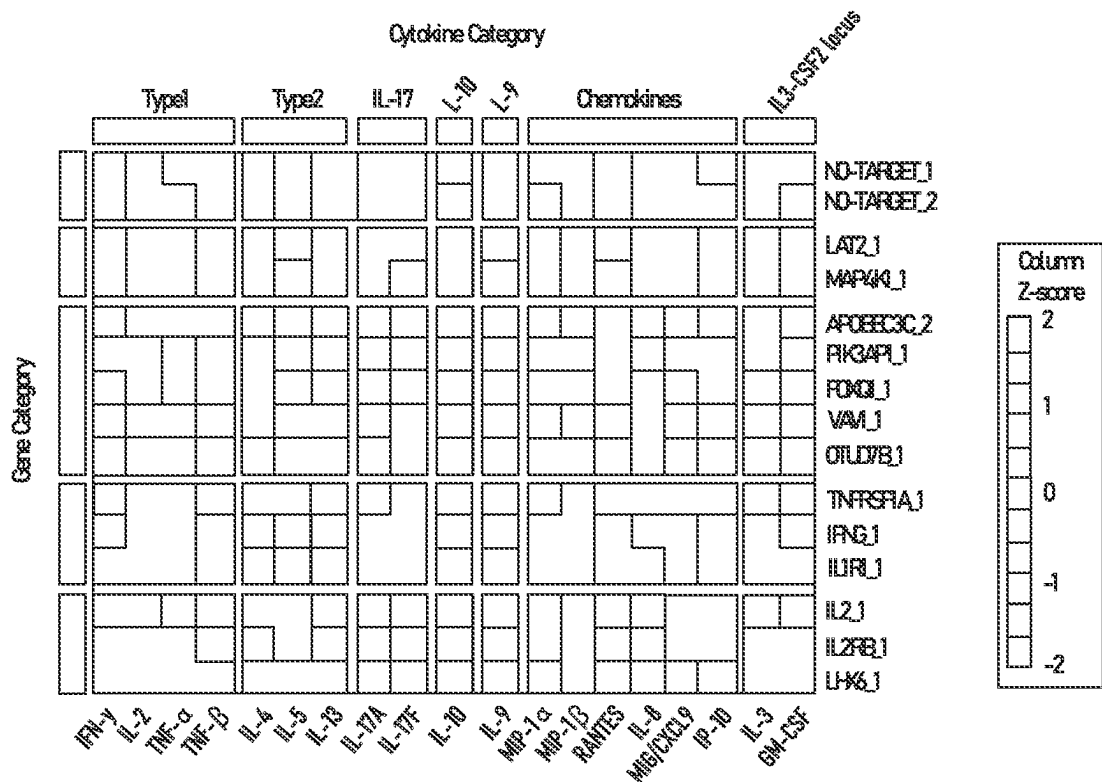


FIG.3H

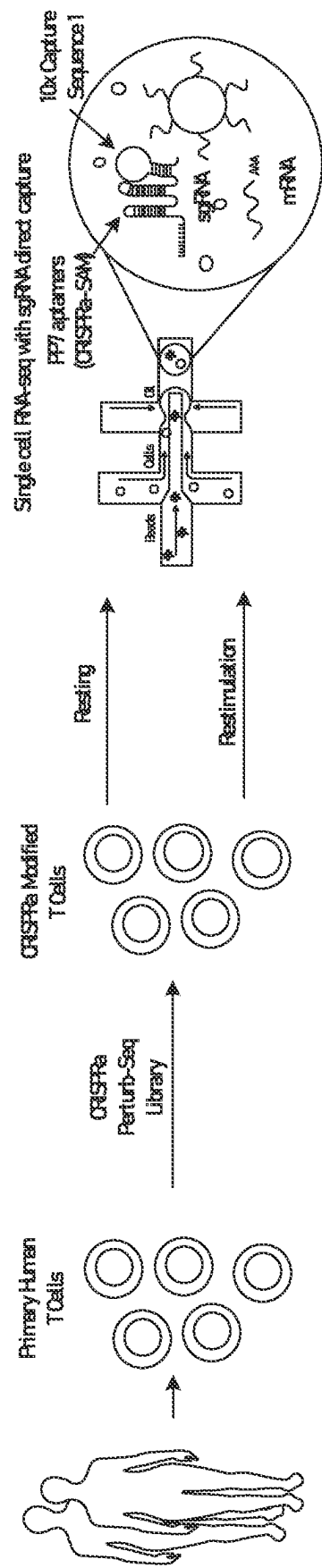
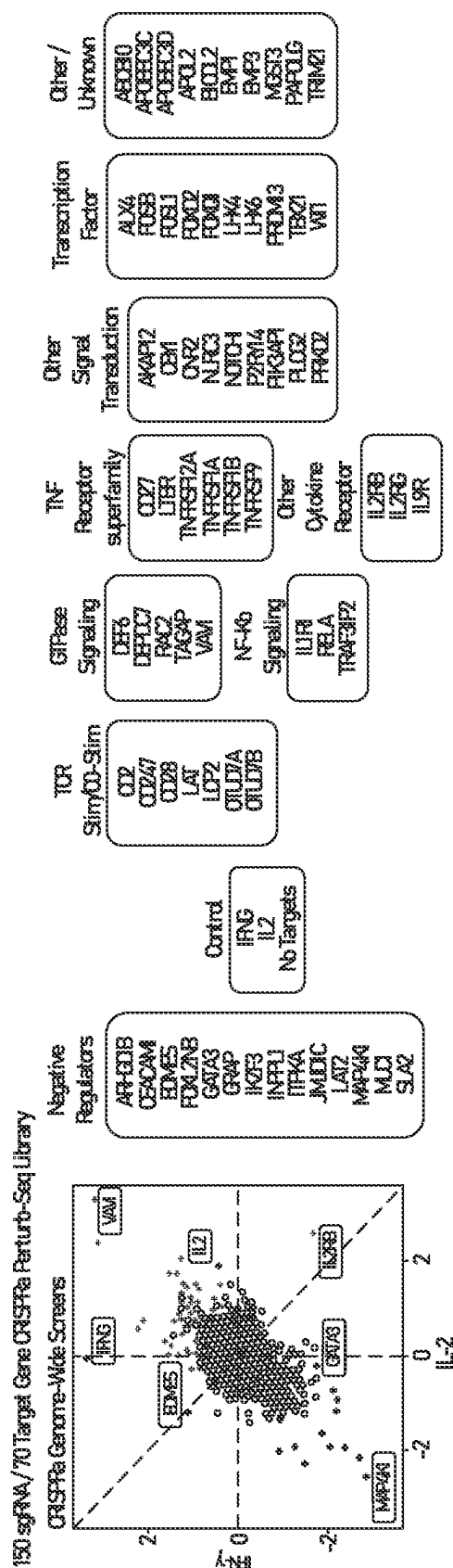


FIG.4A



34. G
L

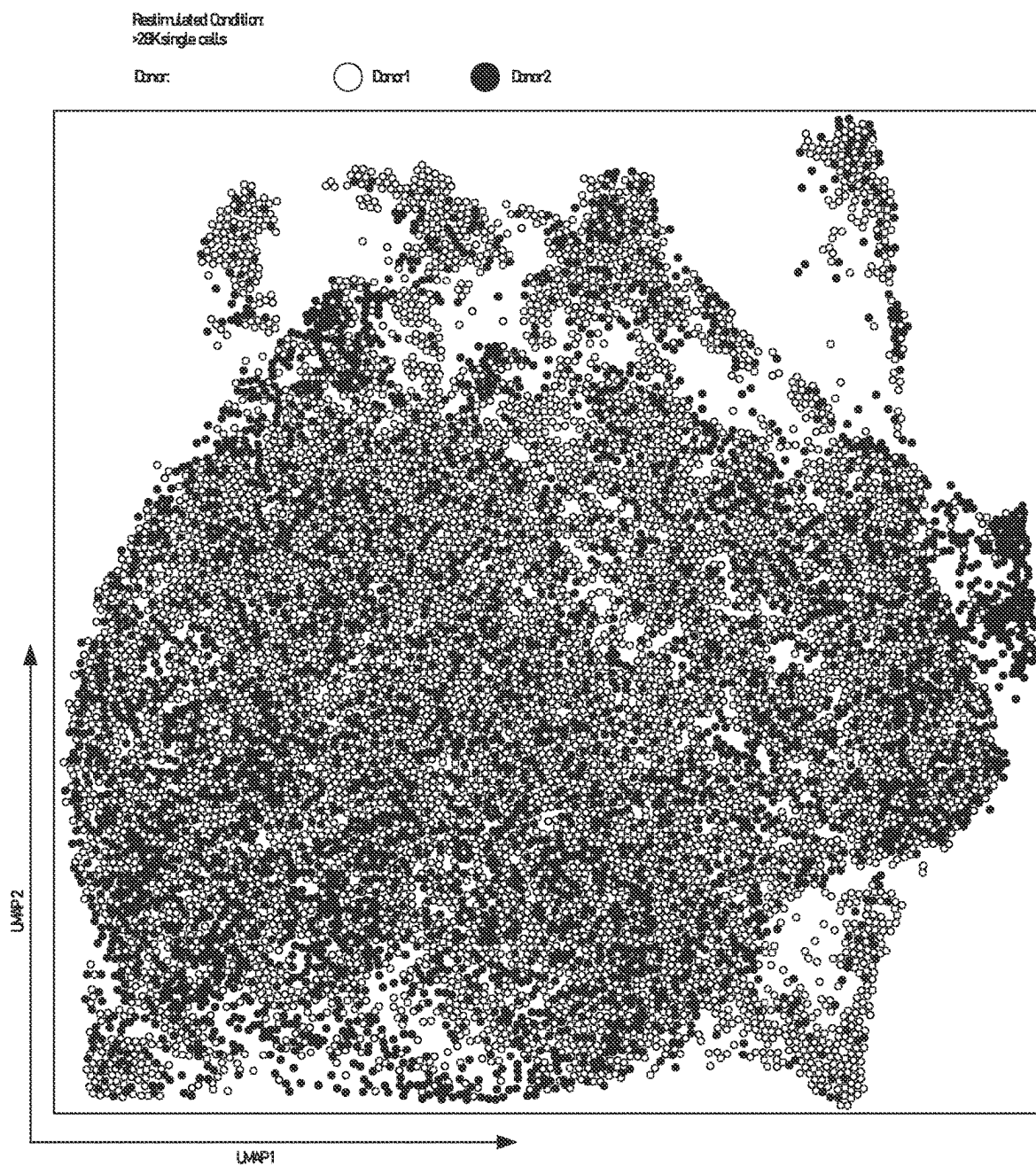


FIG.4C

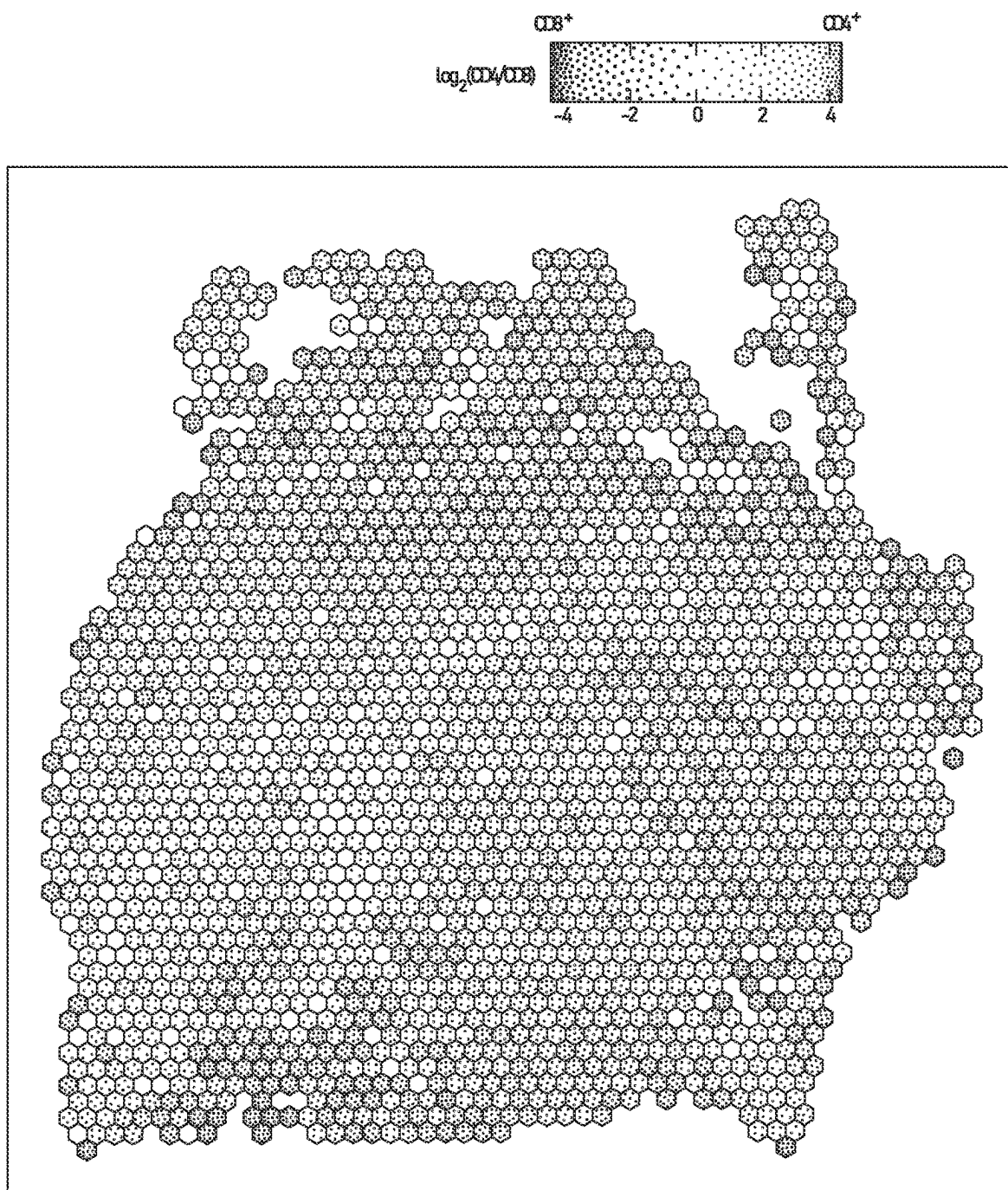


FIG.4D

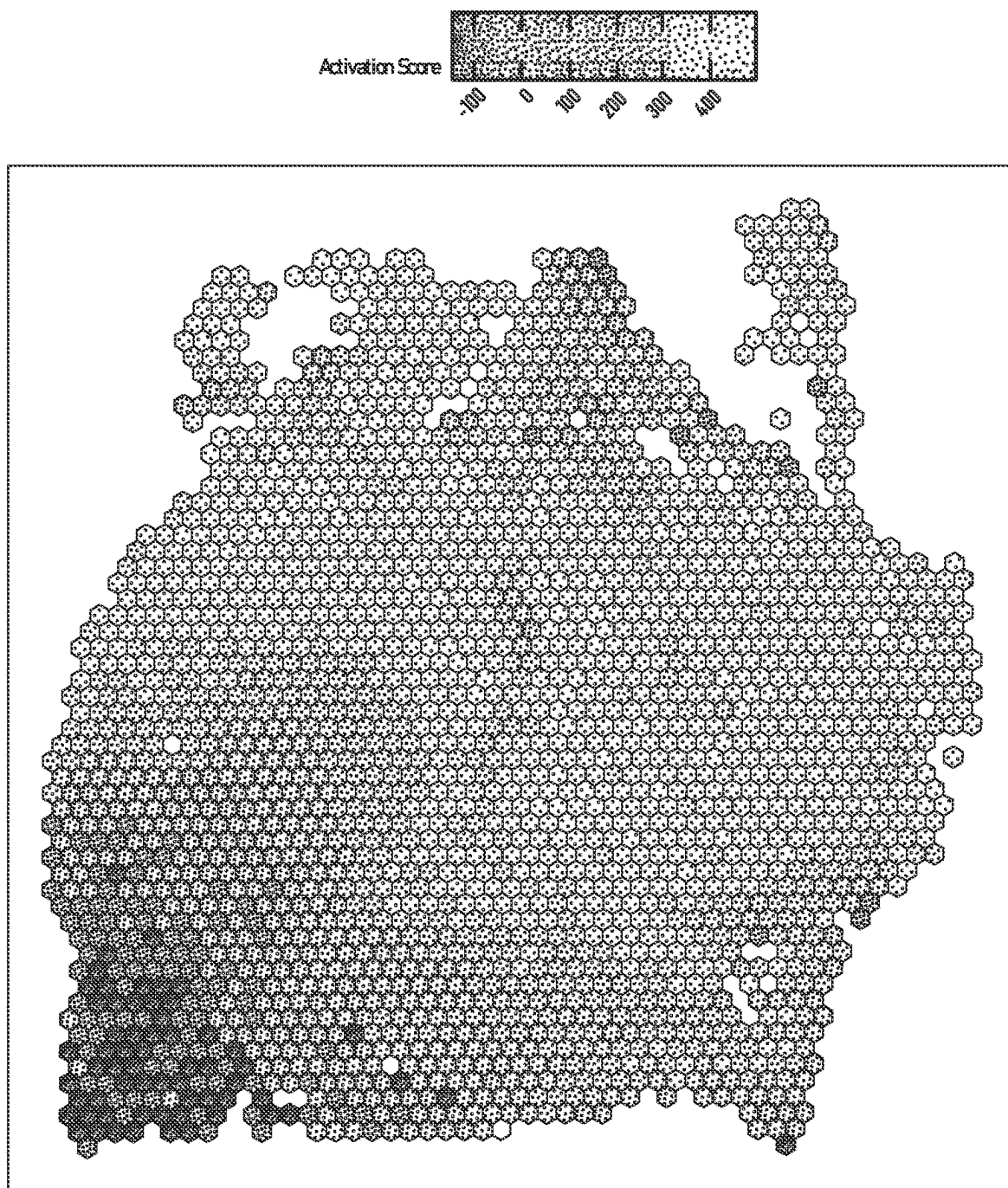


FIG.4E

FIG. 4F

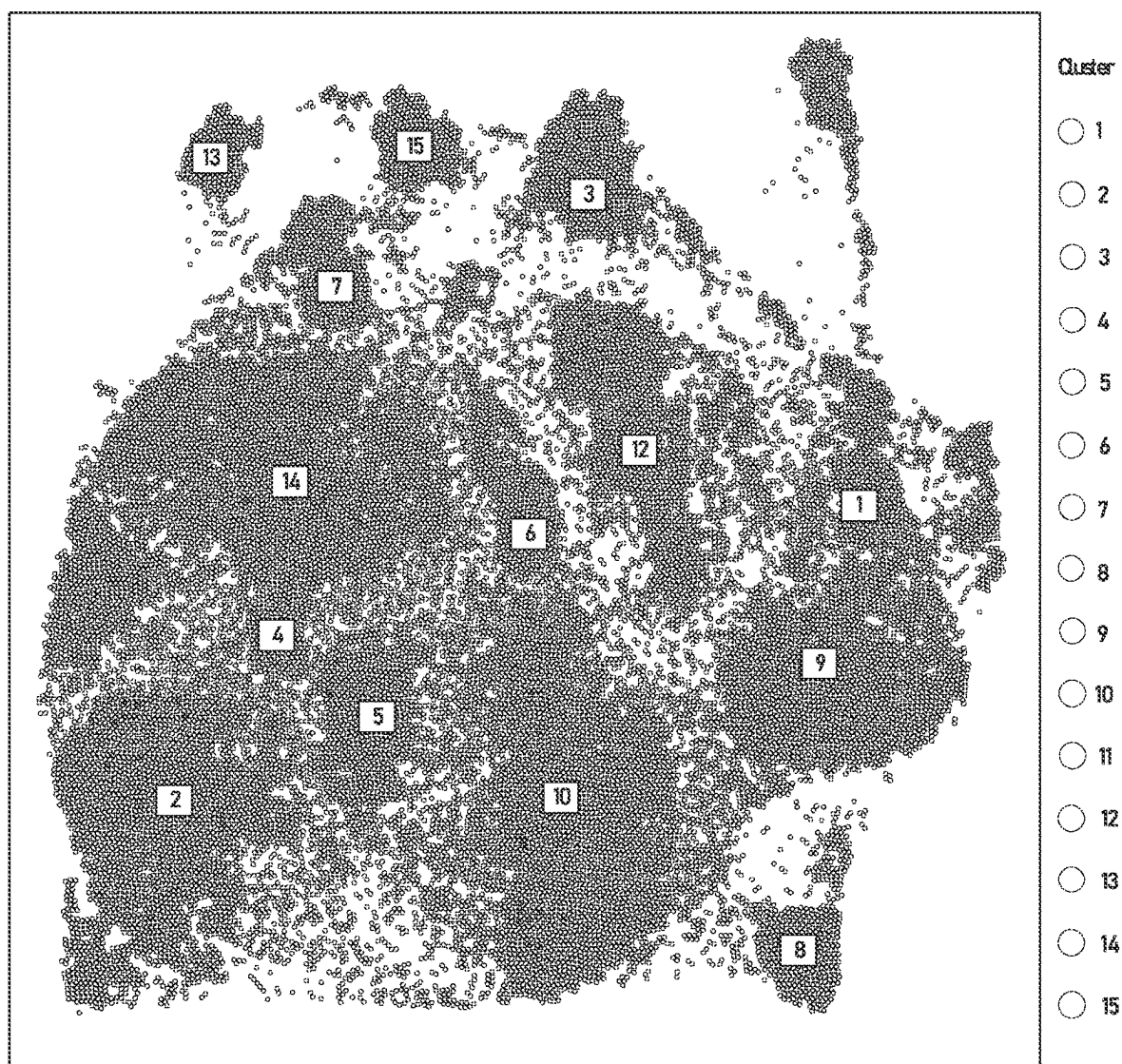


FIG.4G

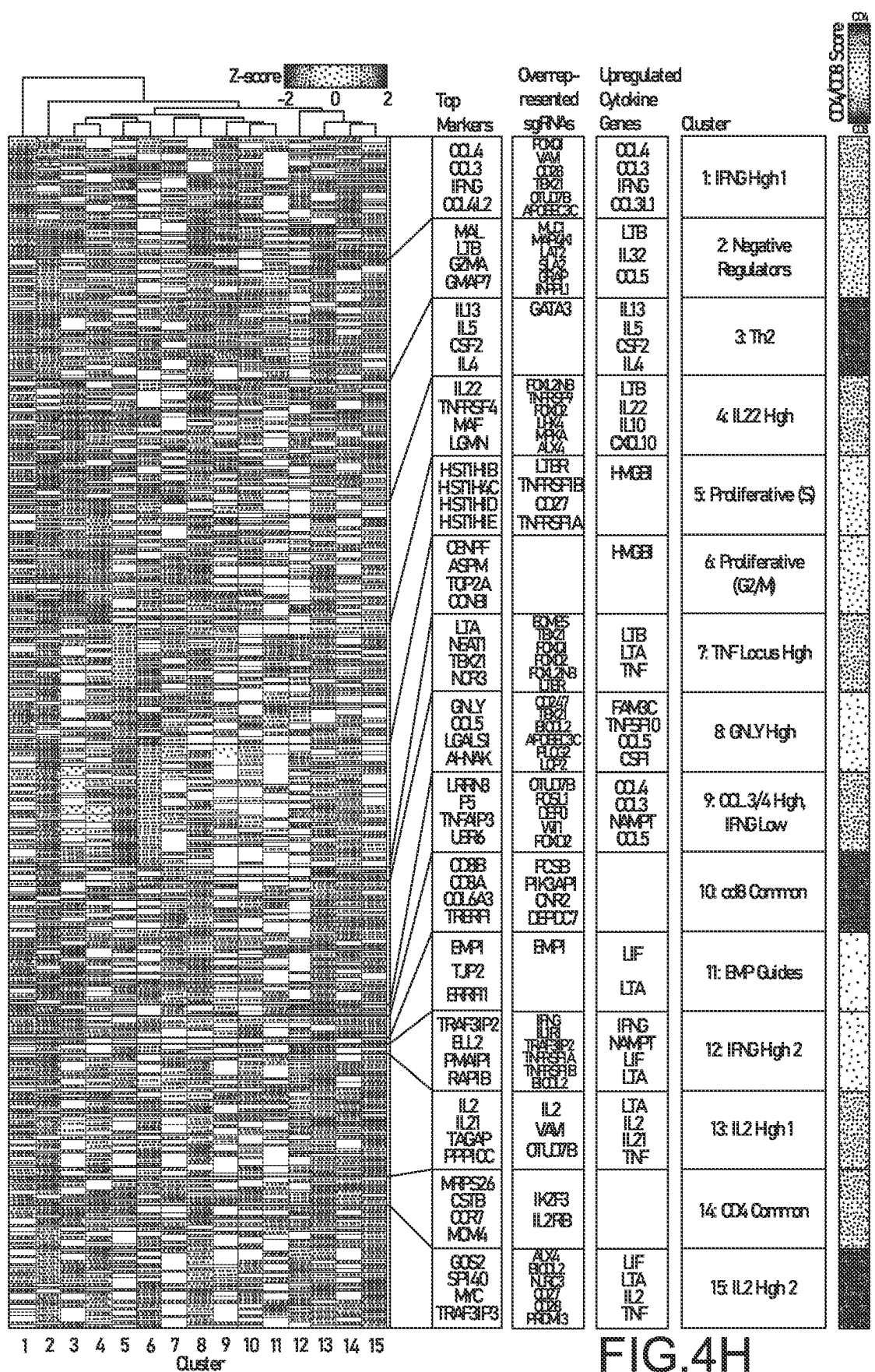


FIG.4H

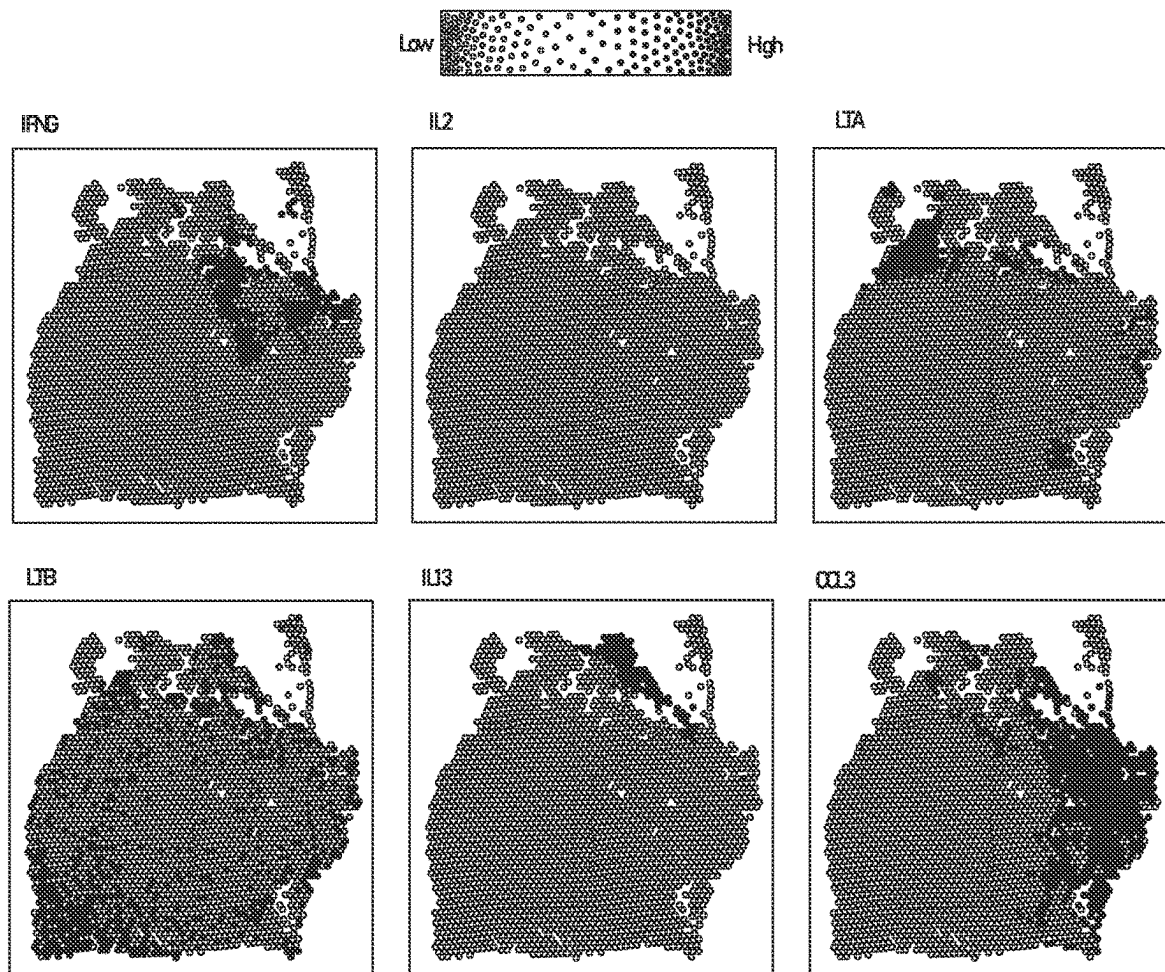


FIG. 4I

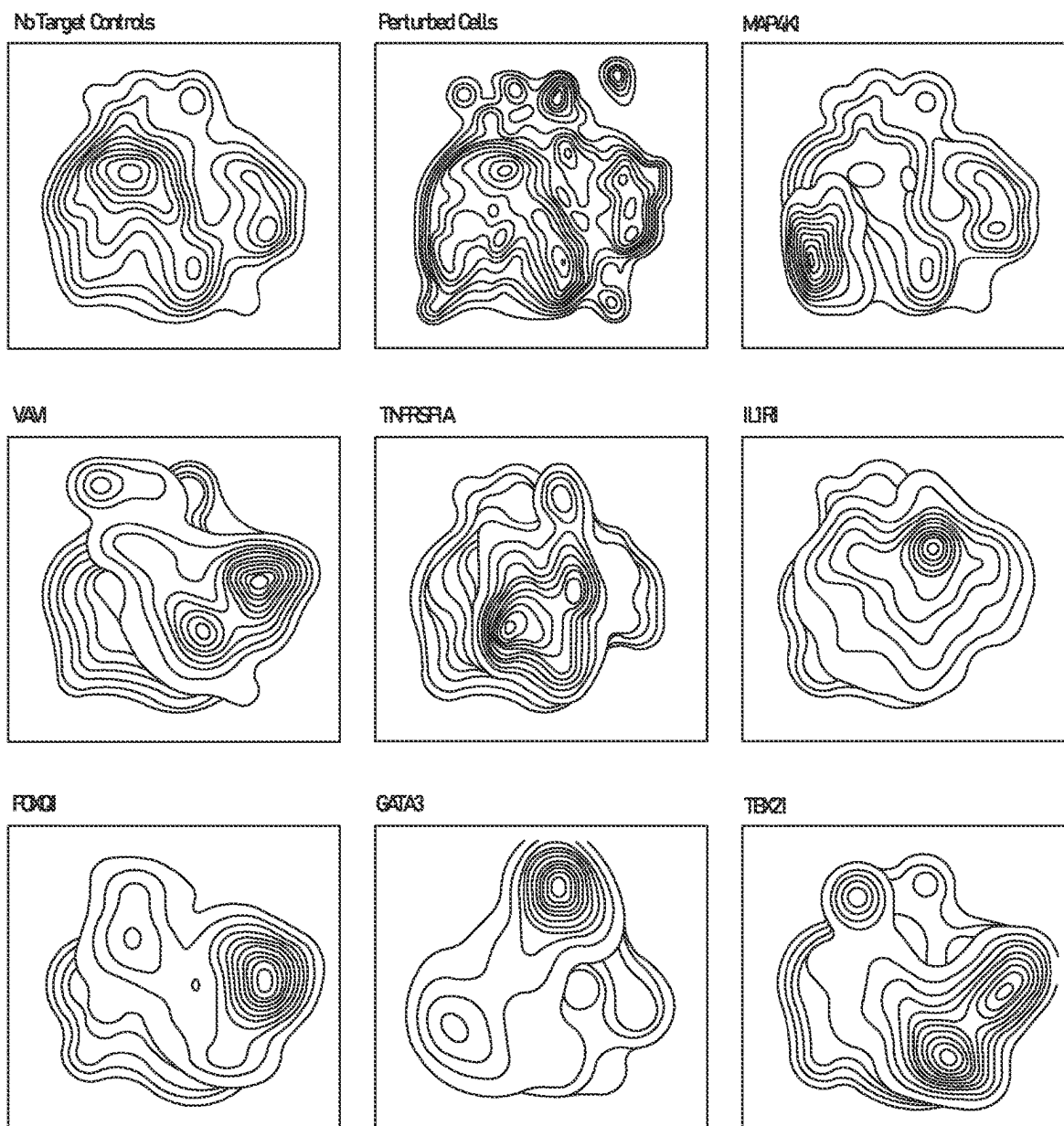


FIG.4J

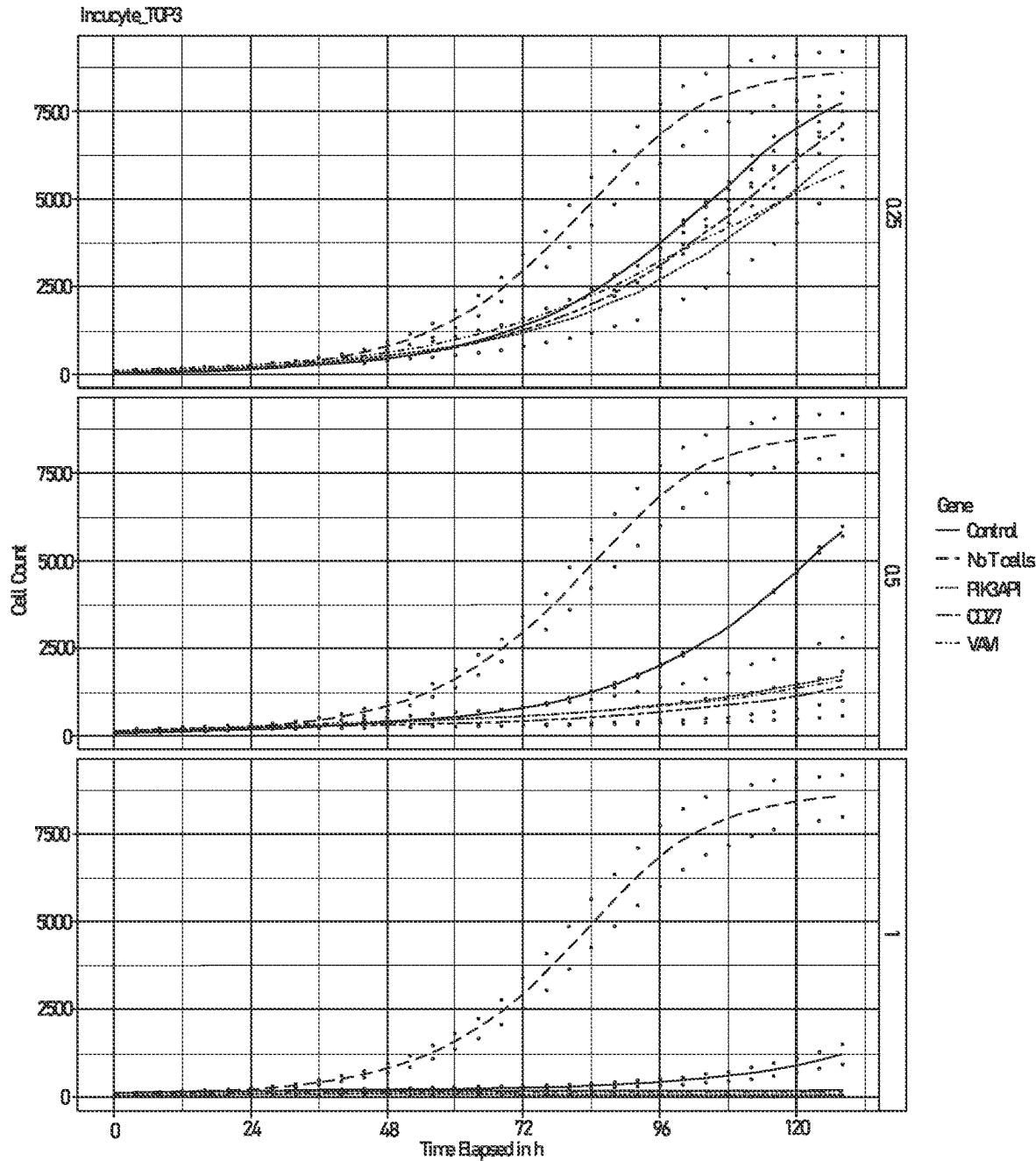


FIG.5

GENE ACTIVATION TARGETS FOR ENHANCED HUMAN T CELL FUNCTION

PRIORITY

[0001] This application claims the benefit of priority of U.S. Provisional Patent Application No. 63/138,841, filed on Jan. 19, 2021, the benefit of priority of which is claimed hereby, and which is incorporated by reference herein in its entirety.

GOVERNMENT SUPPORT

[0002] This invention was made with government support under grant no. DK111914 awarded by The National Institutes of Health, The government has certain rights in the invention,

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

[0003] This application contains a Sequence Listing which has been submitted electronically in ST25 format and is hereby incorporated by reference in its entirety. Said ST25 file, created on Apr. 5, 2024, is named "3730183WO1.txt" and is 314,677 bytes in size.

BACKGROUND

[0004] Examples of cellular therapeutic agents that can be useful as anticancer therapeutics include CD8⁺ T cells, CD4⁺ T cells, NK cells, macrophages, dendritic cells, and chimeric antigen receptor (CAR) T cells. Use of patient-derived immune cells can also be an effective cancer treatment that has little or no side effects. NK cells have cell-killing efficacy and have several side effects due to not having antigen specificity. Dendritic cells are therapeutic agents belonging to the vaccine concept in that they have no function of directly killing cells and are capable of delivering antigen specificity to T cells in the patient's body so that cancer cell specificity is imparted to T cells with high efficiency. In addition, CD4⁺ T cells play a role in promoting productive, antigen-dependent immune responses, and CD8⁺ T cells are known to have antigen specificity and cell-killing function.

[0005] However, most cell therapeutic agents, which have been used or developed to date, have major clinical limitations. For example, cancer cells, on their own, secrete substances that suppress immune responses in the human body, or do not present antigens necessary for production of antibodies against such cancer cells, thereby preventing an appropriate immune response from occurring.

SUMMARY

[0006] Regulators of T cell function are described herein as well as methods of using such regulators. Genome-wide CRISPR activation (CRISPRa) and CRISPR interference (CRISPRi) screens were performed in primary human T cells to identify genetic regulators of therapeutically relevant T cell phenotypes. These screens identified 1074 genes exhibiting significant responses to those phenotypes. The screen identified known genes involved in T cell function, showing that the screens reliably identified genes that actually do affect T cell function. However, the screens also identified novel genes involved in T cell function.

[0007] Methods are described herein that involve ex vivo modification of any of the regulator genes listed in Tables 1-7 or FIGS. 1-4 within at least one lymphoid or myeloid cell, or a combination thereof to generate at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells. For example, the modification can be one or more deletions, substitutions or insertions into one or more endogenous genomic sites of any of the genes listed in Tables 1-7 or FIGS. 1-4. The modification can be reduction of expression or translation of any of the genes listed in Tables 1-7 or FIGS. 1-4. The reduction of expression or translation can be by an inhibitory nucleic acid (e.g., RNAi, shRNA, siRNA). The modification can be increased expression of any of the genes listed in Tables 1-7 or FIGS. 1-4. For example, the increased expression can be by modification of one or more promoters of any of the genes listed in Tables 1-7 or FIGS. 1-4. The modification can be one or more CRISPR-mediated modifications or activations of any of the genes listed in Tables 1-7 or FIGS. 1-4. The modification can involve transformation of at least one lymphoid or myeloid cell, or a combination thereof, with one or more expression cassettes comprising a promoter operably linked to a nucleic acid segment comprising a coding region of any of the genes listed in Tables 1-7 or FIGS. 1-4.

[0008] The methods can also include administering at least one of the modified lymphoid cells, at least one of the modified myeloid cells, or a mixture of modified lymphoid and modified myeloid cells to a subject.

[0009] In some cases, the method can include incubating the at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to form a population of modified cells. Such a population of modified cells can be administered to a subject. In some cases, the subject can have a disease or condition. For example, the disease or condition is an immune condition or cancer.

[0010] Also described are methods that involve contacting at least one test agent with test cells to provide a test assay mixture, and measuring:

[0011] cellular proliferation of the test cells, cytokine release by the test cells, or a combination thereof;

[0012] activation of the test cells;

[0013] expression or activity of any of the regulators listed in Tables 1-7 or FIG. 1-4 in the cells; or

[0014] a combination thereof.

[0015] The methods can also include comparing the measured results to control results. The control results can be results of the test cells measured without any of the test agents.

[0016] For example, the test cells can include lymphoid and/or myeloid cells. Examples of the test cells can include cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, chimeric antigen receptor (CAR) cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof.

[0017] The results so measured can be compared to results of a control cell mixture that includes the T cells and test cells measured without any of the test agents.

DRAWINGS

[0018] FIGS. 1A-E. Genome-wide CRISPRa screens for cytokine production in stimulated primary human T cells. (A) Schematic of CRISPRa screens. (B) sgRNA $\log_2$ -fold changes for genes of interest in IL-2 (left) and IFN- γ (right) screens. Bars represent the mean $\log_2$ -fold change for each sgRNA across two human blood donors. Density plots above represent the distribution of all sgRNAs. (C and D) Scatter plots of median sgRNA $\log_2$ -fold change (high/low sorting bins) for each gene, comparing screens in two donors, for IL-2 (C) and IFN- γ screens (D). (E) Comparison of gene $\log_2$ -fold change (median sgRNA, mean of two donors) in IL-2 and IFN- γ screens.

[0019] FIGS. 2A-1H. Integrated CRLISPRi and CRISPRi screens map the genetic circuits underlying T cell cytokine response in high resolution. (A and B) Median sgRNA $\log_2$ -fold change (high/low sorting bins) for each gene, comparing CRISPRi screens in two donors, for IL-2 (A) and IFN- γ screens (B). (C) Distributions of gene mRNA expression for CRISPRa and CRISPRi cytokine screen hits in resting CD4-T cells (this study). (D) Comparison IL-2 CRISPRi and CRISPRa screens with genes belonging to the T cell receptor signaling pathway (KEGG pathways) indicated in colors other than gray. (E) Comparison IFN- γ CRISPRi and CRISPRa screens with manually selected NF- κ B pathway regulators labeled. All other genes are shown in gray. (F) Map of NF- κ B pathway regulators labeled in (D). (G) Map of screen hits with previous evidence of defined function in T cell stimulation and costimulation signal transduction pathways. Genes shown are significant hits in at least one screen and were selected based on review of literature and pathway databases (e.g., KEGG and Reactome). Tiles represent proteins encoded by indicated genes, with the caveat that due to space constraints, subcellular localization is inaccurate, as many of the components shown in the cytoplasm occur at the plasma membrane. Tiles are colored according to $\log_2$ -fold change Z-score as shown in the sub-panel, with examples of different hits. Large arrows at the top represent stimulation/costimulation sources. (H) Select screen hits with less well-described functions in T cells in the same format as (G). For (H), only significant hits from the top positive and negative ranked genes by $\log_2$ -fold change for each screen were candidates for inclusion.

[0020] FIGS. 3A-1. Characterization of CRISPRa screen hits by arrayed profiling. (A) Schematic of arrayed experiments. (B) Comparison of IL-2 (in CD4 T cells) and IFN- γ (in CD8⁺ T cells) CRISPRa screens, with genes targeted by the arrayed sgRNA panel indicated, as well as their screen hit categorization. Paralogous of arrayed panel genes that were also highly ranked hits are additionally indicated. (C) Representative intracellular cytokine staining flow cytometry for indicated cytokines in control (NO-TARGET_1 sgRNA) or VAV1 (VAV1_1 sgRNA) CRISPRa T cells after 10 hours of stimulation. (D) Intracellular cytokine staining of full arrayed sgRNA panel, showing percent of cells gated positive for indicated cytokines in CD4⁺ or CD8⁺ T cells. Points represent the mean value of four donors, with and without stimulation. Dashed vertical lines represent the mean no-target control sgRNA control value with stimulation. * $q < 0.05$, ** $q < 0.01$, Mann-Whitney U test followed by q -value multiple comparison correction. Medium-stimulation dose is shown for IL-2 and IFN- γ and low-dose stimulation is shown for TNF- α . (E) Scatter plot comparison of $\log_2$ -fold

changes in percent cytokine positive cells for arrayed panel sgRNAs versus the mean of no-target control sgRNAs in stimulated CD4⁺ and CD8⁺ cells, using the same data from (D). (F) Secreted cytokine staining arrayed panel grouped by indicated gene categories, with sgRNAs targeting IL2 and IFNG genes removed. Points represent a single gene and donor measurement. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, Mann-Whitney U test. (G) Principal component analysis of secreted cytokine measurements resulting from indicated CRISPRa sgRNAs. (H) Heatmap of selected secreted cytokine measurements grouped by indicated biological category. Values represent the median of four donors, followed by Z-score scaling for each cytokine.

[0021] FIGS. 4A-J. CRTSPRa perturb-seq captures diverse T cell states driven by genome-wide cytokine screen hits. (A) Schematic of CRISPRa Perturb-seq experiment. (B) Categorical breakdown of genes targeted by sgRNA library, with the library comprising hits from our primary genome wide CRISPRa cytokine screens as indicated. Genes with a summed $\log_2$ -fold change < 0 across both screens (diagonal line) are categorized as negative regulators. (C) UMAP projection of post-quality control filtered restimulated T cells, colored by blood donor. (D) Distribution of CD4 and CD8 T cells across restimulated T cell UMAP projection. Each bin is colored by the average $\log_2$ (CD4/CD8) transcript levels of cells in that bin. (E) Restimulated T cell UMAP colored by average cell activation score in each bin. (F) Boxplots of restimulated T cells' activation scores grouped by sgRNA target genes. Dashed line represents the median activation score of no-target control cells. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ Mann-Whitney U test with Bonferroni correction. (G) Restimulated T cell UMAP with cells colored by cluster. (H) Heatmap of differentially expressed marker genes in each cluster. The top 50 statistically significant (FDR < 0.05) differentially upregulated genes for each cluster are shown, with genes that are upregulated in multiple clusters being given priority to the cluster with the higher $\log_2$ -fold change for the given gene. The top marker genes by $\log_2$ -fold change in each clusters' section are listed to the right. Top overrepresented sgRNAs in each cluster by odds ratio are listed to the next right. Top differentially upregulated cytokine genes in each cluster are listed to the next right. Mean cell $\log_2$ (CD4/CD8) cell transcript values in each cluster are shown on the far right. (I) Restimulated T cell UMAP with the expression of indicated genes shown. (J) Contour density plots of restimulated cells assigned to indicated sgRNA targets in UMAP space. The no-target control contour is shown in grayscale underneath. "Perturbed Cells" represents all cells assigned a single sgRNA other than no-target control sgRNAs.

[0022] FIG. 5 provides In vitro data using the identified hits for T cell cancer therapies.

DETAILED DESCRIPTION

[0023] Methods and compositions are described herein for modulating T cell responses. The T cells can be modulated in vivo or ex vivo. T cells modulated ex vivo can be administered to a subject who may benefit from such administration. Methods are also described herein for evaluating test agents and identifying agents that are useful for modulating T cell functions.

[0024] Regulation of cytokine production in stimulated T cells can be disrupted in autoimmunity, immunodeficiencies, and cancer. Systematic discovery of stimulation-dependent

cytokine regulators requires both loss-of-function and gain-of-function studies, which have been challenging in primary human cells. We now report genome wide CRISPR activation (CRISPRa) and interference (CRISPRi) screens in primary human T cells to identify gene networks controlling interleukin 2 and interferon gamma production. Arrayed CRISPRa confirmed key hits and enabled multiplexed secretome characterization, revealing reshaped cytokine responses. Coupling CRISPRa screening with single-cell RNA-seq enabled deep molecular characterization of screen hits, revealing how perturbations tuned T cell activation and promoted cell states characterized by distinct cytokine expression profiles. Together, these screens reveal genes that reprogram immune cell functions.

Modulating T Cell Responses

[0025] Lists of negative and positive regulators of T cells are provided in Tables 1-7 or FIGS. 1-4. Such regulators can modulate gamma interferon (IFN- γ) production, interleukin 2 (IL-2) production, cellular proliferation of T cells, or a combination thereof. Any of the regulators of T cells can be used in the methods and compositions described herein. Agents that modulate the listed regulators can also be used in the methods and compositions described herein. For example, to positively regulate T cells one or more expression cassettes encoding one or more positive T cell regulators, one or more agents that increase the expression or activity of such positive regulator, or agents that inhibit negative regulators of T cells can be used. To negatively regulate T cells, for example, antibodies, one or more expression cassettes encoding one or more negative T cell regulators, one or more agents that increase the expression or activity of such negative T cell regulator, or agents that inhibit positive regulators of T cells can be used. Agents that can modulate the T cell regulators can include expression vectors, inhibitory nucleic acids, antibodies, small molecules, guide RNAs, nucleases (e.g., one or more cas nucleases), nuclease-dead cas variants (e.g., dCas9-VP64, dCas9-KRAB), or a combination thereof.

[0026] For example, T cells and other types of cells can be modified *ex vivo* to increase or decrease any of the T cell regulators listed in Tables 1-7 or FIGS. 1-4, and the modified cells can be administered to a subject that may benefit from such administration. In another example, the expression or activity of any of the T cell regulators listed in Tables 1-7 or FIGS. 1-4 can be modulated by *in vivo* administration of expression vectors, virus-like particles (VLP), CRISPR-related ribonucleoprotein (RNP) complexes, and combinations thereof that include or target any of the regulators listed in Tables 1-7 or FIGS. 1-4. The regulator nucleic acids, regulator protein, regulator guide RNAs and CRISPR nucleases can be introduced via one or more vehicles such as by one or more expression vectors (e.g., viral vectors), virus like particles, ribonucleoproteins (RNPs), nanoparticles, liposomes, or a combination thereof. The vehicles can include components or agents that can target particular cell types (e.g., antibodies that recognize cell-surface markers), facilitate cell penetration, reduce degradation, or a combination thereof.

[0027] In addition, new agents can be identified by screening methods described herein that include, for example, evaluating assay mixtures containing one or more test agents and a population of T cells after incubation of the assay mixtures for a time and under conditions sufficient for determining whether the test agent can modulate the expression or activities of any of the regulators described herein.

In some cases, the assay mixtures can include T cells and other types of cells, for example, other immune cells such as those that can interact with T cells. Useful test agents identified by such methods can, for example, increase or decrease the expression or activities of any of the regulators listed in any of Tables 1-7 or FIGS. 1-4.

[0028] Hence, any of the regulators of T cells, as well as agents that can modulate those regulators (i.e., modulators), can be used in the methods and compositions described herein.

[0029] The T cell regulators were identified by detecting altered IL-2 cytokine production, IFN- γ production, and cell proliferation of T cell receptor (TCR) stimulated primary T cells isolated from two different donors that were subjected to CRISPR-mediated genetic modification. Both positive and negative regulators of T cells were identified.

[0030] The agents that can modulate T cells or the T cell regulators described herein can be expression systems encoding a regulator or modulating agent, antibodies, small molecules, inhibitory nucleic acids, peptides, polypeptides, guide RNAs, cas nucleases (e.g., a cas9 nuclease), nuclease-dead cas variants (e.g., dCas9-VP64, dCas9-KRAB), and combinations thereof. Examples of such agents are described hereinbelow.

[0031] The regulators and/or the agents that modulate the regulators can be evaluated by various assay procedures. Such assay procedures can also be used to identify new T cell regulators. In some cases, the assay procedures can be used to evaluate the utility of a type (positive or negative effect), quantity, or extent of a regulator or modulating agent activity on T cell activity or T cell numbers.

[0032] For example, the methods for evaluating Applicants' regulators/agents or new regulators/agents can involve contacting one or more T cells (or a T cell population) with a test agent to provide a test assay mixture, and evaluating the test assay mixture for at least one of:

[0033] Detecting and/or quantifying cytokine (e.g., interferon- γ , (IFN- γ , interleukin-2(IL-2)) production;

[0034] Quantifying the numbers of T cells within the test assay mixture;

[0035] Detecting proliferation via quantification of a dye that dilutes with cell divisions;

[0036] Detecting whether T cells in the test assay mixture express one or more of the positive or negative regulators described herein;

[0037] Quantifying the number of cells that express one or more of the positive or negative regulators expressed by a population of T cells; or

[0038] A combination thereof.

[0039] The T cells or T cell populations that are contacted with the test agent/test regulator can also include a variety of lymphoid and/or myeloid immune cells. For example, test agents can be introduced into an assay mixture that contains cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, chimeric antigen receptor (CAR) cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof.

[0040] Test agents that exhibit *in vitro* activity for modulating the T cells or for modulating the amount or activity of any of the regulators described herein can be evaluated in animal disease models. Such animal disease models can include cancer disease animal models, immune system disease models, or combinations thereof.

Positive T Cell Regulators

[0041] The following genes are positive regulators of T cells as detected by interferon- γ production (see Table 1): APOBEC3C, APOBEC3D, APOL2, ASB12, BACE2, BCL9, BICDL2, C1orf52, C1orf94, CD2, CD247, CD28, CNGB1, CTSK, DEAF1, DEF6, DEPDC7, DKK2, EMIP1, EOMES, EP300, FLT3, FOSL1, FOXQ1, GINS3, GLMN, GNA11, HELZ2, -RASLS5, IFNG, IL11R, IL9R, KLHDC3, KLRC4, LAT, LCP2, LDB2, LTBR, MVB12A, NBPf6, NIT1, NLRC3, ORC1, OTUD7A, OTUD7B, PIK3AP1, PLCG2, PRDMI, PRKD2, PROCA1, RELA, RNF217, SAFB2, SLC16A1, SLC5A10, SLC7A3, SPPL2B, TAGAP, TBX21, TMEM150B, TMILGD2, TNFRSF12A, TNFRSF14, TNFRSF1A, TNFRSF1B, TNFRSF8, TNFRSF9, TOR1A, TPGS2, TRADD, TRAF3TP2, TRIM21,

VAV1, WT1, ZNF630, and ZNF717. Example 2 provides additional positive regulators T cells that were detected by interferon- γ production.

[0042] Sequences and other information relating to these genes, and their encoded proteins, is available, for example from the NCBI and UniPROT databases, which are incorporated by reference.

[0043] A few examples of protein sequences encoded by some of the genes detected as positive regulators of T cells by interferon- γ production are provided. For example, an amino acid sequence for the protein encoded by the human BICDL2 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. A1A5D9, shown below as SEQ ID NO:1.

```

      10      20      30      40      50
MSSPDGSPFP SGPLSGGASP SGDEGFFPFV LERRDSFLGG GPGPEEPEDL

      60      70      80      90     100
ALQLQQKEKD LLLAELGKM LLERNEELRR QLETLSAQHL EREERLQQEN

      110     120     130     140     150
HELRRGLAAR GAWEARAVE LEGDVEALRA QLGEQRSEQQ DSGRERARAL

      160     170     180     190     200
SELSEQNLRL SQQLAQASQT EQELQRELD LRGQCQAQAL AGAELRTRLE

      210     220     230     240     250
SLQGENQMLQ SRRQDLEAQI RGLREEVEKG EGRLQTTHEE LLLLRERRE

      260     270     280     290     300
HSELERARS EAGEALSALR RLQRRVSELE EESRLQDADV SAASLQSELA

      310     320     330     340     350
HSLDDGDQGG GADAPGDTPT TRSPKTRKAS SPQSPPEEI LEPPKKRTSL

      360     370     380     390     400
SPAETILEEKE VEVAKLQDEI SLQQABEQSL REELQRQKEL RAQEDPGEAL

      410     420     430     440     450
HSALSDRDEA VNKALELSLQ LNRVSLERDS LSRELLRAIR QKVALTQELE

      460     470     480     490     500
AWQDDMQVVI GQQLRSQRQK ELSASASSST PRRAAPRFSL RLGP GPAGGF

LSNLFRRRT

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A cDNA and a chromosomal sequence encoding the BICDL2 protein is available from the NCBI database as accession no. AL833749 and AC108134, respectively.

[0044] An amino acid sequence for the protein encoded by the human C1orf94 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q6P1IW5, shown below as SEQ ID NO:2.

```

      10      20      30      40      50
MRGGGGCVLA LGGQRGFQKE RRRMASGNGL PSSSALVAKG PCALGPFPRY

      60      70      80      90     100
IWIHQDTPQD SLDKTCHEIW KRVQGLPEAS QPWTSMQELS VPVVGTLRGN

      110     120     130     140     150
ELSFQEEALE LSSGKDEISL LVEQEFLSLT KEHSILVEES SGELEVPGSS

      160     170     180     190     200
PEGTRELAPC ILAPPLVAGS NERPRASIIV GDKLLKQKVA MPVISSRQDC

      210     220     230     240     250
DSATSTVTDI LCAAEVKSSK GTEDRGRILG DSNLQVSKLL SQFPLKSTET

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      260      270      280      290      300
SKVPDNKNVL DKTRVTKDFL QDNLFSGGPGP KEPTGLSPFL LLPPRPPPAR

      310      320      330      340      350
PDKLPPLPAQ KRQLPVFAKI CSKPKADPAV ERHHLMEWSP GTKEPKKGQG

      360      370      380      390      400
SLFLSQWPQS QKDACGEEGC CDAVGTSALT LPPKKPTCPA EKNLLYEFLG

      410      420      430      440      450
ATKNPSGQPR LRNKVEVDGP ELKFNAFVTV ADKNNPKYTG NVFTPHEPTA

      460      470      480      490      500
MTSATLNQPL WLNLYPPPPP VFTNHSTFLQ YQGLYPQQAA RMPYQQALHP

      510      520      530      540      550
QLGCYSQQVM PYNPQQMGQQ IFRSSYTPLL SYIPFVQPNY PYPQRTPPKM

      560      570      580      590
SANPRDPPLM AGDGPQYLFP QGYGFGSTSG GPLMHSPYES SSGNGINF

```

A cDNA and a chromosomal sequence encoding the Q6P1W5 protein is available from the NCBI database as accession no. AK123355 and AC115286, respectively.

[0045] An amino acid sequence for the protein encoded by the human CNGB1 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q14028, shown below as SEQ ID NO: 3.

```

      10      20      30      40      50
MLGWVQRVLP QPPGTPRKTK MQEEEEVEPE PEMEAEEVEPE PNPEEAETES

      60      70      80      90     100
ESMPPEESEK EEEVAVADPS PQETKEAALT STISLRAQGA EISEMNSPSR

      110     120     130     140     150
RVLTWLMKGV EKVIPQPVHS ITEDPAQILG HGSTGDTGCT DEPNEALEAQ

      160     170     180     190     200
DTRPGLRLLL WLEQNLERVL PQPPKSSEVW RDEPAVATGA ASDPAPPGRP

      210     220     230     240     250
QEMGPKLQAR ETPSLPTPIP LQPKEEPKEA PAPEPQPGSQ AQTSSLPPTR

      260     270     280     290     300
DPARLVAVWL HRLEMALPQP VLHGKIGEQE PDSPGICDVQ TISILPGGQV

      310     320     330     340     350
EPDLVLEEVE PPWEDAHQDV STSPQGTEVV PAYEEENKAV EKMPRELSRI

      360     370     380     390     400
EEEEKEEEEE EEEEEEEEE E VTEVLLDSC VVSQVGVGQS EEDGTRPQST

      410     420     430     440     450
SDQKLWEEVG EEAKEAEKEE AKEEAEVVAE EEAKEPQDW AETKEEPEAE

      460     470     480     490     500
AEAASSGVPA TKQHPQVQE DTDADSCPLM AEENPPSTVL PPPSPAKSDT

      510     520     530     540     550
LIVPSSASGT HRKKLPSEDD EAEELKALSP AESPVVAWSD PTTPKDTDGQ

      560     570     580     590
DRAASTASTN SAIINDRLQE LVKLEKERT EKVKEKLIDPD VTSDEESPKP

      610     620     630     640     650
SPAKKAPEPA PDTKPAAEAP VEEHYCDML CCKFKHRPWK KYQFPQSIDP

      660     670     680     690     700
LTNLMYVLWL FFVVMAWNWN CWLIPVRWAF PYQTPDNIHH WLLMDYLCDL

      710     720     730     740     750
IYFLDITVPQ TRLQFVRGGD IITDKKDMRN NYLKSRRFKM DLLSLPLDF

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760	770	780	790	800
LYLKVGVNPL	LRLPRCLKYM	AFPEFNSRLE	SILSKAYVYR	VIRTAYLLY
810	820	830	840	850
SLHLNSCLYY	WASAYQGLGS	THWVYDGVGN	SYIRCYYFAV	KTLLITIGGLP
860	870	880	890	900
DPKTLFEIVF	QLLNYFTGVF	AFSVMIGQMR	DVVGAAATAGQ	TYRSCMDST
910	920	930	940	950
VKYMNFYKIP	KSVQNRVKTW	YEYTWHSQGM	LDESELMVQL	PDKMRLDLAI
960	970	980	990	1000
DVNYNIVSKV	ALFQGCQRQM	IFDMLKRLRS	VVYLPNDYVC	KKGEIGREMY
1010	1020	1030	1040	1050
IIQAGQVQVL	GGPDGKSVLV	TLKAGSVFGE	ISLLAVGGGN	RRTANVVAHG
1060	1070	1080	1090	1100
FTNLFILDKK	DLNEILVHYP	ESQKLLRKKA	RRMLRSNNKP	KEEKSVLILP
1110	1120	1130	1140	1150
PRAGTPKLFN	AALAMTGKMG	GKGAKGGKLA	HLRARLKELA	ALEAAKQQE
1160	1170	1180	1190	1200
LVEQAKSSQD	VKGEEGSAAP	DQHTHPKEAA	TDPPAPRTPP	EPPGSPSPSSP
1210	1220	1230	1240	1250
PPASLGRPEG	EEEGPAEPPE	HSVRICMSPG	PEPGEQILSV	KMPEREKEKA

E

[0046] A cDNA and a chromosomal sequence encoding the Q14028 protein is available from the NCBI database as accession no. UI8945 and L 15296, respectively,

[0047] An amino acid sequence for the protein encoded by the human DEPDC7 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q96QD5. shown below as SEQ ID NO. 4

10	20	30	40	50
MATVQEKA	LNLSALHSPA	HRPPGFSSVAQ	KPFGATYVWS	SIINTLQTQV
60	70	80	90	100
EVKKRRHRLK	RHNDCFVGSE	AVDVIFSHLI	QNKYFGDVIDI	PRAKVVRVCQ
110	120	130	140	150
ALMDYKVFEA	VPTKVFGKDK	KPTFEDSSCS	LYRFTTIPNQ	DSQLGKENKL
160	170	180	190	200
YSPARYADAL	FKSSDIRSAS	LEDLWENLSL	KPANSPHVNI	SATLSPQVIN
210	220	230	240	250
EVWQEETIGR	LLQLVDLPLL	DSLLKQQEAV	PKIPQPKRQS	TMVNSSNYLD
260	270	280	290	300
RGILKAYSDS	QEDEWLSAAI	DCLEYLPDQM	VVEISRSFPE	QPDRTDLVKE
310	320	330	340	350
LLFDAIGRYY	SSREPLLNHL	SDVHNGIAEL	LVNGKTEIAL	EATQLLLKL
360	370	380	390	400
DFQNREEFRR	LLYEMAVAA	PSEFKLQKES	DNRMVVKRIF	SKAIVDNKNL
410	420	430	440	450
SKGKTDLVL	FLMDHQKDV	KIPGTLHKIV	SVKLMAIQNG	RDPNRDAGYI
460	470	480	490	500
YCQRIDQRDY	SNNTEKTKD	ELLNLLKTL	EDSKLSAKEK	KKLLGQFYKC
510				
HPDIFIEHFG	D			

A cDNA and a chromosomal sequence encoding the Q96QD5 protein is available from the NCBI database as accession no. AJ245600 and AC107939, respectively.

[0048] An amino acid sequence for the protein encoded by the human HRASLS5 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q96KN8, shown below as SEQ ID NO:5.

```

      10      20      30      40      50
MGLSPGAEGE YALRLPRIPP PLPKPASRTA STGPKDQPPA LRRSAVPHSG

      60      70      80      90     100
LNSISPLELE ESVGFAALVQ LPAKQPPPGT LEQGRSIQQG EKAVVSLETT

     110     120     130     140     150
PSQKADWSSI PKPENEGKLI KQAAEGKPRP RPGDLIEIFR IGYEHWAIYV

     160     170     180     190     200
EDDCVVHLAP PSEEFVGSII TSIFSNRAVV KYSRLVDVLH GCSWKVNNKL

     210     220     230     240     250
DGTYLPLPVD KIIQRTKKMV NKIVQYSLIE GNCEHFNGL RYGVPRSQQV

     260     270
EHALMEGAKA AGAVISAVVD SIKPKPITA

```

A cDNA and a chromosomal sequence encoding the HRASLS5 protein is available from the NCBI database as accession no. AB298804 and AP000484, respectively.

[0049] An amino acid sequence for the protein encoded by the human KLHDC3 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9BQ90, shown below as SEQ ID NO: 6.

```

      10      20      30      40      50
MLRWTVHLEG GPRRVNHAHV AVGHRVYSFG GYCSGEDYET LRQIDVHIFN

      60      70      80      90     100
AVSLRWTKLP PVKSAIRQA PVVPYMYRGH STVLIDDTVL LWGGRNDTEG

     110     120     130     140     150
ACNVLYAPDV NTHKWFTPRV SGTVPGARDG HSACVLGKIM YIFGGYEQA

     160     170     180     190     200
DCFSNDIHLK DTSTMTWTLI CTKGSPARWR DFHSATMLGS HMYVFGGRAD

     210     220     230     240     250
RFGPFHSNNE IYCNIRIVFD TRTEAWLDCP PTPVLPEGRR SHSAFGYNGE

     260     270     280     290     300
LYIEGGYNAR LNRHEHDLWK FNPVSFTWKK IEPKKGKPCP RRRQCCCIIV

     310     320     330     340     350
DKIVLFGGTS PSPEEGLGDE FDLIDHSDLH ILDFSPSLKT LCKLAVIQYN

     360     370     380
LDQSCPLPHDI RWELNMTIN SNISRPIVSS HG

```

A cDNA and a chromosomal sequence encoding the KLHDC3 protein is available from the NCBI database as accession no. AB055925 and AL136304, respectively, [0050] An amino acid sequence for the protein encoded by the human NBPF6 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q5VWK0, shown below as SEQ ID NO: 7.

```

      10      20      30      40      50
MVVSADPLSS ERAEMNILEI NQELRSQLA E SNQQFRDLKE KFLITQATAY

      60      70      80      90     100
SLANQLKKYK CEEYKDIIDS VLRDELQSM E KLAEKLRQAE ELRQYKALVH

     110     120     130     140     150
SQAKELTQLR EKLREGRDAS RWLNKHIKTL LTPDDPKSQ GQDLREQLAE

     160     170     180     190     200
GHRLAEHLVH KLSPENDEDE DEDEDDKDEE VEKVQESPA P REVQKTEEKE

     210     220     230     240     250
VPQDSLEECA VTCSNSHNPS NSNQPHRSTK ITFKEHEVDS ALVVESEHPH

     260     270     280     290     300
DEEEEALNIP PENQNDHEEE EGKAPVPPRH HDKSNYSYRHR EVSFLALDEQ

     310     320     330     340     350
KVCSAQDVAR DYSNPKWDET SLGFLEKQSD LEEVKGQETV APRLSRGPLR

     360     370     380     390     400
VDKHEIPQES LDGCCLTPSI LPDLIPSYHP YWSTLYSFED KQVSLALVDK

     410     420     430     440     450
IKKDQEEIED QSPPCPRLSQ ELPEVKEQEV PEDSVNEVYL TPSVHHDVSD

     460     470     480     490     500
CHQPYSSSTLS SLEDQLACSA LDVASPTEAA CPQGTWSGDL SHHRSEVQIS

     510     520     530     540     550
QAQLEPSTLV PSCLRLQLDQ GFHCGNGLAQ RGLSSTTCSF SANADSGNQW

     560     570     580     590     600
PFQELVLEPS LGMKNPPQLE DDALEGSASN TQGRQVTGRI RASLVLILKT

     610     620     630
IRRLPFPSKW RLAFRFAGPH AESAEIPNTA ERMQRMIG

```

A cDNA and a chromosomal sequence encoding the Q5VWK0 protein is available from the NCBI database as accession no. BC125161 and AL390038, respectively.

[0051] An amino acid sequence for the protein encoded by the human OTUTD7B gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q5VWK0, shown below as SEQ ID NO:8.

```

      10      20      30      40
MVVSADPLSS ERAEMNILEI NQELRSQLA E SNQQERDLKE

      50      60      70      80
KFLITQATAY SLANQLKKYK CEEYKDIIDS VERDELQSM E

      90     100     110     120
KLAEKLRQAE ELRQYKALVH SQAKELTQLR EKLREGRDAS

     130     140     150     160
RWLNKHLKTL LTPDDPKSQ GQDLREQLAE GHRLAEHLVH

     170     180     190     200
KLSPENDEDE DEDEDDKDEE VEKVQESPA P REVQKTEEKE

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```

      210      220      230      240
VPQDSLEECA VTCSNSHNPS NSNQPHRSTK ITFKEHEVDS

      250      260      270      280
ALVVESEHPH DEEEEALNIP PENQNDHEEE EGKAPVPPRH

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```

      290      300      310      320
HDKSNYSYRHR EVSFLALDEQ KVCSAQDVAR DYSNPKWDET

      330      340      350      360
SLGFLEKQSD LEEVKGQETV APRLSRGPLR VDKHEIPQES

      370      380      390      400
LDGCCLIPSI LPDLIPSYHP YWSTLYSFED KQVSLALVDK

      410      420      430      440
IKKDQEEIED QSPPCPRLSQ ELPEVKEQEV PEDSVNEVYL

      450      460      470      480
TPSVHHDVSD CHQPYSSSTLS SLEDQLACSA LDVASPTEAA

      490      500      510      520
CPQGTWSGDL SHHRSEVQIS QAQLEPSTLV PSCLRLQLDQ

      530      540      550      560
GFHCGNGLAQ RGLSSTICSE SANADSGNQW PFQELVLEPS

      570      580      590      600
LGMKNPPQLE DDALEGSASN TQGRQVTGRI RASLVLILKT

      610      620      630
IRRLPFPSKW RLAFRFAGPH AESAEIPNTA ERMQRMIG

```


A cDNA and a chromosomal sequence encoding the Q5VWK0 protein is available from the NCBI database as accession no. BC125161 and AL390038, respectively.

[0052] An amino acid sequence for the protein encoded by the human TPGS2 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q68CL5, shown below as SEQ ID NO:9.

```

      10      20      30      40
MEEEASSPGL GCSKPHLEKL TIGITRILES SPGVTEVTII

      50      60      70      80
EKPPAERHMI SSWEQKNNCV MPEDVKNFYL MINGFHMTWS

      90     100     110     120
VKLDEHIPL  GSMAINSISK LTQLTQSSMY SLPNAPTLD

     130     140     150     160
LEDDTHEASD DQPEKPHEDS RSVIFELDSC NGSGKVCIVY

     170     180     190     200
KSGKPALAED TEIWELDRAL YWHFLIDTET AYYRLLITHL

     210     220     230     240
GLPQWQYAFT SYGISPQAKQ WESMYKPITY NTNLLTEETD

     250     260     270     280
SFVNKLDPSK VEKSKNKIVI PKKKGPVQPA GGQKGPSGPS

     290     300
GPSTSSTSKS SSGSGNPTRK

```

[0053] A cDNA and a chromosomal sequence encoding the TPGS2 protein is available from the NCBI database as accession no. AK295817 and AC009854, respectively.

[0054] An amino acid sequence for the ZNF630 protein encoded by the human ZNF630 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q2M218, shown below as SEQ ID NO:10.

```

      10      20      30      40
MIESQEPVTF EDVAVDFTQE EWQQLNPAQK TLHRDVMLET

      50      60      70      80
YNHLVSVGCS GIKPDVIPKL EHKGDPWIE SELSRWIYPD

      90     100     110     120
RVKGLESSQQ IISGELLFQR EILERAPKDN SLYSVLKIWH

     130     140     150     160
IDNQMDRYQG NQDRVLRQVT VISRETLTDE MGSKYSAFGK

     170     180     190     200
MENRCTDLAP LSQKFHKEDS CENSLKNSD LLNYNRSYAR

     210     220     230     240
KNPTKRERCG RPPKYNASCS VPEKEGFIHT GMEPYGDSQC

     250     260     270     280
EKVISHQAH  VQYKKFQARE KPNVCSMCGK AFIKKSQILII

     290     300     310     320
HQRIHTGEKP YVCGDCRKAF SEKSHLIVHQ RIHTGEKPYE

     330     340     350     360
CTKYGRAFSR KSPFTVHQRV HTGEKPYECE ECPKAFSQKS

     370     380     390     400
HLIIHQRVHT REKPFECSEC RKAFCEMSHL FIHQIHTGK

```

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                                -continued
      410      420      430      440
KPYECTECGK TFPRTQLII HQRTHTGEKP YKCGECGKTF

      450      460      470      480
CQQSHLIGHQ RIHTGEKPYV CTDCGKAFSQ KSHLTGHQRL

      490      500      510      520
HTGEKPYMCT ECGKSFSQKS PLIIHQRIHT GEKPYQCGEC

      530      540      550      560
GKTFSQKSLI IHLRVHTGE KPYECTECGR AFSLKSHLIL

      570      580      590      600
HQRGHTGEKP YECSECGKAF CGKSPLIIHQ KTHPREKTPE

      610      620      630      640
CAESGMTFEW KSQMITYQRR HTGEKPSRCS DCGKAFQCQHV

      650
YFTGHQNPYR KDTLYIC

```

A cDNA and a chromosomal sequence encoding the Q2M218 protein is available from the NCBI database as accession no. BC112139 and Z98304, respectively.

[0055] An amino acid sequence for the ZNF717 protein encoded by the human ZNF717 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9BY31, shown below as SEQ ID NO:11.

```

      10      20      30      40
MLETYNSLVS LQELVSFEEV AVHFTWEWQ DLDDAQTLYI

      50      60      70      80
RDVMLETYSS LVSLGHGCHT PEMIFKLEQG AEPWIVEETP

      90     100     110     120
NLRLSAVQII DDLIERSHES HDRFEWQIVI TNSNTSTQER

     130     140     150     160
VELGKTENLN SNHVLNLIIN NGNSSGMKPG QENDCQNMLF

     170     180     190     200
PIKPGETQSG EKPHVCDITR RSHRHHEHLT QHHKIQTLLQ

     210     220     230     240
TFQCNEQGKT FNTEAMFFIH KRVHIVQTFG KYNEYEKACN

     250     260     270     280
NSAVIVQVIT QVGQPTCCRK SDFTKHQQTH TGEKPYECVE

     290     300     310     320
CEKPSISKSD LMLQCKMPT EKPACNWCE KLESYKSSLI

     330     340     350     360
IHQRIHTGEK PYGCNECGKT FRRKSFLTIL ERTHTGDKPY

     370     380     390     400
KCIECGKTFH CKSLTLHHR THSGEKPYQC SECCKTESQK

     410     420     430     440
SYLTIHHRTH TGEKPYACDH CEEAFSHKSR LTVHQRTHTG

     450     460     470     480
EKPYECNECG KPFINKSNLR LHQRTHTGEK PYECNECGKT

     490     500     510     520
FHRKSFLTIH QWHTHTGEKPY ECNECGKTER CKSELTVHQR

     530     540     550     560
THAGEKPYAC NECGKTYSHK SYLTVVHHRTH TGEKPYECNE

     570     580     590     600
CGKSFHCKSF LTIHQRTHTAG KKPYECECE KTFINKNLNG

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-continued

610	620	630	640
IHKITHGTGER	PYECNECGKT	FRQKSNISTH	QGTHTGEKPY
650	660	670	680
VCGKTFHRKS	FLTIHQRTHT	GKNRMDVMNV	EKL FVRNHTL
690	700	710	720
LYIRELTPGK	SPMNMVMVEN	PFIRRIQIERS	IKVFTRGRNP
730	740	750	760
MNVANVEKPC	QKSVLIVHHR	THTGEKPYEC	NECGKTFCHK
770	780	790	800
SNLSTHQGTH	SGEKPYECDE	CRKTFYDKTV	LTIHQRTHTG
810	820	830	840
EKPFECKEKR	KTESQKSKLF	VHHRTHTEK	PERCNECKRT
850	860	870	880
FSQKSGLSIH	QRTHTEKPY	ECKECGKTFC	QKSHLSRHQQ
890	900		
THIGKSDVA	EAGYVFPQNH	SFEP	

A cDNA and a chromosomal sequence encoding the Q9BY31 protein is available from the NCBI database as accession no. AF226994 and AC 08724, respectively.

[0056] The following genes are positive regulators of T cells as detected by Interleukin-2 production (see Table 2): ABCB10, ACSS2, ADAM19, ADAM23, ADAMTS5, ALKBH7, ALX4, ANXA2R, AP2AI, APOBEC3C, APOBEC3D, APOL2, ARNT, ART1, ASCL4, BEX4, BTG2, BTNL2, C11orf21, C12orf80 (also called LINC02874), CBX4, CBY1, CCDC183, CCDC71L, CD2, CD28, CD6, CDKNIB, CDKN2C, CHERP, CIPC, CLIP3, CNGB1, CNR2, CREB5, CUL3, DCTN5, DEF6, DEPDC7, DYNLL2, EAPP, EEPDI, ELFN2, EMB, EMP1, EMP3, EP300, ERCC3, ESRP1, F2, FBXL13, FBXO41, FNBPII, FOSB, FOSL1, FOXO4, FOXQ1, FUZ, GABRG1, GIGTLC2, GNPD1, GPR18, GPR20, GPR21, GPR84, GRIN3A, GSDMD, GSTM1, HCST, HELZ2, HEPHL1, IL2, IL2RB, IRX4, ISM1, KLF7, KLRC4, KRT 8, LAT, LCP2, LHX6, LMNA, MAGEA9B, MAP3KI2, MERIK, MTMRI1, NDRG3, N1T1, NLRC3, NLRP2, NPLOC4, ORC1, OSBPL7, OTOP3, OTUD7A, OTUD7B, P2RY14, PAFAH11B2, PCP4, PDE3A, PHF8, PIK3API, PLA2G3, PLCG2, POLK, POU2F2, PPHL2, PRAC1, PRKCB, PRKD2, RAB6A, RAC1, RAC2, RIPK3, RRAS2, RYR1, SAFB2, SCN3A, SDCCA(uG8, SERPINF1, SGTA, SHOC2, SIGLEC1, SIRT1, SLC16A1, SLC44A5, SLC5A5, SMC4, SPPL2B, SSU-12, SWAP70, TAF15, THEMIS, TM4SF4, TMEM79, TNFRSF1(B, TNFSF11, TNRC6A, TPGS2, TRAF3IP2, TRIM21, TRMT5, TRPM4, TRPV5, TSPYL5, UBA52, UBL5, VAV1, WARS2, ZAP70, ZNF141, ZNF296, and ZNF701. Example 2 provides additional positive regulators T cells that were detected by Interleukin-2 production.

[0057] Sequences and other information relating to these genes, and their encoded proteins, is available, for example from the NCBI and UniPROT databases, which are incorporated by reference.

[0058] A few examples of protein sequences encoded by some of the genes detected as positive regulators of T cells by Interleukin-2 production are provided. For example, an amino acid sequence for the protein encoded by the human ADAMTS5 gene that is a positive regulator of T cells as

detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9UNA0, shown below as SEQ ID NO: 12.

10	20	30	40
MLLGWASLLL	CAFRLPLAAV	GPAATPAQDK	AGQPPTAAAA
50	60	70	80
AQPRRRQGE	VQERAEPFGH	PHPLAQRRRS	KGLVQNIDQL
90	100	110	120
YSGGGKVGYL	VYAGGRRELL	DLERDGSVGI	AGFVPAGGGT
130	140	150	160
SAPWRHRSCH	FYRGTVDGSP	RLSLAVEDLCG	GLDGGFAVKH
170	180	190	200
ARYTLKPLLR	GPWAEEEKGR	VYDGSARIL	HVYTREGES
210	220	230	240
EALPPRASCE	TPASTPEAHE	HAPAHNSPSG	RAALASQLLD
250	260	270	280
QSALSPAGGS	GPQTWRRRR	RSISRARQVE	LLLVADASMA
290	300	310	320
RLYGRGLQHY	LLTLASIANR	LYSHASIENH	IRLAVVKVVV
330	340	350	360
IGDKDKSLEV	SKNAATTLKN	FCKWQHQNQ	LGDDHEEHYD
370	380	390	400
AAILFTREDL	CGHHSCTDLG	MADVGTICSP	ERSCAVIDED
410	420	430	440
GLHAAFTVAH	EIGHLIGLSH	DDSKFCEETF	GSTEDKRLMS
450	460	470	480
SILTSIDASK	PWSKCTSATI	TEFLDDGHGN	CLLDLPRKQI
490	500	510	520
LGPEELPGQT	YDATQQCNLT	FGPEYSVCPG	MDVCARLWCA
530	540	550	560
VVRQGMVCL	TKKLPAVEGT	PCGKGRIQLQ	GKCVDKTKKK
570	580	590	600
YYSTSSHGNW	GSWGSWGQCS	RSCGGGVQFA	YRHCNNPAPR
610	620	630	640
NNGRYCTGKR	AIYRSCSIMP	CPPNGKSRFH	EQCEAKNGYQ
650	660	670	680
SDAKGVKTEV	EWVPKYAGVL	PADVCKLTGR	AKGTGYVVVF
690	700	710	720
SPKVTGDTEC	RLYSNSVCVR	GKCVRTGCDG	IIGSKLQYDK
730	740	750	760
CGVCGGDNSS	CIKIVGTENK	KSKGYTDVVR	IPEGATHIKV
770	780	790	800
RQFKAKDQTR	FTAYLALKKK	NGEYLINGKY	MISTSETIID
810	820	830	840
INGTVMNYSG	WSHRDDELHG	MGYSATKEIL	IVQILATDPT
850	860	870	880
KPLDVRYSF	VPKSTPKVN	SVTSHGSKNV	GSHTSQPQWV
890	900	910	920
TGPWLACSRT	CDTGWHTRIV	QCQDGNRKLK	KGCPLSQRPS
930			
AFKQCLKKKC			

A cDNA and a chromosomal sequence encoding the protein is available from the NCBI database as accession no. AF 42099 and AP001698, respectively.

[0059] A nucleotide sequence for human C12orf80 cDNA (also called LINC02874) that is a positive regulator of T cells as detected by Interleukin-2 production is available from the NCBI database as accession no. NR_164127.1, shown below as SEQ ID NO: 13.

```

1  AGAGCGAGAA GATGATGCAT GTGAGCCCTG CCCTTGGGAA
41 GCTTCCAGGT TGGGAAATGA GGAATGAGCC TGACACCCAG
81 GGCCCGAGAG GACCCAGGAC AGAGGCAGGT CAGGAGGCAG
121 ACACGCGCTG CIGGGTAATG ACGACAGCAC CAGTAATCAC
161 GGCTACTCCT TGTTAAGTAC TTAATAAGTG CAAGACTCCA
201 AGCTAAGCAA TTAATAGACC TTTTCTTGTT TAATCCTTAC
241 CACAATTCCA TAGGGITGAG TAGGAAGTCC TTCTTGAAGT
281 CTAATCTCAA GAATTCATGC CATGGATTGG GCCAATGTTC
321 CACCTTTATT GGAGTCTGGG GACTTAGGTG GAAAAGGCAG
361 AACTGGCTGG TTGGAGGAGC CACCTCCCTG CAAGGGGCCA
401 GGAGGGAGAT TACGGAGGCG CCAAGCCCAG AGCTCCAACA
441 TGCACCTGCC ACAGCTCCAG GGGAGATCGG GGGCCTGCCA
481 ATTTACCCCG CCCCATGATC TCATGGCTGT GTGCGCCAGG
521 CACTGGCTCA GGGAGCAGCA TCTCAGAG AAGATACTTG
561 GATGGGCCAC AGGCCAAAAC TGGGCCAAAA CCCTGGAGAG
601 GGGGACCTGG CTCAAGGCCA CACCACATTT CCATGTCATT
641 TTCCAGTGGC ACCAACTGAG CTGGACAGAT GTTCTCACAG
681 AGCTACACAT GCCACAGTCA TCACTAGTTG CGAGAGTCTC
721 CAGTGTTCAT TAAGCCTATT TTGCTGGAAG TTTAGTCCCT
761 GGAGGATTAG TCCCTGCCCT TAAGGAGTGC CAAAGAAGAC
801 TTTGGAATCT AAGTCACATC TCCCCTGCTG CAATTTTTTC
841 CTCTTGATAA TCAAGGAATC TCACTGATAA AACTCTCAAT
881 GAGGAGCAGC TCTCCCCATG AGCAAGTTCT TGCCTTCATT
921 GCCAGGGAGG CCCCAGCTAA GGTATACTG GGAAGGAAT
961 CTCTGGGGAG TACCTGGAAG AGGTCAGCTT CTCCTCAAAG
1001 GAAGCCTCTC CAGCTGGTGT ACTGCAAAGC CTTCCAAGAC
1041 CAAGTTCCCT CTTCTGGGCC TCTGGAACCA GCCTGTGTGC
1081 CAGCGCTGTG TCCCAGTAA CACACATGAG TCTCTCCCTC
1121 AGAAGCCACA GAAGGATGCA GAGAGAGACC AACCCAGGAT
1161 TGATTCTCAG CACTTACTAA TTGTGTGGAC ATAACTGCTC
1201 TGGACCTCAG CTTTGTCTATC TATAAATGA GCACCCTTTT
1241 TATAGATATG AACACTTTAA GAGAGTCATA AAGATTGAAG
1281 TTGATTIGTG CTGCGCCTGG CACAGAGGTC TCCCTCAGTA
1321 AATGGCAGCC ACTGCTATTG GGGTGCCCA ACACCCCTTG

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1361 CCCCTGCCCC CCATTAGCTT TCACTTAGGC CCACACTGAG
1401 GGTGTGGCTG TTGTGTTGGG GGAAGGAAAA AAACCATGCC
1441 TGGGTTGCAG GGCCCCCACC ACCAACTGGT CTGCTTTTGT
1481 GGAAGCATT TTCTGATGCC CTCGCCCTAC CCGTGGGCTG
1521 GTTGACTAAG TGTCCAGCAT CATGAGGAAA AGAGCAGGGT
1561 TCAGGGACAG CTTGGCTCAG CCCTGAAACT CATCTGGGCC
1601 TAGGTCCAGA TGAGAGGCAA GCCTGGGAGG CTTCCGTTG
1641 TACCTTGCC TATCCTGCAG CACCCTCTAG CCGCAGGCC
1681 GCTCCTGGGT GGCATAGCAC CTTGGATGTC AGGTGTGGGC
1721 CTTCCAGGC ACATGGCATG AGGGGGCACT TCTGTGGGTT
1761 GTTGGGGGCA GGAAGGGATG TGTCTCCAGC TGGACTTGGG
1801 CTCTCGCATT TTGGGGCCCA GCCATGCCAG GACAGCACAC
1841 ATGGGCGCTT AGTGCGAATC TCATGATGGA GCAGAGGAGG
1881 AGCAAAGCAA AACCAGGGAG TCCCAGGCC CTGCTTCTG
1921 CCCCGCCCAG AGACAGTGA GCAGGTGTCT CCAGCTTCTT
1961 AACCTCAACG CACAGTAAGA AATACATTTT ACAGCAACCA
2001 ATAGACACAT ACATTACAA AACGA

```

[0060] An amino acid sequence for the protein encoded by the human CCDC183 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q5T5S1, shown below as SEQ ID NO: 14.

```

10      20      30      40
MRRHSETDVE EQTQELKITIT QLQEQCRAQL IQGVKENMDQ

50      60      70      80
NKATLALLRS NIRRGADWA LAKKYDQWTI SKACGKNLPL

90      100     110     120
RLAHCRTSME VVREKLRYV EDRVNMHNL IHLVRRRGQK

130     140     150     160
LESMLQELDS LRSQPDASKE ELRLQLIIRQ LENNIEKIMI

170     180     190     200
KIITSQNIHL LYLDLLDYLK TVLAGYPIEL DKLQNLVVNY

210     220     230     240
CSELSDMKIM SQDAMMITDE VKRNMQRQA SFIEERRARE

250     260     270     280
NRLNQKKLI DKIKTKETSE KYRRGQMDLD FPSNLMSTET

290     300     310     320
LKLRRKETST AEMEYQSGVT AVVEKVS AV RSHVWDITS

330     340     350     360
RELAQRNTEE NLELQMEDCE EWRVQLKALV KQLELEEAVL

370     380     390     400
KFRQKPSSIS EKSVEKMTD MLKEEEERLQ LAHSNMTKQG

410     420     430     440
ELLLTIQMG I DNLVRLMGI NLPATQREV LSNILDINSK

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      450      460      470      480
LAYCEGKLT Y LADRVQMVSR TEEGDTKVRD TLESSTLMEK

      490      500      510      520
YNTRISFENR EEDMIDTEQE PDMDHSYVPS RAEIKRQAQR

      530
LIEGKLKAAK KKKK

```

A cDNA and a chromosomal sequence encoding the protein is available from the NCBI database as accession no. AB075864 and AL355987, respectively.

[0061] An amino acid sequence for the protein encoded by the human CIPC gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9C0C6. shown below as SEQ ID NO: 15,

```

      10      20      30      40
MERKNPSRES PRRLSAKVGK GTEMKKVARQ LGMAAAESDK

      50      60      70      80
DSGESDGSSE CLSSAEQMES EDMLSALGWS REDRPRQNSK

      90     100     110     120
TAKNAFPTLS PMVVMKNVLV KQGSSSSQLQ SWIVQPSFEV

     130     140     150     160
ISAQPQLLFL HPPVPSVPSP CHTGEKKSDS RNYLPILNSY

     170     180     190     200
TKIAPHGPKR GLSLGPKEEG TSGVQKKICT ERLGPSLSSS

     210     220     230     240
EPTKAGAVPS SPSTPAPPSA KLAEDSALQG VPSLVAGGSP

     250     260     270     280
QTLQPVSSSH VAKAPSLIFA SPASPVCSAD STLHGLESNS

     290     300     310     320
PLSPLSANYS SPLWAAEHL C RSPDIFSEQR QSKHRRFQNT

     330     340     350     360
LVVLHKSGLL EITLTKTELI RQNQATQVEL DQLKEQTQLF

     370     380     390
IEATKSRAPQ AWAKLQASLT PGSSNIGSDL EAFSDHPAI

```

A cDNA and a chromosomal sequence encoding the CIPC protein is available from the NCBI database as accession no. AB051524 and AC007686, respectively.

[0062] An amino acid sequence for the protein encoded by the human CUL3 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q13618, shown below as SEQ ID NO: 16.

```

      10      20      30      40
MSNLSKGTGS RKDTKMIRRA FPMTMDEKYV NSIWDLLKNA

      50      60      70      80
IQEIQRKNNS GLSFEELYRN AYTMVLHKGK EKLYTGLREV

      90     100     110     120
VTEHLINKVR EDVINSLNNN FLQTLNQAWN DHQTAMVMIR

     130     140     150     160
DILMYMDRVY VQQNNVENVY NEGLIIERDQ VVRYGCI RDH

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      170      180      190      200
LRQTLLDMIA RERKGEVVD R GAINRACQML MILGLEGRSV

      210      220      230      240
YEEDFEAPFL EMSAEFFQME SQKFLAENSA SVYIKKVEAR

      250      260      270      280
INEEIERVMH CLDKSTEEPI VKVVERELIS KHMKTIVEME

      290      300      310      320
NSGLVHMLKN GKTEDLGCMY KLF SRVPNGL KTMCECMSSY

      330      340      350      360
LREQGKALVS EEGEGKNPVD YIQGLLDLKS RFDRELLESF

      370      380      390      400
NNDRLFKQTI AGDFEYFLNL NSRSPEYLSL FIDDKLKKGV

      410      420      430      440
KGLTEQEVET ILDKAMVLER FMQEKDVFER YYQHLARRI

      450      460      470      480
LINKSVSDDS EKNMISKLKT ECGCQFTSKL EGMRDMSIS

      490      500      510      520
NTIMDEFROH LQATGVSLGG VDLTVRVLT T GYWPTQSATP

      530      540      550      560
KCNIPPAPRH AFEIFRRFYL AKHSGRQLTL QHHMGSADLN

      570      580      590      600
ATFYGPVKKE DGSEVGVGGA QVTGSNTRKH ILQVSTFQMT

      610      620      630      640
ILMLENNREK YTFEEIQQET DIPERELVRA LQSLACGKPT

      650      660      670      680
QRVLTKEPKS KEIENGHIPT VNDQFTSKLH RVKIQTVAAK

      690      700      710      720
QGESPDERKE TRQKVDDDRK HEIEAAIVRI MKSRKMQHNN

      730      740      750      760
VIVAEVTQQL KARFLPSPVV IKKRIEGLIE REYLARTPED

      RKVYTYVA

```

A cDNA and a chromosomal sequence encoding the Q13618 protein is available from the NCBI database as accession no. AF064087 and AC073052, respectively.

[0063] An amino acid sequence for the protein encoded by the human EMB (Embigin) gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q6PCB8, shown below as SEQ ID NO: 17.

```

      10      20      30      40
MRALPGLLEA RARTPRLLLL QCLLAAARPS SADGSAPDSP

      50      60      70      80
FTSPPLREEI MANNFSLESH NISLTHESSM PVEKNITLER

      90     100     110     120
PSNVNLICQF TISGDLNAV N VIWKDGEQL ENNYLVSATG

     130     140     150     160
STLYTQYRFT IINSKQMGSY SCFFREEKEQ RGTNFNFKVPE

     170     180     190     200
LHGKKNPLIS YVG DSTVLTC KCQNCPELNV TWYSSNGSVK

     210     220     230     240
VPVGVQMKNY VINGTYANET KLKITQLLEE DGE SYWCRAL

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250 260 270 280
FQLGESEEH ELVVLSTYLP LKPLVIVAE VILLVATILL

290 300 310 320
CEKYTQKKKK HSDEGKEFEQ IEQLKSDDSN GIENNVPRHR

KNESLGQ

A cDNA and a chromosomal sequence encoding the protein is available from the NCBI database as accession no. AK300860 and AC035145, respectively.

[0064] An amino acid sequence for the protein encoded by the human ESRP1 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q6NXG1. shown below as SEQ ID NO: 18.

10 20 30 40
MTASPDYLVV LFGITAGATG AKLGSDEKEL ILLEWKVVDL

50 60 70 80
ANKKVGQLHE VIVRPDQLEL TEDCKEETKI DVESISSASQ

90 100 110 120
LDQALRQENQ SVSNELNIGV GTSFCLCTDG QLHVRQILHP

130 140 150 160
EASKKNVLLP ECFYSFFDLR KEFKKCCPGS PDIDKLDVAT

170 180 190 200
MTEYLNFEKS SSVSRYGASQ VEDMGNIILA MISEPYNHRE

210 220 230 240
SDPERVNYKF ESGTCSKMEL IDNTVVRAR GLPWQSSDQD

250 260 270 280
IARFFKGLNI AKGGAALCIN AQGRRNGEAL VRFVSEHRD

290 300 310 320
LALQRHKHHM GTRYIEVYKA TGEDELKIAG GTSNEVAQFL

330 340 350 360
SKENQVIVRM RGLPFTATAE EVVAFFGQHC PITGGKEGIL

370 380 390 400
FVTYPDGRPT GDAFVLFACE EYAQNALRKH KDLLGKRYIE

410 420 430 440
LERSTAAEVQ QVLNRESSAP LIPLTPPII PVLPPQFVPP

450 460 470 480
INVRDCIRER GIPYAATIED ILDELGEFAT DIRTHGVHNV

490 500 510 520
LNHQGRPSGD AFIQMSADR AFMAAQKCHK KNMKDRYVEV

530 540 550 560
FQCSAEEMNF VLMGGTLNRN GLSPPPCKLP CLSPPSYTEP

570 580 590 600
APAAVIPTEA AIYQPSVILN PRALQPSTAY YPAGTQLEMN

610 620 630 640
YTAYYPPPG SPNSLGYEPT AANLSGVPPQ PGTVVVMQGL

650 660 670 680
AYNTGVKEIL NFFQGYQYAT EDGLIHINDQ ARTLPKEWVC

I

A cDNA and a chromosomal sequence encoding the Q6NXG1 protein is available from the NCBI database as accession no. BC067098 and AP005660, respectively.

[0065] An amino acid sequence for the protein encoded by the human FBXL13 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q8NEE6, shown below as SEQ ID NO:19.

10 20 30 40
MTEPELMIKAC SFYTGHLVKT HFCTWRDIAR TNENVVLAEK

50 60 70 80
MNRVAVTCYNE RLQKSVFHHW HSYMEDQKEK LKNILLRIQQ

90 100 110 120
IIYCHKLTII LTKWRNTARH KSKKKEDELI LKHELQLKKW

130 140 150 160
KNRLILKRAA AEESNFPERS SSEVELVDET LKCDISLLPE

170 180 190 200
RAILQIFFYL SLKDVIIICGQ VNHAWMLMTQ INSLWNAIDE

210 220 230 240
SSVENVIPDK YIVSTLQRWR INVIRLNERG CLLRPKTERS

250 260 270 280
VSHCRNLQEL NVSDCPTFTD ESMRHISEGC PGVLCNLNSN

290 300 310 320
TTITNRTMRL LPRHFHNLQN LSLAYCRRFT DKGLQYLNLG

330 340 350 360
NGCHKLIYLD LSGCTQISVQ GERYIANSCT GIMHLTINDM

370 380 390 400
PTLTDNCVKA LVEKCSRITS LVETGAPHIS DCTFRALSAC

410 420 430 440
KLRKIRFEGN KRVIDASPKF IDKNYPNISH IYMADCKGIT

450 460 470 480
DSSLRLSLPL KQLTVLNLAN CVRIGDMGLK QFLDGPASMR

490 500 510 520
IRELNLSNCV RLSVASVMKL SERCPNLNLY SLRNCEHLTA

530 540 550 560
QGIGYIVNIF SLVSIIDLSGT DISNEGLNVL SRHKKLKELS

570 580 590 600
VSECYRITDD GIQAFCKSSL ILEHLDVSYC SQLSDMIKA

610 620 630 640
LAIYCINITLS LSIAGCPKIT DSAMEMLSAK CHYLHILDIS

650 660 670 680
GCVLLTDQIL EDLQIGCKQL RILKMQYCIN ISKKAQQRMS

690 700 710 720
SKVQQQEYNT NDPPRWEGYD REGNPVTELD NITSSKGALE

730
LTVKKSTYSS EDQAA

A cDNA and a chromosomal sequence encoding the FBXL13 protein is available from the NCBI database as accession no. AY359238 and AC005250, respectively.

[0066] An amino acid sequence for the protein encoded by the human FBXO41 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q8TF61, shown below as SEQ ID NO:20.

```

      10      20      30      40
MASLDLPYRC PRCGEHKRER SLSSLRAHLE YSHTYETLYI

      50      60      70      80
LSKINSICDG AAAAAAAAAA ASGEPLAPEP AALLAVPGAR

      90     100     110     120
REVFEFSTFQ GKEQAAGPSP AAPHLHHHH HHAPLAHPPG

      130     140     150     160
DLVPASIPCE ELAEPGIVPA AAARYALREI EIPLGELFAR

      170     180     190     200
KSVASSACST PPPGPGPGPC PGPASASPAS PSPADVAYEE

      210     220     230     240
GLARLKIRAL EKLEVDRLAE RLSEEVEQKI AGQVGRLQAE

      250     260     270     280
LERKAAELET ARQESARLGR EKEELEERAS ELSRQVDVSV

      290     300     310     320
ELLASLKQDL VHKEQELSRK QQEVVQIDQF LKETAAREAS

      330     340     350     360
AKLRLQQFIE ELLERADRAE RQLQVISSSC GSTPSASIGR

      370     380     390     400
GGGGGGAGPN ARGPGRMREH HVGPAVPNTY AVSRHGSSPS

      410     420     430     440
TGASSRVPAA SQSSGCYDSD SLELPRPEEG APEDSGPGGL

      450     460     470     480
GTRAQAANGG SERSQPPRSS GLRRQAIQNW QRRPRRHSTE

      490     500     510     520
GEEGDVSDVG SRTTESEAEQ PIDAPRPGPA MAGPLSSCRL

      530     540     550     560
SARPEGGSGR GRRAERVSPS RSNEVISPEI LKMRAALFCI

      570     580     590     600
FTYLDTRILL HAAEVCRDWR FVARHPAVWT RVLLENARVC

      610     620     630     640
SKFLAMLAQW CTQAHSLTLQ NLKPRQRGKK ESKEEYARST

      650     660     670     680
RGCLEAGLES LIKAAGGNLL ILRISHCPNI LTDRSLWLAS

      690     700     710     720
CYCRALQAVT YRSAIDPVGH EVIHALGAGC REIVSLQVAP

      730     740     750     760
LHPCQQPTRF SNRCLQMIGR CWPHLRALGV GGAGCGVQGL

      770     780     790     800
ASLARNCMRL QVLELDHVSE ITQEVAEVC REGLKGLEML

      810     820     830     840
VLTATPVTPK ALLHENSICR NLKSIVVQIG IADYFKEPSS

      850     860     870
PEAQKLFEDM VTKLQALRRR PGESKILHIK VEGGC

```

A cDNA and a chromosomal sequence encoding the FBXO41 protein is available from the NCBI database as accession no. AB075820 and AC010913, respectively.

[0067] An amino acid sequence for the protein encoded by the human FOSL1 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. P15407, shown below as SEQ ID NO:21.

```

      10      20      30      40
MERDEGEGGP SSGNGGGYGG PAQPPAAQA AQQKFHLVPS

      50      60      70      80
INTMSGSQEL QWMVQPHLGL PSSYPRPITY PQYSPPQPRP

      90     100     110     120
GVIRALGPPP GVRRRPCEQI SPEEEERRRV RRERNKLAAA

      130     140     150     160
KCRNRKELT DFLQAETDKL EDEKSLGLRE IEELQKQKER

      170     180     190     200
LELVLEAHRP ICKIPEGAKE GDTGSTSGTS SPPAPCRPVP

      210     220     230     240
CISLSPGPVL EPEALHTPTL MITPSLTPTT PSLVFTYPST

      250     260     270
PEPCASAHRK SSSSSGDPSS DPLGSPTLLA L

```

A cDNA and a chromosomal sequence encoding the FOSL1 protein is available from the NCBI database as accession no. X16707 and AP006287, respectively.

[0068] An amino acid sequence for the protein encoded by the human FOXO4 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. P98177, shown below as SEQ ID NO:22.

```

      10      20      30      40
MDPGNENSAT EAAAIIDLDP DEEPQSRPRS CTWPLPRPEI

      50      60      70      80
ANQPSEPPEV EPDLGEKVHT EGRSEPILLP SRLPEPAGGP

      90     100     110     120
QPGILGAVTG PRKGGSRRNA WGNQSYAELI SQAIESAPEK

      130     140     150     160
RLTLAQIYEW MVRTVPYEKD KGDSNSSAGW KNSIRHNISL

      170     180     190     200
HSKFIKVHNE ATGKSSWML NPEGKSGKA PRRRAASMDS

      210     220     230     240
SSKLLRGRSK APKKKPSVLP APPEGATPTS PVGHEAKWGG

      250     260     270     280
SPCSRNREEA DMWTTFRPRS SSNASSVSTR LSPLRPESEV

      290     300     310     320
LAEEIPASVS SYAGVPPPTL NEGLELDGL NLTSSHSLLS

      330     340     350     360
RSGLSGESLQ HPGVTGPLET YSSSLESPAE GPLSAGEGCF

      370     380     390     400
SSSQALEALL TSDTPPPPAD VLMTQVDPII SQAPTLLLLG

      410     420     430     440
GIPSSSKLAT GVGLCPKPLE APGPSSLVPT LSMIAPPPVM

      450     460     470     480
ASAPIPKALG TPVLIPTTEA ASQDRMPQDL LDLMYMNLE

      490     500
CDMDNIISDL MDEGEGLDEN FEPDP

```

A cDNA and a chromosomal sequence encoding the FOXO4 protein is available from the NCBI database as accession no. X93996 and AL590764, respectively.

[0069] An amino acid sequence for the protein encoded by the human FUZ gene that is a positive regulator of T cells

as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9BT04, shown below as SEQ ID NO:23.

```

      10      20      30      40
MGEEGTGGTV HLLCLAASSG VPLFCRSSRG GAPARQQLPF

      50      60      70      80
SVIGSLNGVH MFGQNLEVQL SSARTENTTV VWKSPHDSIT

      90     100     110     120
LIVLSSEVGI SELRLERLLQ MVFGAMVILV GLEELINIRN

     130     140     150     160
VERLKKDLRA SYCLIDSELG DSELIGDLTQ CVDVIPPEG

     170     180     190     200
SLLQEALSGF AEAAGTTEVS LVVSGRVVAA TEGWVRLGTP

     210     220     230     240
EAVLLPWLVG SLPPQTARDY PVYLPFGSPT VPHRLTLTL

     250     260     270     280
LPSLELCLIC GPSPPLSQLY PQLLRWWQP LLDPLRACL P

     290     300     310     320
LGPRALPSGF PLHTDILGLL LLHLELKRCL ETVEPLGDKE

     330     340     350     360
PSPEQRRRL RNFYTIVIST HEPPEPGPPE KTEDEVYQAQ

     370     380     390     400
LPRACYLVLG TEEP GTGVRL VALQLGERRL LLLSPQSP T

     410
HGLRSLATHT LHALTPLL

```

A cDNA and a chromosomal sequence encoding the FUZ protein is available from the NCBI database as accession no. AK026341 and AC006942, respectively.

[0070] An amino acid sequence for the protein encoded by the human IRX4 gene is available from the UniPROT database as accession no. P78413, shown below as SEQ ID NO:23.

```

      10      20      30      40
MSYPQFGYPY SSAPQELMAT NSISTCCESG GRTLADSGPA

      50      60      70      80
ASAQAPVYCP VYESRLLATA RHEINSAAAL GVGPGPYGGS

      90     100     110     120
QGYGNVVTYG SEASAFYSLN SEDSKDGS GS AHGGLAPAAA

     130     140     150     160
AAYPYEPALG QYPYDRYGT M DSGTRRKNAT RETTSTLKAW

     170     180     190     200
LQEHKPNPYP TKGEKIMLAI ITKMTLTQVS TWFANARRRI

     210     220     230     240
KKENKMTWPP RNKCADEKRP YAEGE EEEEGG EEEAREEPLK

     250     260     270     280
SSKNAEPVGK EEKELELSDL DDEDPLEAEP PACELKPPFH

     290     300     310     320
SLDGGLELVP AAPDGVKKEA SGALRMSLAA GGGAALDEDL

     330     340     350     360
ERARSCLRSA AAGPEPLPGA EGGPQVCEAK LGFVPAGASA

     370     380     390     400
GLEAKPRIWS LAHTATAAAA AATSLSQTEF PSCMLKRQGP

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```

      410      420      430      440
AAPAAVSSAP ATSPSVALPH SGALDRHQDS PVISLRNWVD

      450      460      470      480
GVFHDPILRH STLNQAWATA KGALLDPGPL GRS LGAGANV

      490      500      510
LTAPLARAFP PAVPQDAPAA GAARELLALP KAGGKPECA

```

A cDNA and a chromosomal sequence encoding the IRX4 protein is available from the NCBI database as accession no. AF124733 and AB690778, respectively.

[0071] An amino acid sequence for the protein encoded by the human ISM1 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. B1AKI, shown below as SEQ ID NO:24.

```

      10      20      30      40
MVRLAAELLL LLGLLLTLH ITVLRGSGAA DGPDAAGNA

      50      60      70      80
SQAQLQNNLN VGSDITSETS FSLSKAPE R HLDHQA AHQP

      90     100     110     120
FPRPRERQET GHPSLQ RDEP RSFLLDIPNF PDL SKADING

     130     140     150     160
QNPNIQVTIE VVDGPDSEAD KDQHPENKPS WSVSPDWRA

     170     180     190     200
WWQRSLSLAR ANSGDQDYKY DSTSDSNFL NPPRGWDHTA

     210     220     230     240
PGHRTFETKD QFEYDSTDGE GDWSLWSVCS VTCGNGNQKR

     250     260     270     280
TRSCGYACTA TESRTCDRPN CPGIEDTERT AATEVSLLAG

     290     300     310     320
SEEFNATKLF EVDTDSCERW MSCKSEFLKK YMHKVMNDLP

     330     340     350     360
SCPCSYPTEV AYSTADIEDR IKRKDERWKD ASGPKEKLEI

     370     380     390     400
YKPTARYCIR SMLSLESTIL AAQHCCYGDN MQLITRGKGA

     410     420     430     440
GTPNLISTEF SAELHYKVDV LPWIICKGDW SRYNEARPPN

     450     460
NGQKCTESPS DEDYIKQFQE AREY

```

A cDNA and a chromosomal sequence encoding the ISM1 protein is available from the NCBI database as accession no. BC017997 and AL050320, respectively.

[0072] An amino acid sequence for the protein encoded by the human MTMR-11 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. A4FU01, shown below as SEQ ID NO:25.

```

      10      20      30      40
MWWGGRGQSF NIAPQKEEPE MGSVQENRMP EPRS RQPSSC

      50      60      70      80
LASRCLPGEQ ILAWAPGVRK GLEPELSGTL ICTNERVTFQ

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90	100	110	120
PCGWQWNQDT	PLNSEYDFAL	VNIGRLEAVS	GLSRVQLLRP
130	140	150	160
GSLHKFPIEE	ILIHGRDERL	LRVGFEAGGL	EPQAFQVIMA
170	180	190	200
IVQARAQSNQ	AQQYSGITLS	KAGQSGSRK	PPIPLMETAE
210	220	230	240
DWETERKKQA	ARGWRVSTVN	EREDVATSLP	RYFWVPNRIL
250	260	270	280
DSEVRRAPGH	FHQGRGPRLS	WHHPGSDLL	RCGGFYTASD
290	300	310	320
PNKEDIRAVE	LMLQAGHSDV	VLVDTMDLDP	SLADVQLAHL
330	340	350	360
RIRALCLPDS	SVAEDKWLSA	LEGTRWLDYV	RACLRKASDI
370	380	390	400
SVIVISRVR	VILQERGRDR	LNGLSSLVQ	LISAPEARTL
410	420	430	440
FGFQSLVQRE	WVAAGHPILT	RIGGTGASEE	APVELLELDC
450	460	470	480
VWQLLQGEPA	DEEFSEFELL	ALHDSVRVPD	ILTFLRNTPW
490	500	510	520
ERKGQSGQLN	SYTQVYTPGY	SQPPAGNSEN	LQLSVWDWDL
530	540	550	560
RYSNAQILQF	QNPGYDPEHC	PDSWLPRPQP	SEMVPGPSS
570	580	590	600
VWLFSGALT	PLNQLCPWRD	SPSLAVSSR	WLPRPAISSE
610	620	630	640
SLADQEWGLP	SHWGACPLPP	GLLLPGYLG	QIRLWRRCYL
650	660	670	680
RGRPEVQMG	SAPTISGLQD	ELSHLQELLR	KWTPRISPED
690	700		
HSKKRDPHTI	LNPTEIAGIL	KGRAEGDLG	

A cDNA and a chromosomal sequence encoding the MTMR11 protein is available from the NCBI database as accession no. U78556 and AL590487, respectively.

[0073] An amino acid sequence for the protein encoded by the human NDRG3 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9UGV2. shown below as SEQ ID NO:26.

10	20	30	40
MDELQDVQLT	EIKPLNDKN	GTRNFQDEDC	QEHDITTHG
50	60	70	80
VVHVITIRGLP	KGNRPVILTY	HDIGLNHKSC	FNAFENFEDM
90	100	110	120
QEITQHFAVC	HVDAPGQQEG	APSFTGYQY	PTMDELAEML
130	140	150	160
PPVLTHLSLK	SIIGIGVGAG	AYILSRFALN	HPELVEGLVL
170	180	190	200
INVDPCAKGW	IDWAASKLSG	LTINVVDIIL	AHHFGQEELQ

-continued

210	220	230	240
ANLDLIQTYR	MHIAQDINQD	NLQLFLNSYN	GRRDLEIERP
250	260	270	280
ILGQNDNKS	TLKCSTLLVV	GDNSPAVEAV	VECNSRLNPI
290	300	310	320
NTILLKMADC	GGLPQVVQPG	KLTEAFKYFL	QGMGYIPSAS
330	340	350	360
MTRLARSRTH	STSSSLGSGE	SPESRSVTSN	QSDGIQESCE
370			
SPDVLDHRQT	MEVSC		

A cDNA and a chromosomal sequence encoding the NDRG3 protein is available from the NCBI database as accession no. AB044943 and AL031662, respectively.

[0074] An amino acid sequence for the protein encoded by the human NPLOC4 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q8TAT6, shown below as SEQ ID NO:27.

10	20	30	40
MAESIIIRVQ	SPDGVKRITA	TKRETAATFL	KKVAKEFGFQ
50	60	70	80
NNGESVYINR	NKTGEITASS	NKSLNLLKIK	HGDLFLFPPS
90	100	110	120
SLAGPSSEME	TSVPPGEKVE	GAPNVVEDEI	DQYLSKQDGK
130	140	150	160
IYRSRDPQLC	RHGPLGKCVH	CVPLEPEDED	YLNHLEPPVK
170	180	190	200
HMSFHAYIRK	LIGGADKGF	VALENISCKI	KSGCEGHLWP
210	220	230	240
PNGICTKCP	SAITLNRQKY	RHVDNIMFEN	HTVADRELDF
250	260	270	280
WRKTGNQHEG	YLYGRYTEHK	DIPIGIRAEV	AAIYEPPQTG
290	300	310	320
TQNSLELLED	PKAEVVDEIA	AKLGLRKVGW	IFTDLVSEDT
330	340	350	360
RKGIVRYSRN	KDITYLSSEE	CITAGDBQNK	HPNMCRSPD
370	380	390	400
GHPGSKEVTA	VATGGPDNQV	HFEGYQVSNQ	CMALVRDECL
410	420	430	440
LPCKDAPELG	YAKESSEQY	VPDVFKDQVD	KEGNEITQLA
450	460	470	480
RPLPVEYLII	DITTEPKDP	VYTFSSISQNP	FPIENRDVLG
490	500	510	520
ETQDEHSLAT	YLSQNTSSVE	LDTISDEHLL	LFLVINEVMP
530	540	550	560
LQDSISILLE	AVRIRNEELA	QTKWRSEQWA	TIEQLCSTVG
570	580	590	600
GQLPGLHEYG	AVGGSTHTAT	AAMWACQHCT	FMNQPGTGHC
EMCSLPT			

A cDNA encoding the NPLOC4 protein is available from the NCBI database as accession no. AB040932.

[0075] An amino acid sequence for the protein encoded by the human OTOP3 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q7RTS5. shown below as SEQ ID NO:28.

```

      10      20      30      40
MGRGAAAAA QSRWGRASRA SVSPGRTIRS APAVGEAQET

      50      60      70      80
EAAPEKENRV DVGAEEERAAA TRPRQKSWLV RHESLLLRD

      90     100     110     120
RQAQKAGQLF SGLLALNVVE LGGAFICSMI FNKVAIVLGD

     130     140     150     160
VWILLATLKV LSLIWLIVYV ASTTRRPHAV LYQDPHAGPL

     170     180     190     200
WVRGSLVLFQ SCTFCLNIFR VGYDVSHIRC KSQDLVESV

     210     220     230     240
IEMVEIGVQT WVLWKHKDCD VRVQTNFTRC GLMLTLATNL

     250     260     270     280
LLWVLAVIND SMHREIEAEL GILMEKSTGN ETNTCLCLNA

     290     300     310     320
TACEAFRRGF LMLYPESTFY CLICCAVLEV MWKNVGRHVA

     330     340     350     360
PHMGAHPATA PFHLGGAIFG PLLGLLVLLA GVCVFVLFQI

     370     380     390     400
EASGPAIACQ YFTLYYAFYV AVLPTMSLAC LAGTAIHGLE

     410     420     430     440
ERELDTVKNP TRSLDVLLM GAALGQMGIA YESIVAIVAK

     450     460     470     480
RPHELLNRLI LAYSLLLILQ HIAQNLFIIE GLHRRPLWET

     490     500     510     520
VPEGLAGKQE AEPFRRGSLI EIGQGLQRAS LAYIHSYSHL

     530     540     550     560
NWKRRLKEI SLFLILCNIT LWMMPAFGIH PEFENGLEKD

     570     580     590
FYGYQIWFAT VNFGLPLGVF YRMHVSGLV EVYLGA

```

A cDNA and a chromosomal sequence encoding the OTOP3 protein is available from the NCBI database as accession no. BK000568 and AC087651, respectively.

[0076] An amino acid sequence for the protein encoded by the human OTUD7A gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q8TE49, shown below as SEQ ID NO:29.

```

      10      20      30      40
MVSSVLNPNT SAECWAALLH DPMTLDMDAV LSDEVRSSTGA

      50      60      70      80
EPGLARDLLE GKNWDLTAAL SDYEQLRQVH TANLPHVENE

      90     100     110     120
GRGPKQPERE PQPGHKVERP CLQRQDDIAQ EKRLSRGISH

     130     140     150     160
ASSAIVSLAR SHVASECNNE QFPLEMPIYT FQLPDLVSYS

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     170     180     190     200
EDERSFIERD LIEQATMVAL EQAGRINWWS TVCTSKRLL

     210     220     230     240
PLATTGDGNC LLHAASLGMW GFHDRDEVIR KALYTMRTG

     250     260     270     280
AEREALKRRW RWQQTQQNKE EEWEREWTEL LKLASSEPT

     290     300     310     320
HFSKNGGTGG GVDNSEDVYV ESLEEFHVEV LAHILRRPIV

     330     340     350     360
VVADTMLRDS GGEAFAPIPE GGIYLPLEVP PNRCHCSPLV

     370     380     390     400
LAYDQAHESA LVSMEQRDQQ REQAVIPLTD SEHKLLPLHE

     410     420     430     440
AVDPGKDWEW GKDDNDNARI AHLILSLEAK LNLHLSYMNV

     450     460     470     480
TWIRIPSETR APLAQESPT ASAGEDVQSL ADSLSDRDS

     490     500     510     520
VCSNSNSNNG KNGKDKKEKQ QRKEKDKTRA DSVANKLGSP

     530     540     550     560
SKILGIKLLK NMGGGLGLVH GKMGANSAN GKNGDSAERG

     570     580     590     600
KEKKAKSRKG SKEESGASAS TSPSEKTPPS PTDKAAGASP

     610     620     630     640
AEKGGGPRGD AWKYSTDVKL SLNILRAAMQ GERKFIFAGL

     650     660     670     680
LLTSHRHQFH EEMIGYYLTS AQERFSAEQE QRRRDATAA

     690     700     710     720
AAAAAAAAT AKRPPRRPET EGVFVPERAS PGPPTQLVLK

     730     740     750     760
LKERPSGPA AGRAARAAG GTASPGGAR RASASGPVPG

     770     780     790     800
RSPPAPARQS VIHVQASGAR DEACAPAVGA LRPCATYPQQ

     810     820     830     840
NRSLSQSYS PARAAALRTV NIVESLARAV PGALPGAAGT

     850     860     870     880
AGAAEHKSQT YINGEGALRD GLEFADADAP TARNGECGR

     890     900     910     920
GGPGPVQRRQ QRENCAFYGR AETEHYCSYC YREELRRRE

ARGARP

```

A cDNA sequence encoding the OTUD7A protein is available from the NCBI database as accession no. AJ430383.

[0077] An amino acid sequence for the protein encoded by the human PDE3A gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q14432, shown below as SEQ ID NO:30.

```

      10      20      30      40
MAVPGDAARV RDKPVHSGVS QAPTAGRDCR HRADPASPRD

      50      60      70      80
SGCRGCWGDL VLQPLRSSRK LSSALCAGSL SELLALLVRL

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90	100	110	120
VRGEVGCDE	QCKEAAAAEE	EAAAPGAEGB	VEPGPRGGAP
130	140	150	160
GGGARLSPWL	QPSALLESLL	CAFEWMGLYL	LRAGVRLPLA
170	180	190	200
VALLAACCGG	EALVQIGLGV	GEDHLLSLPA	AGVVLSCLA
210	220	230	240
ATWLIVIRLRL	GVLMIALISA	VRTVSLISLE	REKVAWRPYL
250	260	270	280
AYLAGVLGIL	LARYVEQILP	QSAEAAPREH	LGSQLIAGTK
290	300	310	320
EDIPVFKRRR	RSSSVVSAEM	SGCSSKSHRR	TSLPCIPREQ
330	340	350	360
LMGHSEWDHK	RGPRGSQSSG	TSITVDIAVM	GEAHGLITDL
370	380	390	400
LADPSLPPNV	CISLRAVSNL	LSTQLTFQAI	HKPRVNPVTS
410	420	430	440
ISENYTCSDS	EESSEKDKLA	IPKRIRRSPL	PGLLRVSST
450	460	470	480
WTTTTSATGE	PTLEPAPVRR	DRSTSILQIE	APSSSPDSWN
490	500	510	520
NPVMMTLTKS	RSFTSSYALS	AANHVKAKKQ	SRPGALAKIS
530	540	550	560
PLSSPCSSPL	QGTPASSLVS	KISAVQFPES	ADTTAKQSLG
570	580	590	600
SHRALTYTQS	APDLSPQILT	PPVICSSCGR	PYSQGNPADE
610	620	630	640
PLERSGVATR	TPSRIDDTAQ	VISDYEINNN	SDSSDIVQNE
650	660	670	680
DETECLREPL	RKASACSTYA	PETMMELDKP	ILAPEPLVMD
690	700	710	720
NLDSIMEQLN	TWNFPIDLV	ENIGRKCGRI	LSQVSYRLFE
730	740	750	760
DMGLFEAFKI	PIREEMNYPH	ALEIGYRDIP	YHNRIHATDV
770	780	790	800
LHAVWYLTQ	PIPGLSTVIN	DHGSTDSDS	DSGFTHGHMG
810	820	830	840
YVESKTYNVT	DDKYGCISGN	IPALELMALY	VAAAMHDYDH
850	860	870	880
PGRTNAPLVA	TSAPQAVLYN	DRSVLENHHA	AAAWNLEMSR
890	900	910	920
PEYNFLINLD	HVEFKHFREL	VIEAILATDL	KKHEDFVAKF
930	940	950	960
NGKVNDVGI	DWTNENDRL	VCQMCIKLAD	INGPAKCKEL
970	980	990	1000
HLQWTDGIVN	EFYEQGDEEA	SLGLPISFFM	DRSAPQLANL
1010	1020	1030	1040
QESFISHIVG	PLCNSYDSAG	LMPGKWVEDS	DESGDIDDPE
1050	1060	1070	1080
EEEEAPAPN	EEETCENNES	PKKTEKRRK	IYCQITQHLL

-continued

1090	1100	1110	1120
QNHKMWKKVI	EEEQRLAGIE	NQSLDQTPQS	HSSEIQIAIK
1130	1140		
EEEEKGGKPR	GEEIPTQKPD	Q	

A cDNA sequence encoding the PDE3A protein is available from the NCBI database as accession no. M91667.

[0078] An amino acid sequence for the protein encoded by the human POLK gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database, shown below as SEQ ID NO:31.

10	20	30	40
MDSTKEKCDS	YKDDLLLRMG	INDNKAGMEG	LDKEKINKII
50	60	70	80
MEATKGSRFY	GNELKKEKQV	NQRIENMMQ	KAQITSQQLR
90	100	110	120
KAQLQVDRFA	MELEQSRNLS	NTIVHIDMDA	FYAAYEMRDN
130	140	150	160
PELKDKPIAV	GSMSMLSTSN	YHARREGVRA	AMPGFIAKRL
170	180	190	200
CPQLIIVPPN	FDKYRAVSKE	VKEILADYDP	NEMAMSLDEA
210	220	230	240
YLNITKHLEE	RQNPEDKRR	YFIKMGSSVE	NDNPGKEVKN
250	260	270	280
LSEHERSISP	LIFEESPSDV	QPPGDPFQVN	FEEQNNPQIL
290	300	310	320
QNSVVFQTS	QEVVKEIRER	IEQKTTLTAS	AGIAPNTMLA
330	340	350	360
KVCSKKNKPN	GQYQILPNRQ	AVMDFIKDLP	IRKVSIGIGKV
370	380	390	400
TEKMLKALGI	ITCTELYQQR	ALLSLESET	SWHYFLHISL
410	420	430	440
GLGSTHLIRD	GERKSMSVER	TESEINKAEE	QYSICQELCS
450	460	470	480
ELAQLQKER	LKGRVTIKL	KNVNFVKTR	ASTVSVSVST
490	500	510	520
AEEIFAIAKE	LLKTEIDADE	PHPLRLRIMG	VRISFPNEE
530	540	550	560
DRKHQORSII	GELQAGNQAL	SATECTLEKT	DKDKFVKPLE
570	580	590	600
MSHKKSFEDE	KRSERKWSHQ	DTEKCEAVNK	QSFQTSQPEQ
610	620	630	640
VLKKMNENL	EISENSDDCQ	ILTCPVCFRA	QGCISLEALN
650	660	670	680
KHVDECLDGP	SISENEKMF	CSHVSATKVN	KKENVPASSL
690	700	710	720
CEKQDYEAHP	KIKEISSVDC	IALVDTIDNS	SKAESIDALS
730	740	750	760
NKHSKEECSS	LPSKSFNIEH	CHQNSSTVS	LENEDVGSFR
770	780	790	800
QYRQPYLCE	VKTGQALVCP	VCNVEQKTS	LTLENVHVDV

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810 820 830 840
CINKSFIQEL RKDKENPVNQ PKESSRSTGS SSGVQKAVIR

850 860 870
TKRPGLMTKY STSKKIKPNN PKHTLDIFEK

[0079] A cDNA and a chromosomal sequence encoding the POLK protein is available from the NCBI database as accession no. AB027564 and AY273797, respectively.

[0080] An amino acid sequence for the protein encoded by the human PRACI gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q96KF2, shown below as SEQ ID NO:32.

10 20 30 40
MICAHFSDQG PAHLTTSKSA PLSNKKSTL KHLGETRSD

50
GSACNSGISG GRGRKIP

A cDNA and a chromosomal sequence encoding the PRACI protein is available from the NCBI database as accession no. AF331 165 and CH471 109, respectively.

[0081] An amino acid sequence for the protein encoded by the human SERPINF1 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database, shown below as SEQ ID NO:33.

10 20 30 40
MQALVLLLCI GALLGHSSCQ NPASPPPEGS PDPDSTGALV

50 60 70 80
EEEDPFFKVP VNKLAASVSN FGIDLYRVRS STSPTINVLL

90 100 110 120
SPLSVATALS ALSLGAEQRT ESIIHRALYY DLISSPDIHG

130 140 150 160
TYKELDITVT APQKNLKSAS RIVFEKKLRI KSSFVAPLEK

170 180 190 200
SYGTRPRVLT GNPRDLQEI NNWVQAQMKG KLARSTKEIP

210 220 230 240
DEISILLGV AHFKGQWVIK FDSRKTSLED FYLDEERTVR

250 260 270 280
VPMMSDPKAV LRYGLSDLS CKIAQLPLIG SMSIIFFLPL

290 300 310 320
KVTQNLTLIE ESLTSEFIHD IDRELKTVQA VLTVPKLKLS

330 340 350 360
YEGEVIKSLQ EMKLQSLSDS PDFSKITGKP IKLTQVEHRA

370 380 390 400
GFEWNEDGAG TTPSPGLQPA HLTFFLDYHL NQPFIFVERD

410
TDTGALLFIG KILDPRGP

A cDNA and a chromosomal sequence encoding the SERPINF1 protein is available from the NCBI database as accession no. M76979 and U29953, respectively.

[0082] An amino acid sequence for the protein encoded by the human SSUH-12 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9Y2M2, shown below as SEQ ID NO:34.

10 20 30 40
MPSPVGLIRA LPLPWPQFLA CTLRRLAGPR ESTGPSQKPP

50 60 70 80
PLCSVPCRVP AMTEEVAREA LLSEVDSKCC YSSTVAGDLV

90 100 110 120
IQELKRQTLIC RYRLETESES RISEWTFQPF TNHSVDGPQR

130 140 150 160
GASPRLDWIK VQGPPMFQED TRKFQVPHSS LVKECHKCHG

170 180 190 200
RGRYKCSGCH GAGTVRCPSC CGAKRKAKQS RRCQLCAGSG

210 220 230 240
RRRCSTCSGR GNKICATCKG EKKLLHFIQL VIMWKNSLEE

250 260 270 280
EVSEHRLNCP RELAKAKGE NLFKDENSVV YPIVDFPLRD

290 300 310 320
ISLASQRGIA EHSAAALASRA RVLQQRQTIE LIPLTEVHYW

330 340 350
YQGTKYVYYI YGTDHQVYAV DYPERYCCGC TIV

A cDNA and a chromosomal sequence encoding the SSUH2 protein is available from the NCBI database as accession no. AB024705 and AC034187, respectively.

[0083] An amino acid sequence for the protein encoded by the human TMISF4 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. P48230, shown below as SEQ ID NO:35.

10 20 30 40
MCTGGCARCL GGTLIPLAFF GFLANILLFF PGKVIDDND

50 60 70 80
HLSQEIWEEG GILGSGVLM I FPALVELGLK NNDCCGCCGN

90 100 110 120
EGCGKRFAMF ISTIFAVVGE IGAGYSFIIS AISINKGPKC

130 140 150 160
LMANSTWGYP FHDGDYLNDE ALWNKCREPL NVVPWNILIF

170 180 190 200
SILLVVGIGI MVLCAIQVNN GLLGTLGCGC QCCGCCGGDG

PV

A cDNA and a chromosomal sequence encoding the TM4SF4 protein is available from the NCBI database as accession no. U31449 and C1471052, respectively.

[0084] The following genes are positive regulators of T cells as detected by increased T cell proliferation (see Table 3): ABCB1, ASAP1, ATP10A, DEAF1, FOXP1, ITGAX, LCE6A, LCP2, LEFTY1, MYC, NAT8B, OLFM3, and PLD6. Table 7 provides additional positive regulators of T cells as detected by increased T cell proliferation.

[0085] An amino acid sequence for the protein encoded by the human ATP10A gene that is a positive regulator of T cells as detected by increased cell proliferation is available from the UniPROT database as accession no. 060312, shown below as SEQ ID NO:36.

```

      10      20      30      40
MEREPAGTEE PGPPGRRRRR EGRTRTVRSN LLPPPGAEDP

      50      60      70      80
AAGAAGKERR RRRGCAQHLL DNRLKTKYT LLSFLPKNLF

      90     100     110     120
EQFHRPANVY FVFIALLNFV PAVNAFQPL ALAPVLFILA

     130     140     150     160
ITAFRDLWED YSRHRSDHKL NHLGCLVESR EEKKYVNREW

     170     180     190     200
KEIHVGDEVR LRCNEIFPAD ILLSSSDPD GLCHIETANL

     210     220     230     240
DGETNLKRRQ VVRGESELVS EENPLFTSV IECEKPNNDL

     250     260     270     280
SRERGCIHD NGKKAGLYKE NLLIRGCTLR NTDVVGIVI

     290     300     310     320
YAGHETKALL NNSGPRYKRS KLERQMNCV LWCVLLVCM

     330     340     350     360
SLESVGHGL WIWRYQEKKS LEYVPKSDGS SISPVTAAYV

     370     380     390     400
SELTMIIVIQ VLIPISLYVS IEIVKACQVY FINQDMQLYD

     410     420     430     440
EETDSQLQCR ALNITEDIGQ IQYIFSDKTG TLTENKMVER

     450     460     470     480
RCTVSGVEYS HDANAQLRLR YQEADSEEEE VVPRGGSVSQ

     490     500     510     520
RGSIGSHQSV RVVHRTQSTK SHRRIGSRAE AKRASMLSKH

     530     540     550     560
TAFSSPMEKD ITPDPKLEK VSECDKSLAV ARHQEHLAH

     570     580     590     600
LSPELSDVED FFIALTICNT VVVTSPDQPR TKVRVRFELK

     610     620     630     640
SPVKTIEDEL RRFTPSCLTS GCSSIGSLAA NKSSHKIGSS

     650     660     670     680
FPSTPSSDGM LIRLEERLQG PTAIASNGY SSQADNWASE

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      690      700      710      720
LAQEQESERE LRYEAESPDE AALVYAARAY NCVIVERLHD

      730      740      750      760
QVSVELPHLG RLTFELLHTL GEDSVRKRMS VVIRHPLIDE

      770      780      790      800
INVYTKGADS VVMDLLQPCS SVDARGRHQK KIRSKTQNYL

      810      820      830      840
NVYAABGLRT ICIAKRVISK EYACWLQSH LEAESLENS

      850      860      870      880
EELLFQSAIR LETNIHLIGA TGIEDRLQDG VPETISKLRQ

      890      900      910      920
AGLQIIVLTG DKQETAVNIA YACKLLDHDE EVITLNATSQ

      930      940      950      960
EACAALLDQC LCYVQSRGLQ RAPEKTKGVK SMRFSSLCPP

      970      980      990     1000
STSTASGRRP SLVIDGRSLA YALEKNLEDK FLFLAKQCRS

     1010     1020     1030     1040
VICCRSTPLQ KSMVVKLVRK KLKAMTLAIG DGANDVSMIQ

     1050     1060     1070     1080
VADVGVGISG QEGMQAVMAS DEAVPKERYL ERLLLHGHWH

     1090     1100     1110     1120
CYSRLANMVL YFFYKNTMEV GLLFWQFFC GFSASTMIDQ

     1130     1140     1150     1160
WYLIFENLLF SSLPPLVTGV LDRDVPANVL LINPQLYKSG

     1170     1180     1190     1200
QNMEEYRPRP FWENMADAAQ QSLVCFSSIPY LAYYDSNVDL

     1210     1220     1230     1240
FTWGTPIVTI ALLTFLLHLG IETKIWTWIN WITCGFSVLL

     1250     1260     1270     1280
FFTVALIYNA SCATCYPPSN PYWTMQALLG DPVFYLTCLM

     1290     1300     1310     1320
TPVAALLPRL FERSLQGRVE PTQLQLARQL TRKSPRRCSA

     1330     1340     1350     1360
PKETFAGGRL PKDSGTEHSS GRTVKTSVPL SQPSWHTQQP

     1370     1380     1390     1400
VCSLEASGEP STVDMSPVVR EHTLLEGLSA PAPMSSAPGE

     1410     1420     1430     1440
AVLRSPGGCP EESKVRAAST GRVTPLSSLF SLPTESLLNW

     1450     1460     1470     1480
ISSWSLVSRG GSVLQFSRTE QLADGQAGRG LPVQPHSGRS

     1490
GLQGPDRHLL IGASSRRSQ

```

An amino acid sequence for the protein encoded by the human LCE6A gene that is a positive regulator of T cells as detected by increased cell proliferation is available from the UniPROT database as accession no. A0A183, shown below as SEQ U) NO:37.

```

      10      20      30      40      50
MSQQKQSWK PPNVPKCSPP QRSNPCLAPY STPCGAPHSE GCHSSSRPPE

      60      70      80
VQKPRRARQK LRCLSRGTTY HCKEECEGD

```

A cDNA and a chromosomal sequence encoding the LCE6A protein is available from the NCBI database as accession no. DQ991251 and AL162596, respectively.

[0086] An amino acid sequence for the protein encoded by the human NAT8B gene that is a positive regulator of T cells as detected by increased cell proliferation is available from the UniPROT database as accession no. Q9UHF3, shown below as SEQ ID NO:38.

```

      10      20      30      40      50
MAPYHIRKYQ ESDRKSVVGL LSGGMAEHAP ATFRRLKLP RTLILLGGA
      60      70      80      90     100
LALLLVSGSW ILALVFSLSL LPALWFLAKK PWTRYVDIAL RTDMSDITKS
     110     120     130     140     150
YLSECGSCFW VAEESEKVVG TVGALPVDDP TLREKRLQLF HLSVDNEHRG
     160     170     180     190     200
QGIKALVVRT VLQFARDQGY SEVVDTSNI QLSAMGLYQS LGFKKTGQSF
     210     220
FHVWARLVDL HTVHFIIYHL SAQAGRL

```

A cDNA sequence encoding the NAT8B protein is available from the NCBI database as accession no. AF185571

Negative Regulators of T Cells

[0087] The following genes are negative regulators of T cells as detected by interferon- γ production (see Table 4): ACER2, ADGRVI, AIF1L, ALPL, AMACR, AMZ1, ARHGAP30, ARHGDIB, ARHGEF11, ARL11, ATP2A2, B3GNT5, BAC H2, BLM, BSG, BTBD2, BTLA, BTRC, CAIL, CASTOR2, CBLB, CCNT2, CCSER1, CD37, CD44, CD5, CD52, CD55, CDK6, CEACAMI, CEBPA, CEBPB, CEP164, CKAP2L, CLCN2, CLDN25, COLQ, CST5, CTNNA1, CYP24A, DDIT4L, DENND3, DGKG, DGKK, DGKZ, DSCI, EBF2, ECEL1, EIF3K, EPB41, EPS8L1, FAM35A, FAM53B, FAM83A, FKRP, FOXA3, FOXF1, FOXF2, *FOX3*, FOXJ1, FOXL2, FOXL2NB, GiABRQ,

PIP5K1B, PITPNA, POGZ, POLK, POU2AF1, PSTPI1, PTPN12, PTPRC, PVRIG, RAB14, RBP7, RETREGI, RFC2, RHCE, RN^F19B, RNF2, RUSC2, SELPLG, SETD1IB, SH3KBP1, SIGLEC6, SIPA1L1, SLA, SLA2, SLC26A4, SLC44A5, SLC45A1, SLC6A8, SLC6A9, SMAD9, SMLKGP, SOCS3, SOX13, SPATA31A1, SPN, SPOCK3, SPRED1, STAP1, STK35, SULLT6B1, SYTI5, TEC, TIAMI, TMEMI51A, TMEM87B, TMPRSS1 IE,

TNNT2, TRIB2, TRLM28, TSPAN UBASH3B, UBQLN4, UBXN7, UNC119, UPP1, VPS28, WLS, ZKSCAN4, ZNF445, and ZNF474. Table 7 provides additional negative regulators of T cells as detected by interferon- γ production.

[0088] Sequences and other information relating to these genes, and their encoded proteins, is available, for example from the NCBI and UniPROT databases, which are incorporated by reference.

[0089] A few examples of protein sequences encoded by some of the genes detected as negative regulators of T cells by interferon- γ production are provided. For example, an amino acid sequence for the protein encoded by the human ATI 1L gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9BQI0, shown below as SEQ ID NO:39.

```

      10      20      30      40      50
MSGELSNRFQ GGKAFGLLKA RQERRLAEN REFLCDQKYS DEENLPEKLT
      60      70      80      90     100
AFKEKYMEDF LNNEGEIDLM SLKRMMKLG VPKTHLEMKK MISEVTGGVS
     110     120     130     140     150
DTISYRDFVN MMLGKRSVAVL KLVMMFEGKA NESSPKPVGP PPERDIASLP

```

GATA3, GATA4, GATA6, GCM2, GCSAM, GCSAML, GMFG, GNL3L, GRAP, GRTB2, GRIA1, GTSF1IL, HRI-2, HiYLS1, IKZF1, IKZF3, IL2RB, INPPL1, JMJD1C, KCNV1, KRIT1, LaiBI, LAPIM5, LAT2, LAXI, LCK, LENEPI, LMO4, LRRC25, LRRC4B, LYN, MAB21L2, MAP4K1, MBIP, MBOA Ti, MIETITL23, MIPEP, MIPOLI, MMP2I, MSMB, MUC1, MUC21, MUC8, N4BPI, NAIF1, NDNF, NFFAIC1, NFKB2, NFKBIA, NKX2-1, NKX2-3, NMB, NR2F1, ODF4, OPRD1, ORC5, OTUD4, PASDI, PBK, PCBP2, PDLIM1, PDPN, PECAIM1, PIP5K1 A,

A cDNA and a chromosomal sequence encoding the AIF1L protein is available from the NCBI database as accession no. AL136566 and AL157938, respectively.

[0090] An amino acid sequence for the protein encoded by the human ARHGDIB gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. P52566, shown below as SEQ ID NO:40.

```

      10      20      30      40      50
MTEKAPEPHV EEDDDDELDS KINYKPPQK SLKELQEMDK DDESLIKYKK

      60      70      80      90     100
TLLGDGPVVT DPKAPNVVVT RLTLVCESAP GPITMDLTGD LEALKKETIV

     110     120     130     140     150
LKEGSEYRVK IHFKVNRDIV SGLKYVQHTY RTGVKVDKAT FMVGSYGPRP

     160     170     180     190     200
EEYEFLTPVE EAPKGMLARG TYHNKSFFTD DDKQDHLWE WNLSIKKEWT

```

E

A cDNA and a chromosomal sequence encoding the ARHGDIB protein is available from the NCBI database as accession no. L20688 and CH471094, respectively.

[0091] An amino acid sequence for the protein encoded by the human BLM gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. P54132, shown below as SEQ ID NO:41.

```

      10      20      30      40      50
MAAVPQNNLQ EQLERHSART LNNKLSLSKP KFSGFTFKKK TSSDNNVSVT

      60      70      80      90     100
NVSVAKTPVL RNKDVNVTED FSFSEPLPNT TNQQRVKDFF KNAPAGQETQ

     110     120     130     140     150
RGGSKSLLPD FLQTPKEVVC TTQNTPTVKK SRDTALKKLE FSSSPDSLST

     160     170     180     190     200
INDWDDMDDF DTSETSKSFV TPPQSHFVRV STAQSKKKGK RNFFKAQLYT

     210     220     230     240     250
TNTVKTDLPP PSSESEQIDL TEEQKDDSEW LSSDVICIDD GPIAEVHINE

     260     270     280     290     300
DAQESDSLKT HLEDERDNSE KKKNLEEAEI HSTEKVPCIE FDDDDYDTDF

     310     320     330     340     350
VPPSPPEEII ASSSSSKCLS TLKDLDTSDR KEDVLSTSKD LLSKPEKMSM

     360     370     380     390     400
QELNPETSTD CDARQISLQQ QLIHVMEHIC KLIDTIPDDK LKLLDCGNEL

     410     420     430     440     450
LQQRNIRKRL LTEVDFNKSD ASLLGSLWRY RPSLDGPME GDSCPTGNSM

     460     470     480     490     500
KELNFSHLPS NSVSPGDCLL TTTLGKTGFS ATRKNLFERP LFNTHLQKSF

     510     520     530     540     550
VSSNWAETPR LGKKNESSYF PGNVLTSTAV KDQNKHTASI NDLERETQPS

     560     570     580     590     600
YDIDNFDIDD FDDDDDWEDI MHNLAASKSS TAAYQPIKEG RPIKSVSERL

     610     620     630     640     650
SSAKTDCLPV SSTAQNINFS ESIQNYTDKS AQNLASRNK HERFQSLSFP

     660     670     680     690     700
HTKEMMKIFH KKFGHLNFRT NQLEAINAAL LGEDCFILMP TGGGKSLCYQ

     710     720     730     740     750
LPACVSPGVV VVISPLRSLI VDQVQKLTSL DIPATYLTGD KTDSEATNIY

     760     770     780     790     800
LQLSKKDPII KLLYVTPEKI CASNRLISTL ENLYERKLLA RFVIDEAHCV

     810     820     830     840     850
SQWGHDFRQD YKRMNMLRQK FPSVPVMALT ATANPRVQKD ILTQLKILRP

```

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860	870	880	890	900
QVFSMSFNRH	NLKYYVLPPK	PKKVAFDCLE	WIRKHHPYDS	GIIYCLSRRE
910	920	930	940	950
CDTMADTLQR	DGLAALAYHA	GLSDSARDEV	QQKWINQDGC	QVICATIAFG
960	970	980	990	1000
MGIDKPDVRF	VIHASLPKSV	EGYYQESGRA	GRDGEISHCL	LFYTYHDVTR
1010	1020	1030	1040	1050
LKRLIMMEKD	GNHHTRETHF	NNLYSMVHYC	ENITECRRIQ	LLAYFGENG
1060	1070	1080	1090	1100
NPDFCKKHPD	VSCDNCKTK	DYKTRDVTDD	VKSIVRFVQE	HSSSQGMRNI
1110	1120	1130	1140	1150
KHVGPSGRFT	MNMLVDIFLG	SKSAKIQSGI	FGKGSAYSRH	NAERLPKKLI
1160	1170	1180	1190	1200
LDKILDEDLY	INANDQAIAY	VMLGNKAQTV	LNGNLKVDPM	ETENSSSVKK
1210	1220	1230	1240	1250
QKALVAKVSQ	REEMVKKCLG	ELTEVCKSLG	KVFGVHYFNI	FNTVTLKKLA
1260	1270	1280	1290	1300
ESLSSDPEVL	LQIDGVTEDE	LEKYGAEVIS	VLQKYSEWTS	PAEDSSPGIS
1310	1320	1330	1340	1350
LSSSRGPGRS	AAEELDEEIP	VSSHYPASKT	RNERKRKKMP	ASQSRKRRKT
1360	1370	1380	1390	1400
ASSGSKAKGG	SATCRKISSK	TKSSSIIGSS	SASHTSQATS	GANSKLGIMA
1410				
PPKPINRPFL	KPSYAFS			

A cDNA and a chromosomal sequence encoding the BLM protein is available from the NCBI database as accession no. U39817 and AY886902, respectively.

[0092] An amino acid sequence for the protein encoded by the human BSG gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q7KTJ7, shown below as SEQ ID NO:42.

10	20	30	40	50
MEAKFLASAL	SFLSIFLAII	AQSLANDLSK	ESTEFESPT	IYYGDPVVNL
60	70	80	90	100
GQFPSITCII	PITDQIHWLK	NGEPITRHL	RHGRDDHAYV	LSAIEGK
110	120	130	140	150
HKIEAHLVSR	HALKVHEGRY	QCNRRRGSYI	LHVRDPKGVG	AGAGEPTESG
160	170	180	190	200
YQTIDELTPN	SADDFFTRAW	LEQQQQQQQL	PHQSHKLHKS	HLGYGNASLS
210	220	230	240	250
GSQFWHPSAG	GGGIHRVISA	TPPDFPPPR	NLLEQTVAPP	EPPTILYNPN
260	270	280	290	300
PTHPTASATA	TETSVLLTTA	HHHAHQQQQL	QQSQHTLNA	FQLPLPPRPN
310	320	330	340	350
PGQNERVQTY	APHYVPPVVV	SGAGAGAGAD	PGAGASGEQT	TISAATSTRA
360	370	380	390	400
MMGGGGGVAG	AGFSAGASGP	MLGAGGHMLM	GGQGHQVHLQ	HQTLPLPKMD
410	420	430	440	450
KLVPNYDNAE	HQMKFYDIRS	PLVLSKNVKD	GTPGGVLIWK	KNGTAVTDVP

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```

      460      470      480      490      500
SLRGRFKLIA DENKFIIDKT DTNDDGKYSC EFDGVSKEIE VIARVVVRVP

      510      520      530      540      550
SNTAVVEGEK MSVTCVVVGT KPELTWTFAN VTLTNATDRF ILKPDDNGVP

      560      570      580      590      600
NAILTLDNVT LDDRGEYKCI GRNAANVYGG NTTTPASDVT TVRVKGKFAA

      610      620      630      640      650
LWPFLGICAE VLILCIIILI YEKRRNKSEL EESDTPQEQ KKKRRNYD

```

A cDNA and a chromosomal sequence encoding the BSG protein is available from the NCBI database as accession no. AE014134 and AAN10661.2, respectively.

[0093] An amino acid sequence for the protein encoded by the human BTBD2 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9BX70, shown below as SEQ ID NO: 43.

```

      10      20      30      40      50
MAAGSGGGR SCPPGVGVGP GTGGSPGPSA NAAATPAPGN AAAAAAAAAA

      60      70      80      90     100
AAAAAGPTPP APPGPGTDAQ AAGAERAEEA AGPGAAALQR EAAYNWQASK

      110     120     130     140     150
PTVQERFAPL FNNEVLCDVH FLVGKGLSSQ RIPAHRFVLA VGSVFDAMF

      160     170     180     190     200
NGGMATTSTE IELPDVEPAA FLALLKFLYS DEVQIGPETV MTTLYTAKKY

      210     220     230     240     250
AVPALEAHCV EFLKKNLRAD NAFMLLTQAR LFDEPQLASL CLENIDKNTA

      260     270     280     290     300
DAITAEGFTD IDLDTLVAVL ERDTLGIREV RLFNAVVRWS EAECQRQQLQ

      310     320     330     340     350
VTPENRRKVL GKALGLIRFP LMTIEEFAAG PAQSGILVDR EVVSLFLHFT

      360     370     380     390     400
VNPKEPRVEFI DRPRCCLRGK ECSINRFQQV ESRWGYSGTS DRIRFSVNKR

      410     420     430     440     450
IPVVGFGLYG SIHGPTDYQV NIQIIHTDSN TVLGQNDTGF SCDGSASTFR

      460     470     480     490     500
VMFKEPVEVL PNVNYTACAT LKGPDSHYGT KGLRKVTHES PTTGAKTCFT

      510     520
FCYAAGNNNG TSVEDGQIPE VIFYT

```

A cDNA and a chromosomal sequence encoding the BTBD2 protein is available from the NCBI database as accession no. AF355797 and AC004678, respectively.

[0094] An amino acid sequence for the protein encoded by the human CASTOR2 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. A6NHX0, shown below as SEQ ID NO:44.

```

      10      20      30      40      50
MELHILEHRL QVASVAKESI PLFTYGLIKL AFLSSKTRCK FFSLTETPED

      60      70      80      90     100
YTIIVDEEGF LELPSSEHLS VADATWLALN VVSGGGSFSS SQPIGVTKIA

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110	120	130	140	150
KSVIAPLADQ	NISVFMLSTY	QTDFILVRER	DLPFVTHTLS	SEFTILRVVN
160	170	180	190	200
GETVAAENLG	ITNGFVKPKL	VQRPVIHPLS	SPSNRFCVTS	LDPDTLPAVA
210	220	230	240	250
TLLMDVMFY	NGVKDPMATG	DDCGHIRFFS	FSLIEGYISL	VMDVQTQQR
260	270	280	290	300
PSNLLFTSAS	GELWKVMVRIG	GQPLGFDECG	IVAQISEPLA	AADIPAYYIS
310	320			
TFKFDHALVP	EENINGVISA	LKVSQAEKH		

A cDNA and a chromosomal sequence encoding the CAS-TOR2 protein is available from the NCBI database as accession no. BC147030 and AC245150, respectively.

[0095] An amino acid sequence for the protein encoded by the human CCSER1 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no, Q9C013, shown below as SEQ ID NO:45.

10	20	30	40	50
MGDSGSRST	LVSRLPIFRR	SINRRHDSLP	SSPSSSNTVG	VHSSSPSSTN
60	70	80	90	100
SSSGSTGKRR	SIFRTPSISF	HHKKGSEPKQ	EPTNQNLIS	NGAQPGHSNM
110	120	130	140	150
QKLSLEEHIK	TRGRHSGVFS	SSRNKKITRS	LTEDFEREKE	HSTNKNVFIN
160	170	180	190	200
CLSSGKSEGD	DSGFTEQTR	RSVKQSTRKL	LPKSFESSHYK	FSKPVLSQS
210	220	230	240	250
ISLVQQSEFS	LEVTQYQERE	PVLVRASPSC	SVDVTERAGS	SLQSPLLSAD
260	270	280	290	300
LTTAQTPEF	LALTEDSVSE	MDAFKSGSM	ASHCDNFGHN	DTSQMSLNS
310	320	330	340	350
AAVIKTTTEL	TGTVPCAIS	PGKYRLEGQC	STESNSLPET	SAANQKEVLL
360	370	380	390	400
QIAELPATSV	SHSESNLPAD	SERENIGLQ	NGETMLGTNS	PRKLGIFYEQH
410	420	430	440	450
KAIAEHVKGI	HPISDSKIIP	TSGDHHIFNK	TSHGYEANPA	KVLASSLSPE
460	470	480	490	500
REGRFIERRL	RSSSEGTAGS	SRMILKPKDG	NIEEVNSLRK	QRAGSSSSKM
510	520	530	540	550
NSLDVLNNLG	SCELEDDLM	LDLEFLEEQS	LHPSVCREDS	YHSVVSCAAV
560	570	580	590	600
VLTPMEPMIE	MKKREEPEFP	EPSKQNLCLK	LTQDQDEAR	CSHISRMFNS
610	620	630	640	650
PSADWPLQGV	EENGIDSLP	FRLMLQDCTA	VKTLLKMKR	VLQESADMSP
660	670	680	690	700
ASSTSLPV	PLTEEPVFPK	DIMKDECSML	KLQLEKDEL	ISQLQEELGK
710	720	730	740	750
VRHLQKAFAS	RVDKSTQTEL	LCYDGLNLKR	LETVQGGREA	TYRNRIVSQN

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```

      760      770      780      790      800
LSTRDRKAH TPTEDRFRYS AADQTSFYKN KTCQLPSLCL SNFLKDKELA

      810      820      830      840      850
EVIKHSRGTY ETLTSDVTQN LRATVGQSSL KPTAKTEGLS TFLEKPKDQV

      860      870      880      890      900
ATARQHSTFT GRFGQPPRGP ISLHMYSRKN VFLHHNLHST ELQTLGQQDG

```

A cDNA and a chromosomal sequence encoding the CCSER1 protein is available from the NCBI database as accession no. AB051467 and AC093729, respectively.

[0096] An amino acid sequence for the protein encoded by the human CLCN2 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. P51788, shown below as SEQ ID NO:46,

```

      10      20      30      40      50
MAAAAAEEGM EPRALQYEQT LMYGRYTQDL GAFAKEEAAR IRLGGPEPFWK

      60      70      80      90     100
GPPSSRAAPE LLEYGRSRCA RCRVCSVRCH KFLVSRVGED WIFLVLLGLL

      110     120     130     140     150
MALVSWVMDY AIAACLQAQQ WMSRGLNTSI LLQYLAWVTY PVVLITFSAG

      160     170     180     190     200
FTQILAPQAV GSGIPEMKTI LRGVVLKEYL TLKTFIAKVI GLTCALGSGM

      210     220     230     240     250
PLGKEGPFVH IASMCAALLS KFLSLFGGIY ENESRNTEML AAACAVGVGC

      260     270     280     290     300
CFAAIPGGVL PSIEVTSTFF AVRNYWRGFF AATFSAFIFR VLAVWNRDEE

      310     320     330     340     350
TITALFKTRF RLDFFFDLQE LPFAFVIGIA SGFGGALFVY LNRKIVQVMR

      360     370     380     390     400
KQKTINRFIM RKRLLPALV TLLISTLTFP PGFGQFMAGQ LSQKETLVTL

      410     420     430     440     450
FDNRTWVRQG LVEELEPPST SQAWNPPRAN VFLLTVIFIL MKFWMSALAT

      460     470     480     490     500
TIPVPCGAFM PVEVIGAAFG RLVGESMAAW FPDGIHTDSS TYRIVPGGYA

      510     520     530     540     550
VVGAAALAGA VTHTVSTAVI VFELTGQIAH ILPVMIAVIL ANAVAQSLQP

      560     570     580     590     600
SLYDSIIRIK KLPYLPELGW GRHQYRVRV EDIMVRDVPV VALSCTFRDL

      610     620     630     640     650
RLALHRTKGR MLALVESPEL MILLGSIERS QVVALLGAQL SPARRRQHMV

      660     670     680     690     700
ERRATQTSPL SDQEGPPTPE ASVCFQVNTS DSAPPAARGE THKPLKPALK

      710     720     730     740     750
RGPSVTRNLG ESPTGSAESA GIALRSLFCG SPPPEAAASEK LESCEKRKLK

      760     770     780     790     800
RVRISLASDA DLEGEMSPPE ILEWEEQQLD EPVNFSDCKI DPAPFQLVER

      810     820     830     840     850
TSLHKHTTIF SLLGVDHAYV TSIGRLIGIV TLKELRKAIE GSVTAQGKVK

      860     870     880     890
RPPLASFRDS ATSSSDTETT EVHALWGPHS RHGLPREGSP SDSDDKCQ

```

A cDNA and a chromosomal sequence encoding the CLCN2 protein is available from the NCBI database as accession no. S77770 and AC078797, respectively.

[0097] An amino acid sequence for the protein encoded by the human EBF2 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9 AK2, shown below as SEQ ID NO:47.

```

      10      20      30      40      50
MFGIQDTLGR GPTLKEKSLG AEMDSVRSWV RNVGVVDANV AAQSGVALSR

      60      70      80      90     100
AHFEKQPPSN LRKSNFFHFV LALYDRQGQP VEIERTAFVD EVENDKEQGN

      110     120     130     140     150
EKTNNGTHYK LQLLYSNGVR TEQDLYVRLI DSVTKQPIAY EGQNKNPENC

      160     170     180     190     200
RVLLTHEVMC SRCCEKSCG NRNETPSDPV IIDRFFLKFF LKCNQNCCLKT

      210     220     230     240     250
AGNPRDMRRF QVVLSTTVNV DGHVLAVSDN MFVHNNSKHG RRARRLDPSE

      260     270     280     290     300
ATPCIKAISSP SEGWTGGAM VIIIGDNFFD GLQVVFGTML VWSELITPHA

      310     320     330     340     350
IRVQTPPRHI PGVVEVTLSY KSKQFCKGAP GRFIYTALNE PTIDYGFQRL

      360     370     380     390     400
QKVIPIRHPGD PERLAKEMLL KRAADLVEAL YGTPHNNQDI ILKRAADIAE

      410     420     430     440     450
ALYSVPRNPS QLPALSSSPA HSGMMGINSY GSQLGVSISE STQGNNGGYI

      460     470     480     490     500
RNTSSISPRG YSSSSTPQOS NYSTSSNSMN GYSNVPMANL GVPGSPGFLN

      510     520     530     540     550
GSPTGSPYGI MSSSPTVGSS STSSILPFSS SVFPAVKQKS AFAPVIRPQG

      560     570
SPSPACSSGN GNGFRAMTGL VVPPM

```

A cDNA and a chromosomal sequence encoding the EBF2 (COE2) protein is available from the NCBI database as accession no. AY700779 and AC023566, respectively.

[0098] An amino acid sequence for the protein encoded by the human FAM83A gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q86UY5, shown below as SEQ ID NO:48.

```

      10      20      30      40      50
MSRSRHLGKI RKRLEDVKSQ WVRPARADFS DNESARLATD ALLDGGSEAY

      60      70      80      90     100
WRVLSQEGEV DFLSSVEAQY IQAQAREPPC PPDTLGGAEA GPKGLDSSSL

      110     120     130     140     150
QSGTYFPVAS EGSEPALHLS WASAEKPYLK EKSSATVYFQ TVKHNNIRDL

      160     170     180     190     200
VRRICITRISQ VLVILMDVFT DVEIFCDILE AANKRGVFVC VILDQGGVKL

      210     220     230     240     250
FQEMCDKVQI SDSHLKNISI RSVEGEIYCA KSGRKFAQI REKFIIISDWR

      260     270     280     290     300
FVLSGSYSFT WLCGHVHRNI LSKFTGQAVE LFDEEFRLHY ASSKPVMGLE

```

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310	320	330	340	350
SPRLVAPVPP	GAAPANGRLS	SSSGSASDRT	SSNPFSGRSA	GSHPGTRSVS
360	370	380	390	400
ASSGPCSPAA	PHPPPPPRFQ	PHQGPWGAPS	PQAHLSRPH	DGPPAAVYSN
410	420	430		
LGAYRPTRLQ	LEQLGLVPRL	TPTWRPFLQA	SPHF	

A cDNA sequence encoding the FAM83A protein is available from the NCBI database as accession no. DQ280322.

[0099] An amino acid sequence for the protein encoded by the human FOXF1 gene that is a negative regulator of T cells as detected by interferon- γ , production is available from the UniPROT database, shown below as SEQ ID NO:49.

10	20	30	40	50
MSSAPEKQQP	PHGGGGGGGG	GGAAMPAS	SGPSKAKKTN	AGIRRPEKPP
60	70	80	90	100
YSYIALIVMA	IQSSPTKRLT	LSEIYQFLQS	RFPFFRGSYQ	GWKNSVRHNL
110	120	130	140	150
SLNECFIKLP	KGLGRPGKGH	YWTIDPASEF	MFEEGSFRRR	PRGFRRKQQA
160	170	180	190	200
LKPMYSMMNG	LGFNHLPTY	GFQGSAGGLS	CPPNSLALEG	GLGMMNGHLP
210	220	230	240	250
GNVDGMALPS	HSVPHLPSNG	GHSYMGCGG	AAAGEYPHHD	SSVPASPLLP
260	270	280	290	300
TGAGGVMEPH	AVYSGSAAAW	PPSASAALNS	GASYIKQQPL	SPCNPAANPL
310	320	330	340	350
SGSLSTHSLE	QPYLHQNSHN	APAEQGIPR	YHSQSPSMCD	RKEFVFSFNA
360	370			
MASSMHSAG	GGSYHQQVT	YQDIKPCVM		

A cDNA and a chromosomal sequence encoding the FOXF1 protein is available from the NCBI database as accession no. U13219 and AF085343, respectively.

An amino acid sequence for the protein encoded by the human FOXL3 gene that is a negative regulator of T cells as detected by interferon-production is available from the UniPROT database as accession no. A8MTJ6, shown below as SEQ ID NO:50.

10	20	30	40	50
MALYCGDNFG	VYSQPGLPPP	AATAAAPGAP	PAARAPYGLA	DYAAPPAAAA
60	70	80	90	100
NPYLWLNGPG	VGGPPSAAAA	AAAAYLGAPP	PPPPPGAAAG	PFLQPPPAAG
110	120	130	140	150
TFGCSQRPFA	QPAPAAPASP	AAPAGPGELG	WLSMASREDL	MKMVRPPYSY
160	170	180	190	200
SALIAMAISQ	APERKLTLSH	IYQFVADSFP	FYQRSKAGWQ	NSIRHNLNLSN
210	220	230	240	250
DCFKVKVPRDE	DDPGKGNWT	LDPNCEKMGD	NGNFRKRKR	RSEASNGSTV
260	270	280	290	300
AAGTSKSEEG	LSSGLGSGVG	GKPEEESPST	LLRPSHSPEP	PEGTKSTASS
310	320	330	340	350
PGGPMLTSTP	CLNTFFSSL	SLSVSSSVST	QRALPGSRHL	GIQGAQLPSS

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      360      370      380      390      400
GVFSPTTISE ASADTLQLSN STSNSTGQRS SYSPFPAST SGGQSSPFSS

      410      420
PFHNFMSVNS LIYPREGSEV

```

A cDNA and a chromosomal sequence encoding the *FOXL3* protein is available from the NCBI database as accession no. BN001222 and AC012671, respectively.

[0100] An amino acid sequence for the protein encoded by the human *FOXL2NB* gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q6ZUIU3, shown below as SEQ ID NO:51.

```

      10      20      30      40      50
MTRTPVGSAR TRPKPRKLGQ QRGKALQASS RLSESPALVK KRMPDACTLG

      60      70      80      90     100
RAGIGLPKMC LHMAVRHSA QKTGPGILQQ RQKPPAPRAS GGPALLGKRR

      110     120     130     140     150
GCSEAGSASL EPLSSSRAAA GCLNQVPLSP FLAGPRNTRR LPAPERERIE

      160     170
LAATLCLEGW PLRCLASKGK LHCVY

```

A cDNA and a chromosomal sequence encoding the *FOXL2NB* protein is available from the N-CBI database as accession no. AK125319 and AC092947, respectively.

[0101] An amino acid sequence for the protein encoded by the human *HYLS1* gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q96M11, shown below as SEQ ID NO:52.

```

      10      20      30      40      50
MEELLPDGQI WANMDPEERM LAAATAFTHI CAGQGEGDVR REAQSIQYDP

      60      70      80      90     100
YSKASVAPGK RPALPVQLQY PHVESNPVSE TVSEASQRLR KPVMKRKVLRL

      110     120     130     140     150
RKPDGEVLVT DESIISESES GTENDQDLWD LRQRLMNVQF QEDKESSFDV

      160     170     180     190     200
SQKFNLPHYE QGISQDQLIC SLQREGMGSP AYEQDLIVAS RPKSFILPKL

      210     220     230     240     250
DQLSRNRGKT DRVARYFEYK RDWDSIRLPG EDHRKELRWG VREQMLCRAE

      260     270     280     290
PQSKPQHIIYV PNNYLVPTK KRSALRWGVR CDLANGVIPR KLPFPLSPS

```

A cDNA and a chromosomal sequence encoding the *HYLS1* protein is available from the NCBI database as accession no. AK057477 and AP000842, respectively.

[0102] An amino acid sequence for the protein encoded by the human *LAMB1* gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. P07942, shown below as SEQ ID NO:53,

```

      10      20      30      40      50
MGLLQLLAFS FLALCRARVR AQEPEFSYGC AEGSCYPATG DLLIGRAQKL

      60      70      80      90     100
SVTSTCGLHK PEPYCIIVSHL QEDKKCFICN SQDPYHETLN PDSHLIENVV

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110	120	130	140	150
TTFAPNRLKI	WWQSENGVEN	VTIQLDLEAE	FHFTHLIMTF	KTFRPAAMLI
160	170	180	190	200
ERSSDFGKTW	GVYRYFAYDC	EASFPGISTG	PMKKVDDIIC	DSRYSDIEPS
210	220	230	240	250
TEGEVIFRAL	DPAFKIEDPY	SPRIQNLLKI	TNLRIKFVKL	HTLGDNLDS
260	270	280	290	300
RMEIREKYYY	AVYDMVVRGN	CFCYGHASEC	APVDGFNEEV	EGMVHGHCMC
310	320	330	340	350
RHNTKGLNCE	LCMDFYHDLF	WRPAEGRNSN	ACKKCNCNEH	SISCHFDMAV
360	370	380	390	400
YLATGNVSGG	VCDDCQHNTM	GRNCEQCKPF	YYQHPERDIR	DPNFCERCTC
410	420	430	440	450
DPAGSQNEGI	CDSYTDFTSG	LIAGQCRCKL	NVEGEHCDVC	KEGFYDLSSE
460	470	480	490	500
DPGGKSCAC	NPLGTIPGGN	PCDSETHGCH	CKRLVIGQHC	DQCLPEHWGL
510	520	530	540	550
SNDLDGCRPC	DCDLGGALNN	SCFAESGQCS	CRPHMIGRQC	NEVEPGYYFA
560	570	580	590	600
TLDHYLYEAE	EANIGPGVSI	VERQYIQDRI	PSWTGAGFVR	VPEGAYLEFF
610	620	630	640	650
IDNIPYSMEY	DILIRYEPQL	PDHWEKAVIT	VQRPGRIPTS	SRCGNTIPDD
660	670	680	690	700
DNQVVSLSPG	SRVVLPRPV	CFEKGTYNTV	RLELPQYTSS	DSDVESPYTL
710	720	730	740	750
IDSLVLMPYC	KSLDIFTVGG	SGDGVVTNSA	WETFQRYRCL	ENSRSVVKTP
760	770	780	790	800
MTDVCNRIIF	SISALLHQTG	LACECDPQGS	LSSVCDPNGG	QCQCRPNVVG
810	820	830	840	850
RTCNRCAPGT	FGFGPSGCKP	CECHLQGSVN	AFCNPVTGQC	HCFQGVYARQ
860	870	880	890	900
CDRCLPGHWG	FPSCQPCQCN	GHADDCDPVT	GECLNCQDYT	MGHNCERCLA
910	920	930	940	950
GYGDPPIIGS	GDHCRPCPCP	DGPDSGRQFA	RSCYQDPVTL	QLACVCDPGY
960	970	980	990	1000
IGSRDDCAS	GYFGNPSEVG	GSCQPCQCHN	NIDTTDPEAC	DKETGRCLKC
1010	1020	1030	1040	1050
LYHTEGEHCQ	FCRFGYYGDA	LQQDCRKCVC	NYLGTVQEH	NGSDCQCDKA
1060	1070	1080	1090	1100
TGQCLCLPNV	IGQNCRCAP	NTWQLASGTG	CDPCNCNAAH	SFGPSCNEFT
1110	1120	1130	1140	1150
GQCQCMPGFG	GRTCSECQEL	FWGDPDVECR	ACDCDPRGIE	TPQCDQSTGQ
1160	1170	1180	1190	1200
CVCVEGVEGP	RCDKCTRGYS	GVFPDCTPCH	QCFALWDVII	AELTNRTHRF
1210	1220	1230	1240	1250
LEKAKALKIS	GVIGPYRETV	DSVERKVSEI	KDILAQSPAA	EPLKNIGNLF
1260	1270	1280	1290	1300
EEAEKLIKDV	TEMMAQVEVK	LSDTTSQSNS	TAKELDSLQT	EAESLDNTVK
1310	1320	1330	1340	1350
ELAEQLEFIK	NSDIRGALDS	ITKYFQMSLE	AEERVNASTT	EPNSTVEQSA
1360	1370	1380	1390	1400
LMRDRVEDVM	MERESQFKEK	QEEQARLLDE	LAGKLQSLDL	SAAAEAMTCGT

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      1410      1420      1430      1440      1450
PPGASCSETE CGGPNCRIDE GERKCGGPGC GGLVTVAHNA WQKAMDLDQD

      1460      1470      1480      1490      1500
VLSALAEVEQ LSKMVSEAKL RADEAKQSAE DILLKTNATK EKMDKSNEEL

      1510      1520      1530      1540      1550
RNLIKQIRNF LTQDSADLDS IEAVANEVLK MEMPSTPQQL QNLTEDIRER

      1560      1570      1580      1590      1600
VESLSQVEVI LQHSAADIR AEMLLEEAKR ASKSATDVKV TADMVKEALE

      1610      1620      1630      1640      1650
EAEKAQVAEE KAIKQADEDI QGTQNLTSI ESETAASEET LFNASQRISE

      1660      1670      1680      1690      1700
LERNVEELKR KAAQNSGEAE YIEKVVTYVK QSAEDVKKTL DGELDEKYKK

      1710      1720      1730      1740      1750
VENLIAKKE ESADARRKAE MLQNEAKTLL AQANSKLQLL KDLEKRYEDN

      1760      1770      1780
QRYLEDKAE LARLEGEVRS LIKDISQKVA VYSTCL

```

A cDNA and a chromosomal sequence encoding the LAMB1 protein is available from the NCBI database as accession no. M61916 and M61950, respectively.

[0103] An amino acid sequence for the protein encoded by the human LENEP gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9Y5L5, shown below as SEQ ID NO:54.

```

      10      20      30      40      50
MQPRTQPLAQ TLPFFLGGAP RDTGLRVPVI KMGTWGEFQ RTLKEVAYIL

      60
LCCWCICKELL D

```

A cDNA and a chromosomal sequence encoding the LENEP protein is available from the NCBI database as accession no. AF268478 and AF144412, respectively.

[0104] An amino acid sequence for the protein encoded by the human LRRC4B gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9NT99, shown below as SEQ ID NO:55.

```

      10      20      30      40      50
MARARGSPCP PLPPGRMSWP HGALLFLWLF SPPLGAGGGG VAVTSAAGGG

      60      70      80      90     100
SPPATSCPVA CSCSNQASRV ICTRDIAEV PASIPVNTRY LNLQENGIQV

      110     120     130     140     150
IRTDTFKHLR HLEILQLSKN LVRKIEVGAF NGLPSLNTLE LFDNRLTTPV

      160     170     180     190     200
TQAFEYLSKL RELWLRNNPI ESIPSYAFNR VPSLRRLDLG ELKRLEYISE

      210     220     230     240     250
AAFEGLVNLK YNLGMCNLK DIPNLTALVR LEELELSGNR LDLIRPGSFQ

      260     270     280     290     300
GLTSLRKLWL MHAQVATIER NAFDDLKSL EELNSHNNLM SLPHDLFTPL

      310     320     330     340     350
HRLERVHLNH NPWHCNCVLL WLSWWLKETV PSNTTCCARC HAPAGLKGRY

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360	370	380	390	400
IGELDQSHFT	CYAPVIVEPP	TDLNVTEGMA	AELKCRTGTS	MTSVNWLTPN
410	420	430	440	450
GTLMTGHSYR	VRISVLHDGT	LNFTNVTVQD	TGQYTCMVIN	SAGNTTASAT
460	470	480	490	500
LNVSADVDPVA	AGGTGSGGGG	PGGSGGVGGG	SGGYTYFTTV	TVETLETQPG
510	520	530	540	550
EEALQPRGTE	KEPPGPTTDG	VWGGGRPGDA	AGPASSSTTA	PAPRSSRPTE
560	570	580	590	600
KAFTVPITDV	TENALKDLDD	VMKTTKIIIG	CFVAITFMAA	VMLVAFYKLR
610	620	630	640	650
KQHQLKHHG	PTRTVEIINV	EDELPAASAV	SVAAAAVAS	GGGVGDSHL
660	670	680	690	700
ALPALERDHL	NHHHYVAAAF	KAHYSSNPSG	GGCGKGKPPG	LNSIHEPLL
710				
KSGSKENVQE	TQI			

A cDNA and a chromosomal sequence encoding the LRRC4B protein is available from the NCBI database as accession no. BC019687 and AC008743, respectively.

[0105] An amino acid sequence for the MAB21L2 protein encoded by the human gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9Y586, shown below as SEQ ID NO:56.

10	20	30	40	50
MIAAQAKLVY	QLNKYYTERC	QARKAAIAKT	IREVCKVVS	VLKEVEVQEP
60	70	80	90	100
RFISLSLEID	ARYEGLEVIS	PTEFEVVLVL	NQMGVFNFD	DGSLPGCAVL
110	120	130	140	150
KLSDGRKRS	SLWVEFITAS	GYLSARKIRS	RFQTLVAQAV	DKCSYRDVVK
160	170	180	190	200
MIADTSEVKL	RIRERYVQI	TPAFKCTGIW	PRSAQWPMP	HIPWPGPNRV
210	220	230	240	250
AEVKAEGFNL	LSKECYSLTG	KQSSAESDAW	VLQFGEAENR	LLMGGCRNKC
260	270	280	290	300
LSVLKTLRDR	HLELPGQLN	NYHMKTLLEY	ECEKHPRETD	WDESCLGDR
310	320	330	340	350
NGILLQLISC	LQCRRCPHYF	LPNLDLFQGK	PHSALESAK	QTWRLAREIL
TNPKSLDKL				

A cDNA and a chromosomal sequence encoding the MAB21L2 protein is available from the NCBI database as accession no. AF262032 and A-155219, respectively.

[0106] An amino acid sequence for the protein encoded by the human RETREGI gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9H6L5, shown below as SEQ ID NO:57.

10	20	30	40	50
MASPAPPEHA	EEGCPAPAAE	EQAPPSPPPP	QASPAERQQQ	EEEEQEAGAA
60	70	80	90	100
EGAGLQVEEA	AGRAAAVTV	LLGEPVLWLG	CRADELLSWK	RPLRSLLGFV

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      110      120      130      140      150
AANLLFWFLA LTPWRVYHLI SVMILGRVIM QIIKDMVLSR TRGAQLWRSL

      160      170      180      190      200
SESWEVINSK PDERPRLSHC IAESWMNFSI FLQEMSLFKQ QSPGKFCLLV

      210      220      230      240      250
CSVCTFPTIL GSYIPGVILS YLLLLCAFLC PLEKNDIGQ KIYSKIKSVL

      260      270      280      290      300
LKLDPGIGEY INQKKRERSE ADKEKSHKDD SELDFSALCP KISLTVAAKE

      310      320      330      340      350
LSVSDTDVSE VSWTDNGTFN LSEGYTPQTD TSDDLDRPSE EVFSRDLSDF

      360      370      380      390      400
PSLENGMGTN DEDELSLGLP TELKRKKEQL DSGHRPSKET QSAAGLTPLP

      410      420      430      440      450
NSDQTFHLMS NLAGDVITAA VIAAIKDQLE GVQQALSQAA PIPEEDTDTE

      460      470      480      490
EGDDFELLDQ SELDQIESEL GLTQDQHAEA QQNKKSSGFL SNLLGGH

```

[0107] A cDNA sequence encoding the RETREG1 protein is available from the NCBI database as accession no. AK000159.

[0108] An amino acid sequence for the protein encoded by the human SMAD9 gene that is a negative regulator of T cells as detected by interferon-production is available from the UniPROT database as accession no. 015198, shown below as SEQ ID NO:58.

```

      10      20      30      40      50
MHSTTPISSL FSFTSPAVKR LLGWKQGDEE EKWAEKAVDS LVKKLKKKKG

      60      70      80      90     100
AMDELERALS CPGQPSKCVT IPRSLDGRLLQ VSHRKGLPHV IYCRVWRWPD

      110     120     130     140     150
LQSHHELKPL ECCEFPFGSK QKEVCINPYH YRRVETPVLP PVLVPRHSEY

      160     170     180     190     200
NPQLSLLAKF RSASLHSEPL MPHNTATPDS FQQPPCSALP PSPSHAFSQS

      210     220     230     240     250
PCTASYPHSP GSPSEPESPY QHSVDTPLPL YHATEASETQ SGQPV DATAD

      260     270     280     290     300
RHVVL SIPNG DFRPVCYEPP QHWCSVAYYE LNNRVGETFQ ASSRSVLIDG

      310     320     330     340     350
FTDPSNNRNR FCLGLLSNVN RNSTIENTRR HIGKGVHLYY VGGEVYABCV

      360     370     380     390     400
SDSSIFVQSR NCNYQHGFHP ATVCKIPSGC SLKVFNQLF AQLLAQSVHH

      410     420     430     440     450
GFEVVYELTK MCTIRMSFVK GWGAEYHRQD VTSTPCWIEI HLHGPLQWLD

      460
KVLTMGSPH NPISSVS

```

A cDNA and a chromosomal sequence encoding the SMAD9 protein is available from the NCBI database as accession no. D83760 and AL138706, respectively.

[0109] An amino acid sequence for the protein encoded by the human SPATA31A1 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q5TZJ5, shown below as SEQ ID NO:59.

```

      10      20      30      40      50
MENLPPPLKL LSASSLNAPS STPWVLDIFL TLVFALGFFF LLLPYLSYFR

      60      70      80      90     100
CDDPPSPSPG KRKCPVGRRR RPRGRMKNHS LRAGRECPRG LQETSDLLSQ

      110     120     130     140     150
LQSLLGPHLD KGDFGQLSGP DPPGEVGERA PDGASQSSHE PMEDAAPILS

      160     170     180     190     200
PLASDPDQAK HPQDLASTPS PGPMTTSVSS LSASQPPEPS LPLEHPSPEP

      210     220     230     240     250
PALFPHPPHT PDPLACSPPP PKGFTAPPLR DSTLITPSHC DSVALPLGTV

      260     270     280     290     300
PQSLSPHEDL VASVPAISGL GGSNSHVSAS SRWQETARTS CAFNSSVQQD

      310     320     330     340     350
HLSRHPPETY QMEAGSLFLL SSDGQNAVGI QVTETAKVNI WEEKENVGSF

      360     370     380     390     400
TDRMTPEKHL NSLRNLAKSL DAEQDTTNPK PFWNMGENSK QLPGPQKLSLSD

      410     420     430     440     450
PRLWQESFWK NYSQLFWGLP SLHSESLVAN AWVTDRSYTL QSPPELNFEM

      460     470     480     490     500
SNVCIQRET TMSPLLFAQ PPSHLGPECQ PRISSTPQFR PTPMAQAEAQ

      510     520     530     540     550
AHLQSSFPVL SPAFFSLIKN TGVACPASQN KVQALSLPET QHPEWPLLR

      560     570     580     590     600
QLEGRALALPS RVQKSQDVFS VSTPNLPQES LTSILPENFP VSPELRRQLE

      610     620     630     640     650
QHIKKWIIQH WGNLGRIQES LDLMQLRDES PGTSQAKGKP SPWQSSMSTG

      660     670     680     690     700
ESSKEAQVKV FQLERDPCPH LGQILGETPQ NLSRDMKSFP RKVLGVTSEE

      710     720     730     740     750
SERNLRKPLR SDSGSDLLRC TERTHIENIL KAHMGRNLGQ TNEGLIPVRV

      760     770     780     790     800
RRSWLAVNQA LPVSNTHVKT SNLAAPKSGK ACVNTAQVLS FLEPCTQQGL

      810     820     830     840     850
GAHIVRFWAK HRWGLPLRVL KPIQCFKLEK VSSLSLTQLA GPSSATCESG

      860     870     880     890     900
AGSEVEVDMF LRKPPMASLR KQVLTKASDH MPESLLASSP AWKQFQRAPR

      910     920     930     940     950
GIPSWNDHGP LKPPPAGQEG RWPSKPLTYS LTGSTQQSRS LGAQSSKAGE

      960     970     980     990    1000
TREAVPQCRV PLETCLANL QATSEDMHGF EAPGTSKSSL HPRVSVSQDP

     1010     1020     1030     1040     1050
RKLCLMEEVV NEFEPGMATK SETQPQVCAA VVLLPDGQAS VVPHASENLV

     1060     1070     1080     1090     1100
SQVPQGHLQS MPAGNMRASQ ELHDLMAARR SKLVHEEPRN PNCQGSCKNQ

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1110	1120	1130	1140	1150
RPMFPPIHKS	EKSARKPNLEK	HEERLEGLRT	PQLTPVRKTE	DTHQDEGVQL
1160	1170	1180	1190	1200
LPSKKQPPSV	SHFGGNIKQF	FQWIFSKKKS	KPAPVTAESQ	KTVMNRSCVY
1210	1220	1230	1240	1250
SSSAEAQGLM	TAVGQMLDEK	MSLCHARHAS	KVNQHKQKQFQ	APVCGFPNCN
1260	1270	1280	1290	1300
RHLFYSEHGR	ILSYAASSQQ	ATLKSQGCPCN	RDRQIRNQQP	LKSVCNNEQ
1310	1320	1330	1340	
WGLRHPQILH	PKKAVSPVSP	LQHWPKTSGA	SSHHHCPRH	CLLWEGI

A chromosomal sequence encoding the SPATA31A1 protein is available from the NCBI database as accession no. BX005214.

[0110] An amino acid sequence for the protein encoded by the human ZNF445 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. P59923, shown below as SEQ ID NO:60.

10	20	30	40	50
MPPGRWHAAY	PAQAQSSRER	GRLQTVKKEE	EDESYPVQA	ARPQTLNRPG
60	70	80	90	100
QELFRQLFRQ	LRYHESSGPL	ETLSRLRELC	RWWLRPDVLS	KAQILELLVL
110	120	130	140	150
EQFLSILPGE	LRVWVQLHNP	ESGEEAVALL	EELQRDLDT	SWRDPGPAQS
160	170	180	190	200
PDVHWMGTGA	LRSAQIWSLA	SPLRSSSALG	DHLEPPYEIE	ARDFLAGQSD
210	220	230	240	250
TPAAQMPALF	PREGCPGDQV	TPTRSLTAQL	QETMTFKDVE	VTFSQDEWGW
260	270	280	290	300
LDSAQRNLVR	DVMLENYRNM	ASLVGPFTKP	ALISWLEARE	PWGLNMQAAQ
310	320	330	340	350
PKGNPVAAPT	GDDLQSKTNK	FILNQEPLEE	AETLAVSSGC	PATSVSEGIG
360	370	380	390	400
LRESFQKSR	QKDCENPIQ	VRVKKEETNF	SHRTGKDSEV	SGSNSLDLKH
410	420	430	440	450
VTYLRVSGRK	ESLKHGCGKH	FRMSSHHDY	KKYGKGLRHM	IGGFSLHQRI
460	470	480	490	500
HSGLKGNKKD	VCGKDFSLSS	HHQRGQSLHT	VGVSPKCSDC	GRTFSSSHL
510	520	530	540	550
AYHQRLHTQE	KAFKCRVCGK	AFRWSSNCAR	HEKIHTGVKP	YKCDLCEKAF
560	570	580	590	600
RRLSAYRLHR	ETHAKKKFLE	LNQYRAALTY	SSGFDHHLGD	QSGEKLFDCS
610	620	630	640	650
QCRKSFHCKS	YVLEHQRIHT	QEKPYKCTKC	RKTPRWRSNF	TRHMLHEEE
660	670	680	690	700
KFYKQDECRE	GFRQSPDCSQ	PQGAPAVEKT	FLCQQCGKTF	TRKKTLDVHQ
710	720	730	740	750
RIHTGEKPYQ	CSDCGKDFAY	RSAFIVHKKK	HAMKRKPEGG	PSFSQDTVQF
760	770	780	790	800
VPQSSHSKEE	PYKCSQCGKA	FRNHSFLLIH	QRVHTGEKPY	KCRECGKAPR

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      810      820      830      840      850
WSSNLYRHQR IHSLQKQYDC HESEKTPNVE PKILTGEKRF WCQECGKTFT

      860      870      880      890      900
RKRTLDDHKG IHSGEKRYKC NLCGKSYDRN YRLVNHQRIH STERPFFKCQW

      910      920      930      940      950
CGKEFIGRHT LSSHQRKHTR AAQAERSPPA RSSSQDTKLR LQKLKPSEEM

      960      970      980      990     1000
PLEDCKEACS QSSRLTGLQD ISIGKKCHKC SICGKTFNKS SQLISHKRPH

     1010     1020     1030
TRERPFCCKS CGKTRFWSSN LARHMKNHIR D

```

A cDNA encoding the ZNF445 protein is available from the NCBI database as accession 45 no. AY262260.

[0111] The following genes are negative regulators of T cells as detected by Interleukin-2 production (see Table 5): ABI3BP, AEBP1, AHR, ANTXR2, ARHGAP15, ARHGAP27, ARHGDIB, ARID3A, ARL4D, B4GALNT3, BICD1, C10orf82, C17orf75, C19orf35, C1 RL, C2orf69, C6orf132, C9orf84, CABP1, CBLB, CCSER1, CD34, CD4, CD5, CD52, CEACAM11, CEACAM7, CEBPB, CES3, CGB3, COL11A1, COL4A3, COLQ, CPEB3, CRELD2, CST9L, DDX55, DLG4, DOK1, EBF3, EIF3K, EN2, EOMES, EPB41, ETSI, F5, FAM96A, FH1, FOXA3, FOXE1, FOX13, FOXL2NB, FUS, FUT4, GCSAM, GCSAML, GDAPIL1, GDDP2, (GMIP, GNL3L, GOLPH3, CRAP, GRB2, HAUS7, HERC1, HLA-DQB2, I-ISP17B11, IKZF1, IKZF3, INPPL1, INTS1, ITIH2, ITPKA, IT1 KB, ITPKC, JDP2, JKAMP, JMDIC, KIAA1024, KIF15, KIF5A, KNTC1, LAT2, LAX1, LGR5, LLIME, LMBRD2, LOC401052, LONP2, LRCH3, LRRC23, LRRC25, LRRC52, LYN, LYPDI, MAATS1, MAB21L2, MAGEB17, MAP4K1, MEF2C, METTL9, MICU1, MRPL17, MUC1, NAIF1, NCF2, NDNF, NDLUFB1, NH-P2, NKX2-6, NLGN4Y, NNL, NPIPB9, NR4AA, NR4A3, NRCAM, NRPI, NRSN2, NSUN7, OLFML1, OM1P, OPRD1, ORIK1, OR2BI1, OSBPL11, OTOG, OTUD4, PATL2,

PAX5, PFKL, PHF2, PILBF1, P11P5KIA, PIP5KIB, PITPNC1, PLCL1, PLEKHM2, PPARG, PP1C, PSRC1, PSTPIP1, PTPN12, PTPN22, PTPN6, PTPRC, PVRIG, RBP4, RPL13A, S100A2, SALL4, SAMD8, SENP6, SETDIB, SEZ6L, SFT2DI, SH3TC1, SIGIRR, SIT1, SLA, SLA2, SLC20A2, SLC39A2, SLC6A8, SMAGP, SNRNP48, SOCS2, SORBS1, SOX13, SPN, SPRED1, SPRFD2, SRPK1, STAP1, STK38L, SYPL1, TCF12, TEX35, TFCP2L1, TMEM14C, TME 1223, TMEIEM262, TNTNT2, TPRA1, TREM6-TRIM34, TSPAN1, UBASH3B, UBE2W, UBR4, UBXN7, UCPI, UTMCI, ULLKI, U7PK3B, VPS28, VSTM5, XKR9, YLPM1, ZDHCC7, EB1, ZEB2, ZNF445, ZNF70, and ZN F831. Table 7 provides additional negative regulators of T cells as detected by interleukin-2 production

[0112] Sequences and other information relating to these genes, and their encoded proteins, is available, for example from the NCBI and UniPROT databases.

[0113] A few examples of protein sequences encoded by some of the genes detected as negative regulators of T cells by Interleukin-2 production are provided. For example, an amino acid sequence for the protein encoded by the human ABI3BP gene that is a negative regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q727G0, shown below as SEQ ID NO:61.

```

      10      20      30      40      50
MRGGKCNMLS SLGCLLLCGS ITLALGNAQK LPKGKRPNLK VHINTTSDSI

      60      70      80      90     100
LLKFLRPSPN VKLEGLLLGY GSNVSPNQYF PLPAEGKFTE AIVDAEPKYL

     110     120     130     140     150
IVVRPAPPPS QKKSCSGKTR SRKPLQLVVG TLTSSSVFLS WGFLINPHHD

     160     170     180     190     200
WTLPSHCND RFYTIRYREK DKEKKWIFQI CPATETIVEN LKPNTVYEFQ

     210     220     230     240     250
VKDNVEGGIW SKIFNHKTVV GSKKVNQIKQ STYDQDHTVP AYVPRKLIPI

     260     270     280     290     300
TIIKQVIQNV THKDSAKSPE KAPLGGVILV HLIIPGLNET TVKLPAASLMF

     310     320     330     340     350
EISDALKTQL AKNETLALPA ESKTPEVEKI SARPTTVTPE TVPRSTKPTT

     360     370     380     390     400
SSALDVSETT LASSEKPIW PTAKISEDK VLQPQTATYD VFSSPTTSDE

     410     420     430     440     450
PEISDSYTAT SDRILDSIPP KTSRTLEQPR ATLAPSETPF VPQKLEIFTS

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-continued

460	470	480	490	500
PEMQPTTPAP	QQTTSIPSTP	KRRPRPKPPR	TKPERTTSAG	TITPKISKSP
510	520	530	540	550
EPTWTTAPAG	KTQFISLKPK	IPLSPEVTHT	KPAPKQTPRA	PPKPKTSRP
560	570	580	590	600
RIPQTQPVVK	VPQRVTAQPK	TSPSPEVSYT	TPAPKDVLLP	HKPYPEVSQS
610	620	630	640	650
EPAPLETRGI	PFIPMISFSP	SQEELQTTLE	ETDQSTQEPF	TTKIPRTTEL
660	670	680	690	700
AKTTQAPHRF	YTTVRPRTSD	KPHIRPGVKQ	APRPSGADRN	VSVDSHTPTK
710	720	730	740	750
KPGTRRPPLP	PRPTHPRRKP	LPPNNVTGKP	GSAGIISSGP	ITTPPLRSTP
760	770	780	790	800
RPTGTPLERI	ETDIKQPTVP	ASGEELENIT	DFSSSPTRET	DPLGKPRFKG
810	820	830	840	850
PHVRYIQKPD	NSPCSITDSV	KRFPKEEATE	GNATSPPQNP	PTNLTVVTVE
860	870	880	890	900
GCPSFVILDW	EKPLNDTVTE	YEVISRENGS	FSGKNKSIQM	TNQTFFSTVEN
910	920	930	940	950
LKPNTSYEFQ	VKPKNPLGEG	PVSNTVAFST	ESADPRVSEP	VSAGRDAIWT
960	970	980	990	1000
ERPFNSDSYS	ECKGKQYVKR	TWYKKFVGQV	LCNSLRYKIY	LSDSLTGKFY
1010	1020	1030	1040	1050
NIGDQRGHGE	DHCQFVDSFL	DGRTGQQLTS	DQLPIKEGYF	RAVRQEPVQF
1060	1070			
GEIGGHTQIN	YVQWYECGTT	IPGKW		

A cDNA and a chromosomal sequence encoding the ABI3BP protein is available from the NCBT database as accession no. AB056106 and CH471052, respectively.

[0114] An amino acid sequence for the protein encoded by the human GCSAML gene that is a negative regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. 043741, shown below as SEQ ID NO:62.

10	20	30	40	50
MGNTTSDRVS	GERHGAKAAR	SEGAGGHAPG	KEHKIMVGST	DDPSVFLSPD
60	70	80	90	100
SKLPGDKEFV	SWQQDLEDSV	KPTQQARPTV	IRWSEGGKEV	FISGSFNNWS
110	120	130	140	150
TKIPLIKSHN	DFVAILDLPE	GEHQYKFFVD	GQWVHDPSEP	VVTSQLGTIN
160	170	180	190	200
NLIHVKKSDF	EVFDALKLDS	MESSETSCRD	LSSSPPGPYG	QEMYAFRSEE
210	220	230	240	250
RFKSPILPFP	HLLQVILNKD	TNISCDPALL	PEPNHVMLNH	LYALSIKDSV
260	270			
MVLSATHRYK	KKYVTTLLYK	PI		

A cDNA and a chromosomal sequence encoding the GCSAML protein is available from the NCBI database as accession no. AJ224538 and AL356378, respectively. PGP-83,DNA The following genes are negative regulators of T cells as detected by reduced cellular proliferation (see Table 6): ABCB1, ASAP1, ATP10A, DEAF1, FOXK1, ITGAX, LCE6A, LCP2, LEFTY1, MYC, NAT8B, OLFM3, PLD6,

PREP, SULT1A1, SULT1A4, A-INAK, AR-IODLB, B3GNT5, CASZ1, CD27, CEBPB, CRIBP, FL11, FOSL2, HLX, MAP4K1, MUC21, MXI, NDRG1, NEUROD2, SLC2A1, SLC43A3, SMAGP, SOX13, SP140, TP11, and TTC39C. Table 7 provides additional negative regulators of T cells as detected by reduced cellular proliferation.

[0115] An amino acid sequence for the protein encoded by the human SULT1A4 gene that is a negative regulator of T cells as detected by reduced cellular proliferation is available from the UniPROT database as accession no. POD-MIN0, shown below as SEQ ID NO:63.

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      10      20      30      40      50
MELIQDTSRP PLEYVKGVPL IKYFAEALGP LQSFQARPDD LLINTYPKSG

      60      70      80      90     100
TTWVSQILDM IYQGGDLEKC NRAPIYVRVP FLEVNDPGEP SGLETLKDTF

     110     120     130     140     150
PPRLIKSHLP LALLPQTLLD QKVKKVYVAR NPKDVAVSYY HFHRMEKAHP

     160     170     180     190     200
EPGTWDSFLE KFMAGEVSYG SWYQHVCQEWV ELSRTHPVLY LFYEDMKENP

     210     220     230     240     250
KREIQKILEF VGRSLPEETM DFMVQHTSFK EMKKNPMTNY TTVPQELMDH

     260     270     280     290
SISPFMRKGM AGDWKTTFTV AQNERFDADY AEKMAGCSLS FRSEL

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A chromosomal sequence encoding the SULT1A4 protein is available from the NCBI database as accession no. AC106782.

[0116] An amino acid sequence for the protein encoded by the human SLC43A3 gene that is a negative regulator of T cells as detected by reduced cellular proliferation is available from the UniPROT database as accession no. Q8NB15, shown below as SEQ ID NO:64.

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      10      20      30      40      50
MAGQGLPLHV ATLLTGLEEC LGFAGVLFGW PSLVFVFKNE DYFKDLCGPD

      60      70      80      90     100
AGPIGNATGQ ADCKAQDERF SLIFTLGSGM NNFMTFPTGY IFDRFKTTVA

     110     120     130     140     150
RLIAIFFYTT ATLIIAFTSA GSAVLLFLAM PMLTIGGILF LITNLQIGNL

     160     170     180     190     200
FGQHRSTIIT LYNGAFDSSS AVFLIKLLY EKGISLRASF IFISVCSTWH

     210     220     230     240     250
VARTFLLMPR GHIPIPLPPN YSYGLCPGNG TTKEEKETAE HENRELQSKE

     260     270     280     290     300
FLSAKEETPG AGQKQELRSF WSYAFSRRFA WHLVWLSVIQ LWHYLFIGTL

     310     320     330     340     350
NSLLTNMAGG DMARVSTYTN AFAFTQFGVL CAPWNGLLMD RLKQKYQKEA

     360     370     380     390     400
RKTGSSTLAV ALCSTVPSLA LTSLLCLGFA LCASVPILPL QYLTFILQVI

     410     420     430     440     450
SRSFLYGSNA AFLTLAFPSE HFGKLFGLVM ALSAVVSLQ FPIFTLIKGS

     460     470     480     490     500
LQNDPPYVNV MFMLAILLTF FHPFLVYREC RTWKESPSAI A

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A cDNA and a chromosomal sequence encoding the SLC43A3 protein is available from the NCBI database as accession no. AB028927 and AP000781. Any of these genes or the proteins encoded by these genes that are described herein can regulate T cells.

[0117] The sequences provided herein are exemplary. Isoforms and variants of these sequences and of any of regulators listed in Tables 1-7 or FIGS. 1-4 can also be used in the methods and compositions described herein.

[0118] For example, isoforms and variants of the proteins and nucleic acids can be used in the methods and compositions described herein when they are substantially identical to the genes or the encoded proteins listed in Tables 1-7 or FIGS. 1-4. The terms “substantially identical” indicates that a polypeptide or nucleic acid comprises a sequence with between 55-100% sequence identity to a reference sequence, for example with at least 55% sequence identity, preferably 60%, preferably 70%, preferably 80%, preferably at least

90%, preferably at least 95%, preferably at least 96%, preferably at least 97% sequence, preferably at least 98%, preferably at least 99% identity to a reference sequence over a specified comparison window. Optimal alignment may be ascertained or conducted using the homology alignment algorithm of Needleman and Wunsch, *J. Mol. Biol.* 48:443-53 (1970).

[0119] An indication that two polypeptide sequences are substantially identical is that both polypeptides have the same function acting as a regulator of T cells or T cell activity. The polypeptide that is substantially identical to a regulator sequence and may not have exactly the same level of activity as the regulator. Instead, the substantially 10 identical polypeptide may exhibit greater or lesser levels of regulator activity than the those listed in Tables 1-7 or FIG. 1-4, or any of the sequences recited herein. For example, the substantially identical polypeptide or nucleic acid may have at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90%, or at least about 95%, or at least about 97%, or at least about 98%, or at least about 100%, or at least about 105%, or at least about 110%, or at least about 120%, or at least about 130%, or at least about 140%, or at least about 150%, or at least about 200% of the activity of a regulator described herein a when measured by similar assay procedures.

[0120] Alternatively, substantial identity is present when second polypeptide is immunologically reactive with antibodies raised against the first polypeptide (e.g., a polypeptide with encoded by any of the genes listed in Tables 1-7 or FIGS. 1-4). Thus, a polypeptide is substantially identical to a first polypeptide, for example, where the two polypeptides differ only by a conservative substitution. In addition, a polypeptide can be substantially identical to a first polypeptide when they differ by a non-conservative change if the epitope that the antibody recognizes is substantially identical. Polypeptides that are "substantially similar" share sequences as noted above except that some residue positions, which are not identical, may differ by conservative amino acid changes.

Expression Systems

[0121] Nucleic acid segments encoding one or more regulator proteins, or nucleic acid segments that are inhibitory nucleic acids or such regulators, can be inserted into or employed with any suitable expression system. Nucleic acids segments encoding one or more agents that can modulate a regulator protein expression or activity can be inserted into or employed with any suitable expression system. A therapeutically effective quantity of one or more regulator proteins or modulators of such regulator proteins can be generated from such expression systems. A therapeutically effective of one or more inhibitory nucleic acids can also be generated from such expression systems.

[0122] Recombinant expression of nucleic acids (or inhibitory nucleic acids) is usefully accomplished using a vector, such as a plasmid. The vector can include a promoter operably linked to nucleic acid segment encoding one or more regulator/modulator proteins. In another example, a vector can include a promoter operably linked to nucleic acid segment that encodes a regulator/modulator inhibitory nucleic acid.

[0123] The vector can also include other elements required for transcription and translation. As used herein, vector

refers to any carrier containing exogenous DNA. Thus, vectors are agents that transport the exogenous nucleic acid into a cell without degradation and include a promoter yielding expression of the nucleic acid in the cells into which it is delivered. Vectors include but are not limited to plasmids, viral nucleic acids, viruses, phage nucleic acids, phages, cosmids, and artificial chromosomes. A variety of prokaryotic and eukaryotic expression vectors suitable for carrying, encoding and/or expressing regulator/modulator. A variety of prokaryotic and eukaryotic expression vectors suitable for carrying, encoding and/or expressing regulator/modulator inhibitory nucleic acids can be employed. Such expression vectors include, for example, pET, pET3d, pCR2.1, pBAD, pUC, and yeast vectors. The vectors can be used, for example, in a variety of in vivo and in vitro situations.

[0124] The expression cassette, expression vector, and sequences in the cassette or vector can be heterologous. As used herein, the term "heterologous" when used in reference to an expression cassette, expression vector, regulatory sequence, promoter, or nucleic acid refers to an expression cassette, expression vector, regulatory sequence, or nucleic acid that has been manipulated in some way. For example, a heterologous promoter can be a promoter that is not naturally linked to a nucleic acid of interest, or that has been introduced into cells by cell transformation procedures. A heterologous nucleic acid or promoter also includes a nucleic acid or promoter that is native to an organism but that has been altered in some way (e.g., placed in a different chromosomal location, mutated, added in multiple copies, linked to a non-native promoter or enhancer sequence, etc.). Heterologous nucleic acids may comprise sequences that comprise cDNA forms; the cDNA sequences may be expressed in either a sense (to produce mRNA) or anti-sense orientation (to produce an anti-sense RNA transcript that is complementary to the mRNA transcript). Heterologous coding regions can be distinguished from endogenous coding regions, for example, when the heterologous coding regions are joined to nucleotide sequences comprising regulatory elements such as promoters that are not found naturally associated with the coding region, or when the heterologous coding regions are associated with portions of a chromosome not found in nature (e.g., genes expressed in loci where the protein encoded by the coding region is not normally expressed). Similarly, heterologous promoters can be promoters that at linked to a coding region to which they are not linked in nature.

[0125] Viral vectors that can be employed include those relating to lentivirus, adenovirus, adeno-associated virus, herpes virus, vaccinia virus, polio virus, AIDS virus, neuronal trophic virus, Sindbis and other viruses. Also useful are any viral families which share the properties of these viruses which make them suitable for use as vectors. Retroviral vectors that can be employed include those described in by Verma, T. M., *Retroviral vectors for gene transfer*. In *Microbiology-1985*, American Society for Microbiology, pp. 229-232, Washington, (1985). For example, such retroviral vectors can include Murine Maloney Leukemia virus, MMLV, and other retroviruses that express desirable properties. Typically, viral vectors contain, nonstructural early genes, structural late genes, an RNA polymerase III transcript, inverted terminal repeats necessary for replication and encapsidation, and promoters to control the transcription and replication of the viral genome. When engineered as

vectors, viruses typically have one or more of the early genes removed and a gene or gene/promoter cassette is inserted into the viral genome in place of the removed viral nucleic acid.

[0126] A variety of regulatory elements can be included in the expression cassettes and/or expression vectors, including promoters, enhancers, translational initiation sequences, transcription termination sequences and other elements. A “promoter” is generally a sequence or sequences of DNA that function when in a relatively fixed location in regard to the transcription start site. For example, the promoter can be upstream of the nucleic acid segment encoding a regulator protein. In another example, the promoter can be upstream of an inhibitory nucleic acid segment of a modulating agent for one or more regulators.

[0127] A “promoter” contains core elements required for basic interaction of RNA polymerase and transcription factors and can contain upstream elements and response elements. “Enhancer” generally refers to a sequence of DNA that functions at no fixed distance from the transcription start site and can be either 5' or 3' to the transcription unit. Furthermore, enhancers can be within an intron as well as within the coding sequence itself. They are usually between 10 and 300 bp in length, and they function in cis. Enhancers function to increase transcription from nearby promoters. Enhancers, like promoters, also often contain response elements that mediate the regulation of transcription. Enhancers often determine the regulation of expression.

[0128] Expression vectors used in eukaryotic host cells (yeast, fungi, insect, plant, animal, human or nucleated cells) can also contain sequences for the termination of transcription, which can affect mRNA expression. These regions are transcribed as polyadenylated segments in the untranslated portion of the mRNA encoding tissue factor protein. The 3' untranslated regions also include transcription termination sites. It is preferred that the transcription unit also contains a polyadenylation region. One benefit of this region is that it increases the likelihood that the transcribed unit will be processed and transported like mRNA. The identification and use of polyadenylation signals in expression constructs is well established. It is preferred that homologous polyadenylation signals be used in the transgene constructs.

[0129] The expression of regulator/modulator proteins or inhibitory nucleic acid molecules therefor from an expression cassette or expression vector can be controlled by any promoter capable of expression in prokaryotic cells or eukaryotic cells. Examples of prokaryotic promoters that can be used include, but are not limited to, SP6, T7, T5, tac, bla, frp, gal, lac, or maltose promoters. Examples of eukaryotic promoters that can be used include, but are not limited to, constitutive promoters, e.g., viral promoters such as CMV, SV40 and RSV promoters, as well as regulatable promoters, e.g., an inducible or repressible promoter such as the tet promoter, the hsp70 promoter and a synthetic promoter regulated by CRE. Vectors for bacterial expression include pGEX-5X-3, and for eukaryotic expression include pCIneo-CMV.

[0130] The expression cassette or vector can include nucleic acid sequence encoding a marker product. This marker product is used to determine if the gene has been delivered to the cell and once delivered is being expressed. Marker genes can include the *E. coli* lacZ gene which encodes P-galactosidase, and green fluorescent protein. In some embodiments the marker can be a selectable marker.

When such selectable markers are successfully transferred into a host cell, the transformed host cell can survive if placed under selective pressure. There are two widely used distinct categories of selective regimes. The first category is based on a cell's metabolism and the use of a mutant cell line which lacks the ability to grow independent of a supplemented media. The second category is dominant selection which refers to a selection scheme used in any cell type and does not require the use of a mutant cell line. These schemes typically use a drug to arrest growth of a host cell. Those cells which have a novel gene would express a protein conveying drug resistance and would survive the selection. Examples of such dominant selection use the drugs neomycin (Southern P. and Berg, P, J. Molec. Appl. Genet, 1: 327 (1982)), mycophenolic acid, (Mulligan, R. C. and Berg, P. Science 209: 1422 (1980)) or hygromycin, (Sugden, B. et al., Mol. Cell Biol. 5: 410-413 (1985)).

[0131] Gene transfer can be obtained using direct transfer of genetic material, in but not limited to, plasmids, viral vectors, viral nucleic acids, phage nucleic acids, phages, cosmids, and artificial chromosomes, or via transfer of genetic material in cells or carriers such as cationic liposomes. Such methods are well known in the art and readily adaptable for use in the method described herein. Transfer vectors can be any nucleotide construction used to deliver genes into cells (e.g., a plasmid), or as part of a general strategy to deliver genes, e.g., as part of recombinant retrovirus or adenovirus (Rain et al. Cancer Res. 53:83-88, (1993)). Appropriate means for transfection, including viral vectors, chemical transfectants, or physico-mechanical methods such as electroporation and direct diffusion of DNA, are described by, for example, Wolff, J. A., et al., Science, 247, 1465-1468, (1990); and Wolff, J. A. Nature, 352, 815-818, (1991).

[0132] For example, the nucleic acid molecules, expression cassette and/or vectors encoding regulator/modulator proteins or encoding inhibitory nucleic acid molecules therefor can be introduced to a cell by any method including, but not limited to, calcium-mediated transformation, electroporation, microinjection, lipofection, particle bombardment and the like. The cells can be expanded in culture and then administered to a subject, e.g., a mammal such as a human. The amount or number of cells administered can vary but amounts in the range of about 10^6 to about 10^9 cells can be used. The cells are generally delivered in a physiological solution such as saline or buffered saline. The cells can also be delivered in a vehicle such as a population of liposomes, exosomes or microvesicles.

[0133] In some cases, the transgenic cell can produce exosomes or microvesicles that contain nucleic acid molecules, expression cassettes and/or vectors encoding one or more regulator/modulator. In some cases, the transgenic cell can produce exosomes or microvesicles that contain inhibitory nucleic acid molecules that can target regulator/modulator nucleic acids, one or more nucleic acids for regulator, or a combination thereof. Microvesicles can mediate the secretion of a wide variety of proteins, lipids, mRNAs, and micro RNAs, interact with neighboring cells, and can thereby transmit signals, proteins, lipids, and nucleic acids from cell to cell (see, e.g., Shen et al., J Biol Chem, 286(16): 14383-14395 (2011); Hu et al., Frontiers in Genetics 3 (April 2012); Pegtel et al., Proc. Nat'l Acad Sci 107(14): 6328-6333 (2010); WO/201 3/084000; each of which is incorporated herein by reference in its entirety. Cells pro-

ducing such microvesicles can be used to express the one or more regulator/modulator protein and/or inhibitory nucleic acids for one or more regulator/modulators, or a combination thereof Transgenic vectors or cells with a heterologous expression cassette or expression vector can express one or more regulator, can optionally also express one or more regulator inhibitory nucleic acids, or a combination thereof. Any of these vectors or cells can be administered to a subject. Exosomes produced by transgenic cells can be used to administer regulator/modulator proteins, regulator/modulator nucleic acids, regulator/modulator inhibitory nucleic acids, or a combination thereof to a. subject or to tumor and cancer cells in the subject.

[0134] Methods and compositions that include inhibitors of one or regulators such as inhibitory nucleic acids, antibodies, or any combination thereof.

CRISPR Modifications

[0135] In some cases, clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated (Cas) systems can be used to create one or more modifications in genomic regulator genes. Such CRISPR modifications can reduce or activate the expression or functioning of the regulator gene products. CRISPR/Cas systems are useful, for example, for RNA-programmable genome editing (see e.g., Marraffini and Sontheimer. Nature Reviews Genetics 11: 181-190 (2010); Sorek et al. Nature Reviews Microbiology 2008 6: 181-6; Karginov and Hannon. Mol Cell 2010 1:7-19; Hale et al. Mol Cell 2010:45:292-302; Jinek et al. Science 2012 337:815-820; Bikard and Marraffini Curr Opin Immunol 2012 24:15-20; Bikard et al. Cell Host & Microbe 2012 12: 177-186; all of which are incorporated by reference herein in their entirety).

[0136] A CRISPR guide RNA can be used that can target a Cas enzyme to the desired location in the genome, where it can cleave the genomic DNA for generation of a genomic modification. This technique is described, for example, by Mali et al. Science 2013 339:823-6; which is incorporated by reference herein in its entirety. Kits for the design and use of CRISPR-mediated genome editing are commercially available, e.g. the PRECISION X CAS9 SMART NUCLEASE™ System (Cat No. CAS900A-1) from System Biosciences, Mountain View, CA.

[0137] In some cases, transcriptional activators can be linked to defective Cas9 or to one or more guide RNAs to target the transcriptional activator. Such transcriptional activators include protein domains or whole proteins that assist in the recruitment of co-factors and RNA Polymerase to increase transcription of one or more of the regulator gene(s) listed in Tables 1-7 or FIGS. 1-4.

[0138] In some cases, a cre-lox recombination system of bacteriophage P1, described by Abremski et al. 1983. Cell 32:1301 (1983), Sternberg et al., Cold Spring Harbor Symposium on Quantitative Biology, Vol. XLV 297 (1981) and others, can be used to promote recombination and alteration of the regulator genomic site(s). The cre-lox system utilizes the cre recombinase isolated from bacteriophage P1 in conjunction with the DNA sequences that the recombinase recognizes (termed lox sites). This recombination system has been effective for achieving recombination in plant cells (see, e.g., U.S. Pat. No. 5,658,772), animal cells (U.S. Pat. Nos. 4,959,317; 5,801,030), and in viral vectors (Hardy et al., J. Virology 71:1842 (1997).

[0139] The genomic mutations so incorporated can alter one or more amino acids in the encoded regulator gene products. For example, genomic sites modified so that the encoded regulator protein is more prone to degradation, is less stable so that the half-life of such protein(s) is reduced, or so that the regulator has improved expression or functioning. In another example, genomic sites can be modified so that at least one amino acid of a regulator polypeptide is deleted or mutated to alter its activity. For example, a conserved amino acid or a conserved domain can be modified to improve or reduce of the activity of the regulator polypeptide. For example, a conserved amino acid or several amino acids in a conserved domain of the regulator polypeptide can be replaced with one or more amino acids having physical and/or chemical properties that are different from the conserved amino acid(s). For example, to change the physical and/or chemical properties of the conserved amino acid(s), the conserved amino acid(s) can be deleted or replaced by amino acid(s) of another class, where the classes are identified in the following table.

Classification	Genetically Encoded
Hydrophobic	A, G, F, I, L, M, P, V, W
Aromatic	F, Y, W
Apolar	M, G, P
Aliphatic	A, V, L, I
Hydrophilic	C, D, E, H, K, N, Q, R, S, T, Y
Acidic	D, E
Basic	H, K, R
Polar	Q, N, S, T, Y
Cysteine-Like	C

[0140] The guide RNAs and nuclease can be introduced via one or more vehicles such as by one or more expression vectors (e.g., viral vectors), virus like particles, ribonucleo-proteins (RNPs), via nanoparticles, liposomes, or a combination thereof. The vehicles can include components or agents that can target particular cell types (e.g., antibodies that recognize cell-surface markers), facilitate cell penetration, reduce degradation, or a combination thereof.

Inhibitory Nucleic Acids

[0141] The expression of one or more regulators/modulators can be inhibited, for example by use of an inhibitory nucleic acid that specifically recognizes a nucleic acid that encodes the regulator or modulator.

[0142] An inhibitory nucleic acid can have at least one segment that will hybridize to a regulator nucleic acid or modulator under intracellular or stringent conditions. The inhibitory nucleic acid can reduce expression of a regulator/modulator nucleic acid. A nucleic acid may hybridize to a genomic DNA, a messenger RNA, or a combination thereof. An inhibitory nucleic acid may be incorporated into a plasmid vector or viral DNA. It may be single stranded or double stranded, circular or linear.

[0143] An inhibitory nucleic acid is a polymer of ribose nucleotides or deoxyribose nucleotides having more than 13 nucleotides in length. An inhibitory nucleic acid may include naturally occurring nucleotides; synthetic, modified, or pseudo-nucleotides such as phosphorothiolates; as well as nucleotides having a detectable label such as P³², biotin or digoxigenin. An inhibitory nucleic acid can reduce the expression and/or activity of a regulator/modulator nucleic acid. Such an inhibitory nucleic acid may be completely

complementary to a segment of an endogenous regulator/modulator nucleic acid (e.g., an RNA). Alternatively, some variability is permitted in the inhibitory nucleic acid sequences relative to regulator/modulator sequences. An inhibitory nucleic acid can hybridize to a regulator/modulator nucleic acid under intracellular conditions or under stringent hybridization conditions and is sufficiently complementary to inhibit expression of the endogenous regulator/modulator nucleic acid. Intracellular conditions refer to conditions such as temperature, pH and salt concentrations typically found inside a cell, e.g., an animal or mammalian cell. One example of such an animal or mammalian cell is a myeloid progenitor cell. Another example of such an animal or mammalian cell is a more differentiated cell derived from a myeloid progenitor cell. Generally, stringent hybridization conditions are selected to be about 5° C. lower than the thermal melting point (T_m) for the specific sequence at a defined ionic strength and pH. However, stringent conditions encompass temperatures in the range of about 1° C. to about 20° C. lower than the thermal melting point of the selected sequence, depending upon the desired degree of stringency as otherwise qualified herein. Inhibitory oligonucleotides that comprise, for example, 2, 3, 4, or 5 or more stretches of contiguous nucleotides that are precisely complementary to a regulator/modulator coding sequence, each separated by a stretch of contiguous nucleotides that are not complementary to adjacent coding sequences, can inhibit the function of one or more nucleic acids for any of the regulators or modulators described herein. In general, each stretch of contiguous nucleotides is at least 4, 5, 6, 7, or 8 or more nucleotides in length. Non-complementary intervening sequences may be 1, 2, 3, or 4 nucleotides in length. One skilled in the art can easily use the calculated melting point of an inhibitory nucleic acid hybridized to a sense nucleic acid to estimate the degree of mismatching that will be tolerated for inhibiting expression of a particular target nucleic acid. Inhibitory nucleic acids of the invention include, for example, a short hairpin RNA, a small interfering RNA, a ribozyme or an antisense nucleic acid molecule.

[0144] The inhibitory nucleic acid molecule may be single or double stranded (e.g., a small interfering RNA (siRNA)) and may function in an enzyme-dependent manner or by steric blocking. Inhibitory nucleic acid molecules that function in an enzyme-dependent manner include forms dependent on RNase H activity to degrade target mRNA. These include single-stranded DNA, RNA, and phosphorothioate molecules, as well as the double-stranded RNAi/siRNA system that involves target mRNA recognition through sense-antisense strand pairing followed by degradation of the target mRNA by the RNA-induced silencing complex. Steric blocking inhibitory nucleic acids, which are RNase-H independent, interfere with gene expression or other mRNA-dependent cellular processes by binding to a target mRNA and getting in the way of other processes. Steric blocking inhibitory nucleic acids include 2'-O alkyl (usually in chimeras with RNase-H dependent antisense), peptide nucleic acid (PNA), locked nucleic acid (LNA) and morpholino antisense.

[0145] Small interfering RNAs, for example, may be used to specifically reduce translation of regulator/Modulator such that translation of the encoded regulator/modulator polypeptide is reduced. SiRNAs mediate post-transcriptional gene silencing in a sequence-specific manner. See, for

example, website at invitrogen.com/lsite/us/en/home/Products-and-Services/Applications/mai.html. Once incorporated into an RNA-Induced silencing complex, siRNA mediate cleavage of the homologous endogenous mRNA transcript by guiding the complex to the homologous mRNA transcript, which is then cleaved by the complex. The siRNA may be homologous and/or complementary to any region of the regulator/modulator transcript and/or any of the transcripts of the regulators/modulators. The region of homology may be 30 nucleotides or less in length, preferable less than 25 nucleotides, and more preferably about 21 to 23 nucleotides in length. SiRNA is typically double stranded and may have two-nucleotide 3' overhangs, for example, 3' overhanging UU dinucleotides. Methods for designing siRNAs are known to those skilled in the art. See, for example, Elbashir et al. *Nature* 411: 494-498 (2001); Harborth et al. *Antisense Nucleic Acid Drug Dev.* 13: 83-106 (2003).

[0146] The pSuppressorNeo vector for expressing hairpin siRNA, commercially available from IMGENEX (San Diego, California), can be used to generate siRNA for inhibiting expression of regulators/modulators. The construction of the siRNA expression plasmid involves the selection of the target region of the mRNA, which can be a trial-and-error process. However, Elbashir et al. have provided guidelines that appear to work ~80% of the time. Elbashir, S. M., et al., *Analysis of gene function in somatic mammalian cells using small interfering RNAs*. *Methods*, 2002. 26(2): p. 199-213. Accordingly, for synthesis of synthetic siRNA, a target region may be selected preferably 50 to 100 nucleotides downstream of the start codon. The 5' and 3' untranslated regions and regions close to the start codon should be avoided as these may be richer in regulatory protein binding sites. As siRNA can begin with AA, have 3' UU overhangs for both the sense and antisense siRNA strands, and have an approximate 50% G/C content. An example of a sequence for a synthetic siRNA is 5'-A A(N 9)UU, where N is any nucleotide in the mRNA sequence and should be approximately 50% G-C content. The selected sequence(s) can be compared to others in the human genome database to minimize homology to other known coding sequences (e.g., by Blast search, for example, through the NCBI website).

[0147] SiRNAs may be chemically synthesized, created by in vitro transcription, or expressed from an siRNA expression vector or a PCR expression cassette. See, e.g., website at invitrogen.com/site/us/en/home/Products-and-Services/Applications/mai.html. When an siRNA is expressed from an expression vector or a PCR expression cassette, the insert encoding the siRNA may be expressed as an RNA transcript that folds into an siRNA hairpin. Thus, the RNA transcript may include a sense siRNA sequence that is linked to its reverse complementary antisense siRNA sequence by a spacer sequence that forms the loop of the hairpin as well as a string of U's at the 3' end. The loop of the hairpin may be of any appropriate lengths, for example, 3 to 30 nucleotides in length, preferably, 3 to 23 nucleotides in length, and may be of various nucleotide sequences including, AUG, CCC, UUCG, CCACC, CTCGAG, AAGCUU, CCACACC and UUCAAGAGA (SEQ ID NO:61). SiRNAs also may be produced in vivo by cleavage of double-stranded RNA introduced directly or via a transgene or virus. Amplification by an RNA-dependent RNA polymerase may occur in some organisms.

[0148] An inhibitory nucleic acid such as a short hairpin RNA siRNA or an antisense oligonucleotide may be prepared using methods such as by expression from an expression vector or expression cassette that includes the sequence of the inhibitory nucleic acid. Alternatively, it may be prepared by chemical synthesis using naturally occurring nucleotides, modified nucleotides or any combinations thereof. In some embodiments, the inhibitory nucleic acids are made from modified nucleotides or non-phosphodiester bonds, for example, that are designed to increase biological stability of the inhibitory nucleic acid or to increase intracellular stability of the duplex formed between the inhibitory nucleic acid and the target regulators/modulators nucleic acids.

[0149] An inhibitory nucleic acid may be prepared using available methods, for example, by expression from an expression vector encoding a complementarity sequence of the regulator/modulator nucleic acids described herein. Alternatively, it may be prepared by chemical synthesis using naturally occurring nucleotides, modified nucleotides or any mixture of combination thereof. In some embodiments, the nucleic acids of the regulators/modulators described herein are made from modified nucleotides or non-phosphodiester bonds, for example, that are designed to increase biological stability of the nucleic acids or to increase intracellular stability of the duplex formed between the inhibitory nucleic acids and other (e.g., endogenous) nucleic acids.

[0150] For example, the regulator/modulator nucleic acids can be peptide nucleic acids that have peptide bonds rather than phosphodiester bonds.

[0151] Naturally occurring nucleotides that can be employed in the regulator/modulator nucleic acids include the ribose or deoxyribose nucleotides adenosine, guanine, cytosine, thymine and uracil. Examples of modified nucleotides that can be employed in the regulator/modulator nucleic acids include 5-fluorouracil, 5-bromouracil, 5-chlorouracil, 5-iodouracil, hypoxanthine, xanthine, 4-acetylcytosine, 5-(carboxyhydroxymethyl) uracil, 5-carboxymethylaminomethyl-2-thiouridine, 5-carboxymethylaminomethyluracil, dihydrouracil, beta-D-galactosylqueosine, inosine, N6-isopentenyladenine, 1-methylguanine, 1-methylinosine, 2,2-dimethylguanine, 2-methyladenine, 2-methylguanine, 3-methylcytosine, 5-methylcytosine, N6-adenine, 7-methylguanine, 5-methylaminomethyluracil, 5-methoxyaminomethyl-2-thiouracil, beta-D-mannosylqueosine, 5'-methoxycarboxymethyluracil, 5-methoxyuracil, 2-methylthio-N6-isopentenyladenine, uracil-5-oxoacetic acid, wybutosine, pseudouracil, queosine, 2-thiocytosine, 5-methyl-2-thiouracil, 2-thiouracil, 4-thiouracil, 5-methyluracil, uracil-5-oxoacetic acid methylester, uracil-5-oxoacetic acid, 5-methyl-2-thiouracil, 3-(3-amino-3-N-2-carboxypropyl) uracil, (acp3)w, and 2,6-diaminopurine.

[0152] Thus, inhibitory nucleic acids of the regulators/modulators described herein may include modified nucleotides, as well as natural nucleotides such as combinations of ribose and deoxyribose nucleotides. The inhibitory nucleic acids may be of same length as wild type regulators/modulators described herein. The inhibitory nucleic acids of the regulators described herein can also be longer and include other useful sequences. In some embodiments, the inhibitory nucleic acids of the regulators/modulators described herein are somewhat shorter. For example, inhibitory nucleic acids of the regulators/modulators described

herein can include a segment that has a nucleic acid sequence that can be missing up to 5 nucleotides, or missing up to 10 nucleotides, or missing up to 20 nucleotides, or missing up to 30 nucleotides, or missing up to 50 nucleotides, or missing up to 100 nucleotides from the 5' or 3' end.

Antibodies

[0153] Antibodies can be used as inhibitors or activators of any of the regulators/modulators described herein. For example, in some cases, antibody preparations can target one or more of the regulators or modulators described herein to block interactions by the regulators/modulators described herein or to reduce the activities of the regulators/modulators. In other cases, for example, antibodies can activate one or more of the regulator or modulators described herein that are cell surface receptors. One example of such activation is Varlilumab (a CD27 activating antibody) currently in clinical trial and that has been shown to increase anti-tumor T cell function Ansell et al. (2020) Blood Adv. 4(9): 1917-1926.

[0154] Antibodies can be raised against various epitopes of the regulators/modulator described herein. Some antibodies for regulators/modulators described herein may also be available commercially. However, the antibodies contemplated for treatment pursuant to the methods and compositions described herein are preferably human or humanized antibodies and are highly specific for their targets.

[0155] In one aspect, the present disclosure relates to use of isolated antibodies that bind specifically to regulators/modulators described herein. Such antibodies may be monoclonal antibodies. Such antibodies may also be humanized or fully human monoclonal antibodies. The antibodies can exhibit one or more desirable functional properties, such as high affinity binding to one or more regulators/modulators described herein, or the ability to inhibit functioning of any of the regulators/modulators described herein.

[0156] Methods and compositions described herein can include antibodies that bind any of the regulators/modulators described herein, or a combination of antibodies where each antibody type can separately bind one of the regulators/modulators described herein.

[0157] The term "antibody" as referred to herein includes whole antibodies and any antigen binding fragment (i.e., "antigen-binding portion") or single chains thereof. An "antibody" refers to a glycoprotein comprising at least two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds, or an antigen binding portion thereof. Each heavy chain is comprised of a heavy chain variable region (abbreviated herein as V_H) and a heavy chain constant region. The heavy chain constant region is comprised of three domains, C_{H1} , C_{H2} and C_{H3} . Each light chain is comprised of a light chain variable region (abbreviated herein as V_L) and a light chain constant region. The light chain constant region is comprised of one domain, C_L . The V_H and V_L regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDR), interspersed with regions that are more conserved, termed framework regions (FR). Each V_H and V_L is composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. The variable regions of the heavy and light chains contain a binding domain that interacts with an antigen. The constant regions of the antibodies may mediate the binding of the immuno-

globulin to host tissues or factors, including various cells of the immune system (e.g., effector cells) and the first component (C1q) of the classical complement system.

[0158] The term “antigen-binding portion” of an antibody (or simply “antibody portion”), as used herein, refers to one or more fragments of an antibody that retain the ability to specifically bind to an antigen (e.g., a peptide or domain of an of the regulators/modulators described herein). It has been shown that the antigen-binding function of an antibody can be performed by fragments of a full-length antibody.

[0159] Examples of binding fragments encompassed within the term “antigen-binding portion” of an antibody include (i) a Fab fragment, a monovalent fragment consisting of the V_L , V_H , C_L and C_H1 domains; (ii) a $F(ab')_2$ fragment, a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region; (iii) a Fd fragment consisting of the V_H and C_H1 domains; (iv) a Fv fragment consisting of the V_L and V_H domains of a single arm of an antibody, (v) a dAb fragment (Ward et al., (1989) Nature 341:544-546), which consists of a V_H domain; and (vi) an isolated complementarity determining region (CDR). Furthermore, although the two domains of the Fv fragment, V_L and V_H , are coded for by separate genes, they can be joined, using recombinant methods, by a synthetic linker that enables them to be made as a single protein chain in which the V_L and V_H regions pair to form monovalent molecules (known as single chain Fv (scFv); see e.g., Bird et al. (1988) Science 242:423-426; and Huston et al. (1988) Proc. Natl. Acad. Sci. USA 85:5879-5883). Such single chain antibodies are also intended to be encompassed within the term “antigen-binding portion” of an antibody. These antibody fragments are obtained using conventional techniques known to those with skill in the art, and the fragments are screened for utility in the same manner as are intact antibodies.

[0160] An “isolated antibody,” as used herein, is intended to refer to an antibody that is substantially free of other antibodies having different antigenic specificities (e.g., an isolated antibody that specifically binds any of the regulators/modulators described herein is substantially free of antibodies that specifically bind antigens other than any of the regulators/modulators described herein). An isolated antibody that specifically binds regulators/modulators described herein may, however, have cross-reactivity to other antigens, such as isoforms or related forms of the regulators/modulators proteins from other species. Moreover, an isolated antibody may be substantially free of other cellular material and/or chemicals.

[0161] The terms “monoclonal antibody” or “monoclonal antibody composition” as used herein refer to a preparation of antibody molecules of single molecular composition. A monoclonal antibody composition displays a single binding specificity and affinity for a particular epitope.

[0162] The term “human antibody,” as used herein, is intended to include antibodies having variable regions in which both the framework and CDR regions are derived from human germline immunoglobulin sequences. Furthermore, if the antibody contains a constant region, the constant region also is derived from human germline immunoglobulin sequences. The human antibodies of the invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo). However, the term “human antibody,” as

used herein, is not intended to include antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

[0163] The term “human monoclonal antibody” refers to antibodies displaying a single binding specificity which have variable regions in which both the framework and CDR regions are derived from human germline immunoglobulin sequences. In one embodiment, the human monoclonal antibodies are produced by a hybridoma which includes a B cell obtained from a transgenic nonhuman animal, e.g., a transgenic mouse, having a genome comprising a human heavy chain transgene and a light chain transgene fused to an immortalized cell.

[0164] The term “recombinant human antibody,” as used herein, includes all human antibodies that are prepared, expressed, created or isolated by recombinant means, such as (a) antibodies isolated from an animal (e.g., a mouse) that is transgenic or transchromosomal for human immunoglobulin genes or a hybridoma prepared therefrom (described further below), (b) antibodies isolated from a host cell transformed to express the human antibody, e.g., from a transfectoma, (c) antibodies isolated from a recombinant, combinatorial human antibody library, and (d) antibodies prepared, expressed, created or isolated by any other means that involve splicing of human immunoglobulin gene sequences to other DNA sequences. Such recombinant human antibodies have variable regions in which the framework and CDR regions are derived from human germline immunoglobulin sequences. In certain embodiments, however, such recombinant human antibodies can be subjected to in vitro mutagenesis (or, when an animal transgenic for human Ig sequences is used, in vivo somatic mutagenesis) and thus the amino acid sequences of the V_H and V_L regions of the recombinant antibodies are sequences that, while derived from and related to human germline V_L and V_H sequences, may not naturally exist within the human antibody germline repertoire in vivo.

[0165] As used herein, “isotype” refers to the antibody class (e.g., IgM or IgG1) that is encoded by the heavy chain constant region genes.

[0166] The phrases “an antibody recognizing an antigen” and “an antibody specific for an antigen” are used interchangeably herein with the term “an antibody which binds specifically to an antigen.”

[0167] The term “human antibody derivatives” refers to any modified form of the human antibody, e.g., a conjugate of the antibody and another agent or antibody.

[0168] The term “humanized antibody” is intended to refer to antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences. Additional framework region modifications may be made within the human framework sequences.

[0169] The term “chimeric antibody” is intended to refer to antibodies in which the variable region sequences are derived from one species and the constant region sequences are derived from another species, such as an antibody in which the variable region sequences are derived from a mouse antibody and the constant region sequences are derived from a human antibody.

[0170] As used herein, an antibody that “specifically binds to a human regulator/modulator protein described herein” is intended to refer to an antibody that binds to the human

regulator/modulator protein described herein with a K_D of 1×10^{-7} M or less, more preferably 5×10^{-8} M or less, more preferably 1×10^{-8} M or less, more preferably 5×10^{-9} M or less, even more preferably between 1×10^{-8} M and 1×10^{-10} M or less.

[0171] The term “ K_{assoe} ” or “ K_a ,” as used herein, is intended to refer to the association rate of a particular antibody-antigen interaction, whereas the term “ K_{dis} ” or “ K_d ,” as used herein, is intended to refer to the dissociation rate of a particular antibody-antigen interaction. The term “ K_D ,” as used herein, is intended to refer to the dissociation constant, which is obtained from the ratio of K_d to K_a (i.e., K_d/K_a) and is expressed as a molar concentration (M). K_D values for antibodies can be determined using methods well established in the art. A preferred method for determining the K_D of an antibody is by using surface plasmon resonance, preferably using a biosensor system such as a Biacore™ system.

[0172] The antibodies of the invention are characterized by particular functional features or properties of the antibodies. For example, the antibodies bind specifically to a human regulator/modulator described herein. Preferably, an antibody of the invention binds to a regulator/modulator described herein with high affinity, for example with a K_D of 1×10^{-7} M or less. The antibodies can exhibit one or more of the following characteristics:

[0173] (a) binds to a human regulators/modulators described herein with a K_r of 1×10^{-7} M or less;

[0174] (b) inhibits the function or activity of a human regulators/modulators described herein;

[0175] (c) inhibits cancer (e.g., metastatic cancer); or

[0176] (d) a combination thereof.

[0177] Assays to evaluate the binding ability of the antibodies toward a human regulators/modulators described herein can be used, including for example, ELISAs, Western blots and RIAs. The binding kinetics (e.g., binding affinity) of the antibodies also can be assessed by standard assays known in the art, such as by Biacore™ analysis.

[0178] Given that each of the subject antibodies can bind to a human regulator/modulator described herein, the V_L and V_H sequences can be “mixed and matched” to create other binding molecules that bind to a human regulators/modulators described herein. The binding properties of such “mixed and matched” antibodies can be tested using the binding assays described above and assessed in assays described in the examples. When V_L and V_H chains are mixed and matched, a V_H sequence from a particular V_H/V_L pairing can be replaced with a structurally similar V_H sequence. Likewise, preferably a V_L sequence from a particular V_H/V_L pairing is replaced with a structurally similar V_L sequence.

[0179] Accordingly, in one aspect, the invention provides an isolated monoclonal antibody, or antigen binding portion thereof comprising:

[0180] (a) a heavy chain variable region comprising an amino acid sequence; and

[0181] (b) a light chain variable region comprising an amino acid sequence; wherein the antibody specifically binds a human regulators/modulators described herein.

[0182] In some cases, the CDR3 domain, independently from the CDR1 and/or CDR2 domain(s), alone can determine the binding specificity of an antibody for a cognate antigen and that multiple antibodies can predictably be generated having the same binding specificity based on a common CDR3 sequence. See, for example, Klimka et al.,

British 1 of Cancer 83(2):252-260 (2000) (describing the production of a humanized anti-CD30 antibody using only the heavy chain variable domain CDR3 of murine anti-CD30 antibody Ki-4), Beiboer et al., J. Mol. Biol. 296:833-849 (2000) (describing recombinant epithelial glycoprotein-2 (EGP-2) antibodies using only the heavy chain CDR3 sequence of the parental murine MOC-31 anti-EGP-2 antibody); Rader et al., Proc. Natl. Acad. Sci. U.S.A. 95:8910-8915 (1998) (describing a panel of humanized anti-integrin α 5 β 1 antibodies using a heavy and light chain variable CDR3 domain. Hence, in some cases a mixed and matched antibody or a humanized antibody contains a CDR3 antigen binding domain that is specific for regulators/modulators described herein.

Assays for Drug Development

[0183] Methods are also described herein for evaluating whether test agents can modulate the expression or activity of any of the regulators/modulators described herein. T cells, cancer cells, and combinations thereof can be evaluated for susceptibility to treatment with candidate compounds.

[0184] specifically, the methods can include assay steps for identifying a candidate test agent that selectively modulates the proliferation, functioning, or visibility of T cells or cancer cells, or for increasing or decreasing the levels or functioning of regulators described herein. For example, if the proliferation, cytokine production, activity, or viability of T cells is increased or decreased in the presence of one or more of the regulators described herein but the proliferation, cytokine production, activity, or the proliferation, activity, or viability of the T cells in the T cell-regulator assay mixture changes in the presence of a test agent then that test agent has utility for modulating the regulator of the T cells. Such a test agent is referred to as a modulator.

[0185] An assay can include determining whether a test agent can specifically cause decreased or increased numbers of T cells or whether a compound can specifically cause decreased or increased functioning of T cells. If the test agent does cause altered T cell numbers or T cell functioning, then the test agent can be selected/identified for further study, such as for its suitability as a therapeutic agent to treat a cancer or an immune condition or disease. For example, the test agent identified by the selection methods featured in the invention can be further examined for their ability to target a tumor, target an immune cell, or to treat cancer by, for example, administering the test agent (modulator) to an animal model.

[0186] The cells that are evaluated can include cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 cells, CD8 T cells, metastatic cells, benign cell samples, cell lines (including as cancer cell lines), or a combination thereof. The cells that are evaluated can also include cells from a patient with cancer (including a patient with metastatic cancer), or cells from a known cancer type or cancer cell line, or cells exhibiting an overproduction of any of the regulators described herein. A test agent that can modulate the production or activity of any of these cell types can be administered to an animal, including a patient.

[0187] For example one method can include (a) obtaining a cell sample from a patient; (b) measuring the amount or concentration of T cells/regulators/modulators in a known number of weight of cells from the sample to generate a reference value; (c) mixing the known number or weight of

cells from the sample with a test agent to generate a test assay; to generate a test assay T cell/regulator/modulator value; (e) optionally repeating steps (c) and (d) with separate samples; and (f) selecting a test agent with a lower or a test assay T cell/regulator/modulator value than the reference value. The method can further include administering a test agent to an animal model, for example, to further evaluate the toxicity and/or efficacy of the test agent. In some cases, the method can further include administering the test agent to the patient from whom the cell or tissue sample as obtained.

[0188] Test agent or modulators (e.g., top hits identified by any method described herein) can be used in a cell-based assay using T cells or cells that express any of the regulators described herein as a readout of the efficacy of the test agents or modulators.

[0189] Assay methods are also described herein for identifying and assessing the potency of agents that may modulate T cells any of the regulators listed in tables 1-7 of FIGS. 1-4.

[0190] For example, T cells can release cytokines, such as Interferon γ or Interleukin-2, T cells or T cells expressing any of the modulators described herein can be contacted with a test agent and the release of cytokines by the T-cells can be measured. Such a test agent-related level of cytokines can be compared to the level observed for T cells not contacted with a test agent..

[0191] Useful test regulators, modulators, and test agents can be administered to a test animal or a patient.

[0192] “Treatment” or “treating” refers to both therapeutic treatment and to prophylactic or preventative measures. Those in need of treatment include those already with the disorder as well as those prone to have the disorder, or those in whom the disorder is to be prevented.

[0193] “Subject” for purposes of administration of a regulator, modulator, test agent or composition described herein refers to administration to any animal classified as a mammal or bird, including humans, domestic animals, farm animals, zoo animals, experimental animals, pet animals, such as dogs, horses, cats, cows, etc. The experimental animals can include mice, rats, guinea pigs, goats, dogs, monkeys, or a combination thereof. In some cases, the subject is human.

[0194] As used herein, the term “cancer” includes solid animal tumors as well as hematological malignancies. The terms “tumor cell(s)” and “cancer cell(s)” are used interchangeably herein.

[0195] “Solid animal tumors” include cancers of the head and neck, lung, mesothelioma, mediastinum, lung, esophagus, stomach, pancreas, hepatobiliary system, small intestine, colon, colorectal, rectum, anus, kidney, urethra, bladder, prostate, urethra, penis, testis, gynecological organs, ovaries, breast, endocrine system, skin central nervous system; sarcomas of the soft tissue and bone; and melanoma of cutaneous and intraocular origin. In addition, a metastatic cancer at any stage of progression can be treated, such as micrometastatic tumors, megametastatic tumors, and recurrent cancers.

[0196] In some cases, a hematological cancer or hematological malignancy can be treated. The term “hematological malignancies” includes adult or childhood leukemia and lymphomas, Hodgkin’s disease, lymphomas of lymphocytic and cutaneous origin, acute and chronic leukemia, plasma cell neoplasm, and cancers associated with AIDS.

[0197] The inventive methods and compositions can also be used to treat leukemias, lymph nodes, thymus tissues, tonsils, spleen, cancer of the breast, cancer of the lung, cancer of the adrenal cortex, cancer of the cervix, cancer of the endometrium, cancer of the esophagus, cancer of the head and neck, cancer of the liver, cancer of the pancreas, cancer of the prostate, cancer of the thymus, carcinoid tumors, chronic lymphocytic leukemia, Ewing’s sarcoma, gestational trophoblastic tumors, hepatoblastoma, multiple myeloma, non-small cell lung cancer, retinoblastoma, or tumors in the ovaries. A cancer at any stage of progression can be treated or detected, such as primary, metastatic, and recurrent cancers. In some cases, metastatic cancers are treated but primary cancers are not treated. Information regarding numerous types of cancer can be found, e.g., from the American Cancer Society (cancer.org), or from, e.g., Wilson et al. (1991) Harrison’s Principles of Internal Medicine, 12th Edition, McGraw-Hill, Inc.

[0198] In some embodiments, the cancer and/or tumors to be treated are hematological malignancies, or those of lymphoid origin such as cancers or tumors of lymph nodes, thymus tissues, tonsils, spleen, and cells related thereto. In some embodiments, the cancer and/or tumors to be treated are those that have been resistant to T cell therapies.

[0199] Treatment of, or treating, metastatic cancer can include the reduction in cancer cell migration or the reduction in establishment of at least one metastatic tumor. The treatment also includes alleviation or diminishment of more than one symptom of metastatic cancer such as coughing, shortness of breath, hemoptysis, lymphadenopathy, enlarged liver, nausea, jaundice, bone pain, bone fractures, headaches, seizures, systemic pain and combinations thereof. The treatment may cure the cancer, e.g., it may prevent metastatic cancer, it may substantially eliminate metastatic tumor formation and growth, and/or it may arrest or inhibit the migration of metastatic cancer cells.

[0200] Anti-cancer activity can reduce the progression of a variety of cancers (e.g., breast, lung, pancreatic, or prostate cancer) using methods available to one of skill in the art. Anti-cancer activity, for example, can be determined by identifying the lethal dose (LD_{100}) or the 50% effective dose (ED_{50}) or the dose that inhibits growth at 50% (GI_{50}) of an agent of the present invention that prevents the migration of cancer cells. In one aspect, anti-cancer activity is the amount of the agent that reduces 50%, 60%, 70%, 80%, 90%, 95%, 97%, 98%, 99% or 100% of cancer cell migration, for example, when measured by detecting expression of a cancer cell marker at sites proximal or distal from a primary tumor site, or when assessed using available methods for detecting metastases.

[0201] In another example, agents that increase or decrease regulator/modulator expression or function can be administered to sensitize tumor cells to immune therapies. Hence, by administering an agent that increase or regulators/modulator expression or function, tumor cells can become more sensitive to the immune system and to various immune therapies.

Compositions

[0202] The invention also relates to compositions containing one or more active agents such as any of the regulators described herein, modulators described herein, or combinations thereof. Such active agents can be a polypeptide, a nucleic acid encoding a polypeptide (e.g., within an expres-

sion cassette or expression vector), a modified cell, an inhibitory nucleic acid, a small molecule, a compound identified by a method described herein, or a combination thereof. The compositions can be pharmaceutical compositions.

[0203] In some embodiments, the compositions can include a pharmaceutically acceptable carrier. By “pharmaceutically acceptable” it is meant that a carrier, diluent, excipient, and/or salt is compatible with the other ingredients of the formulation, and not deleterious to the recipient thereof.

[0204] The composition can be formulated in any convenient form. In some embodiments, the compositions can include a protein or polypeptide encoded by any of the genes listed in Tables 1-7 or FIGS. 1-4. In other embodiments, the compositions can include at least one nucleic acid or expression cassette encoding a polypeptide listed in Tables 1-7 or FIGS. 1-4. In other embodiments, the compositions can include at least one nucleic acid or expression cassette that includes a nucleic acid segment complementarity to gene listed in Table 1-2 (e.g., an inhibitory nucleic acid). In other embodiments, the compositions can include at least one nucleic acid or expression cassette that includes a nucleic acid segment encoding a cas nuclease and at least one guide RNA that can target a regulator or modulator described herein. In other embodiments, the compositions can include at least one antibody that binds at least one protein encoded by at least one gene listed in Tables 1-7 or FIGS. 1-4. In other embodiments, the compositions can include at least one small molecule that binds, that activates, or that inhibits at least one gene listed in Tables 1-7 or FIG. 1-4, or at least one small molecule that binds, that activates, or that inhibits at least one protein encoded by at least one gene listed in Tables 1-7 or FIGS. 1-4. In other embodiments, the compositions can include cells with at least one modified genomic regulator or modulator genetic site, cells that express one or more of the regulators described herein, cells that express a cas nuclease and at least one guide RNA that can target at least one regulator or modulator gene, cells that express one or more inhibitory nucleic acids, or a combination thereof. The cells can be immune cells. In some cases, the cells can be one or more types of lymphoid cells, myeloid cells, cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, chimeric antigen receptor (CAR) cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof.

[0205] The amount or number of cells administered can vary but amounts in the range of about 10^6 to about 10^9 cells can be used. The cells are generally delivered in a physiological solution such as saline or buffered saline. The cells can also be delivered in a vehicle such as within a population of liposomes, exosomes or microvesicles.

[0206] In some embodiments, the active agents of the invention (e.g., polypeptide, a nucleic acid encoding a polypeptide (e.g., within an expression cassette or expression vector), an antibody, an inhibitory nucleic acid, a small molecule, a compound identified by a method described herein, modified cells, or a combination thereof), are administered in a “therapeutically effective amount.” Such a therapeutically effective amount is an amount sufficient to obtain the desired physiological effect, such a reduction of at least one symptom of disease.

[0207] The disease can be cancer or an immune disease or condition. For example, active agents can reduce the symptoms of disease by 5%, or 10%, or 15%, or 20%, or 25%, or 30%, or 35%, or 40%, or 45%, or 50%, or 55%, or 60%, or 65%, or 70%, or 80% or 90%, 95%, or 97%, or 99%, or any numerical percentage between 5% and 100%. For example, symptoms of cancer can also include tumor cachexia, tumor-induced pain conditions, tumor-induced fatigue, tumor growth, and metastatic spread. Hence, the active agents may also reduce tumor cachexia, tumor-induced pain conditions, tumor-induced fatigue, tumor growth, or a combination thereof by 5%, or 10%, or 15%, or 20%, or 25%, or 30%, or 35%, or 40%, or 45%, or 50%, or 55%, or 60%, or 65%, or 70%, or 80%, or 90%, 95%, or 97%, or 99%, or any numerical percentage between 5% and 100%.

[0208] To achieve the desired effect(s), the active agents may be administered as single or divided dosages. For example, active agents can be administered in dosages of at least about 0.01 mg/kg to about 500 to 750 mg/kg, of at least about 0.01 mg/kg to about 300 to 500 mg/kg, at least about 0.1 mg/kg to about 100 to 300 mg/kg or at least about 1 mg/kg to about 50 to 100 mg/kg of body weight, although other dosages may provide beneficial results. The amount administered will vary depending on various factors including, but not limited to, the type of small molecules, compounds, peptides, or nucleic acid chosen for administration, the disease, the weight, the physical condition, the health, and the age of the mammal. Such factors can be readily determined by the clinician employing animal models or other test systems that are available in the art.

[0209] Administration of the active agents in accordance with the present invention may be in a single dose, in multiple doses, in a continuous or intermittent manner, depending, for example, upon the recipient's physiological condition, whether the purpose of the administration is therapeutic or prophylactic, and other factors known to skilled practitioners. The administration of the active agents and compositions of the invention may be essentially continuous over a preselected period of time or may be in a series of spaced doses. Both local and systemic administration is contemplated.

[0210] To prepare the composition, small molecules, compounds, polypeptides, nucleic acids, expression cassettes, ribonucleoprotein complexes, and other agents are synthesized or otherwise obtained, purified as necessary or desired. These small molecules, compounds, polypeptides, nucleic acids, expression cassettes, ribonucleoprotein complexes, and other agents can be suspended in a pharmaceutically acceptable carrier and/or lyophilized or otherwise stabilized. The small molecules, compounds, polypeptides, nucleic acids, expression cassettes, ribonucleoprotein complexes, other agents, and combinations thereof can be adjusted to an appropriate concentration, and optionally combined with other agents. The absolute weight of a given small molecule, compound, polypeptide, nucleic acid, ribonucleoprotein complex, and/or other agents included in a unit dose can vary widely. For example, about 0.01 to about 2 g, or about 0.1 to about 500 mg, of at least one molecule, compound, polypeptide, nucleic acid, ribonucleoprotein complexes, and/or other agent, or a plurality of molecules, compounds, polypeptides, nucleic acids, ribonucleoprotein complexes, and/or other agents can be administered. Alternatively, the unit dosage can vary from about 0.01 g to about 50 g, from about 0.01 g to about 35 g, from about, 1 g to about 25 g,

from about 0.5 g to about 12 g, from about 0.5 g to about 8 g, from about 0.5 g to about 4 g, or from about 0.5 g to about 2 g.

[0211] Daily doses of the active agents of the invention can vary as well. Such daily doses can range, for example, from about 0.1 g/day to about 50 g/day, from about 0.1 g/day to about 25 g/day, from about 0.1 g/day to about 12 g/day, from about 0.5 g/day to about 8 g/day, from about 0.5 g/day to about 4 g/day, and from about 0.5 g/day to about 2 g/day.

[0212] It will be appreciated that the amount of active agent for use in treatment will vary not only with the particular carrier selected but also with the route of administration, the nature of the cancer condition being treated and the age and condition of the patient. Ultimately the attendant health care provider can determine proper dosage. In addition, a pharmaceutical composition can be formulated as a single unit dosage form.

[0213] Thus, one or more suitable unit dosage forms comprising the active agent(s) can be administered by a variety of routes including parenteral (including subcutaneous, intravenous, intramuscular and intraperitoneal), oral, rectal, dermal, transdermal, intrathoracic, intrapulmonary and intranasal (respiratory) routes. The active agent(s) may also be formulated for sustained release (for example, using microencapsulation, see WO 94/07529, and U.S. Pat. No. 4,962,091). The formulations may, where appropriate, be conveniently presented in discrete unit dosage forms and may be prepared by any of the methods well known to the pharmaceutical arts. Such methods may include the step of mixing the active agent with liquid carriers, solid matrices, semi-solid carriers, finely divided solid carriers or combinations thereof, and then, if necessary, introducing or shaping the product into the desired delivery system. For example, the active agent(s) can be linked to a convenient carrier such as a nanoparticle, albumin, polyalkylene glycol, or be supplied in prodrug form. The active agent(s), and combinations thereof can be combined with a carrier and/or encapsulated in a vesicle such as a liposome.

[0214] The compositions of the invention may be prepared in many forms that include aqueous solutions, suspensions, tablets, hard or soft gelatin capsules, and liposomes and other slow-release formulations, such as shaped polymeric gels. Administration of inhibitors can also involve parenteral or local administration of the in an aqueous solution or sustained release vehicle.

[0215] Thus, while the active agent(s) and/or other agents can sometimes be administered in an oral dosage form, that oral dosage form can be formulated so as to protect the small molecules, compounds, polypeptides, nucleic acids, expression cassettes, ribonucleoprotein complexes, and combinations thereof from degradation or breakdown before the small molecules, compounds, polypeptides, nucleic acids encoding such polypeptides, expression cassettes, ribonucleoprotein complexes, and combinations thereof provide therapeutic utility. For example, in some cases the small molecules, compounds, polypeptides, nucleic acids encoding such polypeptide, expression cassettes, ribonucleoprotein complexes, and/or other agents can be formulated for release into the intestine after passing through the stomach. Such formulations are described, for example, in U.S. Pat. No. 6,306,434 and in the references contained therein.

[0216] Liquid pharmaceutical compositions may be in the form of, for example, aqueous or oily suspensions, solutions,

emulsions, syrups or elixirs, dry powders for constitution with water or other suitable vehicle before use. Such liquid pharmaceutical compositions may contain conventional additives such as suspending agents, emulsifying agents, non-aqueous vehicles (which may include edible oils), or preservatives. The pharmaceutical compositions may take such forms as suspensions, solutions, or emulsions in oily or aqueous vehicles, and may contain formulatory agents such as suspending, stabilizing and/or dispersing agents. Suitable carriers include saline solution, encapsulating agents (e.g., liposomes), and other materials. The active agent(s) and/or other agents can be formulated in dry form (e.g., in freeze-dried form), in the presence or absence of a carrier.

[0217] If a carrier is desired, the carrier can be included in the pharmaceutical formulation, or can be separately packaged in a separate container, for addition to the inhibitor that is packaged in dry form, in suspension or in soluble concentrated form in a convenient liquid.

[0218] An active agent(s) and/or other agents can be formulated for parenteral administration (e.g., by injection, for example, bolus injection or continuous infusion) and may be presented in unit dosage form in ampoules, prefilled syringes, small volume infusion containers or multi-dose containers with an added preservative.

[0219] The compositions can also contain other ingredients such as active agents, anti-viral agents, antibacterial agents, antimicrobial agents and/or preservatives. Examples of additional therapeutic agents that may be used include, but are not limited to: alkylating agents, such as nitrogen mustards, alkyl sulfonates, nitrosoureas, ethylenimines, and triazenes; antimetabolites, such as folate antagonists, purine analogues, and pyridine analogues; antibiotics, such as anthracyclines, bleomycins, mitomycin, dactinomycin, and plicamycin; enzymes, such as L-asparaginase; farnesyl-protein transferase inhibitors; hormonal agents, such as glucocorticoids, estrogens/antiestrogens, androgens/antiandrogens, progestins, and luteinizing hormone-releasing hormone antagonists, octreotide acetate; microtubule-disruptor agents, such as ecteinascidins or their analogs and derivatives; microtubule-stabilizing agents such as paclitaxel (Taxol®), docetaxel (Taxotere®), and epothilones A-F or their analogs or derivatives; plant-derived products, such as *vinca* alkaloids, epipodophyllotoxins, taxanes; and topoisomerase inhibitors; prenyl-protein transferase inhibitors, and miscellaneous agents such as, hydroxyurea, procarbazine, mitotane, hexamethylmelamine, platinum coordination complexes such as cisplatin and carboplatin; and other agents used as anti-cancer and cytotoxic agents such as biological response modifiers, growth factors; immune modulators, and monoclonal antibodies. The compositions can also be used in conjunction with radiation therapy.

[0220] The present description is further illustrated by the following examples, which should not be construed as limiting in any way. The contents of all cited references (including literature references, issued patents, published patent applications as cited throughout this application) are hereby expressly incorporated by reference.

Example 1: CRISPRa Screening Primary Human T cells to Identify Genetic Regulators

[0221] This Example describes use of CRISPRa for screening of primary human T cells to identify genetic regulators of therapeutically relevant T cell phenotypes.

[0222] T cells were isolated from two separate donors. The two populations of T cells were transduced with a dCas9-VP64-expressing lentivirus (CRISPRa), or KRAB-dCas9-expressing lentivirus (CRISPRi) and T cells that stably expressed dCas9 were selected with mCherry. The dCas9-VP64 or KRAB-dCas9 expressing T cell populations were then transfected with two genome-wide sgRNA libraries each to initiate CRISPR activation or interference of the T cells' genomes. For CRISPR activation Calabrese Sets A & B were used (see website at addgene.org/pooled-library/broadgpp-hurnan-crispra-calabrese-p65hsf/). For CRISPR interference, Dolcetto Sets A & B were used (see, addgene.org/pooled-library/broadgpp-hurnan-crispri-dolcetto/).

[0223] The T cell populations were stimulated with ImmunoCult[®] CD3/CD28/CD2 T cell activator (Stemcell Technologies, Vancouver, Canada), and the stimulated CRISPRa/i edited T cells from the two donors were sorted using fluorescent activated cell sorting (FACS) for the following markers: IL-2 cytokine production, IFN- γ production, and CellTraceM Violet for cell proliferation. Sorted cells were subjected to genomic DNA extraction, and sgRNAs were PCR amplified, followed by next-generation-sequencing, to determine sgRNA frequencies in each population. Data was analyzed using MaGeck version 0.5.9.2 Li et al. Genome Biol 15:544 (2014).

[0224] These screens identified 1074 unique genes with significant responses to those phenotypes (FDR<−0.01), including both known and novel genes in T cell function.

[0225] Table 1 below lists positive regulators of T cell functions as detected by IFN- γ production.

TABLE 1		
Positive Regulators of T Cell Functions As detected by Interferon- γ Production		
Positive Regulator Gene		Positive Interferon- γ Production
APOBEC3C		IFNG
APOBEC3D		IFNG
APOL2		IFNG
ASB12		IFNG
BACE2		IFNG
BCL9		IFNG
BICDL2		IFNG
C15orf52		IFNG
C1orf94		IFNG
CD2		IFNG
CD247		IFNG
CD28		IFNG
CNGB1		IFNG
CTSK		IFNG
DEAF1		IFNG
DEF6		IFNG
DEPDC7		IFNG
DKK2		IFNG
EMP1		IFNG
EOMES		IFNG
EP300		IFNG
FLT3		IFNG
FOSL1		IFNG
FOXQ1		IFNG
GINS3		IFNG
GLMN		IFNG
GNA11		IFNG
HELZ2		IFNG
HRASLS5		IFNG
IFNG		IFNG
IL1R1		IFNG

TABLE 1-continued		
Positive Regulators of T Cell Functions As detected by Interferon- γ Production		
Positive Regulator Gene		Positive Interferon- γ Production
IL9R		IFNG
KLHDC3		IFNG
KLRC4		IFNG
LAT		IFNG
LCP2		IFNG
LDB2		IFNG
LTBR		IFNG
MVB12A		IFNG
NBPF6		IFNG
NTT1		IFNG
NLRC3		IFNG
ORC1		IFNG
OTUD7A		IFNG
OTUD7B		IFNG
PIK3AP1		IFNG
PLCG2		IFNG
PRDM1		IFNG
PRKD2		IFNG
PROCA1		IFNG
RELA		IFNG
RNF217		IFNG
SAFB2		IFNG
SLC16A1		IFNG
SLC5A10		IFNG
SLC7A3		IFNG
SPPL2B		IFNG
TAGAP		IFNG
TBX21		IFNG
TMEM150B		IFNG
TMIGD2		IFNG
TNFRSF12A		IFNG
TNFRSF14		IFNG
TNFRSF1A		IFNG
TNFRSF1B		IFNG
TNFRSF8		IFNG
TNFRSF9		IFNG
TOR1A		IFNG
TPGS2		IFNG
TRADD		IFNG
TRAF3IP2		IFNG
TRIM21		IFNG
VAV1		IFNG
WT1		IFNG
ZNF630		IFNG
ZNF717		IFNG

[0226] Table 2 below lists positive regulators of T cell functions as detected by Interleukin-2 production.

TABLE 2		
Positive Regulators of T Cells as Detected by Interleukin-2 Production		
Positive Regulator Gene		Positive IL2 Production
ABCB10		IL2
ACSS2		IL2
ADAM19		IL2
ADAM23		IL2
ADAMTSS		IL2
ALKBH7		IL2
ALX4		IL2
ANXA2R		IL2
AP2A1		IL2
APOBEC3C		IL2

TABLE 2-continued

Positive Regulators of T Cells as Detected by Interleukin-2 Production	
Positive Regulator Gene	Positive IL2 Production
APOBEC3D	IL2
APOL2	IL2
ARNT	IL2
ART1	IL2
ASCL4	IL2
BEX4	IL2
BTG2	IL2
BTNL2	IL2
C11orf21	IL2
C12orf80	IL2
CBX4	IL2
CBY1	IL2
CCDC183	IL2
CCDC71L	IL2
CD2	IL2
CD28	IL2
CD6	IL2
CDKN1B	IL2
CDKN2C	IL2
CHERP	IL2
CIPC	IL2
CLIP3	IL2
CNGB1	IL2
CNR2	IL2
CREB5	IL2
CUL3	IL2
DCTN5	IL2
DEF6	IL2
DEPDC7	IL2
DYNLL2	IL2
EAPP	IL2
EEPD1	IL2
ELFN2	IL2
EMB	IL2
EMP1	IL2
EMP3	IL2
EP300	IL2
ERCC3	IL2
ESRP1	IL2
F2	IL2
FBXL13	IL2
FBXO41	IL2
FBNP1L	IL2
FOSB	IL2
FOSL1	IL2
FOXO4	IL2
FOXQ1	IL2
FUZ	IL2
GABRG1	IL2
GGTLC2	IL2
GNPDA1	IL2
GPR18	IL2
GPR20	IL2
GPR21	IL2
GPR84	IL2
GRIN3A	IL2
GSDMD	IL2
GSTM1	IL2
HCST	IL2
HELZ2	IL2
HEPHL1	IL2
IL2	IL2
IL2RB	IL2
IRX4	IL2
ISM1	IL2
KLF7	IL2
KLRC4	IL2
KRT18	IL2
LAT	IL2
LCP2	IL2

TABLE 2-continued

Positive Regulators of T Cells as Detected by Interleukin-2 Production	
Positive Regulator Gene	Positive IL2 Production
LHX6	IL2
LMNA	IL2
MAGEA9B	IL2
MAP3K12	IL2
MERTK	IL2
MTMR11	IL2
NDRG3	IL2
NTI1	IL2
NLR3	IL2
NLRP2	IL2
NPLOC4	IL2
ORC1	IL2
OSBPL7	IL2
OTOP3	IL2
OTUD7A	IL2
OTUD7B	IL2
P2RY14	IL2
PAFAH1B2	IL2
PCP4	IL2
PDE3A	IL2
PHF8	IL2
PIK3AP1	IL2
PLA2G3	IL2
PLCG2	IL2
POLK	IL2
POU2F2	IL2
PPIL2	IL2
PRAC1	IL2
PRKCB	IL2
PRKD2	IL2
RAB6A	IL2
RAC1	IL2
RAC2	IL2
RIPK3	IL2
RRAS2	IL2
RYR1	IL2
SAFB2	IL2
SCN3A	IL2
SDCCAG8	IL2
SERPINF1	IL2
SGTA	IL2
SHOC2	IL2
SIGLEC1	IL2
SIRT1	IL2
SLC16A1	IL2
SLC44A5	IL2
SLC5A5	IL2
SMC4	IL2
SPPL2B	IL2
SSUH2	IL2
SWAP70	IL2
TAF15	IL2
THEMIS	IL2
TM4SF4	IL2
TMEM79	IL2
TNFRSF10B	IL2
TNFSF11	IL2
TNRC6A	IL2
TPGS2	IL2
TRAF3IP2	IL2
TRIM21	IL2
TRMT5	IL2
TRPM4	IL2
TRPV5	IL2
TSPYL5	IL2
UBA52	IL2
UBL5	IL2
VAV1	IL2
WARS2	IL2
ZAP70	IL2

TABLE 2-continued

Positive Regulators of T Cells as Detected by Interleukin-2 Production	
Positive Regulator Gene	Positive IL2 Production
ZNF141	IL2
ZNF296	IL2
ZNF701	IL2

[0227] Table 3 below lists positive regulators of T cell functions as detected by T cell proliferation.

TABLE 3

Positive Regulators of T Cells as Detected Cell Proliferation	
Positive Regulator Gene	Increased Cell Proliferation
ABCB1	Proliferation
ASAP1	Proliferation
ATP10A	Proliferation
DEAF1	Proliferation
FOKK1	Proliferation
ITGAX	Proliferation
LCE6A	Proliferation
LCP2	Proliferation
LEFTY1	Proliferation
MYC	Proliferation
NAT8B	Proliferation
OLFM3	Proliferation
PLD6	Proliferation

[0228] Table 4 below lists negative regulators of T cell functions as detected by reduced IFN- γ production.

TABLE 4

Negative Regulators of T Cell Functions As detected by Less Interferon- γ Production	
Negative Regulator Genes	Negative Interferon- γ Production
ACER2	IFNG
ADGRV1	IFNG
AIF1L	IFNG
ALPL	IFNG
AMACR	IFNG
AMZ1	IFNG
ARHGAP30	IFNG
ARHGDIB	IFNG
ARHGEF11	IFNG
ARL11	IFNG
ATP2A2	IFNG
B3GNT5	IFNG
BACH2	IFNG
BLM	IFNG
BSG	IFNG
BTBD2	IFNG
BTLA	IFNG
BTRC	IFNG
CA11	IFNG
CASTOR2	IFNG
CBLB	IFNG
CCNT2	IFNG
CCSER1	IFNG
CD37	IFNG
CD44	IFNG

TABLE 4-continued

Negative Regulators of T Cell Functions As detected by Less Interferon- γ Production	
Negative Regulator Genes	Negative Interferon- γ Production
CD5	IFNG
CD52	IFNG
CD55	IFNG
CDK6	IFNG
CEACAM1	IFNG
CEBPA	IFNG
CEBPB	IFNG
CEP164	IFNG
CKAP2L	IFNG
CLCN2	IFNG
CLDN25	IFNG
COLQ	IFNG
CST5	IFNG
CTNNA1	IFNG
CYP24A1	IFNG
DDIT4L	IFNG
DENND3	IFNG
DGKG	IFNG
DGKK	IFNG
DGKZ	IFNG
DSC1	IFNG
EBF2	IFNG
ECEL1	IFNG
EIF3K	IFNG
EPB41	IFNG
EPS8L1	IFNG
FAM35A	IFNG
FAM53B	IFNG
FAM83A	IFNG
FKRP	IFNG
FOXA3	IFNG
FOXF1	IFNG
FOXF2	IFNG
FOXI3	IFNG
FOXJ1	IFNG
FOXL2	IFNG
FOXL2NB	IFNG
GABRQ	IFNG
GATA3	IFNG
GATA4	IFNG
GATA6	IFNG
GCM2	IFNG
GCSAM	IFNG
GCSAML	IFNG
GMFG	IFNG
GNL3L	IFNG
GRAP	IFNG
GRB2	IFNG
GRIA1	IFNG
GTSF1L	IFNG
HRH2	IFNG
HYLS1	IFNG
IKZF1	IFNG
IKZF3	IFNG
IL2RB	IFNG
INPPL1	IFNG
JMJD1C	IFNG
KCNV1	IFNG
KRIT1	IFNG
LAMB1	IFNG
LAPTM5	IFNG
LAT2	IFNG
LAX1	IFNG
LCK	IFNG
LENEP	IFNG
LMO4	IFNG
LRRC25	IFNG
LRRC4B	IFNG
LYN	IFNG
MAB21L2	IFNG

TABLE 4-continued

Negative Regulators of T Cell Functions As detected by Less Interferon- γ Production	
Negative Regulator Genes	Negative Interferon- γ Production
MAP4K1	IFNG
MBIP	IFNG
MBOAT1	IFNG
METTL23	IFNG
MIPEP	IFNG
MIPOL1	IFNG
MMP21	IFNG
MSMB	IFNG
MUC1	IFNG
MUC21	IFNG
MUC8	IFNG
N4BP1	IFNG
NAIF1	IFNG
NDNF	IFNG
NFATC1	IFNG
NFKB2	IFNG
NFKBIA	IFNG
NKX2-1	IFNG
NKX2-3	IFNG
NMB	IFNG
NR2F1	IFNG
ODF4	IFNG
OPRD1	IFNG
ORC5	IFNG
OTUD4	IFNG
PASD1	IFNG
PBK	IFNG
PCBP2	IFNG
PDLIM1	IFNG
PDPN	IFNG
PECAM1	IFNG
PIP5K1A	IFNG
PIP5K1B	IFNG
PITPNA	IFNG
POGZ	IFNG
POLK	IFNG
POU2AF1	IFNG
PSTPIP1	IFNG
PTPN12	IFNG
PTPRC	IFNG
PVRIG	IFNG
RAB14	IFNG
RBP7	IFNG
RETREG1	IFNG
RFC2	IFNG
RHCE	IFNG
RNF19B	IFNG
RNF2	IFNG
RUSC2	IFNG
SELPLG	IFNG
SETD1B	IFNG
SH3KBP1	IFNG
SIGLEC6	IFNG
SIPA1L1	IFNG
SLA	IFNG
SLA2	IFNG
SLC26A4	IFNG
SLC44A5	IFNG
SLC45A1	IFNG
SLC6A8	IFNG
SLC6A9	IFNG
SMAD9	IFNG
SMAGP	IFNG
SOCS3	IFNG
SOX13	IFNG
SPATA31A1	IFNG
SPN	IFNG
SPOCK3	IFNG
SPRED1	IFNG
STAP1	IFNG

TABLE 4-continued

Negative Regulators of T Cell Functions As detected by Less Interferon- γ Production	
Negative Regulator Genes	Negative Interferon- γ Production
STK35	IFNG
SULT6B1	IFNG
SYT15	IFNG
TEC	IFNG
TIAM1	IFNG
TMEM151A	IFNG
TMEM87B	IFNG
TMPRSS11E	IFNG
TNNT2	IFNG
TRIB2	IFNG
TRIM28	IFNG
TSPAN1	IFNG
UBASH3B	IFNG
UBQLN4	IFNG
UBXN7	IFNG
UNC119	IFNG
UPP1	IFNG
VPS28	IFNG
WLS	IFNG
ZKSCAN4	IFNG
ZNF445	IFNG
ZNF474	IFNG

[0229] Table 5 below lists negative regulators of T cell functions as detected by reduced Interleukin-2 production.

TABLE 5

Negative Regulators of T Cell Functions As detected by Less Interleukin-2 Production	
Negative Regulator Gene	Negative IL2 Production
ABI3BP	IL2
AEBP1	IL2
AHR	IL2
ANTXR2	IL2
ARHGAP15	IL2
ARHGAP27	IL2
ARHGDIB	IL2
ARID3A	IL2
ARL4D	IL2
B4GALNT3	IL2
BICD1	IL2
C10orf82	IL2
C17orf75	IL2
C19orf35	IL2
C1RL	IL2
C2orf69	IL2
C6orf132	IL2
C9orf84	IL2
CABP1	IL2
CBLB	IL2
CCSER1	IL2
CD34	IL2
CD4	IL2
CD5	IL2
CD52	IL2
CEACAM1	IL2
CEACAM7	IL2
CEBPB	IL2
CES3	IL2
CGB3	IL2
COL11A1	IL2
COL4A3	IL2
COLQ	IL2

TABLE 5-continued

Negative Regulators of T Cell Functions As detected by Less Interleukin-2 Production	
Negative Regulator Gene	Negative IL2 Production
CPEB3	IL2
CRELD2	IL2
CST9L	IL2
DDX55	IL2
DLG4	IL2
DOK1	IL2
EBF3	IL2
EIF3K	IL2
EN2	IL2
EOMES	IL2
EPB41	IL2
ETS1	IL2
F5	IL2
FAM96A	IL2
FHL1	IL2
FOXA3	IL2
FOXE1	IL2
FOXI3	IL2
FOXL2NB	IL2
FUS	IL2
FUT4	IL2
GCSAM	IL2
GCSAML	IL2
GDAP1L1	IL2
GDPD2	IL2
GMP	IL2
GNL3L	IL2
GOLPH3	IL2
GRAP	IL2
GRB2	IL2
HAUS7	IL2
HERC1	IL2
HLA-DQB2	IL2
HSD17B11	IL2
IKZF1	IL2
IKZF3	IL2
INPPL1	IL2
INTS10	IL2
ITIH2	IL2
ITPKA	IL2
ITPKB	IL2
ITPKC	IL2
JDP2	IL2
JKAMP	IL2
JMJD1C	IL2
KIAA1024	IL2
KIF15	IL2
KIF5A	IL2
KNTC1	IL2
LAT2	IL2
LAX1	IL2
LGR5	IL2
LIME1	IL2
LMBRD2	IL2
LOC401052	IL2
LONP2	IL2
LRCH3	IL2
LRRC23	IL2
LRRC25	IL2
LRRC52	IL2
LYN	IL2
LYPD1	IL2
MAATS1	IL2
MAB21L2	IL2
MAGEB17	IL2
MAP4K1	IL2
MEF2C	IL2
METTL9	IL2
MICU1	IL2
MRPL17	IL2

TABLE 5-continued

Negative Regulators of T Cell Functions As detected by Less Interleukin-2 Production	
Negative Regulator Gene	Negative IL2 Production
MUC1	IL2
NAIF1	IL2
NCF2	IL2
NDNF	IL2
NDUFB1	IL2
NHP2	IL2
NKX2-6	IL2
NLGN4Y	IL2
NNT	IL2
NPIP9	IL2
NR4A1	IL2
NR4A3	IL2
NRCAM	IL2
NRP1	IL2
NRSN2	IL2
NSUN7	IL2
OLFML1	IL2
OMP	IL2
OPRD1	IL2
OR1K1	IL2
OR2B11	IL2
OSBPL11	IL2
OTOG	IL2
OTUD4	IL2
PATL2	IL2
PAX5	IL2
PFKL	IL2
PHF2	IL2
PIBF1	IL2
PIP5K1A	IL2
PIP5K1B	IL2
PITPNC1	IL2
PLCL1	IL2
PLEKHM2	IL2
PPARG	IL2
PPIC	IL2
PSRC1	IL2
PSTPIP1	IL2
PTPN12	IL2
PTPN22	IL2
PTPN6	IL2
PTPRC	IL2
PVRIG	IL2
RBP4	IL2
RPL13A	IL2
S100A2	IL2
SALL4	IL2
SAMD8	IL2
SENP6	IL2
SETD1B	IL2
SEZ6L	IL2
SFT2D1	IL2
SH3TC1	IL2
SIGIRR	IL2
SIT1	IL2
SLA	IL2
SLA2	IL2
SLC20A2	IL2
SLC39A2	IL2
SLC6A8	IL2
SMAGP	IL2
SNRNP48	IL2
SOCS2	IL2
SORBS1	IL2
SOX13	IL2
SPN	IL2
SPRED1	IL2
SPRED2	IL2
SRPK1	IL2
STAP1	IL2

TABLE 5-continued

Negative Regulators of T Cell Functions As detected by Less Interleukin-2 Production	
Negative Regulator Gene	Negative IL2 Production
STK38L	IL2
SYPL1	IL2
TCF12	IL2
TEX35	IL2
TFCP2L1	IL2
TMEM14C	IL2
TMEM223	IL2
TMEM262	IL2
TNNT2	IL2
TPRA1	IL2
TRIM6	IL2
TRIM34	
TSPAN1	IL2
UBASH3B	IL2
UBE2W	IL2
UBR4	IL2
UBXN7	IL2
UCP1	IL2
UIMC1	IL2
ULK1	IL2
UPK3B	IL2
VPS28	IL2
VSTM5	IL2
XKR9	IL2
YLP1M1	IL2
ZDHHC7	IL2
ZEB1	IL2
ZEB2	IL2
ZNF445	IL2
ZNF70	IL2
ZNF831	IL2

[0230] Table 6 below lists negative regulators of T cell functions as detected by reduced cell proliferation.

TABLE 6

Negative Regulators of T Cell Functions As detected by Less Cell Proliferation	
Negative Regulator Gene	Decreased Cell Proliferation
ABCB1	Proliferation
ASAP1	Proliferation
ATP10A	Proliferation
DEAF1	Proliferation
FOXK1	Proliferation
ITGAX	Proliferation
LCE6A	Proliferation
LCP2	Proliferation
LEFTY1	Proliferation
MYC	Proliferation
NAT8B	Proliferation
OLFM3	Proliferation
PLD6	Proliferation
PREP	Proliferation
SULT1A1	Proliferation
SULT1A4	Proliferation
AHNAK	Proliferation
ARHGD1B	Proliferation
B3GNT5	Proliferation
CASZ1	Proliferation
CD27	Proliferation
CEBPB	Proliferation
CRHBP	Proliferation
FLI1	Proliferation

TABLE 6-continued

Negative Regulators of T Cell Functions As detected by Less Cell Proliferation	
Negative Regulator Gene	Decreased Cell Proliferation
FOSL2	Proliferation
HLX	Proliferation
MAP4K1	Proliferation
MUC21	Proliferation
MXI1	Proliferation
NDRG1	Proliferation
NEUROD2	Proliferation
SLC2A1	Proliferation
SLC43A3	Proliferation
SMAGP	Proliferation
SOX13	Proliferation
SP140	Proliferation
TP11	Proliferation
TTC39C	Proliferation

[0231] These regulators and agents that modulate these regulators can be used as T cell related immunotherapies for cancer or autoimmune diseases

Example 2: CRISPRi Identification of Genes that Regulate T Cells

[0232] This Example describes use of CRISPRi for screening of primary human T cells to identify genetic regulators of therapeutically relevant T cell phenotypes.

[0233] The two populations of T cells were transduced with a KRAB-dCas9-expressing lentivirus (CRISPRi) and T cells that stably expressed dCas9 were selected with mCherry. The KRAB-dCas9 expressing T cell populations were then transfected with two genome-wide sgRNA libraries each to initiate CRISPR interference of the T cells' genomes. For CRISPR interference, Dolcetto Sets A & B were used (see, addgene.org/pooled-library/broadgpp-human-crispri-dolcetto/).

[0234] The T cell populations were stimulated with ImmunoCult™ CD3/CD28/CD2 T cell activator (Stemcell Technologies, Vancouver, Canada), and the stimulated CRISPRi edited T cells from the two donors were sorted using fluorescent activated cell sorting (FACS) for the following markers: IL-2 cytokine production, IFN-γ production, and CellTrace™ Violet for cell proliferation. Sorted cells were subjected to genomic DNA extraction, and sgRNAs were PCR amplified, followed by next-generation-sequencing, to determine sgRNA frequencies in each population, Data was analyzed using MafGeck version 0.5.9.2 Li et al. Genome Biol 15:544 (2014). Table 7 lists the genes that modulated T cell functions.

TABLE 7

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
ANKRD17	IFNG	positive
AQP3	IFNG	positive
ARID4B	IFNG	positive
ATP6V1C1	IFNG	positive
ATPAF1	IFNG	positive
ATXN7	IFNG	positive
BCAT2	IFNG	positive

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
BCL10	IFNG	positive
CASD1	IFNG	positive
CBFB	IFNG	positive
CD2	IFNG	positive
CD247	IFNG	positive
CD28	IFNG	positive
CD3D	IFNG	positive
CD3E	IFNG	positive
CD3G	IFNG	positive
CD4	IFNG	positive
CHUK	IFNG	positive
CNP	IFNG	positive
COG3	IFNG	positive
CREBBP	IFNG	positive
CUL1	IFNG	positive
DDA1	IFNG	positive
DDX60L	IFNG	positive
DEF6	IFNG	positive
DHDDS	IFNG	positive
DHX29	IFNG	positive
DPP9-AS1	IFNG	positive
ELOF1	IFNG	positive
ERC1	IFNG	positive
ETNK1	IFNG	positive
EXOC4	IFNG	positive
FAM133B	IFNG	positive
FGFR1OP	IFNG	positive
FTTM2	IFNG	positive
FLVCR2	IFNG	positive
FNDC4	IFNG	positive
GOSR1	IFNG	positive
GPX7	IFNG	positive
GRAP2	IFNG	positive
HARS	IFNG	positive
HNRNPL	IFNG	positive
HOXD13	IFNG	positive
IFNG	IFNG	positive
IFNGR1	IFNG	positive
IFNGR2	IFNG	positive
IKBKB	IFNG	positive
IKBKG	IFNG	positive
IL21R	IFNG	positive
INPP1	IFNG	positive
ITK	IFNG	positive
JAK1	IFNG	positive
JUN	IFNG	positive
KAT7	IFNG	positive
KCNIP3	IFNG	positive
KIAA1109	IFNG	positive
KIDINS220	IFNG	positive
LAT	IFNG	positive
LCK	IFNG	positive
LCP2	IFNG	positive
LIMS2	IFNG	positive
LOC101927322	IFNG	positive
LRIG1	IFNG	positive
MALT1	IFNG	positive
MAP3K7	IFNG	positive
MBD2	IFNG	positive
MEAF6	IFNG	positive
MEN1	IFNG	positive
MMP24	IFNG	positive
MOB4	IFNG	positive
MYLIP	IFNG	positive
NDFIP2	IFNG	positive
NSD2	IFNG	positive
NSFL1C	IFNG	positive
NYNRIN	IFNG	positive
OSBP	IFNG	positive
PCYT2	IFNG	positive
PGBD5	IFNG	positive
PI4KB	IFNG	positive
PLCG1	IFNG	positive

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
PRDM1	IFNG	positive
PRKAR1A	IFNG	positive
PRKD2	IFNG	positive
PRRC2B	IFNG	positive
PTPRC	IFNG	positive
RAC2	IFNG	positive
RAET1L	IFNG	positive
RBCK1	IFNG	positive
RDX	IFNG	positive
RHOA	IFNG	positive
RHOG	IFNG	positive
ROPN1B	IFNG	positive
RRAS2	IFNG	positive
RTP2	IFNG	positive
SAE1	IFNG	positive
SCRIB	IFNG	positive
SEC61A1	IFNG	positive
SEC62	IFNG	positive
SEH1L	IFNG	positive
SEL1L	IFNG	positive
SH2D1A	IFNG	positive
SHOC2	IFNG	positive
SLC38A6	IFNG	positive
SLC3A2	IFNG	positive
SPCS2	IFNG	positive
SPTLC2	IFNG	positive
SPTSSA	IFNG	positive
SRD5A2	IFNG	positive
SRP19	IFNG	positive
SRP68	IFNG	positive
SRP72	IFNG	positive
SRPRB	IFNG	positive
SSB	IFNG	positive
STAT3	IFNG	positive
SUGT1	IFNG	positive
SULT2B1	IFNG	positive
SUPT5H	IFNG	positive
SYT15	IFNG	positive
TADA1	IFNG	positive
TADA2B	IFNG	positive
TAF11	IFNG	positive
TAF13	IFNG	positive
TAF2	IFNG	positive
TAF6L	IFNG	positive
TARS	IFNG	positive
TBX21	IFNG	positive
TLN1	IFNG	positive
TMX1	IFNG	positive
TNFRSF1A	IFNG	positive
TRAF6	IFNG	positive
TRIM21	IFNG	positive
TXK	IFNG	positive
UBA2	IFNG	positive
VAV1	IFNG	positive
VPS29	IFNG	positive
VPS35	IFNG	positive
VPS37C	IFNG	positive
VPS41	IFNG	positive
WAS	IFNG	positive
XPO6	IFNG	positive
ZAP70	IFNG	positive
ARHGAP15	IFNG	negative
BRD9	IFNG	negative
BRIP1	IFNG	negative
CAD	IFNG	negative
CBLB	IFNG	negative
CBLL1	IFNG	negative
CD5	IFNG	negative
CDK12	IFNG	negative
CHERP	IFNG	negative
CPSF2	IFNG	negative
CPSF6	IFNG	negative
CSTF3	IFNG	negative

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
CTDSPL2	IFNG	negative
DGKZ	IFNG	negative
E2F1	IFNG	negative
EIF3B	IFNG	negative
EIF3D	IFNG	negative
EIF3K	IFNG	negative
EIF4E2	IFNG	negative
GCN1	IFNG	negative
GIGYF2	IFNG	negative
GNAI2	IFNG	negative
HIF1AN	IFNG	negative
IKBKE	IFNG	negative
LARGE2	IFNG	negative
LAT2	IFNG	negative
MAP4K1	IFNG	negative
MCM2	IFNG	negative
METAP2	IFNG	negative
METTL3	IFNG	negative
MTF1	IFNG	negative
MYB	IFNG	negative
NFKB2	IFNG	negative
NIT1	IFNG	negative
NMT1	IFNG	negative
NRF1	IFNG	negative
NUDC	IFNG	negative
PCBP2	IFNG	negative
PDGFRA	IFNG	negative
PITPNB	IFNG	negative
PNISR	IFNG	negative
PPP1R8	IFNG	negative
PRMT1	IFNG	negative
PSMD13	IFNG	negative
PSMD4	IFNG	negative
PTMA	IFNG	negative
RAB4A	IFNG	negative
RBPJ	IFNG	negative
RIOK2	IFNG	negative
RNF20	IFNG	negative
RNF40	IFNG	negative
RPL19	IFNG	negative
RPL26	IFNG	negative
RPL35A	IFNG	negative
RPL38	IFNG	negative
RPL6	IFNG	negative
RPS13	IFNG	negative
RPS17	IFNG	negative
RPS8	IFNG	negative
SCRN3	IFNG	negative
SF3A1	IFNG	negative
SLA2	IFNG	negative
SLAMF6	IFNG	negative
SMC3	IFNG	negative
SP1	IFNG	negative
SPN	IFNG	negative
SYMPK	IFNG	negative
THOC3	IFNG	negative
TONSL	IFNG	negative
TSC1	IFNG	negative
U2AF2	IFNG	negative
UBASH3A	IFNG	negative
UNCX	IFNG	negative
USP5	IFNG	negative
ZC3H18	IFNG	negative
BCL10	IL2	positive
CASD1	IL2	positive
CD2	IL2	positive
CD247	IL2	positive
CD28	IL2	positive
CD3D	IL2	positive
CD3E	IL2	positive
CD3G	IL2	positive
CHD7	IL2	positive
DEF6	IL2	positive

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
DNTTIP1	IL2	positive
ELOF1	IL2	positive
GRAP2	IL2	positive
IFNGR2	IL2	positive
IL2	IL2	positive
ITK	IL2	positive
KIDINS220	IL2	positive
LAT	IL2	positive
LCP2	IL2	positive
NDFIP2	IL2	positive
NDUFB1	IL2	positive
NYNRIN	IL2	positive
PGBD5	IL2	positive
PLCG1	IL2	positive
PRKD2	IL2	positive
RAC2	IL2	positive
RHOG	IL2	positive
RPN2	IL2	positive
SCRIB	IL2	positive
SHOC2	IL2	positive
SIN3B	IL2	positive
SPRYD3	IL2	positive
SRP19	IL2	positive
SRP68	IL2	positive
SRP72	IL2	positive
SULT2B1	IL2	positive
TAF11	IL2	positive
TAF13	IL2	positive
TAF2	IL2	positive
TAF8	IL2	positive
TLN1	IL2	positive
TRIM21	IL2	positive
VAV1	IL2	positive
VPS29	IL2	positive
VPS35	IL2	positive
WAS	IL2	positive
ZAP70	IL2	positive
ACTL6A	IL2	negative
ADSS	IL2	negative
ANLN	IL2	negative
ARHGAP15	IL2	negative
ARID2	IL2	negative
ATP1A1	IL2	negative
AUNIP	IL2	negative
BECN1	IL2	negative
BMS1	IL2	negative
BOP1	IL2	negative
C21orf62	IL2	negative
CAD	IL2	negative
CBFB	IL2	negative
CBLB	IL2	negative
CD5	IL2	negative
CDC23	IL2	negative
CDK12	IL2	negative
CENPE	IL2	negative
CENPI	IL2	negative
CEP192	IL2	negative
CHAF1B	IL2	negative
CHERP	IL2	negative
CHMP3	IL2	negative
CHMP5	IL2	negative
CHMP6	IL2	negative
CNOT1	IL2	negative
CPSF4	IL2	negative
CPSF6	IL2	negative
CTDSPL2	IL2	negative
CTPS1	IL2	negative
DDX47	IL2	negative
DGKZ	IL2	negative
DHODH	IL2	negative
DLST	IL2	negative
DNTTIP2	IL2	negative
DPY19L3	IL2	negative

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
E2F1	IL2	negative
EDC4	IL2	negative
EFTUD2	IL2	negative
EIF3B	IL2	negative
EIF3D	IL2	negative
EIF3E	IL2	negative
EIF3K	IL2	negative
EIFSA	IL2	negative
EP400	IL2	negative
ESF1	IL2	negative
FADD	IL2	negative
FAM49B	IL2	negative
FAM60A	IL2	negative
FAU	IL2	negative
GCN1	IL2	negative
GIGYF2	IL2	negative
GINS3	IL2	negative
HGS	IL2	negative
IL2RA	IL2	negative
IL2RB	IL2	negative
IL2RG	IL2	negative
ILF2	IL2	negative
INTS3	IL2	negative
JPH1	IL2	negative
KANSL3	IL2	negative
KAT5	IL2	negative
KLF2	IL2	negative
LAMTOR2	IL2	negative
LOC401052	IL2	negative
MAD2L1BP	IL2	negative
MAK16	IL2	negative
MAP4K1	IL2	negative
MAU2	IL2	negative
MCM2	IL2	negative
MCM3AP	IL2	negative
MEMO1	IL2	negative
METAP2	IL2	negative
MMP16	IL2	negative
MRPL22	IL2	negative
MST1L	IL2	negative
MYCBP2	IL2	negative
NARFL	IL2	negative
NEPRO	IL2	negative
NFKB2	IL2	negative
NMT1	IL2	negative
NRF1	IL2	negative
NUDC	IL2	negative
OTUB1	IL2	negative
PCBP1	IL2	negative
PCBP2	IL2	negative
PDGFRA	IL2	negative
PFA5	IL2	negative
PITPNB	IL2	negative
PNISR	IL2	negative
POLE	IL2	negative
POLR1B	IL2	negative
PPAN	IL2	negative
PPIH	IL2	negative
PPP1R8	IL2	negative
PRMT1	IL2	negative
PRPF4B	IL2	negative
PRR12	IL2	negative
PSMD2	IL2	negative
PTPN23	IL2	negative
PUM1	IL2	negative
RAB4A	IL2	negative
RASA2	IL2	negative
RBM14	IL2	negative
RBM25	IL2	negative
RBM42	IL2	negative
RBSN	IL2	negative
RCL1	IL2	negative
RMND5A	IL2	negative

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
RNF20	IL2	negative
RNF40	IL2	negative
RPL10	IL2	negative
RPL10A	IL2	negative
RPL13	IL2	negative
RPL14	IL2	negative
RPL15	IL2	negative
RPL18	IL2	negative
RPL19	IL2	negative
RPL23A	IL2	negative
RPL24	IL2	negative
RPL26	IL2	negative
RPL27	IL2	negative
RPL34	IL2	negative
RPL35	IL2	negative
RPL36	IL2	negative
RPL37A	IL2	negative
RPL38	IL2	negative
RPL6	IL2	negative
RPL7A	IL2	negative
RPL8	IL2	negative
RPL9	IL2	negative
RPLP1	IL2	negative
RPS11	IL2	negative
RPS13	IL2	negative
RPS16	IL2	negative
RPS17	IL2	negative
RPS20	IL2	negative
RPS23	IL2	negative
RPS24	IL2	negative
RPS25	IL2	negative
RPS3	IL2	negative
RPS3A	IL2	negative
RPS4X	IL2	negative
RPS5	IL2	negative
RPS7	IL2	negative
RPS8	IL2	negative
RUUBL2	IL2	negative
SART1	IL2	negative
SETD1A	IL2	negative
SLAMF6	IL2	negative
SLBP	IL2	negative
SMARCE1	IL2	negative
SMC1A	IL2	negative
SMC3	IL2	negative
SNRNP27	IL2	negative
SNRNP70	IL2	negative
SNRPC	IL2	negative
SNRPF	IL2	negative
SP1	IL2	negative
SRFBP1	IL2	negative
SRSF1	IL2	negative
STAT5B	IL2	negative
SURF6	IL2	negative
SYMPK	IL2	negative
TBL1X	IL2	negative
THOC3	IL2	negative
TNPO3	IL2	negative
TRAF2	IL2	negative
TRAP	IL2	negative
TSC1	IL2	negative
TSG101	IL2	negative
TUBGCP5	IL2	negative
TYMS	IL2	negative
U2AF1	IL2	negative
U2AF2	IL2	negative
UBASH3A	IL2	negative
UBASH3B	IL2	negative
UPF1	IL2	negative
UTP14A	IL2	negative
UTP15	IL2	negative
VPS28	IL2	negative
WDR45	IL2	negative

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
WDR5	IL2	negative
YEATS4	IL2	negative
ZMAT2	IL2	negative
AAMP	Proliferation	positive
AARS	Proliferation	positive
AATF	Proliferation	positive
AK2	Proliferation	positive
ALDH18A1	Proliferation	positive
AP2M1	Proliferation	positive
ATIC	Proliferation	positive
ATP1A1	Proliferation	positive
ATP5O	Proliferation	positive
ATP6V1B2	Proliferation	positive
ATP6V1F	Proliferation	positive
ATXN10	Proliferation	positive
BMS1	Proliferation	positive
BOP1	Proliferation	positive
BUD23	Proliferation	positive
C12orf60	Proliferation	positive
CAD	Proliferation	positive
CARS	Proliferation	positive
CCDC86	Proliferation	positive
CCT6A	Proliferation	positive
CD247	Proliferation	positive
CD3D	Proliferation	positive
CD3E	Proliferation	positive
CD3EAP	Proliferation	positive
CD3G	Proliferation	positive
CINP	Proliferation	positive
CLNS1A	Proliferation	positive
CPSF4	Proliferation	positive
CRCP	Proliferation	positive
CTPS1	Proliferation	positive
DAD1	Proliferation	positive
DDX27	Proliferation	positive
DDX52	Proliferation	positive
DGCR8	Proliferation	positive
DHODH	Proliferation	positive
DHX29	Proliferation	positive
DHX37	Proliferation	positive
DICER1	Proliferation	positive
DNAJA3	Proliferation	positive
DNM2	Proliferation	positive
DNTTIP2	Proliferation	positive
DPH6	Proliferation	positive
DROSHA	Proliferation	positive
EIF2B2	Proliferation	positive
EIF2B3	Proliferation	positive
EIF2B4	Proliferation	positive
EIF5A	Proliferation	positive
ELP4	Proliferation	positive
ESF1	Proliferation	positive
EXOSC4	Proliferation	positive
EXOSC5	Proliferation	positive
EXOSC7	Proliferation	positive
EXOSC9	Proliferation	positive
FAM149B1	Proliferation	positive
FARSB	Proliferation	positive
FBL	Proliferation	positive
FCF1	Proliferation	positive
FH	Proliferation	positive
FLVCR1	Proliferation	positive
FTSJ3	Proliferation	positive
GFER	Proliferation	positive
GMPS	Proliferation	positive
GNL2	Proliferation	positive
GNL3L	Proliferation	positive
GNPAT	Proliferation	positive
GTF3C1	Proliferation	positive
HARS	Proliferation	positive
HAUS4	Proliferation	positive
HCCS	Proliferation	positive
HEATR3	Proliferation	positive

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
HSD17B10	Proliferation	positive
HSD17B12	Proliferation	positive
HSPA9	Proliferation	positive
IL2RG	Proliferation	positive
IMPDH2	Proliferation	positive
ISG20L2	Proliferation	positive
KARS	Proliferation	positive
LAGE3	Proliferation	positive
LAT	Proliferation	positive
LCP2	Proliferation	positive
LETM1	Proliferation	positive
LONP1	Proliferation	positive
MARS2	Proliferation	positive
MDN1	Proliferation	positive
METTL16	Proliferation	positive
MMACHC	Proliferation	positive
MRPL16	Proliferation	positive
MRPL22	Proliferation	positive
MRPL35	Proliferation	positive
MRPL36	Proliferation	positive
MRPL37	Proliferation	positive
MRPL41	Proliferation	positive
MRPL42	Proliferation	positive
MRPL45	Proliferation	positive
MRPL54	Proliferation	positive
MRPS11	Proliferation	positive
MRPS14	Proliferation	positive
MRPS17	Proliferation	positive
MRPS18A	Proliferation	positive
MRPS2	Proliferation	positive
MRPS23	Proliferation	positive
MRPS33	Proliferation	positive
MRPS5	Proliferation	positive
MRPS9	Proliferation	positive
MTHED1L	Proliferation	positive
MTOR	Proliferation	positive
MYBBP1A	Proliferation	positive
NAT10	Proliferation	positive
NCL	Proliferation	positive
NEPRO	Proliferation	positive
NIFK	Proliferation	positive
NOC2L	Proliferation	positive
NOL10	Proliferation	positive
NOL6	Proliferation	positive
NOL8	Proliferation	positive
NOP14	Proliferation	positive
NOP2	Proliferation	positive
NOP56	Proliferation	positive
NOP58	Proliferation	positive
NUBP1	Proliferation	positive
NUFIP1	Proliferation	positive
ORAOV1	Proliferation	positive
PAM16	Proliferation	positive
PCYT2	Proliferation	positive
PDCD11	Proliferation	positive
PDGFRA	Proliferation	positive
PDHA1	Proliferation	positive
PDSS2	Proliferation	positive
PELP1	Proliferation	positive
PGD	Proliferation	positive
PGM3	Proliferation	positive
PHB	Proliferation	positive
PHB2	Proliferation	positive
PISD	Proliferation	positive
PITRM1	Proliferation	positive
PMPCA	Proliferation	positive
PNO1	Proliferation	positive
PNPT1	Proliferation	positive
POLG2	Proliferation	positive
POLR1A	Proliferation	positive
POLR1C	Proliferation	positive
POLR1D	Proliferation	positive
POLR2E	Proliferation	positive

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
POLR3B	Proliferation	positive
POLRMT	Proliferation	positive
POP1	Proliferation	positive
POP4	Proliferation	positive
POP5	Proliferation	positive
POT1	Proliferation	positive
PPAN	Proliferation	positive
PPAT	Proliferation	positive
PSMG1	Proliferation	positive
QARS	Proliferation	positive
RAC2	Proliferation	positive
RBM19	Proliferation	positive
RCL1	Proliferation	positive
RIOK1	Proliferation	positive
RIOK2	Proliferation	positive
ROMO1	Proliferation	positive
RPF1	Proliferation	positive
RPF2	Proliferation	positive
RPL28	Proliferation	positive
RPL30	Proliferation	positive
RPL39	Proliferation	positive
RPLP2	Proliferation	positive
RPN2	Proliferation	positive
RPP21	Proliferation	positive
RPP30	Proliferation	positive
RPS11	Proliferation	positive
RPS12	Proliferation	positive
RPS15	Proliferation	positive
RPS17	Proliferation	positive
RPS19BP1	Proliferation	positive
RPS27	Proliferation	positive
RPS4X	Proliferation	positive
RPS6	Proliferation	positive
RPSA	Proliferation	positive
RRP12	Proliferation	positive
RRP36	Proliferation	positive
RRP7A	Proliferation	positive
RRP9	Proliferation	positive
RSL24D1	Proliferation	positive
SAMM50	Proliferation	positive
SARS	Proliferation	positive
SDHC	Proliferation	positive
SEH1L	Proliferation	positive
SLC35B1	Proliferation	positive
SLC38A6	Proliferation	positive
SLC7A11	Proliferation	positive
SPOUT1	Proliferation	positive
SRFBP1	Proliferation	positive
SSB	Proliferation	positive
SURF6	Proliferation	positive
TAF1A	Proliferation	positive
TAF1C	Proliferation	positive
TAF1D	Proliferation	positive
TAF8	Proliferation	positive
TAMM41	Proliferation	positive
TARS	Proliferation	positive
TEX10	Proliferation	positive
TIMM44	Proliferation	positive
TNKS1BP1	Proliferation	positive
TOMM20	Proliferation	positive
TOMM40	Proliferation	positive
TP53I13	Proliferation	positive
TRMT112	Proliferation	positive
TRMT5	Proliferation	positive
TRNT1	Proliferation	positive
TSEN2	Proliferation	positive
TSEN54	Proliferation	positive
TSR1	Proliferation	positive
TTI1	Proliferation	positive
TWISTNB	Proliferation	positive
TWNK	Proliferation	positive
UMPS	Proliferation	positive
UQCR10	Proliferation	positive

TABLE 7-continued

Genes from CRISPRi Screen that Modulate T Cells		
Gene	Screen	Positive or Negative Regulator
UQCRB	Proliferation	positive
UQCRC1	Proliferation	positive
UTP11	Proliferation	positive
UTP14A	Proliferation	positive
UTP3	Proliferation	positive
UTP6	Proliferation	positive
VAV1	Proliferation	positive
VPS29	Proliferation	positive
VPS72	Proliferation	positive
WAS	Proliferation	positive
WDR12	Proliferation	positive
WDR3	Proliferation	positive
WDR36	Proliferation	positive
WDR55	Proliferation	positive
XRCC5	Proliferation	positive
XRCC6	Proliferation	positive
YAE1D1	Proliferation	positive
ZAP70	Proliferation	positive
ZCCHC9	Proliferation	positive
ZNHIT3	Proliferation	positive
ZNHIT6	Proliferation	positive
ZNRD1	Proliferation	positive

[0235] This screen identified quite a few of the same genes as were identified in the screen described in Example 1. The following genes were new genes identified by this CRISPRi screen: HNRNPL, IHOXD13, IFNGR1, IFNGR2, IKBKB, IKBKG, IL21R, INPPI, ITKJAK1, JUN, KA T7, KCNIP3, KIAA1 A109, KID[NS220, LIMS2, LOC101927322, LRIG1, MALTI, MAP3K7, IViBD2, MEAF6, MENI, MMP4, MOB4, MYLIP, NDFIP2, NSD2, NSFL1C, NYN-RIN, OSBP-, PCY T2, PGBD5, PI4 KB, PLCG1, PRKAR1A, PRRC2B, RAET1 L, RBCK1, RDX, RHOA, RIOG1, ROPNIB, RTP2, SAE1, SCRIB, SEC61A1, SEC62, SEH1L, SEL1L, SI-2D11A, SLC38A6, SLC3A2, SPCS2, SPTLC2, SPTSSA, SRD5A2, SRP19, SRP68, SRP72, SRPRB, SSB, STAT3, SUGTI, SULT2B1, SUPTIH1, TADA1, TADA2B, TAF 1, TAF13, TAF2, TAF6L, TARS, TLN1, TMNIX1, TRAF6, TXK, UBA2, VPS29, VPS35, VPS37C, VPS41, WAS, XPO6, BRD9, BIRIP1, CAD, CBLI1, CDK12, CPSF2, CPSF6, CSTF3, CTDSPL2, E2F1, EIF3B, EIF3D, EIF4E2, GCN1, GIGYF2, iNA12, HIFIAN, IKBKE, LARGE2, MCCM2, METAP2, METTL3, M-TF1, MYB, NMT1, NRFI, NUDC, PDGFRA, PITPNB, PNISR, PPP1R8, PRMT1, PSMD13, PSMD4, PTMA, RAB4A, RBPJ, RIOK2, RNF20, RNF40, RPL19, RPL26, RPL35A, RPL38, RPL6, RPS13, RPS17, RPS8, SCRN3, SF3A1, SLAMF6, SMC3, SPI, SYMHPK, THOC3, TONSL, TSCI, U2AF2, UBASH3A, 10 UNCX, USP5, ZC3H18, BCL10, CASD1, CD3D, CD3E, CD3G, CHD7, DNTTIP1, ELOF1, GRAP2, IFNGR2, ITK, KIDINS220, NDFIP2, NYN-RIN, PGBD5, PLCG1, RHOG, RPN2, SCRIB, SIN3B, SPRYD3, SRP19, SRP68, SRP72, SULT2B1, TAF11, TAF13, TAF2, TAF8, TLN1, VPS29, VPS35, WAS, ACTL6A, ADSS, ANLN, ARID2, ATP1A1, AUNIP, BECN1, BMS1, BOP1, C2Iorf62, CAD, CBFB, CDC23, CDK12, CEN⁺E, CENPI, CEP192, CHAFIB, CIHMIP3, CHMP5, CHMP6, CNOT1, CPSF4, CPSF6, CTDSPL2, CTPS1, DDX4.7, DHODH, DLST, DNTTIP2, DPY19L3, E2FI, EDC4, EFTUID2, EIF3B, EIF3D, EIF3E, EF5A, EP400, ESF1, FADD, FAM-149B, FAM60A, FAU, GCN1, GIGYF2, HGS, IL2RA, IL2EG, ILF2, INTS3, JPH1, KANSL3, KAT5, KLF2, LAMTOR2, MAD21BP, MAK16, MAU2, MC 12, MCM3AP, MEMO1, T MT P2, MMIVP16, MRPL22, MSTIL, MYCBP2, NAREL, NEPRO, IMT, N⁺FI, NUDC,

OTUB1, PCBPI, PDGFRA, PEAS, PITPN3, PNISR, POLE, POLRIB, PPAN, PPI-H, PPIR8, PRMT1, PRPF4B, PRR12, PSMD2, PTPN23, PUM1, RAB4A, RASA2, RBM14, RBM25, RBM42, RBSN, RCL1, RMND5A, RNF20, RNF40, RPL10, RPL10A, RPL13, RPL14, RPL15, RPL18, RPL19, RPL23A, RPL24, RPL26, RPL27, RPL34, RPL35, RPL36, RPL37A, RPL38, RPL6, RPL7A, RPL8, RPL9, RPLP1, RPLP11, RPS13, RPS16, RPS17, RPS20, RPS23, RPS24, RPS25, RPS3, RPS3A, RPS4X, RPS5, RPS7, RPS8, RUVBL2, SART1, SETD1A, SLAMF6, SLBP, SMARCE1, SMC1A, SMC3, SNRNP27, SNRNP70, SNRPC, SNRPF, SPI, SRFBP1, SRSF1, STAT5B, SURF6, SYMPK, TBLIX, THOC3, TNPO3, TRAF2, TRAFIP, TSCI, TSGH01, 30⁺ T1CP51³⁵, TYMS, U2AF1, U2AF2, UBASL3A, UPF1, UTP14A, UTP15, WDR45, WDR5, YEATS4, ZMAT2, AAMP, AARS, AATF, AK2, ALDH18A1, AP2M1, A TIC, ATP1A1, ATP5O, ATP6V1B2, ATP6V1F, ATXN10, BMS1, BOPI, BUD23, C12orf60, CAD, CARS, CCDC86, CCT6A, CD3D, CD3E, CD3EAP, CD3G, CINP, CLNS1A, CPSF4, CRCP, CTPS1, DAD1, DDX27, DDX52, DGCR8, DHOD1-1, D-1X29, DUX37, DICER1, DNAJA3, DNM2, DNMT1P2, DPH6, DROSHA, EIF2B2, EIF2B3, EIF2B4, EF5A, ELP4, ESF1, EXOSC4, EXOSC5, EXOSC7, EXOSC9, FAM149B1, FARSB, FBL, FCF1, FH, FLNVCR-1, FTSJ3, GFER, GMPS, GNL2, GNPAT, GTF3C1, H ARS, HAUS4, HCCS, HEATR3, USD17B1 0, HSD17B12, HSPA9, IL2RG, IMPD2, ISG20L2, KARS, LAGE3, LETMI, LONPI1 MARS2, MDN1 MIETTL6, MMACHC, MRPI16, MRPL22, MRPL5, MRPL36, MRPL37, MRPL41, MRPL42, MRPL45, MRPL54, MRPS11, MRPS14, MRPS17, MRPS18A, MRPS2, MRPS23, MRPS33, MRPS5, MRPS9, MTHFD1L, MOR, MYBBPIA, NATO1 NCL, NEPRO, NIFK, NOC2L, NOLI, NOL6, NOL8, NOP14, NOP2, NOP56, NOP58, NUBPI, NLUFIP1, ORAOV1, PAM16, PCYT2, PDCD11, PDGFRA, PDHAI, PDSS2, PELPI, PGD, PGI3, PHB, PHB2, PISD, PVTRMI, PMPCA, PNOI, PNPTI, POLG2, POLRI A, POLRIC, POLRID, POLR2E, POLR3B, POLRMT, POP1, POP4, POP5, POTI, PPAN, PPAT, PSMGI, QARS, RBM19, RCCLI, RIOKI, RJOK2, ROMOI, RPF1, RPF2, RPL28, RPL30, RPL39, RPLP2, RPN2, RPP21, RPP30, RPS11, RPS12, RPS15, RPS17, RPS19BPI, RPS27, RPS4X, RPS6, RPSA, RRP12, RRP36, RRP7A, RRP9, RSL24-D1, SAMM50, SARS, SDHC, SEH1L, SLC35B1, SLC38A6, SLC7A11, SPOUT1, SRFBP1, SSB, SURF6, TAFIA, TAF1C, TAFID, TAF8, TAMM41, TARS., TEX10, TIMM44, TNKS1BPI, TOMM20, TOMM40, TP53113, TRMT 12, TRNT1, TSEN2, TSEN54, TSRI, TT 1, TWJSTNB, TWNK, UMPS, UQCR10, UQCRB, UQCRC1, UTP1 1, UTP14A, UTP3, UTP6, VPS29, VPS72, WAS, WDRI2, W²DR3, WDR36, WDR55, XRCC5, XRCC6, YAEID1, ZCCHC9, ZNHIT3, ZNHIT6, and ZNRD1.

[0236] These regulators and agents that modulate these regulators can be used as T cell related immunotherapies for cancer or autoimmune diseases.

Example 3: CRISPRa Screening Primary fHuman T cells to Identify Genetic Regulators

INTRODUCTION

[0237] Regulated T cell cytokine production in response to stimulation plays a role in balanced immune responses. Cytokine dysregulation can lead to autoimmunity, immunodeficiency, and immune evasion in cancer (1-4). Interleukin 2 (IL-2), secreted predominantly by CD4⁺ cells, drives T cell expansion (5) and is therapeutically applied in autoimmunity

and cancer at different doses (6). Interferon gamma (IFN- γ) is a cytokine secreted by both CD4⁺ and CD8⁺ cells that promotes a type I immune response against intracellular pathogens including viruses (4) and correlates with positive cancer immunotherapy responses (7-9). Much of our current understanding of the pathways leading to cytokine production in humans originates from studies in transformed T cell lines, which often are not representative of primary human cell biology (10-12). Comprehensive understanding of pathways that control cytokine production in primary human T cells would facilitate the development of next-generation immunotherapies.

[0238] Unbiased forward genetic approaches can uncover the components of regulatory networks systematically but challenges with efficient Cas9 delivery have limited their application in primary cells. Genome-wide CRISPR knockout screens have been completed using primary mouse immune cells from Cas9-expressing transgenic mice (13-15), including a screen for regulators of innate cytokine production in dendritic cells (13). Genome-scale CRISPR studies in human primary cells have recently been accomplished using transient Cas9 electroporation to introduce gene knockouts (16, 17). However, comprehensive discovery of regulators requires both gain-of-function and loss-of-function studies. For example, CRISPRa gain-of-function screens can discover genes that may not normally be active in the tested conditions, but which can promote phenotypes of interest (18, 19). In contrast to a CRISPR knockout, CRISPRa or CRISPRi require the sustained expression of an activator-linked endonuclease-dead Cas9 (dCas9) and due to poor lentiviral delivery has been limited to small scale experiments in primary cells (20, 21). Here we developed a CRISPRa and CRISPRi screening platform in primary human T cells, which allowed for the systematic discovery of genes and pathways that can be perturbed to tune stimulation-dependent cytokine responses.

Materials and Methods

Isolation and Culture of Human IT cells

[0239] Human T cells were sourced from PBMC-enriched leukapheresis products (Leukopaks, Stemcell Technologies cat 70500.2) from healthy donors, following IRB approved informed written consent (Stemcell Technologies). Bulk T cells were isolated from Leukopaks using EasySep magnetic selection following manufacturers' recommended protocol (Stemcell Technologies cat 17951). Unless stated otherwise, bulk T cells were frozen in Bamberker Cell Freezing Medium at 5×10^7 cells/ml (Bulldog Bio cat BB01) and stored at -80° C. for short-term or in liquid nitrogen for long-term storage immediately after isolation. Unless otherwise noted, thawed T cells were cultured in X-VIVO 15 (Lonza Bioscience cat 04-418Q) supplemented with 5% FCS, 55 mM 2-mercaptoethanol, 4 mM N-acetyl L-cysteine, and 500 IU/ml of recombinant human IL-2 (Amersource Bergen cat 10101641). Primary T cells were activated using anti-humanCD3/CD28 CTS dynabeads (Fisher Scientific cat 40203D) at a 1:1 cell-to-bead ratio at 10^6 cells/ml.

Cell line maintenance

[0240] Lenti-X HEK293T cells (Takara Bio cat 632180) were maintained in DMEM high glucose with Gluta-MAIXTM (Fisher Scientific cat 10566024), supplemented with 10% FCS, 100 U/ml of PenStrep (Fisher Scientific cat 15140122), 1 mM sodium pyruvate (Fisher Scientific cat 11360070), 1X MEM non-essential amino acids (Fisher Scientific cat 11140050), and 10 mM HEPES solution (Sigma cat H0887-1 00ML). Cells were passaged every 2 days using Tryple Express (Fisher Scientific cat 12604013)

for dissociation and maintained at <60% confluency. NALM6 cells were engineered to express NY-ESO-1 peptide in an HLA-A0201 background, recognizable with the 1G4 TCR by the Fyquem lab at UCSF and provided for TCR stimulation coculture experiments. For simplicity, these cells are referred to as NALM6. NALM6 cells were cultured in RPML (Gibco cat 21870092) supplemented with 10% FCS, 100 U/ml PenStrep (Fisher Scientific cat 15140122), 1 mM sodium pyruvate (Fisher Scientific cat 11360070), and 1X MEM non-essential amino acids (Fisher Scientific cat 11140050), 10 mM HEPES solution (Sigma cat 140887-100ML), and 2 mM L-glutamine (Lonza Bioscience cat 17-605E).

Plasmids

[0241] dCas9-VP64 originated from lentiSAMv2 (addgene 75112) and cloned into the lentiCRISPRv2-dCas9 backbone (addgene 112233) with Gibson Assembly. The

promoter was switched to SFFV and mCherry was introduced upstream of dCas9-VP64, separated by a P2A sequence resulting in the pZRI 12 plasmid. The LTR-LTR range was minimized to enhance lentiviral titer. For CRISPRi, BFP in pHR-SFFV-dCas9-BFP-KRAB (addgene 46911) was switched to mCherry with Gibson Assembly resulting in pZRO 71.

[0242] Single sgRNAs for arrayed experiments have been introduced by Golden Gate Cloning as described before (22). Briefly, DNA oligomers with Golden Gate overhangs were annealed and subsequently cloned into the non-digested target plasmid using the NEB®Golden Gate Assembly Kit (BsmBI-v2, New England Biolabs cat E1602L). sgRNAs have been cloned into pXPR_502 (addgene 96923) for CRISPRa and into CROPseq-Guide-Puro (43) (addgene 86708) for CRISPRi. All single sgRNAs used in this study are found in Table 8.

#	Target	Target-Guide-Number	CRISPR system	Original Guide Sequence
guideRS002	EGFR	1	CRISPRa	CCACCGCTGTCCACGCTC
guideRS003	EGFR	2	CRISPRa	GACCCAAGGCCAGCGCCGC
guideRS004	EGFR	3	CRISPRa	GGAGGGAGGAGAACCAGCAG
guideRS015	GARP	1	CRISPRa	AAATTGCAGCCGGAGCGCGG
guideRS017	GARP	2	CRISPRa	TCCGGATAAACCGAGGCACG
guideRS019	GARP	3	CRISPRa	GCGAAGCATCTTCACACCC
guideRS005	IL1R2	1	CRISPRa	GACCCAGCACTGCAGCCTGG
guideRS006	IL1R2	2	CRISPRa	AAACTTATGCGGCGTTTCCT
guideRS007	IL1R2	3	CRISPRa	ATCACTTTAAACCACTCT
guideRS001	NT-CTRL	1	CRISPRa	CTGAAAAAGGAAGGAGTTGA
guideRS022	B2M	1	CRISPRi	CGCGAGCACAGCTAAGGCCA
guideRS023	B2M	2	CRISPRi	GAGTAGCGGAGCACAGCTA
guideRS024	B2M	3	CRISPRi	GGCCGAGATGTCTCGCTCCG
guideRS025	CD4	1	CRISPRi	AACAAAGCACCTCCCCACT
guideRS026	CD4	2	CRISPRi	CAACAGGCGTATCTGTGTG
guideRS027	CD4	3	CRISPRi	CTCTGCAACCAGGAGCCAG
guideRS028	CD45	1	CRISPRi	CACTGTTGTCTTATCAGACG
guideRS029	CD45	2	CRISPRi	CTCGTCTGATAAGACAACAG
guideRS030	CD45	3	CRISPRi	GTTTGTCTTAGGGTAACAG
guideRS021	NT-Ctr	1	CRISPRi	ACTCAGCCATTTTATTAGAA
pZR073	APOBEC3C	1	CRISPRa	GAGCAGCCTGTCTTTATCGG
pZR074	APOBEC3C	2	CRISPRa	GTTCTCCGGGCCCTCTAC
pZR077	FOXQ1	1	CRISPRa	CGCCTGGTGCGCCCGTTG
pZR078	FOXQ1	2	CRISPRa	GAGGCCACACTGCAGCGCGG
pZR079	IFNG	1	CRISPRa	TGGGTCTGTCTCATCGTCAA
pZR080	IFNG	2	CRISPRa	GTGGCACAGGTGGGCATAAT

-continued

#	Target	Target-Guide- Number	CRISPR system	Original Guide Sequence
pZR081	IL1R1	1	CRISPRa	GGGTGGAGAGTTGGGACACC
pZR082	IL1R1	2	CRISPRa	GCTCGGCTGGGCCAGTCCGC
pZR083	IL2	1	CRISPRa	TCCATTCACTCAGTCTTTGG
pZR084	IL2	2	CRISPRa	GAGAGCTATCACCTAAGTGT
pZR085	IL2RB	1	CRISPRa	TATCTGGCCCTGGGTGCTTG
pZR086	IL2RB	2	CRISPRa	GGTGCCGCCCCAGCGTAGG
pZR087	LAT2	1	CRISPRa	GACAGGCTCAGCTATGAAGA
pZR088	LAT2	2	CRISPRa	GCGGCAGTGCGCCGGATGTA
pZR089	LHX6	1	CRISPRa	TCCCCCTCCAGCTGCAACGG
pZR090	LHX6	2	CRISPRa	GGAGGACTACCAAGAGGGGG
pZR091	MAP4K1	1	CRISPRa	ACAGTCGTGCAGTGCAGCTG
pZR092	MAP4K1	2	CRISPRa	GGGGCTCTGAGAGCCTCTGA
pZR093	NT-CTRL	1	CRISPRa	GAGTCAACGGGAATACCAT
pZR094	NT-CTRL	2	CRISPRa	GAACCATTAGATCAATGCGA
pZR095	OTUD7B	1	CRISPRa	GGGGAGCGCGCTAAAGGCG
pZR096	OTUD7B	2	CRISPRa	GAAAACACGGGGTCACGCGC
pZR097	PIK3AP1	1	CRISPRa	ACCTGCACCCGCGGCCGTTG
pZR098	PIK3AP1	2	CRISPRa	GCCGAGTCCCGCAGGCGGGG
pZR099	TNFRSF1A	1	CRISPRa	TTGGGAGTGGTCGGATTGGT
pZR100	TNFRSF1A	2	CRISPRa	GGCACAGGCAGCCAGATCT
pZR101	TRIM21	1	CRISPRa	AAAGGGTGTGTGGAGAAATG
pZR102	TRIM21	2	CRISPRa	GAGCGCGCAACCAGGACCAC
pZR103	VAV1	1	CRISPRa	CCAGGCCTGTGTCGAGTGGG
pZR104	VAV1	2	CRISPRa	GAGGAGGAGCCATGGGCGG

[0243] The genome wide CRISPRa (Calabrese A, cat 92379 and Calabrese B, cat 92380) and CRISPRi libraries (Dolcetto A, cat 92385 and Dolcetto B, cat 92386) (22) were obtained from addgene. Forty nanograms of each library were transformed into Endura™ ElectroCompetent Cells (Lucigen cat 60242-2) following the manufacturer's instructions. After transformation, Endura cells were grown in a shaking incubator for 16 hours at 30° C. in the presence of ampicillin. Library plasmid has been isolated using the Qiagen Plasmid Plus MaxiKit (Qiagen 12963) and sequenced for sgRNA representation as described under "Genome-wide CRISPRa and CRISPRi screens".

[0244] For cDNA mediated target overexpression, the lentiCRISPRv2 (addgene 75112) backbone was rebuilt to a lentiviral cDNA cloning plasmid with an SFV promoter followed by BsmBT restriction sites and P2A-Puro. Transgene cDNAs were purchased from Genscript, choosing the canonical (longest) isoform for each gene, and BsmBI restriction sites were introduced by PCR. The final lentiviral

transfer plasmids were assembled using the NEB®Golden Gate Assembly Kit (BsmBI-v2, New England Biolabs cat E1602L).

[0245] To clone direct-capture compatible CRISPRa-SAM plasmids for Perturb-seq, different sgRNA designs were synthesized as G-Blocks (integrated DNA technologies) and cloned into pXPR_502 (addgene 96923) by Gibson assembly, replacing its sgRNA cassette.

Lentivirus production

[0246] Unless otherwise stated, HEK293T cells were seeded in Opti-MEM™ Reduced Serum Medium (OPTI-MEM) with GlutaMAvX™ Supplement (Gibco cat 31985088) supplemented with 5% FCS, 1 mM Sodium Pyruvate (Fisher Scientific) and IX MEM non-essential amino acids (Fisher Scientific) (cOPTI-MEM) at 3.6×10⁷ cells per T225 flask in 45 ml of medium overnight to achieve confluency between 85% and 95% at the time point of transfection. The following morning, HEK293 Ts cells were transfected with second generation lentiviral packaging plasmids and transfer plasmid using Lipofectamine 3000

transfection reagent (Fisher Scientific cat 1.3000075). Briefly, 165 μ l of Lipofectamine 3000 reagent was added to 5 ml of room temperature OPTI-M-TEM without supplements. Forty-two micrograms of Cas9 transfer plasmid, 30 μ g of psPAX2 (addgene 12260), 13 μ g of pMD2 G (addgene 12259), and 145 μ l of p3000 reagent were added to 5 ml of room temperature OPTI-MEM without supplements and mixed by gentle inversion. The plasmid and Lipofectamine 3000 mixes were combined, mixed by gentle inversion, and incubated for 15 min at room temperature. Following incubation, 20 ml of medium was removed from the T225 flask and the 10 ml transfection mixture was carefully added without detaching HFEK293T cells. After 6 hours, the transfection medium was replaced with 45 ml of cOPTI-MEM supplemented with IX ViralBoost (Alstern Bio cat VB100). Lentiviral supernatant was harvested 24 hours after transfection (first harvest) and replaced with 45 ml of fresh cOPTI-MEM. A second harvest was performed 48 hours after transfection. Immediately after collection, the media was centrifuged at 500g, 5 min, and 4° C. to clear cellular debris. Unless otherwise noted, *Lenti-X-Concentrator* (Takara Bio 631232) was added to the collected supernatant and lentivirus was concentrated following the manufacturer's instructions and resuspended in OPTI-MEM in 1% of the original culture volume without supplements. Lentiviral particles were subsequently aliquoted and frozen at -80C.

Flow cytometry

[0247] Aria 2, Aria 3 and Aria Fusion cell sorters (BD Biosciences) at the UCSF Parnassus Flow Core and the Gladstone Institute Flow Core were used for sorting. The Attune NxT flow cytometer (Thermo Fisher) and LSR-Fortessa X-20 (BD Biosciences) was used for flow cytometry. Antibodies used for flow cytometric analyses and sorting are summarized in Table 9.

Antigen Name	Target Species	Fluorochrome	Clone	Vendor
EGFR	Human	BV421	EGFR.1	BD
IL1R2	Human	APC	34141	Thermo
IL1R2	Human	FTTC	34141	Thermo
GARP	Human	APC	7B11	Biolegend
IFN-gamma	Human	FTTC	4S.B3	Biolegend
TNF-alpha	Human	APC	Mab11	Biolegend
			MQ1-	
IL-2	Human	Pacific Blue	17H12	Biolegend
IFN-gamma	Human	Pacific Blue	B27	Biolegend
CD45	Human	PE	H130	Biolegend
B2M	Human	APC	2M2	Biolegend
CD45	Human	AF488		Biolegend
CD4	Human	PE	RPA-T4	Biolegend
CD4	Human	BV421	38261	Biolegend
GARP	Human	PE	7B11	Biolegend
CD45	Human	APC	H130	Biolegend
IFN-gamma	Human	BV421	B27	Biolegend
			MQ1-	
IL-2	Human	APC	17H12	Biolegend
CD4	Human	FTTC	38261	Biolegend
IFN-gamma	Human	BV605	B27	Biolegend
TNF-alpha	Human	BV421	Mab11	Biolegend
TNF-alpha	Human	BV711	Mab11	Biolegend
EGFR	Human	PE	AY13	Biolegend
CD4	Human	PE-Cy7	38261	Biolegend
CD22	Human	PerCp-Cy5.5	HIB22	Biolegend
CD45RA	Human	BV711	HI100	Biolegend
CD62L	Human	FTTC	DREG-56	Biolegend
CD4	Human	BUV395	SK3	BD
CD8a	Human	BUV496	SK1	BD

Intracellular Cytokine Staining

[0248] Unless indicated otherwise, T cells were stimulated with ImmunoCult™ Human CD3/CD28/CD2 T Cell Activator (Stemcell Technologies cat 10990) with 6.25 μ l per milliliter of culture media at 2×10^6 cells/ml. One hour after restimulation, Golgi Plug protein transport inhibitor (BD Biosciences, cat 555029) was added at a 1/1000 dilution. Nine hours after addition of Golgi Plug, T cells were stained for surface antigens prior to fixation and subsequently processed for intracellular cytokine staining following BD Cytofix/Cytoperm™ kit (BD Biosciences cat 554714) instructions.

Genome-wide CRISPRa and CRISPRi screens

[0249] One day after activation, T cells from two human blood donors were infected with 2% v/v concentrated dCas9-VP64 lentivirus. Two days after activation, T cells were split into two populations and infected with 1% v/v (MOI-0.5) Calabrese Set A (addgene 92379) or 0.8% v/v (MOI ~0.5) Calabrese Set B (addgene 92380) lentivirus. These two sets were independently cultured and processed in parallel until analysis. Three days following activation, fresh media with IL-2 (final concentration 500 TU/ml) and puromycin (final concentration 2 μ g/ml) was added to bring cells to 3×10^5 cells/ml. Cells were split two days later and fresh media with IL-2 was added to bring cells to 3×10^5 cells/ml. Two days later, fresh media without IL-2 was added to bring the concentration to 10^6 /ml. Eight days after initial activation, cells were harvested, centrifuged at 500g for 5 min and resuspended at 2×10^6 cells/ml X-VIVO 15 without supplements. The following day, cells were restimulated and stained for FACS as described under "intracellular cytokine staining". Over subsequent 2 days, cells were sorted at the Parnassus Flow Cytometry Core Facility (PFCC) into IL-21 and IL-2' CD4+ T cell and IFN-1.1 and IFN-thi CD4 T cell populations. Sorted cells were stored in EasySep Buffer (PBS with 2% FCS and 1 mM EDTA) overnight until genomic DNA isolation.

[0250] The same experimental procedure using T cells from the same donors was followed for the CRISPRi screens. T cells were infected with dCas9-mCherry-KRAB at 2% v/v and Dolcetto A (addgene 92385) and B (addgene 92386) sgRNA libraries at 10% v/v or 25% v/v unconcentrated virus, respectively (-0.5 M1 O).

[0251] Genomic DNA was extracted from fixed cells as described previously (44). Integrated sgRNA sequences were amplified as previously described (22), and sequencing libraries were subsequently agarose gel purified using NucleoSpin Gel and PCR Clean-up Mini kit (Machery Nagel cat 740609.50). Libraries were sequenced on a Next-Seq500 instrument to a targeted depth of 100-fold coverage.

[0252] For the supplementary CD4' T cell set of genome wide CRISPRa screens, CD4' T cells were isolated from Leukopaks using magnetic negative selection (Stemcell Technologies, cat 17952) and subsequently stimulated as described under "Isolation and culture of human 'T cells'". T cells were then cultured and infected with lentivirus as described for the primary CRISPRa screens above. For library lentivirus production, Calabrese Set A and Set B plasmid were mixed at equimolar ratios before transfection and the pooled lentiviral particles from both sets was used for transduction. CD4 flow cytometry staining on day 7 after T cell activation confirmed >98% purity. T cells were further processed and restimulated as described above. T cells were separately stained for IL-2, IFN- γ , or TNF- γ for FACS. After

our initial analysis, it appeared the IFN- γ screen was potentially under-sampled due to lower hit resolution than the other screens. To address this, additional fixed cells from the same experiment were stained and sorted as an additional technical replicate and then computationally merged (described below).

CRISPR screen analysis

[0253] Reads were aligned to the appropriate reference library using MAGECK version 0.5.9.2 (45) using-trim-5 22,23,24,25,26,28,29,30 argument to remove the staggered 5' adapter. Next, raw read counts across both library sets were normalized to the total read count in each sample and each of the matching samples across two sets were merged to generate a single normalized read count table. Normalized read counts in high versus low bins were compared using mageck test with-norm-method none, paired, and control-sgrna options, pairing samples by donor and using non-targeting sgRNAs as controls, respectively. Gene hits were classified as having a median absolute $\log_2$ -fold change value greater than 0.5 and an FDR <0.05. For supplemental CD4⁺ screens, reads were aligned to the full Calabrese A and B library in a single reference file. For the supplemental CD4⁺ IFN- γ screen, which was sorted and sequenced as two technical replicates, normalized counts were averaged across technical replicates before analyzing with mageck test.

Gene set-enrichment analysis (GSEA)

[0254] Gene set-enrichment analysis was completed with the fgsea Bioconductor R package using default settings (46). KEGG pathways v7.4 were obtained from GSEA mSigDB <http://www.gsea-msigdb.org/gsea/downloads.jsp>. The KEGG NF- κ B signaling pathway (entry hsa04064) was missing from this dataset and added manually from <https://www.genome.jp/entry/pathway+hsa04064>.

s-LDSC analysis

[0255] GWAS summary statistics were downloaded from the Price lab website (<https://alkesgroup.broadinstitute.org/sumstats/formatted/and> <https://alkesgroup.broadinstitute.org/UKBB/>). LD scores were created for each screen (corresponding to a set of SNPs within 100 kb of genes identified as significant hits in each screen or their corresponding matched background sets) using the 1000G Phase 3 population reference. Each annotation's heritability enrichment for a given trait was computed by adding the annotation to the baselineLD model and regressing against trait chi-squared statistics using HapMap3 SNPs with the stratified LD score regression package (47). Heritability enrichments were then meta-analyzed across immune or non-immune traits using inverse variance weighting. The sets of background genes were sampled from the set of all genes that were expressed in the control sgRNA, stimulated bulk RNA-Seq data. For each screen, the background genes were sampled to match the significant screen hits in number and based on deciles of gene expression. Immune traits used for analysis were: "Eosinophil Count", "Lymphocyte Count", "Monocyte Count", "White Count", "Autoimmune Disease

All", "Allergy Eczema Diagnosed", "Asthma Diagnosed", "Celiac", "Crohn's Disease", "Inflammatory Bowel Disease", "Lupus", "Multiple Sclerosis", "Primary Biliary Cirrhosis", "Rheumatoid Arthritis", "Type I Diabetes", "Ulcerative Colitis". Non-Immune traits used were: "Heel Tscore", "Balding1", "Balding4", "Bmi", "Height", "Type 2 Diabetes", "Neuroticism", "Anorexia", "Autism", "Bipolar Disorder", "Depressive Symptoms", "Fasting Glucose", "ldl", "Ldl", "Triglycerides", "Fasting Glucose"

Arrayed CRISPRa experiments For each gene chosen to target in follow up experiments, one sgRNA was chosen from the Calabrese library used in screens. The first sgRNAs ("1") were manually chosen for consistent $\log_2$ fold-change observed in both donors. The second sgRNA ("2") was picked from the hCRISPRa-v2 genome-wide library (48), choosing the top ranked sgRNA not present in Calabrese libraries for each gene. sgRNAs were cloned into the pXPR 502 vector as described in the plasmid section.

[0256] Primary human T cells were transduced with 2% v/v inCherry-2A-dCas9-VP64 lentivirus (pZR112) 1-day post-activation. The following day (day 2), the dCas9-VP64 transduced cells were split into 96-well flat-bottom plates, avoiding edge wells, and transduced with a different sgRNA lentivirus in each well (5% v/v). One day after sgRNA transduction, fresh medium was added with IL-2 (500 IU/ml) and 2 μ g/ml puromycin (final culture concentrations). Cells were passaged 2 days later, adding fresh medium with 500 IU/ml of IL-2 and maintaining a concentration of 3×10^5 to 1×10^6 cells/ml with 96-well plates copied as needed to maintain this concentration. On day 8, cells from copied plates were pooled and samples were counted. Cells were pelleted and resuspended at a concentration of 2×10^6 cells/ml in fresh X-VIVO-15 without additives. On day 9, cells were restimulated with anti-CD3/CD28/CD2 ImmunoCult T Cell Activator (as described in "Intracellular cytokine staining") or left resting.

RT-gcR

[0257] T cells were prepared as described under Arrayed CRISPRa experiments. Seven days post sgRNA transduction 100,000 T cells per well were pelleted at 500g, 5 mins, and 4° C. Cells were lysed and RNA was extracted using Quick-RNA 96 kit (Zymo Research), following manufacturer's protocol, skipping the option of in-well DNase treatment. DNase treatment and cDNA synthesis were subsequently completed with Maxima First Strand cDNA Synthesis Kit for RT-qPCR, with dsDNase (ThermoFisher Scientific). qPCR was performed with PrimeTime PCR Master Mix (Integrated DNA technologies) and PrimeTime qPCR probe assays (Integrated DNA Technologies, list of probes used in Table 10) on an Applied Biosystems Quantstudio 5 real-time PCR system. Data was analyzed using the deltaDeltaCt method. The mean Ct values of two house-keeping genes, PPIA and (GUSB, to calculate the deltaCt, and the mean deltaCt of non-targeting controls to calculate deltaDeltaCt.

TABLE 10

Primer 1	Primer 2
GGA AGT AGA ATG TGC CTG GAT	GTA AGC AGG AAG AGA AGC CA
GAG ACC ACA GTT AGA GAA CCA C	TCT TGC TAT TGA CCG ATG CTT

TABLE 10-continued

CGA CAG TTC AGC CAT CAC TT	GCA ACA AAA AGA AAC GAG ATG AC
CAC TGT TTT TCC AAG ACC TCA	TTC CTG CTA TGA TTT TCT CCC A
CTC CAG AGG TTT GAG TTC TTC T	AAA CTC ACC AGG ATG CTC AC
GCG AAG AGA GCC ACT TCT G	GTG TAC TTG CTG ATC AAC TGC
CTG GTG TTG CCT CTT GTG AT	AGT GTC AGT GGT GTT GGC
CAG CTT GGA CAC TGG ATC TC	CCT GCA CGG CTA CAT TGA G
AAG CAG ACG GAA AGT GAG G	GTG GCT ATG GTT GGA GGT C
GAT CCT CAA GTA CTT TCA GCC A	CAC TGC CGA GGA ATG AAG AG
TGA TCT CCA AGT CTG TCT GC	GAG TCC TTT CGT TTC CAG CA
ACA ACT TCG TGC ACT CCA	CAG CCT CTG CCT CAA TGG
TCT GCG TGA ATC CTA GAT TTC TG	GCT GAG AAG TTG GAA GTG GAA
GCC GAA CTT CTC ACA GCA	CTT AAC AAC CTG CTA CCC CAT
GTT TTT GAT CCA GAC CCA GAT G	GCC CAT TAT TCA GAG CGA GTA
CAA GAC TGA GAT GCA CAA GTG	GTG GCG GAT TTG ATC ATT TGG
Probe	
/56-FAM/CCA CAG ATC/ZEN/AGA AAC CCG ATG AAG GC/3IABKFQ/	
/56-FAM/TAA AGC TGT/ZEN/AGC CCG TTG CCT GC/3IABKFQ/	
/56-FAM/TCG GTA ACT/ZEN/GAC TTG AAT GTC CAA CGC/3IABKFQ/	
/56-FAM/TCT ACC TCT/ZEN/GAC TGT GAT ATT TTT GTG TTT AAA GTC T/3IABKFQ/	
/56-FAM/TTA CAT GCC/ZEN/CAA GAA GGC CAC AGA/3IABKFQ/	
/56-FAM/TGT AAC ACC/ZEN/CCA GAC CCC TCG AA/3IABKFQ/	
/56-FAM/AGG TGT GCG/ZEN/GGC TCA GGA T/3IABKFQ/	
/56-FAM/CAC TTC CGC/ZEN/ATC TGC CCG TG/3IABKFQ/	
/56-FAM/TGT AGC AGC/ZEN/TGA TCC GAG CCT AGA/3IABKFQ/	
/56-FAM/CGA GGG TGG/ZEN/AGG CCT GAA TTT TGA/3IABKFQ/	
/56-FAM/TGC TTC TCT/ZEN/CTC TGT CTT CGG GTG A/3IABKFQ/	
/56-FAM/ACG GTG TTC/ZEN/TGT TTC TCC TGG CA/3IABKFQ/	
/56-FAM/AGA GAG CAG/ZEN/ACT GGA AGA AAA CAG TGG/3IABKFQ/	
/56-FAM/CAG ATG TCC/ZEN/CAG TTC CTG TGC CTT/3IABKFQ/	
/5Cy5/TGC AGG GTT TCA CCA GGA TCC AC/3IAbRQSp/	
/5Cy5/AAT TCA CGC AGA AGG AAC CAG ACA GT/3IAbRQSp/	

cDNA experiments

[0258] One day after activation, T cells were transduced with the 1G4 TCR lentivirus recognizing the NY-ESO-1 antigen or non-transduced for immunocult assay. One day later, cells were transduced with the transgenes in cDNA format. Three days after initial activation, puromycin was added to obtain a final concentration of 2 μ g/ml along with fresh X-VIVO 15 media with 500 IU/ml of HL-2 and further cultured and expanded analogous to the genome wide CRISPR screens. Nine days after initial activation, T cells were centrifuged and resuspended at 2×10^6 cells/ml in

X-Vivo 15 without supplements. On the same day, 1G4 TCR expression was assessed by flow cytometry following dextran staining (Immudex cat WB3247-PE) to ensure even expression across different cDNA constructs. The following day, T cells were restimulated with either 6.25 μ l per milliliter of Immunocult or NALM6 cells at an effector-target ratio of 1:2 for 1G4 TCR-transduced cells. Cells were further processed as described under "intracellular cytokine staining". CD22 was used as a marker for NALM6 cells to discriminate them from T cells in the coculture. Overexpression of OTUD7B cDNA together with the 1G4 TCR (but not alone) caused toxicity and was therefore excluded from

analyses. Two donors were excluded from the 1G4 TCR assay due to poor TCR transduction.

Cytokine Luminex assay

[0259] T cells were prepared as explained under “Arrayed CRISPRa experiments.” On day 9 after activation, T cells at a concentration of 2×10^6 cells/ml were restimulated with ImmunoCult™ Human CD3/CD28/CD2 (Stemcell Technologies cat 10970) at 6.25 μ l per milliliter. Twenty-four hours after restimulation, supernatant was collected and frozen at -20° C. Following a serial pilot titration, cytokine analyses were performed at a 1/200 dilution by Eve Technologies with the Luminex xMAP technology on the Luminex 200 system (Luminex). To remove very-low-expressed cytokines for downstream analysis, any group where three of four donors had undetectable cytokines, the cytokine was removed. Additionally, the sgfLIR1-1-Donor 4 measurement for IL-1 was removed manually, as this was an extremely high outlier.

Bulk RNA-seq sample preparation

[0260] FOXQ1 and non-targeting sgRNA control primary human T cells from four donors were transduced and expanded as described in “Arrayed CRISPRa experiments” section. On day 8, mCherry+CD4+ populations were sorted and resuspended in X-VIVO-15 without additives at 2×10^6 cells/ml. On day 9, cells were restimulated with 6.25 μ l per milliliter of anti CD3/CD28/CD2 ImmunoCult™ or left unperturbed for resting (non-stimulated) condition. Twenty-four hours later, cells were lysed for RNA.

[0261] RNA was purified using Quick-RNA Microprep kit (Zymo Research) without the optional in-well DNase treatment step. Purified RNA was treated with TURBO DNase (ThermoFisher Scientific) to remove potential contaminating DNA. RNA was subsequently purified using RNA Clean & Concentrator-5 kit (Zymo Research). RNA quality control was performed using an RNA ScreenTape assay (Agilent), with all samples having an RNA integrity number >7 . RNA-seq libraries were prepared using the Illumina Stranded mRNA Prep kit, with 100 ng of input RNA. Libraries were sequenced using paired-end 72-bp reads on a NextSeq500 instrument to an average depth of 3.2×10^7 clusters per sample.

Bulk RNA-seq data analysis

[0262] Adapters were trimmed from fastq files using cutadapt version 2.10 (49) with default settings keeping a minimum read length of 20 bp. Reads were mapped to the human genome GRCh38 keeping only uniquely mapping reads using STAR version 2.7.5b (50) with the following settings “-outFilterMultimapNmax 1”. Reads overlapping genes were then counted using featureCounts version 2.0.1 (51) with the following settings “-s 2” and using the Genecode version 35 basic transcriptome annotation. The count matrix was imported into R. Only genes with at least 1 count per million (CPM) across at least four samples were kept. TMM normalized counts were used for heatmaps. Differentially expressed genes between FOXQ1 overexpression and control samples were then identified using limma version 3.44.3 (52) while controlling for any differences between donors. Significant differentially expressed genes were defined as having an FDR-adjusted P-value <0.05 .

Perturb-seq Library Design and Cloning

[0263] The CRISPRa Perturb-seq target genes were selected from the primary IL-2 and IFN- γ CRISPRa screen results. First, genes that had a significant fitness defect removed from the gene list. Next, genes were ranked by median sgRNA $\log_2$ -fold change and the top ranked, not previously selected gene, was picked in the following order:

(1) IL-2-positive hit, (2) IFN- γ positive hit, (3) IL-2-positive hit, (4) IFN- γ -positive hit, and (5) IL-2- or IFN- γ -positive hit (alternating each round), such that positive hits outnumbered negative hits at a 4:1 ratio. Only hits that were significant (FDR <0.05) were selected in each round. The one exception was TCF7, which was added manually as we considered it worthwhile to analyze due to its known effects on T cell function. To select sgRNAs, the top two enriched sgRNAs by $\log_2$ fold-change in the screen for which the gene was selected were used. The library was ordered as pooled single stranded oligos, PCR amplified, and cloned into the CRISPRa-SAM direct-capture design I cloning vector (pZR158).

Perturb-seq Sample Preparation and Sequencing

[0264] Bulk CD3 $^+$ primary human T cells from two donors were transduced and cultured as described in the “Genome-wide CRISPRa and CRISPRi screens” section, except library transduction was completed at lower MOI, of 0.3. Cells in the stimulated condition were stimulated with 6.25 μ l per milliliter of anti-CD3/CD28/CD2 immunocult. Twenty-four hours later, cells from both the stimulated and non-stimulated condition were sorted for mCherry™ (marking dCas9-VP64). Sorted cells were processed to single-cell RNA-seq and sgRNA sequencing libraries by the Institute for Human Genetics (IHG) Genomics Core using Chromium Next GEM Single Cell 3' Reagent Kit v3.1 with Feature Barcoding technology for CRISPR screening, following manufacturer's protocol. Before loading the Chromium chip, sorted cells from two blood donors were normalized to 1000 cells/ μ l and mixed at a 1:1 ratio, for each condition. Twenty microliters of cell suspension was loaded into four replicate wells per condition, for a total 80,000 cells loaded per condition. Final sgRNA sequencing libraries were further purified for the correct size fragment by 4% agarose F-Gel EX Gels (ThermoFisher Scientific) and gel extracted. Libraries were sequenced over two NovaSeq S4 lanes (2 stimulated wells, two non-stimulated wells per lane), at a 2:1 molar ratio of the gene expression libraries to sgRNA libraries.

Perturb-seq Analysis Alignments and count aggregation of gene expression and sgRNA reads were completed with Cell Ranger version 6.1.1. Gene expression and sgRNA reads were aligned using cellranger count, with default settings. Gene expression reads were aligned to the “refdata-gex-GRCh38-2020-A” human transcriptome reference downloaded from 10x Genomics. sgRNA reads were aligned to the Perturb-seq library using the pattern (BC)GTT-TAAGAGCTATG. Counts were aggregated with cellranger aggr with default arguments. To assign sgRNAs to cells, cellranger count output files “protospacer_calls_per_cell.csv” were used, filtering out droplets with >1 sgRNA called, returning a median of 133 sgRNA UMIs in sgRNA singlets. For increased stringency, only droplets with >5 sgRNA UMIs were used in further analysis. Cell donors were genetically demultiplexed using SoupCell (53) (<https://github.com/wheaton5/souporcell>). The input for each run was the bam file and barcodes.tsv file from the cellranger count output, and the reference fasta. Donor calls across wells were harmonized using the vcf file outputs from SoupCell using a publically available python script (<https://github.com/nihyunminkang/apigenome/blob/master/scripts/vcf-match-sample-ids>).

[0265] Gene expression data were imported and analyzed in R with Seurat version 4.0.3 ReadIOX function (54). Cells were initially quality filtered for percent mitochondrial reads $<25\%$, number of detected RNA features >400 and <6000 , removing 4% of cells.

[0266] After filtering, we recovered a median of 401 cells per sgRNA target gene per condition (median of 127 sgRNA unique molecular indices (UMIs) per singlet), and ~2000 cells with no-target control guides per condition. Four sgRNA targets (HELZ2, TCF7, PRDM1, and IRX4) were removed from downstream analysis due to low cell counts (<100).

[0267] Gene-expression counts were normalized and transform-ed using the Seurat SCTransform function (55), with the following variables regressed: percent mitochondrial reads, S-phase score, and G2M-phase score, performing the regression as described on the Satija Lab website (https://satijalab.org/seurat/articles/cell_cycle_vignette.html). Normalized and transformed counts were used for all downstream analysis. To call CD4⁺ and CD8⁺ T cells, a CD4/CD8 score for each cell using following formula was used: $\log_2(\text{CD4}/\text{mean}(\text{CD8A}, \text{CD8B}))$, with a score <-0.9 called as a CD3⁺ cell, and >1.4 called a CD4 cell.

[0268] For both restimulated and resting conditions, UMAP reduction was performed with dimensions 1-20, and otherwise default settings of the RunUMAP Seurat function. For clustering, FindClusters was run using algorithm 3, and resolution 0.4 for restimulated, and 0.5 for resting condition. Two clusters in the restimulated condition were manually merged to form “Cluster 2: Negative Regulators”. The merged clusters showed highly similar gene expression patterns, with one cluster containing the bulk of cells containing negative regulator sgRNAs, and the other cluster containing sgRNAs targeting the negative regulator, MUC1. Cluster trees shown were generated using the Seurat BuildClusterTree function with default arguments. For pseudobulk differential expression analyses the Seurat FindMtlarkers function was used with the default method, Wilcoxon Rank Sum test.

[0269] To generate the T cell activation score, pseudobulk differential expression analysis was first performed on restimulated versus resting no-target control sgRNAs and log 2-fold change outputs were used as gene weights. Only genes with an absolute log-fold change >0.25 and which were detected in 10% of restimulated or resting cells were used for gene weights. For a given cell, the activation score is calculated as $\text{sum}(\text{GE} \times \text{GW}/\text{GM})$, where GE is a gene’s normalized/transformed expression count, GW is the gene’s weight, and GM is the gene’s mean expression in no-target control cells (to correct for differential levels of baseline expression).

Statistical analysis

[0270] All statistical analyses were performed in R version 4.0.2, unless otherwise noted. In order to deal with ties in non-parametric tests, Mann-Whitney U tests were performed using the wilcox_test function of the Coin R package (version 1.4-1), with default arguments. For q-value based multiple comparison correction, the R qvalue package (version 2.0.0) was used with default arguments.

Results

Genome-wide CRISPRa screens identify regulators of IL-2 and IFN- γ production in T cells

[0271] To enable scalable CRISPRa in primary human T cells, we developed an optimized high-titer lentiviral production protocol with a minimal dCas9-VP64 vector (pZRI12), allowing for transduction efficiencies up to 80%. A second-generation CRISPRa synergistic activation mediator (SAM) system (22, 23) induced robust increases in target expression of established surface markers. Next, we scaled up our platform to perform pooled genome wide CRISPRa screens targeting >18,800 protein coding genes with >112,000 sgRNAs (22). We used fluorescence-activated cell sort-

ing (FACS) to separate IL-2-producing CD4-T cells and IFN- γ -producing CD8-T cells into high and low bins (FIG. 1A). Subsequent sgRNA quantification confirmed sgRNAs targeting IL-2 (IL2) and IFN- γ (IFNG) were strongly enriched in the respective cytokine high populations and non-targeting control sgRNAs were not enriched in either bin (FIG. 1B). Both CRISPRa screens were highly reproducible in two different human blood donors (FIGS. 1, C and D). Gene level statistical analysis of the IL-2 and IFN- γ CRISPRa screens revealed 444 and 471 hits, respectively, including 171 shared hits (FIG. 1E). Thus, CRISPRa screens provide a robust platform to discover gain-of-function regulators of stimulation-dependent responses in primary cells.

[0272] CRISPRa hits included components of the TCR signaling pathway and T cell transcription factors. Activation of TBX21 (encoding T-bet), which promotes both memory CD8⁺ T cell and CD4⁺ T helper 1 (Th1) cell differentiation (24-26), selectively enhanced the signature type I cytokine IFN- γ (FIG. 1E). By contrast, sgRNAs activating GATA3, which promotes type II differentiation by antagonizing T-bet (25, 27), had opposite effects (FIG. 1E). Overexpression of members of the proximal TCR signaling complex, such as VAV1, CD28, LCP2 (encoding SLP 76), and LAT (28, 29) reinforced T cell activation and were enriched in both cytokine-high bins. Conversely, negative TCR signaling regulators MAP4KI and SLA2 were depleted in these bins (FIGS. 1, B and E) (30, 31). Thus, CRISPRa identifies critical “bottlenecks” in signals leading to cytokine production.

Complementary CRISPRa and CRISPRi screens comprehensively reveal circuits of cytokine production in T cells

[0273] CRISPRa screens were effective in identifying limiting factors in cytokine production, but they could miss necessary components that would only be identified through loss-of-function studies. We therefore performed reciprocal genome wide CRISPRi screens, adapting our optimized lentiviral protocols (FIGS. 2, A and B) Drop out of gold-standard essential genes (32) and reproducibility across two human donors confirmed the screen quality. The CRISPRi IL-2 and IFN- γ screens identified 226 and 203 gene hits, respectively, including 92 shared hits (FIGS. 2, A and B). As expected, the CRISPRi hits were biased towards genes with high mRNA expression including members of the CD3 complex, whereas CRISPRa additionally identified regulators that are expressed either at low levels or not at all in T cells under the screened conditions. (FIGS. 2, C and D). For example, PIK3AP1 and IL1R1 were expressed at low levels under the screened conditions (FIG. S7A). They are potentially inducible in some T cell contexts however were detected as hits by CRISPRa but not CRISPRi.

[0274] The power of coupling activation and interference screening was exemplified further by the identification of two IFN- γ -regulating circuits. CRISPRi screens identified components of the NF- κ B pathway that are required for IFN- γ production (and to a lesser extent IL-2 production). CRISPRi detected a circuit of T cell stimulation signaling through MALTI, BCL10, TRAF6, and TAK1 (encoded by MAP3K7) to the inhibitor of NF- κ B complex (I κ B complex, encoded by CHUK, I κ BKIB, and I κ BKG) that promotes IFN- γ production (FIGS. 2, E and F). By contrast, CRISPRa revealed a set of positive IFN- γ regulators that included members of the tumor necrosis factor receptor superfamily (TNFRSF) and IL1R1. These regulators also signal through NF- κ B, even though they are not individually required and therefore not detected by CRISPRi (FIGS. 2, E and F). Thus, CRISPRa and CRISPRi complement each other for the comprehensive discovery of functional cytokine regulators.

To gain insights into functional pathways enriched across CRISPRi and CRISPRa screens, we completed gene set enrichment analysis (GSEA) of KEGG pathways, identifying multiple immune-related pathways as enriched across screens. Furthermore, we analyzed data from numerous genome-wide association studies (GWAS) to ask if the heritability of complex immune traits was enriched in genomic regions harboring our screen hits by stratified linkage disequilibrium score (s-LDSC) regression. Both CRISPRi and CRISPRa regulators of IFN- γ and CRISPRa regulators of IL-2 were in regions enriched for immune trait heritability compared to non-immune traits or an expression matched background set. Thus, these forward genetic screens may serve as a resource to help prioritize candidate functional genes in genomic regions associated with complex immune diseases.

[0275] We next completed integrative analyses of gene hits across CRISPRa and CRISPRi screens for both cytokines. We found that a handful of genes were identified across all screens (e.g., ZAP70 as a positive regulator and CBLB as a negative regulator), representing core regulators of stimulation-responsive cytokine production in T cells. The majority of hits however were either cytokine-(IL-2 in CD4⁺ T cells or IFN- γ in CD8⁺ T 30 cells) or perturbation-(activation or interference) specific. For a few target genes including PTPRC (CD45), CRISPRa and CRISPRi both influenced cytokine production in the same direction, suggesting that for some genes activation and interference both impair optimal levels. The striking overlap in regulators between IL-2 in CD4⁺ T cells and IFN- γ in CD8-T cells led us to perform additional genome-wide CRISPRa screens for IL-2, IFN- γ , and TNF- α in CD4⁺ T cells, allowing for direct comparisons of type 1 cytokine regulators in CD4⁺ T cells. Many of the strongest positive (e.g., VAV1, CD28, and LCP2) and negative hits (e.g., MAP4K1, LAT2, and GRAP) overlapped across all CRISPRa screens, likely representing core regulators of type I cytokine production in response to stimulation/costimulation. Additionally, these screens identified hits that could potentially increase or decrease individual cytokines selectively. Thus, CRISPRi and CRISPRa hits reveal both core and context-specific regulators of cytokine production.

[0276] We used our integrated dataset combined with literature review to build a high-resolution map of tunable regulators of signal transduction pathways leading to cytokine production (FIG. 2G). This included calcium pathway signaling genes (e.g., PLCG1, PLCG2, PRKCB, PRKD2, and NFATC2), and cytokine signaling genes (e.g., STAT3, JAK1, JAK3, and SOCS3), the latter suggesting feedback circuits among cytokine signals. In particular, CRISPRa identified regulators absent from previous literature (e.g., APOBEC3A/D/C, FOXQ1, and EMIPI) (FIG. 2H), underscoring the need for gain-of-function screens for comprehensive discovery. Thus, CRISPRa and CRISPRi screens complement one another to map the tunable genetic circuits controlling T cell stimulation-responsive cytokine production.

Arrayed characterization of selected CRISPRa screen hits

[0277] We next performed arrayed CRISPRa experiments for deeper phenotypic characterization of screen hits (FIG. 3A). We selected 14 screen hits (from different screen categories) (FIG. 3B) including the established regulators VAV1, MAP4K1, and positive controls IL2 and IFNG. Notably, we included genes with relatively low expression in T cells under our experimental conditions, FOXQ1, IL1 R1,

LHX6, and PIK3API. First, we validated that selected sgRNAs increased the expression of target gene mRNA. Next, we assessed IL-2, IFN γ , and TNF- α by intracellular staining in both CD4- and CD8⁺ T cells. Thirteen of 14 target genes caused significant changes in the proportion of cells positive for the relevant cytokine(s), with at least one sgRNA (FIG. 3, C and D). Furthermore, we observed effects on both IL-2 and IFN- γ double- and single-positive populations. With the exception of TNFRSF1 A (and IL2 or IFNG), positive regulators did not cause spontaneous cytokine production without stimulation (FIG. 3D).

[0278] Although IL-2 was screened in CD4⁺ T cells and IFN- γ in CD8⁺ T cells, CRISPRa sgRNA effects were highly correlated across both lineages (FIG. 3F). We also assessed T cell differentiation and observed that FOXQ1 and TNFRSF1A significantly decreased the percentage of CD62L⁺-cells, indicating a shift towards effector T cell states as a potential mechanism. Thus, these studies validate the pooled CRISPRa screens and begin to characterize cytokine production and cell differentiation states promoted by activation of key target genes.

[0279] We next tested if genes identified by CRISPRa could also regulate cytokines when overexpressed as cDNA transgenes, because continuous expression of CRISPRa would present challenges in cell therapies due to Cas9 immunogenicity (33). cDNA transgene overexpression of CRISPRa hits affected cytokine production in T cells stimulated with antibodies or antigen-positive cancer cells. Thus, this strategy could potentially be used to implement CRISPRa discoveries in engineered T cell therapies.

[0280] We next assessed how individual CRISPRa perturbations reprogram cytokine production by measuring a broad panel of 48 secreted cytokines and chemokines, 32 of which were detected in control samples. After confirming that the effects on IL-2, IFN- γ , and TNF- α measurements were consistent with intracellular staining (FIG. 3F), we performed principal component analysis (PCA) and hierarchical clustering on all cytokines. We observed sgRNA categorical grouping consistent with that observed in the screens, with sgRNAs targeting genes identified as regulators of both cytokines causing broad increases or decreases in cytokine concentration (FIG. 3G). Notably, there were distinct patterns in the classes of cytokines increased by different regulators (FIG. 3H). VAV1 and FOXQ1 (a transcription factor that has not been well characterized in T cells) led to preferential increases in type 1 signature cytokines and dampened type 2 cytokines. Surprisingly, OTUDB, a positive regulator of proximal TCR signaling (34), had a distinct effect and increased type 2 cytokines. We next asked if modulations in the secretome correlated with transcriptional control of the corresponding genes. Taking FOXQ1 as an example, we performed bulk RNA-seq on FOXQ1 and control sgRNA CD4-T cells and found it correlated highly with the secretome effects. Thus, the identified regulators may not only modulate TCR stimulation and signaling, but also tune the T cell secretome towards specific signatures.

CRISPRa Perturb-seq characterizes the molecular phenotypes of cytokine regulators

[0281] To assess the global molecular signatures resulting from each CRISPRa gene induction we developed a platform to couple pooled CRISPRa perturbations with bar-coded single-cell RNA sequencing (scRNA-seq) read-outs (CRISPRa Perturb-seq) (FIG. 4A). As similar CRISPRa

Perturb-seq approaches have been powerful in cell lines and animal models (35-37), we incorporated a direct-capture sequence into the CRISPRa-SAM modified sgRNA scaffold to enable compatibility with droplet-based scRNA-seq methods.

[0282] We performed CRISPRa Perturb-seq characterization of regulators of stimulation responses in 56,000 primary human T cells targeting 70 hits and controls from our genome wide CRISPRa cytokine screens (FIGS. 4, A and B). First, we confirmed that sgRNAs led to significant increases in the expression of their target genes. Next, uniform manifold approximation and projection (UMAP) dimensionality reduction revealed discrete separation of the resting and restimulated cells and showed relatively even distribution of cells from two donors (FIG. 4C). Gene signatures allowed us to resolve most T cells as either CD4 or CD8⁺ (FIG. 4D). Thus, we generated a high-quality CRISPRa Perturb-seq dataset.

[0283] Cytokine production can be tuned by reinforced TCR signaling. To identify CRISPRa gene perturbations that tune the general strength of stimulation-responsive genes, we calculated a scRNA-seq “activation” score based on a gene signature we derived from comparing resting and restimulated cells within the non-targeting control sgRNA group. Projecting activation scores on the stimulated cell UMAP revealed discrete regions of higher and lower activation scores among the restimulated cells (FIG. 4E). We next examined activation scores across CRISPRa perturbations (FIG. 4F). Strikingly, negative regulators except IKZF3 (encoding the transcription factor Aiolos) decreased activation scores, suggesting they act to broadly dampen stimulation strength. By contrast, IKZF3 reduced IFNG expression without reducing the overall activation score (FIG. 4F), indicative of a possible distinct mechanism of cytokine gene regulation. Many of the positive regulators significantly increased activation score, with VAV1 causing the strongest activation potentiation (FIG. 4F). Thus, many, but not all, hits act by tuning overall T cell activation to varying degrees.

[0284] We next asked how different perturbations affected the expression of cytokine and other effector genes in stimulated cells. We analyzed pseudobulk differential gene expression under restimulated conditions for each sgRNA target cell group, compared with no-target control cells. IFNG was differentially expressed in 29 different sgRNA targets, with only sgRNAs targeting negative regulators causing decreased expression. IL2, however, was barely detectable by scRNA-seq. Only IL2 and VAV1 sgRNAs caused its increased expression, consistent with our observations that VAV1 activation caused the greatest level of IL2 release (FIG. 3H). Many of the negative regulators drove a stereotyped pattern of differential cytokine gene expression, whereas positive regulators generally promoted more diverse cytokine expression patterns than negative regulators. Notably, TBX21 (T-bet) modulated the expression of most detectable cytokine genes. Furthermore, unlike most perturbations, it altered cytokine expression independently of stimulation.

[0285] We next used clustering analysis to characterize CRISPRa-driven cell states in restimulated and resting T cells (FIG. 4G). For each cluster, we identified the top upregulated gene expression markers and cytokine genes, contributions of CD4/CD8 T cells, and overrepresented sgRNAs revealing a diverse landscape of T cell states

promoted by CRISPRa (FIG. 4, H to J). Negative cytokine regulators (e.g., MAP4KI) were highly enriched in cluster 2, marked by LTB expression and low activation score. Notably, only GATA3 promoted a Th2 phenotype (cluster 3) suggesting that altered T helper differentiation was not a common mechanism among negative LFNG regulators. Thus, Perturb seq reveals cell states promoted by the over-expression of different key regulators.

[0286] We identified two 112-expressing clusters, despite poor capture of the transcript, with both of the clusters consisting primarily of CD4⁺ T cells. Cluster 13 had higher 112 expression of the two and was promoted by VAX1 and OTUD7B sgRNAs. VAV1 sgRNAs were strongly enriched in both IFNG- and IL²-expressing clusters, suggesting that VAV1-mediated potentiation of T cell stimulation may drive differentiation towards multiple distinct cytokine-producing populations.

[0287] We also identified two distinct clusters of cells expressing IFNG (clusters 1 and 12) containing both CD4- and CD8⁺ T cells. Cluster 1 was marked by high expression of CCL3 and CCL4 and was enriched for sgRNAs with strong activation score potentiation, such as VAV1, CD28, and FOXQ1. By contrast, Cluster 12 was enriched for sgRNAs known to activate the NF- κ B pathway, such as IL1RI, TRAF3IP2, TNFRSF1A, and TNFRSF1B. These observations suggest that potentiated stimulation/costimulation may drive T cells to an activated IFNG expressing state distinct from more specific signaling through the NF- κ B pathway. Activation of a subset of TNFRSF receptor genes (TNFRSF1A, TNFRSF1B, LTBR, and CD27), also promoted cell states (clusters 5 and 6) marked by the high expression of cell-cycle genes. LTBR and CD27 sgRNAs were almost exclusively found in cells of this cluster, whereas TNFRSF1A/B sgRNAs appeared to push cells to both proliferative and IFNG-expressing states. Thus, CRISPRa Perturb seq reveals how regulators of cytokine production both tune T cell activation and program cells into different stimulation-responsive states.

Discussion

[0288] Paired CRISPRa and CRISPRi screens complement one another to decode the genetic programs regulating stimulation-responsive cytokine production in primary human T cells. CRISPRi identified required cytokine regulators, whereas CRISPRa uncovered key signaling bottlenecks in pathway function as well as regulators that are not necessarily active in ex vivo-cultured T cells. Future screens performed in various other experimental conditions have potential to identify additional regulators of T cell states and functions.

[0289] The technologies developed in this study enable screening approaches in primary human T cells and other primary cell types, such as screens for functional noncoding regions of the human genome (18, 38, 39). Furthermore, this screening framework is adaptable to other non-heritable editing applications of the CRISPR toolkit (40), continuing to expand opportunities to interrogate complex biological questions in primary cells, especially when CRISPR perturbations are coupled with single-cell analyses.

BIBLIOGRAPHY

- [0290]** 1. A. K. Abbas, E. Trotta, D. R. Simeonov, A. Marson, J. A. Bluestone, Revisiting IL-2: Biology and therapeutic prospects. *Sci Immunol.* 3 (2018), doi:10.1126/sciimmunol.aat1482.

- [0291] 2. L. Ni, J. Lu, Interferon gamma in cancer immunotherapy. *Cancer Med.* 7, 4509-4516 (2018).
- [0292] 3. F. Castro, A. P. Cardoso, R. M. Giongalves, K. Serre, M. J. Oliveira, Interferon-Gamma at the Crossroads of Tumor Immune Surveillance or Evasion. *Front. Immunol.* 9, 847 (2018).
- [0293] 4. L. B. Ivashkiv, IFN γ : signalling, epigenetics and roles in immunity, metabolism, disease and cancer immunotherapy. *Nat. Rev. Immunol.* 18, 545-558 (2018).
- [0294] 5. T. R. Malek, The biology of interleukin-2. *Annu. Rev. Immunol.* 26, 453-479 (2008).
- [0295] 6. D. A. Boardman, M. K. Levings, Cancer immunotherapies repurposed for use in autoimmunity. *Nat. Biomed. Eng.* 3, 259-263 (2019).
- [0296] 7. J. Gao, L. Z. Shi, H. Zhao, J. Chen, L. Xiong, Q. He, T. Chen, J. Roszik, C. Bernatchez, S. E. Woodman, P.-L. Chen, P. Hwu, J. P. Allison, A. Futreal, J. A. Wargo, P. Sharma, Loss of IFN γ Pathway Genes in Tumor Cells as a Mechanism of Resistance to Anti-CTLA-4 Therapy. *Cell.* 167, 397-404.e9 (2016).
- [0297] 8. J. M. Zaretsky, A. Garcia-Diaz, D. S. Shin, H. Escuin-Ordinas, W. Hugo, S. Hu-Lieskovan, D. Y. Torrejon, G. Abril-Rodriguez, S. Sandoval, L. Barthly, J. Saco, B. Hornet Moreno, R. Mezzadra, B. Chmielowski, K. Ruchalski, I. P. Shintakur, P. J. Sanchez, C. Puig-Saus, C. Cherry, E. Seja, X. Kong, J. Pang, B. Berent-Maoz, B. Comin-Anduix, T. G. Graeber, P. C. Tumeh, T. N. M. Schumacher, R. S. Lo, A. Ribas, Mutations Associated with Acquired Resistance to PD-1 Blockade in Melanoma. *N. Engl. J. Med.* 375, 819-829 (2016).
- [0298] 9. M. Ayers, J. Lunceford, M. Nebozhyn, E. Murphy, A. Loboda, D. R. Kaufman, A. Albright, J. D. Cheng, S. P. Kang, V. Shankaran, S. A. Piha-Paul, J. Yearley, T. Y. Seiwert, A. Ribas, T. K. McClanahan, IFN γ -related mRNA profile predicts clinical response to PD-1 blockade. *J. Clin. Invest.* 127, 2930-2940 (2017).
- [0299] 10. R. R. Bartelt, N. Cruz-Orcutt, M. Collins, J. C. D. Houtman, Comparison of T cell receptor induced proximal signaling and downstream functions in immortalized and primary T cells. *PLoS One.* 4, e5430 (2009).
- [0300] 11. H. Colin-York, S. Kumari, L. Barbieri, L. Cords, M. Fritzsche, Distinct actin cytoskeleton behaviour in primary and immortalised T-cells. *J. Cell Sci.* 133 (2019), doi:10.1242/jcs.232322.
- [0301] 12. F. Astoul, C. Edmunds, D. A. Cantrell, S. G. Ward, PI 3-K and T-cell activation: limitations of T-leukemic cell lines as signaling models. *Trends Immunol.* 22, 490-496 (2001).
- [0302] 13. & Parnas, M. Jovanovic, T. M. Fisenhaure, R. H. Herbst, A. Dixit, C. J. Ye, D. Przybylski, R. J. Platt, I. Tirosh, N. F. Sanjana, O. Shalem, R. Satija, R. Raychowdhury, P. Mertins, S. A. Carr, F. Zhang, N. Hacohen, A. Regev, A Genome-wide CRISPR Screen in Primary Immune Cells to Dissect Regulatory Networks. *Cell.* 162, 675-686 (2015).
- [0303] 14. M. B. Dong, G. Wang, R. D. Chow, L. Ye, L. Zhu, X. Dai, J. J. Park, H. R. Kim, Y. Errami, C. D. Guzman, X. Zhou, K. Y. Chen, P. A. Renauer, Y. Du, J. Shen, S. Z. Lam, J. J. Zhou, D. R. Lannin, R. S. Herbst, S. Chen, Systematic immunotherapy Target Discovery Using Genome Scale In Vivo CRISPR Screens in CD8 T Cells. *Cell.* 178, 1189-1204.e23 (2019).
- [0304] 15. J. Henriksson, X. Chen, T. Comes, U. Ullah, K. B. Meyer, R. Hiragaia, G. Duddy, J. Pramanik, K. Yusa, R. Lahesmaa, S. A. Teichmann, Genome-wide CRISPR Screens in T Helper Cells Reveal Pervasive Crosstalk between Activation and Differentiation. *Cell.* 176, 882-896.e18 (2019).
- [0305] 16. E. Shifrut, J. Carnevale, V. Tobin, T. L. Roth, J. M. Woo, C. T. Biui, P. J. Li, M. E. Diolaiti, A. Ashworth, A. Marson, Genome-wide CRISPR Screens in Primary Human T Cells Reveal Key Regulators of Immune Function. *Cell.* 175, 1958-1971.e15 (2018).
- [0306] 17. P. Y. Ting, A. E. Parker, J. S. Lee, C. Trussell, O. Sharif, F. Luna, G. Federe, S. W. Barnes, J. R. Walker, J. Vance, M.-Y. Gao, H. E. Klock, S. Clarkson, C. Russ, L. J. Miraglia, M. P. Cooke, A. E. Boitano, P. McNamara, J. Lamb, C. Schmedt, J. L. Snead, Guide Swap enables genome-scale pooled CRISPR-Cas9 screening in human primary cells. *Nat. Methods.* 15, 941-946 (2018).
- [0307] 18. D. R. Simeonov, B. G. Gowen, M. Boontanart, T. L. Roth, J. D. Gagnon, M. R. Mumbach, A. T. Satpathy, Y. Lee, N. L. Bray, A. Y. Chan, D. S. Lituev, M. L. Nguyen, R. E. Gate, M. Subramaniam, Z. Li, J. M. Woo, T. Mitros, G. J. Ray, G. L. Curie, N. Naddaf, J. S. Chu, H. Ma, E. Boyer, F. Van Gool, H. Huang, R. Liu, V. R. Tobin, K. Schumann, M. J. Daly, K. K. Farh, K. M. Ansel, C. J. Ye, W. J. Greenleaf, M. S. Anderson, J. A. Bluestone, H. Y. Chang, J. E. Corn, A. Marson, Discovery of stimulation-responsive immune enhancers with CRISPR activation. *Nature.* 549, 111-115 (2017).
- [0308] 19. Y. Liu, C. Yu, T. P. Daley, F. Wang, W. S. Cao, S. Bhate, X. Lin, C. Still 2nd, H. Liu, D. Zhao, H. Wang, X. S. Xie, S. Ding, W. H. Wong, M. Wernig, L. S. Qi, CRISPR Activation Screens Systematically Identify Factors that Drive Neuronal Fate and Reprogramming. *Cell Stem Cell.* 23, 758-771.e8 (2018).
- [0309] 20. X. Chen, L. Kozhaya, C. Tastan, L. Placek, M. Dogan, M. Horne, R. Abblett, E. Karhan, M. Vaeth, S. Feske, D. Unutmaz, Functional Interrogation of Primary Human T Cells via CRISPR Genetic Editing. *J. Immunol.* 201, 1586-1598 (2018).
- [0310] 21. R. Nasrallah, C. I. Imnianowski, L. Bossini-Castillo, F. M. Grant, M. Dogan, L. Placek, L. Kozhaya, P. Kuo, F. Sadiyah, S. K. Whiteside, M. R. Mumbach, D. Glinos, P. Vardaka, C. E. Whyte, T. Lozano, T. Fujita, H. Fujii, A. Liston, S. Andrews, A. Cozzani, I. Yang, S. Mitra, E. Lugli, H. Y. Chang, D. Unutmaz, G. Trynka, R. Roychoudhuri, A distal enhancer at risk locus Il12 β promotes suppression of colitis by Treg cells. *Nature.* 583, 447-452 (2020).
- [0311] 22. K. R. Sanson, R. E. Hanna, M. Hegde, K. F. Donovan, C. Strand, M. E. Sullender, E. W. Vaimberg, A. Goodale, D. E. Root, F. Piccioni, J. G. Doench, Optimized libraries for CRISPR Cas9 genetic screens with multiple modalities. *Nat. Commun.* 9, 1-15 (2018).
- [0312] 23. S. Konermann, M. D. Brigham, A. E. Trevino, J. Joung, O. O. Abudayyeh, C. Barcena, P. D. Hsu, N. Habib, J. S. Gootenberg, H. Nishimasu, O. Nureki, F. Zhang, Genome-scale transcriptional activation by an engineered CRISPR-Cas9 complex. *Nature.* 517, 583-588 (2014).
- [0313] 24. S. M. Kaech, W. Cui, Transcriptional control of effector and memory CD8 $^{+}$ T cell differentiation. *Nat. Rev. Immunol.* 12, 749-761 (2012).
- [0314] 25. J. Zhu, H. Yamane, W. E. Paul, Differentiation of effector CD4 T cell populations (*). *Annu. Rev. Immunol.* 28, 445-489 (2010).

- [0315] 26. V. Lazarevic, L. H. Glimcher, G. M. Lord, T-bet: a bridge between innate and adaptive immunity. *Nat. Rev. Immunol.* 13, 777-789 (2013).
- [0316] 27. C. M. Evans, R. G. Jenner, Transcription factor interplay in T helper cell differentiation. *Brief. Funct. Genomics.* 12, 499-511 (2013).
- [0317] 28. A. H. Courtney, W.-L. Lo, A. Weiss, TCR Signaling: Mechanisms of Initiation and Propagation. *Trends Biochem. Sci.* 43, 108-123 (2018).
- [0318] 29. G. Gaud, R. Lesourne, P. E. Love, Regulatory mechanisms in T cell receptor signalling. *Nat. Rev. Immunol.* 18, 485-497 (2018).
- [0319] 30. S. J. Holland, X. C. Liao, M. K. Mendenhall, X. Zhou, J. Pardo, P. Chu, C. Spencer, A. Fu, N. Sheng, P. Yu, E. Pali, A. Nagin, M. Shen, S. Yu, E. Chan, X. Wu, C. Li, M. Woisetschlager, G. Aversa, F. Kolbinger, M. K. Bennett, S. Molineaux, Y. Luo, D. G.
- [0320] Payan, H. S. Mancebo, J. Wi, Functional cloning of Src-like adapter protein-2 (SLAP-2), a novel inhibitor of antigen receptor signaling. *J. Exp. Med.* 194, 1263-1276 (2001).
- [0321] 31. J.-W. Shui, J. S. Boomer, J. Han, J. Xu, G. A. Dement, G. Zhou, T.-H. Tan, Hematopoietic progenitor kinase I negatively regulates T cell receptor signaling and T cell-mediated immune responses. *Nat. Immunol.* 8, 84-91 (2006).
- [0322] 32. T. Hart, A. H. Y. Tong, K.-. Chan, J. Van Leeuwen, A. Seetharaman, M. Aregger, M. Chandrasekhar, N. Hustedt, S. Seth, A. Noonan, A. Habsid, O. Sizova, L. Nedyalkova, R.-Climie, L. Tworzyanski, K. Lawson, M. A. Sartori, S. Alibeh, D. Tieu, S. Masud, P.
- Mero, A. Weiss, K. R-Brown, M. Ujsaj, M. Billmann, M. Rahman, M. Constanzo, C. L.
- [0323] Myers, B. I. Andrews, C. Boone, D. Durocher, J. Moffat, Evaluation and Design of Genome-Wide CRISPR/SpCas9 Knockout Screens. *G3.* 7, 2719-2727 (2017).
- [0324] 33. S. R. Ferdosi, R. Ewaisha, F. Moghadam, S. Krishna, J. (G. Park, M. R. Ebrahimkhani, S. Kiani, K. S. Anderson, Multifunctional CRISPR-Cas9 with engineered immunosilenced human T cell epitopes. *Nat. Commun.* 10, 1842 (2019).
- [0325] 34. H. Hu, H. Wang, Y. Xiao, J. Jin, J.-H. Chang, Q. Zou, X. Xie, X. Cheng, S.-C. Sun, Otud7b facilitates T cell activation and inflammatory responses by regulating Zap70 ubiquitination. *J. Exp. Med.* 213, 399-414 (2016).
- [0326] 35. E. P. Miritou, A. Cheng, A. Montalbano, S. Hao, M. Stoeckius, M. Legut, T. Roush, A. Herrera, E. Papalex, Z. Ouyang, R. Satija, N. E. Sanjana, S. B. Koralov, P. Smibert, Multiplexed detection of proteins, transcriptomes, clonotypes and CRISPR perturbations in single cells. *Nat. Methods.* 16, 409-412 (2019).
- [0327] 36. C. Alda-Catalinas, D. Bredikhin, I. Hernandez-Herraez, F. Santos, O. Kubinyecz, M. A. Eckersley-Maslin, O. Stegle, W. Reik, A Single-Cell Transcriptomics CRISPR-Activation Screen Identifies Epigenetic Regulators of the Zygotic Genome Activation Program. *Cell Syst.* 11, 25-41.e9 (2020).
- [0328] 37. J. M. Replogle, T. M. Norman, A. Xu, J. A. Hussmann, J. Chen, J. Z. Cogan, E. J. Meer, J. M. Terry, D. P. Riordan, N. Srinivas, I. T. Fiddes, J. G. Arthur, L. J. Alvarado, K. A. Pfeiffer, T. S. Mikkelsen, J. S. Weissman, B. Adamson, Combinatorial single-cell CRISPR screens by direct guide RNA capture and targeted sequencing. *Nat. Biotechnol.* 38, 954-961 (2020).
- [0329] 38. P. I Thakore, A. M. D'Ippolito, L. Song, A. Safi, N. K. Shivakumar, A. M. Kabadi, T. E. Reddy, G. E. Crawford, C. A. Gersbach, Highly specific epigenome editing by CRISPR-Cas9 repressors for silencing of distal regulatory elements. *Nat. Methods.* 12, 1143-1149 (2015).
- [0330] 39. C. P. Fulco, M. Munschauer, R. Anyoha, G. Munson, S. R. Grossman, E. M. Perez, M. Kane, B. Cleary, E. S. Lander, J. M. Engreitz, Systematic mapping of functional enhancer-promoter connections with CRISPR interference. *Science.* 354, 769-773 (2016).
- [0331] 40. X. Xu, L. S. Qi, A CRISPR-dCas Toolbox for Genetic Engineering and Synthetic Biology. *J. Mol. Biol.* 431, 34-47 (2019).
- [0332] 41. I. Chun, K. H. Kim, Y. H. Chiang, W. Xie, Y. C. G. Lee, R. Pajarillo, A. Rotolo, O. Shestova, S. J. [long, M. Abdel-Mohsen, M. Wysocka, H. J. Ballard, D. M. Barrett, A. D. Posey, D. Powell Jr, S. J. Gill, S. J. Schuster, S. K. Barta, A. H. Rook, C. H. June, M. Ruella, CRISPR-Cas9 knock out of CD5 enhances the anti-tumor activity of chimeric Antigen Receptor T cells. *Blood.* 136, 51-52 (2020).
- [0333] 42 R. C. Lynn, E. W. Weber, E. Sotillo, D. Gennert, P. Xu, Z. Good, H. Anbunathan, J. Lattin, R. Jones, V. Tieu, S. Nagaraja, J. Granja, C. F. A. de Bourcy, R. Majzner, A. T. Satpathy, S. R. Quake, M. J. V. Monje, H. Y. Chang, C. L. Mackall, c-Jun overexpression in CAR T cells induces exhaustion resistance. *Nature.* 576, 293-300 (2019).
- [0334] 43. P. Datlinger, A. F. Rendeiro, C. Schmidl, T. Krausgruber, P. Traxler, J. Klughammer, L. C. Schuster, A. Kuchler, D. Alpar, C. Bock, Pooled CRISPR screening with single-cell transcriptome readout. *Nat. Methods.* 14, 297-301 (2017).
- [0335] 44. J. W. Freimer, O. Shaked, S. Naqvi, N. Sinnott-Armstrong, A. Kathiria, A. F. Chen, O. I. T. Cortez, W. J. Greenleaf, J. K. Pritchard, A. Marson, Systematic discovery and perturbation of regulatory genes in human T cells reveals the architecture of immune networks. *bioRxiv* (2021), p. 2021.04.18.440363.
- [0336] 45. W. Li, H. Xu, T. Xiao, L. Cong, M. I. Love, F. Zhang, R. A. Irizarry, J. S. Liu, M. Brown, X. S. Liu, MAGeCK enables robust identification of essential genes from genome-scale CRISPR/Cas9 knockout screens. *Genome Biol.* 15, 554 (2014).
- [0337] 46. G. Korotkevich, V. Sukhov, N. Budin, B. Shpak, M. N. Artyomov, A. Sergushichev, Fast gene set enrichment analysis. *bioRxiv* (2016), , doi:10.1101/060012.
- [0338] 47. H. K. Finucane, B. Bulik-Sullivan, A. Gusev, C. Trynka, Y. Reshef, P.-R. Loh, V. Anttila, H. Xu, C. Zang, K. Fart, S. Ripke, F. R. Day, ReproGen Consortium, Schizophrenia Working Group of the Psychiatric Genomics Consortium, RACI Consortium, S. Purcell, E. Stahl, S. Lindstrom, J. R. B. Perry, Y. Okada, S. Raychaudhuri, M. J. Daly, N. Patterson, B. M. Neale, A. L. Price, Partitioning heritability by functional annotation using genome-wide association summary statistics. *Nat. Genet.* 47, 1228-1235 (2015).
- [0339] 48. M. A. Horlbeck, L. A. Gilhert, J. E. Villalta, B. Adamson, R. A. Pak, Y. C. Then, A. P. Fields, C. Y. Park, J. E. Corn, M. Kampmann, J. S. Weissman, Compact and

- highly active next-generation libraries for CRISPR-mediated gene repression and activation. *Elife*. 5 (2016), doi:10.7554/eLife.19760.
- [0340] 49. M. Martin, Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet.journal*. 17, 10-12 (2011).
- [0341] 50. A. Dobin, C. A. Davis, F. Schlesinger, J. Drenkow, C. Zaleski, S. Jha, P. Batut, M. Chaisson, T. R. Gingeras, STAR: ultrafast universal RNA-seq aligner. *Bioinformatics*. 29, 15-21 (2013).
- [0342] 51. Y. Lia.o, G. K. Smyth, W. Shi, featureCounts: an efficient general-purpose program for assigning sequence reads to genomic features. *Bioinformatics*. 30, 923-930 (2014).
- [0343] 52. M. E. Ritchie, B. Phipson, D. Wu, Y. Hu, C. W. Law, W. Shi, G. K. Smyth, limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* 43, e47 (2015).
- [0344] 53. H. Heaton, A. M. Talman, A. Knights, M. Imaz, D. J. Gaffney, R. Durbin, M. Henberg, M. K. N. Lawnczak, Souporecell: robust clustering of single-cell RNA-seq data by genotype without reference genotypes. *Nat. Methods*. 17, 615-620 (2020).
- [0345] 54. R. Satija, J. A. Farrell, D. Gennert, A. F. Schier, A. Regev, Spatial reconstruction of single-cell gene expression data. *Nat. Biotechnol.* 33, 495-502 (2015).
- [0346] 55. C. Hafemeister, R. Satija, Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression. *Cenome Biol.* 20, 296 (2019).
- [0347] 56. Z. Steinhart, Code repository for CRISPR activation and interference screens decode stimulation responses in primary human T cells, Zenodo (2022); 10.5281/zenodo.578465
- [0348] 57. B. J. Schmiedel, D. Singh, A. Madrigal, A. G. Valdovino-Gonzalez, B. M. White, J. Zapardiel Gonzalo, B. Ha, G. Altay, J. A. Greenbaum, CG. McVicker, G. Seumois, A. Rao, M. Kronenberg, B. Peters, P. Vijayanand, Impact of Genetic Polymorphisms on Human Immune Cell Gene Expression, *Cell*. 175, 1701-1715.e16 (2018).
- [0349] 58. Immunological Genome Project, ImmGen at 15. *Nat. Immunol.* 21, 700-703 (2020).
- [0350] 59. B. P. Nicolet, A. Guislain, F. P. J. van Alphen, R. Gomez-Eerland, T. N. M. Schumacher, M. van den Biggelaar, M. C. Wolkers, CD29 identifies IFN- γ -producing human CD8 $^{+}$ T cells with an increased cytotoxic potential. *Proc. Natl. Acad. Sci. U.S.A.* 117, 6686-6696 (2020).

Example 3

- [0351] In vitro data using the identified hits for T cell cancer therapies. For this assay, T cells from two human blood donors were virally transduced with the 1G4 anti-cancer T cell receptor as well as the respective gene from the CRISPRa screens (or “empty” virus as control) and cocultured with NYESO expressing A375 melanoma cells. A live imaging system recorded the cancer cell counts every 4h. T cells transduced with the target genes VAV1, PIK3API and CD27 showed enhanced cancer killing (FIG. 5).
- [0352] All patents and publications referenced or mentioned herein are indicative of the levels of skill of those skilled in the art to which the invention pertains, and each such referenced patent or publication is hereby specifically

incorporated by reference to the same extent as if it had been incorporated by reference in its entirety individually or set forth herein in its entirety. Applicants reserve the right to physically incorporate into this specification any and all materials and information from any such cited patents or publications.

[0353] The following statements are intended to describe and summarize various embodiments of the invention according to the foregoing description in the specification.

Statements:

- [0354] 1. A method comprising contacting one or more test agents with one or more T cells to form an assay mixture, and detecting or quantifying interferon- γ production, interleukin-2 production, cellular proliferation, or a combination thereof in the assay mixture or within one or more T cells, to generate a detected or quantified level of interferon- γ production, interleukin-2 production, cellular proliferation, or a combination thereof.
- [0355] 2. The method of statement 1, further comprising comparing the detected or quantified level of interferon-f production, the detected or quantified level of interleukin-2 production, the detected or quantified level of cellular proliferation, or a combination thereof with a control.
- [0356] 3. The method of statement 1 or 2, further comprising measuring the quantity of one or more of the regulators listed in Tables 1-7 or FIGS. 1-4 in the assay mixture or in one or more of the T cells.
- [0357] 4. The method of statement 1, 2, or 3, wherein one or more of the T cells initially contacted with the test agent naturally expresses any of the regulators listed in Tables 1-7 or FIGS. 1-4.
- [0358] 5. The method of statement 1-3 or 4, wherein one or more of the T cells initially contacted with the test agent do not express one or more of the regulators listed in any of Tables 1-7 or FIGS. 1-4.
- [0359] 6. The method of statement 1-4 or 5, wherein one or more of T cells initially contacted with the test agent have the potential to express one or more of the regulators but when initially mixed with a test agent the cells do not express detectable amounts of one or more of the regulators.
- [0360] 7. The method of statement 1-5 or 6, wherein at least one of the T cells is a mutant T cell comprising a knock-down or knockout mutation that reduces expression or activity of one or more of the regulators listed in any of Tables 1-7 or FIGS. 1-4.
- [0361] 8. The method of statement 7, further comprising: modifying the one or more mutant T cells to express or over-express the one or more regulators listed in Tables 1-7 or FIGS. 1-4; and detecting or quantifying interferon- γ production, interleukin-2 production, cellular proliferation, or a combination thereof in a second assay mixture or within the one or more of the mutant T cells.
- [0362] 9. The method of statement 1-7 or 8, wherein the one or more of the T cells is in a population of T cells.
- [0363] 10. The method of statement 1-8 or 9, wherein one or more of T cells comprises one or more cytotoxic T cells, chimeric antigen receptor T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, natural killer

- (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof.
- [0364] 11. The method of statement 1-9 or 10, further comprising adding at least one second cell type to the assay mixture before detecting or quantifying the interferon- γ production, interleukin-2 production, cellular proliferation, or a combination thereof.
- [0365] 12. The method of statement 11, wherein the second cell type is one or more types of cancer cells, one or more types of immune cells, or a combination thereof.
- [0366] 13. The method of statement 12, wherein one or more of the cancer cells comprise leukemia cells, lymphoma cells, Hodgkin's disease cells, sarcomas of the soft tissue and bone, lung cancer cells, mesothelioma, esophagus cancer cells, stomach cancer cells, pancreatic cancer cells, hepatobiliary cancer cells, small intestinal cancer cells, colon cancer cells, colorectal cancer cells, rectum cancer cells, kidney cancer cells, urethral cancer cells, bladder cancer cells, prostate cancer cells, testis cancer cells, cervical cancer cells, ovarian cancer cells, breast cancer cells, endocrine system cancer cells, skin cancer cells, central nervous system cancer cells, melanoma cells of cutaneous and/or intraocular origin, cancer cells associated with AIDS, or a combination thereof.
- [0367] 14. The method of statement 12 or 13, wherein one or more of cancer cells comprise metastatic cancer cells.
- [0368] 15. The method of statement 12, 13 or 14, wherein one or more of cancer cells comprise micrometastatic tumor cells, megametastatic tumor cells, recurrent cancer cells, or a combination thereof.
- [0369] 16. The method of statement 12-14 or 15, wherein one or more of the immune cells comprises macrophages, natural killer cells, dendritic cells, B cells, chimeric antigen receptor cells, cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof.
- [0370] 17. The method of statement 12-15 or 16, further comprising measuring the cellular proliferation of at least one of the second cell types.
- [0371] 18. The method of statement 1-16 or 17, further comprising identifying one or more of the test agents that modulates the level of interferon- γ production, the level of interleukin-2 production, the level of cellular proliferation, or a combination thereof of one or more of the T cells, to thereby identify one or more useful test agents.
- [0372] 19. The method of statement 1-17 or 18, further comprising identifying one or more of the test agents that modulates expression or activity of one or more of the regulators listed in any of Tables 1-7 or FIGS. 1-4, to thereby identify one or more useful test agents.
- [0373] 20. The method of statement 1-9, further comprising measuring binding of one or more useful test agents to a protein encoded by, or nucleic acid comprising, one or more the regulators listed in any of Tables 1-7 or FIGS. 1-4.
- [0374] 21. The method of statement 19 or 20, further comprising administering one or more useful test agents to one or more experimental animals.
- [0375] 22. The method of statement 21, wherein one or more of the experimental animals has a disease, infection, or medical condition.
- [0376] 23. The method of statement 22, wherein the disease or condition is cancer, an immune disorder, or an immune condition.
- [0377] 24. The method of statement 23, wherein the immune disorder or immune condition is an autoimmune disorder, Graves disease, arthritis, psoriasis, Celiac disease, vitiligo, rheumatoid arthritis, lupus, Crohn's disease, multiple sclerosis, type 1 diabetes, alopecia, inflammatory bowel disease (IBD), Guillain-Barre syndrome, chronic inflammatory demyelinating polyneuropathy, or a combination thereof.
- [0378] 25. The method of statement 21-23 or 24, further comprising monitoring the one or more experimental animals for symptoms of the disease or condition, for toxic side effects of the useful test agents, or a combination thereof.
- [0379] 26. The method of statement 21-24 or 25, further comprising monitoring immune cell numbers and/or types in the one or more experimental animals.
- [0380] 27. The method of statement 22-25 or 26, further comprising identifying one or more useful test agents as a therapeutic agent useful for treatment of the disease or condition.
- [0381] 28. A composition comprising a useful test agent or a therapeutic agent identified by the method of any of claims 1-27.
- [0382] 29. A method comprising ex vivo modification of any of the genes listed in Tables 1-7 or FIGS. 1-4 within at least one lymphoid or myeloid cell, or a combination thereof to generate at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells.
- [0383] 30. The method of statement 29, wherein the modification is one or more deletion, substitution or insertion into one or more genomic sites of any of the genes listed in Tables 1-7 or FIGS. 1-4.
- [0384] 31. The method of statement 29 or 30, wherein the modification is one or more CRISPR-mediated modifications or activations of any of the genes listed in Tables 1-7 or FIGS. 1-4.
- [0385] 32. The method of statement 29, 30 or 31, further comprising administering at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to a subject.
- [0386] 33. The method of statement 29, 30 or 31 further comprising incubating the at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to form a population of modified cells.
- [0387] 34. The method of statement 33, further comprising administering the population of modified cells to a subject.
- [0388] 35. The method of statement 32 or 34, wherein the subject has a disease or condition.
- [0389] 36. The method of statement 35, wherein the disease or condition is an immune condition or cancer.

[0390] The specific methods and compositions described herein are representative of preferred embodiments and are exemplary and not intended as limitations on the scope of the invention. Other objects, aspects, and embodiments will occur to those skilled in the art upon consideration of this specification and are encompassed within the spirit of the invention as defined by the scope of the claims. It will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention.

[0391] The invention illustratively described herein suitably may be practiced in the absence of any element or elements, or limitation or limitations, which is not specifically disclosed herein as essential. The methods and processes illustratively described herein suitably may be practiced in differing orders of steps, and the methods and processes are not necessarily restricted to the orders of steps indicated herein or in the claims.

[0392] As used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to “a nucleic acid” or “a protein” or “a cell” includes a plurality of such nucleic acids, proteins, or cells (for example, a solution or dried preparation of nucleic acids or expression cassettes, a solution of proteins, or a population of cells), and so forth. In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated.

[0393] Under no circumstances may the patent be interpreted to be limited to the specific examples or embodiments

or methods specifically disclosed herein. Under no circumstances may the patent be interpreted to be limited by any statement made by any Examiner or any other official or employee of the Patent and Trademark Office unless such statement is specifically and without qualification or reservation expressly adopted in a responsive writing by Applicants.

[0394] The terms and expressions that have been employed are used as terms of description and not of limitation, and there is no intent in the use of such terms and expressions to exclude any equivalent of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention as claimed. Thus, it will be understood that although the present invention has been specifically disclosed by preferred embodiments and optional features, modification and variation of the concepts herein disclosed may be resorted to by those skilled in the art, and that such modified cations and variations are considered to be within the scope of this invention as defined by the appended claims and statements of the invention.

[0395] The invention has been described broadly and generically herein. Each of the narrower species and sub-generic groupings falling within the generic disclosure also form part of the invention. This includes the generic description of the invention with a proviso or negative limitation removing any subject matter from the genus, regardless of whether or not the excised material is specifically recited herein. In addition, where features or aspects of the invention are described in terms of Markush groups, those skilled in the art will recognize that the invention is also thereby described in terms of any individual member or subgroup of members of the Markush group.

SEQUENCE LISTING

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Leu	Ser	Leu	Gln	Leu	Asn	Arg	Val	Ser	Leu	Glu	Arg	Asp	Ser	Leu	Ser
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Pro	Arg	Arg	Ala	Ala	Pro	Arg	Phe	Ser	Leu	Arg	Leu	Gly	Pro	Gly	Pro
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Ala	Glu	Lys	Glu	Pro	Gln	Asp	Trp	Ala	Glu	Thr	Lys	Glu	Glu	Pro	Glu	435	440	445	
Ala	Glu	Ala	Glu	Ala	Ala	Ser	Ser	Gly	Val	Pro	Ala	Thr	Lys	Gln	His	450	455	460	
Pro	Glu	Val	Gln	Val	Glu	Asp	Thr	Asp	Ala	Asp	Ser	Cys	Pro	Leu	Met	465	470	475	480
Ala	Glu	Glu	Asn	Pro	Pro	Ser	Thr	Val	Leu	Pro	Pro	Pro	Ser	Pro	Ala	485	490	495	
Lys	Ser	Asp	Thr	Leu	Ile	Val	Pro	Ser	Ser	Ala	Ser	Gly	Thr	His	Arg	500	505	510	
Lys	Lys	Leu	Pro	Ser	Glu	Asp	Asp	Glu	Ala	Glu	Glu	Leu	Lys	Ala	Leu	515	520	525	
Ser	Pro	Ala	Glu	Ser	Pro	Val	Val	Ala	Trp	Ser	Asp	Pro	Thr	Thr	Pro	530	535	540	

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Lys 545	Asp	Thr	Asp	Gly 550	Gln	Asp	Arg	Ala	Ala	Ser 555	Thr	Ala	Ser	Thr	Asn 560
Ser	Ala	Ile	Ile	Asn 565	Asp	Arg	Leu	Gln	Glu 570	Leu	Val	Lys	Leu	Phe 575	Lys
Glu	Arg	Thr	Glu 580	Lys	Val	Lys	Glu	Lys 585	Leu	Ile	Asp	Pro	Asp 590	Val	Thr
Ser	Asp 595	Glu	Glu	Ser	Pro	Lys 600	Pro	Ser	Pro	Ala	Lys	Lys 605	Ala	Pro	Glu
Pro	Ala 610	Pro	Asp	Thr	Lys 615	Pro	Ala	Glu	Ala	Glu	Pro 620	Val	Glu	Glu	Glu
His 625	Tyr	Cys	Asp	Met 630	Leu	Cys	Cys	Lys	Phe	Lys 635	His	Arg	Pro	Trp	Lys 640
Lys	Tyr	Gln	Phe 645	Pro	Gln	Ser	Ile	Asp 650	Pro	Leu	Thr	Asn	Leu	Met 655	Tyr
Val	Leu	Trp 660	Leu	Phe	Phe	Val	Val	Met 665	Ala	Trp	Asn	Trp	Asn 670	Cys	Trp
Leu	Ile 675	Pro	Val	Arg	Trp	Ala	Phe 680	Pro	Tyr	Gln	Thr	Pro 685	Asp	Asn	Ile
His 690	His	Trp	Leu	Leu	Met 695	Asp	Tyr	Leu	Cys	Asp 700	Leu	Ile	Tyr	Phe	Leu
Asp 705	Ile	Thr	Val	Phe 710	Gln	Thr	Arg	Leu	Gln	Phe 715	Val	Arg	Gly	Gly	Asp 720
Ile	Ile	Thr	Asp 725	Lys	Lys	Asp	Met	Arg	Asn 730	Asn	Tyr	Leu	Lys	Ser 735	Arg
Arg	Phe	Lys 740	Met	Asp	Leu	Leu	Ser	Leu 745	Leu	Pro	Leu	Asp	Phe 750	Leu	Tyr
Leu	Lys 755	Val	Gly	Val	Asn	Pro	Leu 760	Leu	Arg	Leu	Pro	Arg 765	Cys	Leu	Lys
Tyr 770	Met	Ala	Phe	Phe	Glu 775	Phe	Asn	Ser	Arg	Leu 780	Glu	Ser	Ile	Leu	Ser
Lys 785	Ala	Tyr	Val	Tyr 790	Arg	Val	Ile	Arg	Thr	Thr 795	Ala	Tyr	Leu	Leu	Tyr 800
Ser	Leu	His 805	Leu	Asn	Ser	Cys	Leu	Tyr	Tyr 810	Trp	Ala	Ser	Ala	Tyr	Gln 815
Gly	Leu	Gly 820	Ser	Thr	His	Trp	Val	Tyr 825	Asp	Gly	Val	Gly	Asn 830	Ser	Tyr
Ile 835	Arg	Cys	Tyr	Tyr	Phe	Ala 840	Val	Lys	Thr	Leu	Ile	Thr 845	Ile	Gly	Gly
Leu 850	Pro	Asp	Pro	Lys	Thr 855	Leu	Phe	Glu	Ile	Val	Phe 860	Gln	Leu	Leu	Asn
Tyr 865	Phe	Thr	Gly	Val	Phe 870	Ala	Phe	Ser	Val	Met 875	Ile	Gly	Gln	Met	Arg 880
Asp	Val	Val	Gly 885	Ala	Ala	Thr	Ala	Gly	Gln 890	Thr	Tyr	Tyr	Arg	Ser	Cys 895
Met	Asp	Ser 900	Thr	Val	Lys	Tyr	Met	Asn 905	Phe	Tyr	Lys	Ile	Pro 910	Lys	Ser
Val	Gln 915	Asn	Arg	Val	Lys	Thr	Trp 920	Tyr	Glu	Tyr	Thr	Trp 925	His	Ser	Gln
Gly 930	Met	Leu	Asp	Glu	Ser 935	Glu	Leu	Met	Val	Gln	Leu	Pro	Asp	Lys	Met
Arg	Leu	Asp	Leu	Ala	Ile	Asp	Val	Asn	Tyr	Asn	Ile	Val	Ser	Lys	Val

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945	950	955	960
Ala Leu Phe Gln Gly Cys Asp Arg Gln Met Ile Phe Asp Met Leu Lys			
	965	970	975
Arg Leu Arg Ser Val Val Tyr Leu Pro Asn Asp Tyr Val Cys Lys Lys			
	980	985	990
Gly Glu Ile Gly Arg Glu Met Tyr Ile Ile Gln Ala Gly Gln Val Gln			
	995	1000	1005
Val Leu Gly Gly Pro Asp Gly Lys Ser Val Leu Val Thr Leu Lys Ala			
	1010	1015	1020
Gly Ser Val Phe Gly Glu Ile Ser Leu Leu Ala Val Gly Gly Gly Asn			
	1025	1030	1035
Arg Arg Thr Ala Asn Val Val Ala His Gly Phe Thr Asn Leu Phe Ile			
	1045	1050	1055
Leu Asp Lys Lys Asp Leu Asn Glu Ile Leu Val His Tyr Pro Glu Ser			
	1060	1065	1070
Gln Lys Leu Leu Arg Lys Lys Ala Arg Arg Met Leu Arg Ser Asn Asn			
	1075	1080	1085
Lys Pro Lys Glu Glu Lys Ser Val Leu Ile Leu Pro Pro Arg Ala Gly			
	1090	1095	1100
Thr Pro Lys Leu Phe Asn Ala Ala Leu Ala Met Thr Gly Lys Met Gly			
	1105	1110	1115
Gly Lys Gly Ala Lys Gly Gly Lys Leu Ala His Leu Arg Ala Arg Leu			
	1125	1130	1135
Lys Glu Leu Ala Ala Leu Glu Ala Ala Ala Lys Gln Gln Glu Leu Val			
	1140	1145	1150
Glu Gln Ala Lys Ser Ser Gln Asp Val Lys Gly Glu Glu Gly Ser Ala			
	1155	1160	1165
Ala Pro Asp Gln His Thr His Pro Lys Glu Ala Ala Thr Asp Pro Pro			
	1170	1175	1180
Ala Pro Arg Thr Pro Pro Glu Pro Pro Gly Ser Pro Pro Ser Ser Pro			
	1185	1190	1195
Pro Pro Ala Ser Leu Gly Arg Pro Glu Gly Glu Glu Glu Gly Pro Ala			
	1205	1210	1215
Glu Pro Glu Glu His Ser Val Arg Ile Cys Met Ser Pro Gly Pro Glu			
	1220	1225	1230
Pro Gly Glu Gln Ile Leu Ser Val Lys Met Pro Glu Glu Arg Glu Glu			
	1235	1240	1245
Lys Ala Glu			
	1250		

<210> SEQ ID NO 4

<211> LENGTH: 511

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 4

Met Ala Thr Val Gln Glu Lys Ala Ala Ala Leu Asn Leu Ser Ala Leu			
1	5	10	15
His Ser Pro Ala His Arg Pro Pro Gly Phe Ser Val Ala Gln Lys Pro			
	20	25	30
Phe Gly Ala Thr Tyr Val Trp Ser Ser Ile Ile Asn Thr Leu Gln Thr			
	35	40	45

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Gln	Val	Glu	Val	Lys	Lys	Arg	Arg	His	Arg	Leu	Lys	Arg	His	Asn	Asp
50						55					60				
Cys	Phe	Val	Gly	Ser	Glu	Ala	Val	Asp	Val	Ile	Phe	Ser	His	Leu	Ile
65					70					75					80
Gln	Asn	Lys	Tyr	Phe	Gly	Asp	Val	Asp	Ile	Pro	Arg	Ala	Lys	Val	Val
			85						90					95	
Arg	Val	Cys	Gln	Ala	Leu	Met	Asp	Tyr	Lys	Val	Phe	Glu	Ala	Val	Pro
			100					105					110		
Thr	Lys	Val	Phe	Gly	Lys	Asp	Lys	Lys	Pro	Thr	Phe	Glu	Asp	Ser	Ser
	115						120					125			
Cys	Ser	Leu	Tyr	Arg	Phe	Thr	Thr	Ile	Pro	Asn	Gln	Asp	Ser	Gln	Leu
130						135					140				
Gly	Lys	Glu	Asn	Lys	Leu	Tyr	Ser	Pro	Ala	Arg	Tyr	Ala	Asp	Ala	Leu
145					150					155					160
Phe	Lys	Ser	Ser	Asp	Ile	Arg	Ser	Ala	Ser	Leu	Glu	Asp	Leu	Trp	Glu
				165					170					175	
Asn	Leu	Ser	Leu	Lys	Pro	Ala	Asn	Ser	Pro	His	Val	Asn	Ile	Ser	Ala
			180					185					190		
Thr	Leu	Ser	Pro	Gln	Val	Ile	Asn	Glu	Val	Trp	Gln	Glu	Glu	Thr	Ile
	195						200					205			
Gly	Arg	Leu	Leu	Gln	Leu	Val	Asp	Leu	Pro	Leu	Leu	Asp	Ser	Leu	Leu
210						215					220				
Lys	Gln	Gln	Glu	Ala	Val	Pro	Lys	Ile	Pro	Gln	Pro	Lys	Arg	Gln	Ser
225					230					235					240
Thr	Met	Val	Asn	Ser	Ser	Asn	Tyr	Leu	Asp	Arg	Gly	Ile	Leu	Lys	Ala
				245					250					255	
Tyr	Ser	Asp	Ser	Gln	Glu	Asp	Glu	Trp	Leu	Ser	Ala	Ala	Ile	Asp	Cys
		260						265					270		
Leu	Glu	Tyr	Leu	Pro	Asp	Gln	Met	Val	Val	Glu	Ile	Ser	Arg	Ser	Phe
	275						280					285			
Pro	Glu	Gln	Pro	Asp	Arg	Thr	Asp	Leu	Val	Lys	Glu	Leu	Leu	Phe	Asp
290						295					300				
Ala	Ile	Gly	Arg	Tyr	Tyr	Ser	Ser	Arg	Glu	Pro	Leu	Leu	Asn	His	Leu
305					310					315					320
Ser	Asp	Val	His	Asn	Gly	Ile	Ala	Glu	Leu	Leu	Val	Asn	Gly	Lys	Thr
				325					330					335	
Glu	Ile	Ala	Leu	Glu	Ala	Thr	Gln	Leu	Leu	Leu	Lys	Leu	Leu	Asp	Phe
		340						345					350		
Gln	Asn	Arg	Glu	Glu	Phe	Arg	Arg	Leu	Leu	Tyr	Phe	Met	Ala	Val	Ala
		355					360						365		
Ala	Asn	Pro	Ser	Glu	Phe	Lys	Leu	Gln	Lys	Glu	Ser	Asp	Asn	Arg	Met
370						375					380				
Val	Val	Lys	Arg	Ile	Phe	Ser	Lys	Ala	Ile	Val	Asp	Asn	Lys	Asn	Leu
385					390					395					400
Ser	Lys	Gly	Lys	Thr	Asp	Leu	Leu	Val	Leu	Phe	Leu	Met	Asp	His	Gln
				405					410					415	
Lys	Asp	Val	Phe	Lys	Ile	Pro	Gly	Thr	Leu	His	Lys	Ile	Val	Ser	Val
		420						425					430		
Lys	Leu	Met	Ala	Ile	Gln	Asn	Gly	Arg	Asp	Pro	Asn	Arg	Asp	Ala	Gly
	435						440					445			
Tyr	Ile	Tyr	Cys	Gln	Arg	Ile	Asp	Gln	Arg	Asp	Tyr	Ser	Asn	Asn	Thr

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450	455	460
Glu Lys Thr Thr Lys Asp	Glu Leu Leu Asn Leu Leu Lys Thr Leu Asp	
465	470	475 480
Glu Asp Ser Lys Leu Ser Ala Lys Glu Lys Lys Lys Leu Leu Gly Gln		
	485	490 495
Phe Tyr Lys Cys His Pro Asp Ile Phe Ile Glu His Phe Gly Asp		
	500	505 510

<210> SEQ ID NO 5

<211> LENGTH: 279

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 5

Met Gly Leu Ser Pro Gly Ala Glu Gly Glu Tyr Ala Leu Arg Leu Pro		
1	5	10 15
Arg Ile Pro Pro Pro Leu Pro Lys Pro Ala Ser Arg Thr Ala Ser Thr		
	20	25 30
Gly Pro Lys Asp Gln Pro Pro Ala Leu Arg Arg Ser Ala Val Pro His		
	35	40 45
Ser Gly Leu Asn Ser Ile Ser Pro Leu Glu Leu Glu Glu Ser Val Gly		
	50	55 60
Phe Ala Ala Leu Val Gln Leu Pro Ala Lys Gln Pro Pro Pro Gly Thr		
65	70	75 80
Leu Glu Gln Gly Arg Ser Ile Gln Gln Gly Glu Lys Ala Val Val Ser		
	85	90 95
Leu Glu Thr Thr Pro Ser Gln Lys Ala Asp Trp Ser Ser Ile Pro Lys		
	100	105 110
Pro Glu Asn Glu Gly Lys Leu Ile Lys Gln Ala Ala Glu Gly Lys Pro		
	115	120 125
Arg Pro Arg Pro Gly Asp Leu Ile Glu Ile Phe Arg Ile Gly Tyr Glu		
	130	135 140
His Trp Ala Ile Tyr Val Glu Asp Asp Cys Val Val His Leu Ala Pro		
145	150	155 160
Pro Ser Glu Glu Phe Glu Val Gly Ser Ile Thr Ser Ile Phe Ser Asn		
	165	170 175
Arg Ala Val Val Lys Tyr Ser Arg Leu Glu Asp Val Leu His Gly Cys		
	180	185 190
Ser Trp Lys Val Asn Asn Lys Leu Asp Gly Thr Tyr Leu Pro Leu Pro		
	195	200 205
Val Asp Lys Ile Ile Gln Arg Thr Lys Lys Met Val Asn Lys Ile Val		
	210	215 220
Gln Tyr Ser Leu Ile Glu Gly Asn Cys Glu His Phe Val Asn Gly Leu		
225	230	235 240
Arg Tyr Gly Val Pro Arg Ser Gln Gln Val Glu His Ala Leu Met Glu		
	245	250 255
Gly Ala Lys Ala Ala Gly Ala Val Ile Ser Ala Val Val Asp Ser Ile		
	260	265 270
Lys Pro Lys Pro Ile Thr Ala		
	275	

<210> SEQ ID NO 6

<211> LENGTH: 382

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<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 6

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Met Leu Arg Trp Thr Val His Leu Glu Gly Gly Pro Arg Arg Val Asn
1      5      10      15
His Ala Ala Val Ala Val Gly His Arg Val Tyr Ser Phe Gly Gly Tyr
20      25      30
Cys Ser Gly Glu Asp Tyr Glu Thr Leu Arg Gln Ile Asp Val His Ile
35      40      45
Phe Asn Ala Val Ser Leu Arg Trp Thr Lys Leu Pro Pro Val Lys Ser
50      55      60
Ala Ile Arg Gly Gln Ala Pro Val Val Pro Tyr Met Arg Tyr Gly His
65      70      75      80
Ser Thr Val Leu Ile Asp Asp Thr Val Leu Leu Trp Gly Gly Arg Asn
85      90      95
Asp Thr Glu Gly Ala Cys Asn Val Leu Tyr Ala Phe Asp Val Asn Thr
100     105     110
His Lys Trp Phe Thr Pro Arg Val Ser Gly Thr Val Pro Gly Ala Arg
115     120     125
Asp Gly His Ser Ala Cys Val Leu Gly Lys Ile Met Tyr Ile Phe Gly
130     135     140
Gly Tyr Glu Gln Gln Ala Asp Cys Phe Ser Asn Asp Ile His Lys Leu
145     150     155     160
Asp Thr Ser Thr Met Thr Trp Thr Leu Ile Cys Thr Lys Gly Ser Pro
165     170     175
Ala Arg Trp Arg Asp Phe His Ser Ala Thr Met Leu Gly Ser His Met
180     185     190
Tyr Val Phe Gly Gly Arg Ala Asp Arg Phe Gly Pro Phe His Ser Asn
195     200     205
Asn Glu Ile Tyr Cys Asn Arg Ile Arg Val Phe Asp Thr Arg Thr Glu
210     215     220
Ala Trp Leu Asp Cys Pro Pro Thr Pro Val Leu Pro Glu Gly Arg Arg
225     230     235     240
Ser His Ser Ala Phe Gly Tyr Asn Gly Glu Leu Tyr Ile Phe Gly Gly
245     250     255
Tyr Asn Ala Arg Leu Asn Arg His Phe His Asp Leu Trp Lys Phe Asn
260     265     270
Pro Val Ser Phe Thr Trp Lys Lys Ile Glu Pro Lys Gly Lys Gly Pro
275     280     285
Cys Pro Arg Arg Arg Gln Cys Cys Cys Ile Val Gly Asp Lys Ile Val
290     295     300
Leu Phe Gly Gly Thr Ser Pro Ser Pro Glu Glu Gly Leu Gly Asp Glu
305     310     315     320
Phe Asp Leu Ile Asp His Ser Asp Leu His Ile Leu Asp Phe Ser Pro
325     330     335
Ser Leu Lys Thr Leu Cys Lys Leu Ala Val Ile Gln Tyr Asn Leu Asp
340     345     350
Gln Ser Cys Leu Pro His Asp Ile Arg Trp Glu Leu Asn Ala Met Thr
355     360     365
Thr Asn Ser Asn Ile Ser Arg Pro Ile Val Ser Ser His Gly
370     375     380

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<210> SEQ ID NO 7

<211> LENGTH: 638

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 7

Met Val Val Ser Ala Asp Pro Leu Ser Ser Glu Arg Ala Glu Met Asn
1 5 10 15
Ile Leu Glu Ile Asn Gln Glu Leu Arg Ser Gln Leu Ala Glu Ser Asn
20 25 30
Gln Gln Phe Arg Asp Leu Lys Glu Lys Phe Leu Ile Thr Gln Ala Thr
35 40 45
Ala Tyr Ser Leu Ala Asn Gln Leu Lys Lys Tyr Lys Cys Glu Glu Tyr
50 55 60
Lys Asp Ile Ile Asp Ser Val Leu Arg Asp Glu Leu Gln Ser Met Glu
65 70 75 80
Lys Leu Ala Glu Lys Leu Arg Gln Ala Glu Glu Leu Arg Gln Tyr Lys
85 90 95
Ala Leu Val His Ser Gln Ala Lys Glu Leu Thr Gln Leu Arg Glu Lys
100 105 110
Leu Arg Glu Gly Arg Asp Ala Ser Arg Trp Leu Asn Lys His Leu Lys
115 120 125
Thr Leu Leu Thr Pro Asp Asp Pro Asp Lys Ser Gln Gly Gln Asp Leu
130 135 140
Arg Glu Gln Leu Ala Glu Gly His Arg Leu Ala Glu His Leu Val His
145 150 155 160
Lys Leu Ser Pro Glu Asn Asp Glu Asp Glu Asp Glu Asp Glu Asp Asp
165 170 175
Lys Asp Glu Glu Val Glu Lys Val Gln Glu Ser Pro Ala Pro Arg Glu
180 185 190
Val Gln Lys Thr Glu Glu Lys Glu Val Pro Gln Asp Ser Leu Glu Glu
195 200 205
Cys Ala Val Thr Cys Ser Asn Ser His Asn Pro Ser Asn Ser Asn Gln
210 215 220
Pro His Arg Ser Thr Lys Ile Thr Phe Lys Glu His Glu Val Asp Ser
225 230 235 240
Ala Leu Val Val Glu Ser Glu His Pro His Asp Glu Glu Glu Glu Ala
245 250 255
Leu Asn Ile Pro Pro Glu Asn Gln Asn Asp His Glu Glu Glu Glu Gly
260 265 270
Lys Ala Pro Val Pro Pro Arg His His Asp Lys Ser Asn Ser Tyr Arg
275 280 285
His Arg Glu Val Ser Phe Leu Ala Leu Asp Glu Gln Lys Val Cys Ser
290 295 300
Ala Gln Asp Val Ala Arg Asp Tyr Ser Asn Pro Lys Trp Asp Glu Thr
305 310 315 320
Ser Leu Gly Phe Leu Glu Lys Gln Ser Asp Leu Glu Glu Val Lys Gly
325 330 335
Gln Glu Thr Val Ala Pro Arg Leu Ser Arg Gly Pro Leu Arg Val Asp
340 345 350
Lys His Glu Ile Pro Gln Glu Ser Leu Asp Gly Cys Cys Leu Thr Pro

-continued

355	360	365
Ser Ile Leu Pro Asp Leu Thr	Pro Ser Tyr His Pro	Tyr Trp Ser Thr
370	375	380
Leu Tyr Ser Phe Glu Asp Lys	Gln Val Ser Leu Ala	Leu Val Asp Lys
385	390	395 400
Ile Lys Lys Asp Gln Glu Glu	Ile Glu Asp Gln Ser	Pro Pro Cys Pro
405	410	415
Arg Leu Ser Gln Glu Leu Pro	Glu Val Lys Glu Gln	Glu Val Pro Glu
420	425	430
Asp Ser Val Asn Glu Val Tyr	Leu Thr Pro Ser Val	His His Asp Val
435	440	445
Ser Asp Cys His Gln Pro Tyr	Ser Ser Thr Leu Ser	Ser Leu Glu Asp
450	455	460
Gln Leu Ala Cys Ser Ala Leu	Asp Val Ala Ser Pro	Thr Glu Ala Ala
465	470	475 480
Cys Pro Gln Gly Thr Trp Ser	Gly Asp Leu Ser His	His Arg Ser Glu
485	490	495
Val Gln Ile Ser Gln Ala Gln	Leu Glu Pro Ser Thr	Leu Val Pro Ser
500	505	510
Cys Leu Arg Leu Gln Leu Asp	Gln Gly Phe His Cys	Gly Asn Gly Leu
515	520	525
Ala Gln Arg Gly Leu Ser Ser	Thr Thr Cys Ser Phe	Ser Ala Asn Ala
530	535	540
Asp Ser Gly Asn Gln Trp Pro	Phe Gln Glu Leu Val	Leu Glu Pro Ser
545	550	555 560
Leu Gly Met Lys Asn Pro Pro	Gln Leu Glu Asp Asp	Ala Leu Glu Gly
565	570	575
Ser Ala Ser Asn Thr Gln Gly	Arg Gln Val Thr Gly	Arg Ile Arg Ala
580	585	590
Ser Leu Val Leu Ile Leu Lys	Thr Ile Arg Arg Arg	Leu Pro Phe Ser
595	600	605
Lys Trp Arg Leu Ala Phe Arg	Phe Ala Gly Pro His	Ala Glu Ser Ala
610	615	620
Glu Ile Pro Asn Thr Ala Glu	Arg Met Gln Arg Met	Ile Gly
625	630	635

<210> SEQ ID NO 8

<211> LENGTH: 638

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 8

Met Val Val Ser Ala Asp Pro	Leu Ser Ser Glu Arg Ala	Glu Met Asn
1	5	10 15
Ile Leu Glu Ile Asn Gln Glu	Leu Arg Ser Gln Leu Ala	Glu Ser Asn
20	25	30
Gln Gln Phe Arg Asp Leu Lys	Glu Lys Phe Leu Ile Thr	Gln Ala Thr
35	40	45
Ala Tyr Ser Leu Ala Asn Gln	Leu Lys Lys Tyr Lys	Cys Glu Glu Tyr
50	55	60
Lys Asp Ile Ile Asp Ser Val	Leu Arg Asp Glu Leu	Gln Ser Met Glu
65	70	75 80

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Lys	Leu	Ala	Glu	Lys	Leu	Arg	Gln	Ala	Glu	Glu	Leu	Arg	Gln	Tyr	Lys	85	90	95
Ala	Leu	Val	His	Ser	Gln	Ala	Lys	Glu	Leu	Thr	Gln	Leu	Arg	Glu	Lys	100	105	110
Leu	Arg	Glu	Gly	Arg	Asp	Ala	Ser	Arg	Trp	Leu	Asn	Lys	His	Leu	Lys	115	120	125
Thr	Leu	Leu	Thr	Pro	Asp	Asp	Pro	Asp	Lys	Ser	Gln	Gly	Gln	Asp	Leu	130	135	140
Arg	Glu	Gln	Leu	Ala	Glu	Gly	His	Arg	Leu	Ala	Glu	His	Leu	Val	His	145	150	155
Lys	Leu	Ser	Pro	Glu	Asn	Asp	Glu	Asp	Glu	Asp	Glu	Asp	Glu	Asp	Asp	165	170	175
Lys	Asp	Glu	Glu	Val	Glu	Lys	Val	Gln	Glu	Ser	Pro	Ala	Pro	Arg	Glu	180	185	190
Val	Gln	Lys	Thr	Glu	Glu	Lys	Glu	Val	Pro	Gln	Asp	Ser	Leu	Glu	Glu	195	200	205
Cys	Ala	Val	Thr	Cys	Ser	Asn	Ser	His	Asn	Pro	Ser	Asn	Ser	Asn	Gln	210	215	220
Pro	His	Arg	Ser	Thr	Lys	Ile	Thr	Phe	Lys	Glu	His	Glu	Val	Asp	Ser	225	230	235
Ala	Leu	Val	Val	Glu	Ser	Glu	His	Pro	His	Asp	Glu	Glu	Glu	Glu	Ala	245	250	255
Leu	Asn	Ile	Pro	Pro	Glu	Asn	Gln	Asn	Asp	His	Glu	Glu	Glu	Glu	Gly	260	265	270
Lys	Ala	Pro	Val	Pro	Pro	Arg	His	His	Asp	Lys	Ser	Asn	Ser	Tyr	Arg	275	280	285
His	Arg	Glu	Val	Ser	Phe	Leu	Ala	Leu	Asp	Glu	Gln	Lys	Val	Cys	Ser	290	295	300
Ala	Gln	Asp	Val	Ala	Arg	Asp	Tyr	Ser	Asn	Pro	Lys	Trp	Asp	Glu	Thr	305	310	315
Ser	Leu	Gly	Phe	Leu	Glu	Lys	Gln	Ser	Asp	Leu	Glu	Glu	Val	Lys	Gly	325	330	335
Gln	Glu	Thr	Val	Ala	Pro	Arg	Leu	Ser	Arg	Gly	Pro	Leu	Arg	Val	Asp	340	345	350
Lys	His	Glu	Ile	Pro	Gln	Glu	Ser	Leu	Asp	Gly	Cys	Cys	Leu	Thr	Pro	355	360	365
Ser	Ile	Leu	Pro	Asp	Leu	Thr	Pro	Ser	Tyr	His	Pro	Tyr	Trp	Ser	Thr	370	375	380
Leu	Tyr	Ser	Phe	Glu	Asp	Lys	Gln	Val	Ser	Leu	Ala	Leu	Val	Asp	Lys	385	390	395
Ile	Lys	Lys	Asp	Gln	Glu	Glu	Ile	Glu	Asp	Gln	Ser	Pro	Pro	Cys	Pro	405	410	415
Arg	Leu	Ser	Gln	Glu	Leu	Pro	Glu	Val	Lys	Glu	Gln	Glu	Val	Pro	Glu	420	425	430
Asp	Ser	Val	Asn	Glu	Val	Tyr	Leu	Thr	Pro	Ser	Val	His	His	Asp	Val	435	440	445
Ser	Asp	Cys	His	Gln	Pro	Tyr	Ser	Ser	Thr	Leu	Ser	Ser	Leu	Glu	Asp	450	455	460
Gln	Leu	Ala	Cys	Ser	Ala	Leu	Asp	Val	Ala	Ser	Pro	Thr	Glu	Ala	Ala	465	470	475
Cys	Pro	Gln	Gly	Thr	Trp	Ser	Gly	Asp	Leu	Ser	His	His	Arg	Ser	Glu			

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485					490					495					
Val	Gln	Ile	Ser	Gln	Ala	Gln	Leu	Glu	Pro	Ser	Thr	Leu	Val	Pro	Ser
			500					505					510		
Cys	Leu	Arg	Leu	Gln	Leu	Asp	Gln	Gly	Phe	His	Cys	Gly	Asn	Gly	Leu
		515					520					525			
Ala	Gln	Arg	Gly	Leu	Ser	Ser	Thr	Thr	Cys	Ser	Phe	Ser	Ala	Asn	Ala
		530					535				540				
Asp	Ser	Gly	Asn	Gln	Trp	Pro	Phe	Gln	Glu	Leu	Val	Leu	Glu	Pro	Ser
		545				550					555				560
Leu	Gly	Met	Lys	Asn	Pro	Pro	Gln	Leu	Glu	Asp	Asp	Ala	Leu	Glu	Gly
			565						570					575	
Ser	Ala	Ser	Asn	Thr	Gln	Gly	Arg	Gln	Val	Thr	Gly	Arg	Ile	Arg	Ala
			580				585						590		
Ser	Leu	Val	Leu	Ile	Leu	Lys	Thr	Ile	Arg	Arg	Arg	Leu	Pro	Phe	Ser
		595					600					605			
Lys	Trp	Arg	Leu	Ala	Phe	Arg	Phe	Ala	Gly	Pro	His	Ala	Glu	Ser	Ala
		610				615					620				
Glu	Ile	Pro	Asn	Thr	Ala	Glu	Arg	Met	Gln	Arg	Met	Ile	Gly		
		625				630					635				
<210> SEQ ID NO 9															
<211> LENGTH: 300															
<212> TYPE: PRT															
<213> ORGANISM: Homo sapiens															
<400> SEQUENCE: 9															
Met	Glu	Glu	Glu	Ala	Ser	Ser	Pro	Gly	Leu	Gly	Cys	Ser	Lys	Pro	His
1				5					10					15	
Leu	Glu	Lys	Leu	Thr	Leu	Gly	Ile	Thr	Arg	Ile	Leu	Glu	Ser	Ser	Pro
		20						25					30		
Gly	Val	Thr	Glu	Val	Thr	Ile	Ile	Glu	Lys	Pro	Pro	Ala	Glu	Arg	His
		35					40					45			
Met	Ile	Ser	Ser	Trp	Glu	Gln	Lys	Asn	Asn	Cys	Val	Met	Pro	Glu	Asp
		50				55					60				
Val	Lys	Asn	Phe	Tyr	Leu	Met	Thr	Asn	Gly	Phe	His	Met	Thr	Trp	Ser
		65				70					75				80
Val	Lys	Leu	Asp	Glu	His	Ile	Ile	Pro	Leu	Gly	Ser	Met	Ala	Ile	Asn
			85						90					95	
Ser	Ile	Ser	Lys	Leu	Thr	Gln	Leu	Thr	Gln	Ser	Ser	Met	Tyr	Ser	Leu
			100					105					110		
Pro	Asn	Ala	Pro	Thr	Leu	Ala	Asp	Leu	Glu	Asp	Asp	Thr	His	Glu	Ala
		115					120					125			
Ser	Asp	Asp	Gln	Pro	Glu	Lys	Pro	His	Phe	Asp	Ser	Arg	Ser	Val	Ile
		130				135					140				
Phe	Glu	Leu	Asp	Ser	Cys	Asn	Gly	Ser	Gly	Lys	Val	Cys	Leu	Val	Tyr
		145				150					155				160
Lys	Ser	Gly	Lys	Pro	Ala	Leu	Ala	Glu	Asp	Thr	Glu	Ile	Trp	Phe	Leu
			165						170					175	
Asp	Arg	Ala	Leu	Tyr	Trp	His	Phe	Leu	Thr	Asp	Thr	Phe	Thr	Ala	Tyr
			180					185					190		
Tyr	Arg	Leu	Leu	Ile	Thr	His	Leu	Gly	Leu	Pro	Gln	Trp	Gln	Tyr	Ala
		195					200						205		

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Phe	Thr	Ser	Tyr	Gly	Ile	Ser	Pro	Gln	Ala	Lys	Gln	Trp	Phe	Ser	Met
210						215					220				
Tyr	Lys	Pro	Ile	Thr	Tyr	Asn	Thr	Asn	Leu	Leu	Thr	Glu	Glu	Thr	Asp
225					230				235						240
Ser	Phe	Val	Asn	Lys	Leu	Asp	Pro	Ser	Lys	Val	Phe	Lys	Ser	Lys	Asn
			245						250					255	
Lys	Ile	Val	Ile	Pro	Lys	Lys	Lys	Gly	Pro	Val	Gln	Pro	Ala	Gly	Gly
			260					265					270		
Gln	Lys	Gly	Pro	Ser	Gly	Pro	Ser	Gly	Pro	Ser	Thr	Ser	Ser	Thr	Ser
		275				280						285			
Lys	Ser	Ser	Ser	Gly	Ser	Gly	Asn	Pro	Thr	Arg	Lys				
	290					295					300				

<210> SEQ ID NO 10

<211> LENGTH: 657

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 10

Met	Ile	Glu	Ser	Gln	Glu	Pro	Val	Thr	Phe	Glu	Asp	Val	Ala	Val	Asp
1				5					10					15	
Phe	Thr	Gln	Glu	Glu	Trp	Gln	Gln	Leu	Asn	Pro	Ala	Gln	Lys	Thr	Leu
		20						25					30		
His	Arg	Asp	Val	Met	Leu	Glu	Thr	Tyr	Asn	His	Leu	Val	Ser	Val	Gly
		35					40					45			
Cys	Ser	Gly	Ile	Lys	Pro	Asp	Val	Ile	Phe	Lys	Leu	Glu	His	Gly	Lys
	50					55					60				
Asp	Pro	Trp	Ile	Ile	Glu	Ser	Glu	Leu	Ser	Arg	Trp	Ile	Tyr	Pro	Asp
65					70					75				80	
Arg	Val	Lys	Gly	Leu	Glu	Ser	Ser	Gln	Gln	Ile	Ile	Ser	Gly	Glu	Leu
			85					90						95	
Leu	Phe	Gln	Arg	Glu	Ile	Leu	Glu	Arg	Ala	Pro	Lys	Asp	Asn	Ser	Leu
		100						105					110		
Tyr	Ser	Val	Leu	Lys	Ile	Trp	His	Ile	Asp	Asn	Gln	Met	Asp	Arg	Tyr
		115					120					125			
Gln	Gly	Asn	Gln	Asp	Arg	Val	Leu	Arg	Gln	Val	Thr	Val	Ile	Ser	Arg
	130					135					140				
Glu	Thr	Leu	Thr	Asp	Glu	Met	Gly	Ser	Lys	Tyr	Ser	Ala	Phe	Gly	Lys
145					150					155					160
Met	Phe	Asn	Arg	Cys	Thr	Asp	Leu	Ala	Pro	Leu	Ser	Gln	Lys	Phe	His
			165						170					175	
Lys	Phe	Asp	Ser	Cys	Glu	Asn	Ser	Leu	Lys	Ser	Asn	Ser	Asp	Leu	Leu
		180						185					190		
Asn	Tyr	Asn	Arg	Ser	Tyr	Ala	Arg	Lys	Asn	Pro	Thr	Lys	Arg	Phe	Arg
		195					200					205			
Cys	Gly	Arg	Pro	Pro	Lys	Tyr	Asn	Ala	Ser	Cys	Ser	Val	Pro	Glu	Lys
	210					215					220				
Glu	Gly	Phe	Ile	His	Thr	Gly	Met	Glu	Pro	Tyr	Gly	Asp	Ser	Gln	Cys
225					230					235					240
Glu	Lys	Val	Leu	Ser	His	Lys	Gln	Ala	His	Val	Gln	Tyr	Lys	Lys	Phe
			245					250						255	
Gln	Ala	Arg	Glu	Lys	Pro	Asn	Val	Cys	Ser	Met	Cys	Gly	Lys	Ala	Phe
		260						265					270		

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Ile Lys Lys Ser Gln Leu Ile Ile His Gln Arg Ile His Thr Gly Glu
 275 280 285
 Lys Pro Tyr Val Cys Gly Asp Cys Arg Lys Ala Phe Ser Glu Lys Ser
 290 295 300
 His Leu Ile Val His Gln Arg Ile His Thr Gly Glu Lys Pro Tyr Glu
 305 310 315 320
 Cys Thr Lys Tyr Gly Arg Ala Phe Ser Arg Lys Ser Pro Phe Thr Val
 325 330 335
 His Gln Arg Val His Thr Gly Glu Lys Pro Tyr Glu Cys Phe Glu Cys
 340 345 350
 Pro Lys Ala Phe Ser Gln Lys Ser His Leu Ile Ile His Gln Arg Val
 355 360 365
 His Thr Arg Glu Lys Pro Phe Glu Cys Ser Glu Cys Arg Lys Ala Phe
 370 375 380
 Cys Glu Met Ser His Leu Phe Ile His Gln Ile Thr His Thr Gly Lys
 385 390 395 400
 Lys Pro Tyr Glu Cys Thr Glu Cys Gly Lys Thr Phe Pro Arg Lys Thr
 405 410 415
 Gln Leu Ile Ile His Gln Arg Thr His Thr Gly Glu Lys Pro Tyr Lys
 420 425 430
 Cys Gly Glu Cys Gly Lys Thr Phe Cys Gln Gln Ser His Leu Ile Gly
 435 440 445
 His Gln Arg Ile His Thr Gly Glu Lys Pro Tyr Val Cys Thr Asp Cys
 450 455 460
 Gly Lys Ala Phe Ser Gln Lys Ser His Leu Thr Gly His Gln Arg Leu
 465 470 475 480
 His Thr Gly Glu Lys Pro Tyr Met Cys Thr Glu Cys Gly Lys Ser Phe
 485 490 495
 Ser Gln Lys Ser Pro Leu Ile Ile His Gln Arg Ile His Thr Gly Glu
 500 505 510
 Lys Pro Tyr Gln Cys Gly Glu Cys Gly Lys Thr Phe Ser Gln Lys Ser
 515 520 525
 Leu Leu Ile Ile His Leu Arg Val His Thr Gly Glu Lys Pro Tyr Glu
 530 535 540
 Cys Thr Glu Cys Gly Arg Ala Phe Ser Leu Lys Ser His Leu Ile Leu
 545 550 555 560
 His Gln Arg Gly His Thr Gly Glu Lys Pro Tyr Glu Cys Ser Glu Cys
 565 570 575
 Gly Lys Ala Phe Cys Gly Lys Ser Pro Leu Ile Ile His Gln Lys Thr
 580 585 590
 His Pro Arg Glu Lys Thr Pro Glu Cys Ala Glu Ser Gly Met Thr Phe
 595 600 605
 Phe Trp Lys Ser Gln Met Ile Thr Tyr Gln Arg Arg His Thr Gly Glu
 610 615 620
 Lys Pro Ser Arg Cys Ser Asp Cys Gly Lys Ala Phe Cys Gln His Val
 625 630 635 640
 Tyr Phe Thr Gly His Gln Asn Pro Tyr Arg Lys Asp Thr Leu Tyr Ile
 645 650 655

Cys

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<210> SEQ ID NO 11
<211> LENGTH: 904
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 11
Met Leu Glu Thr Tyr Asn Ser Leu Val Ser Leu Gln Glu Leu Val Ser
1          5          10          15
Phe Glu Glu Val Ala Val His Phe Thr Trp Glu Glu Trp Gln Asp Leu
20          25          30
Asp Asp Ala Gln Arg Thr Leu Tyr Arg Asp Val Met Leu Glu Thr Tyr
35          40          45
Ser Ser Leu Val Ser Leu Gly His Cys Ile Thr Lys Pro Glu Met Ile
50          55          60
Phe Lys Leu Glu Gln Gly Ala Glu Pro Trp Ile Val Glu Glu Thr Pro
65          70          75          80
Asn Leu Arg Leu Ser Ala Val Gln Ile Ile Asp Asp Leu Ile Glu Arg
85          90          95
Ser His Glu Ser His Asp Arg Phe Phe Trp Gln Ile Val Ile Thr Asn
100         105         110
Ser Asn Thr Ser Thr Gln Glu Arg Val Glu Leu Gly Lys Thr Phe Asn
115         120         125
Leu Asn Ser Asn His Val Leu Asn Leu Ile Ile Asn Asn Gly Asn Ser
130         135         140
Ser Gly Met Lys Pro Gly Gln Phe Asn Asp Cys Gln Asn Met Leu Phe
145         150         155         160
Pro Ile Lys Pro Gly Glu Thr Gln Ser Gly Glu Lys Pro His Val Cys
165         170         175
Asp Ile Thr Arg Arg Ser His Arg His His Glu His Leu Thr Gln His
180         185         190
His Lys Ile Gln Thr Leu Leu Gln Thr Phe Gln Cys Asn Glu Gln Gly
195         200         205
Lys Thr Phe Asn Thr Glu Ala Met Phe Phe Ile His Lys Arg Val His
210         215         220
Ile Val Gln Thr Phe Gly Lys Tyr Asn Glu Tyr Glu Lys Ala Cys Asn
225         230         235         240
Asn Ser Ala Val Ile Val Gln Val Ile Thr Gln Val Gly Gln Pro Thr
245         250         255
Cys Cys Arg Lys Ser Asp Phe Thr Lys His Gln Gln Thr His Thr Gly
260         265         270
Glu Lys Pro Tyr Glu Cys Val Glu Cys Glu Lys Pro Ser Ile Ser Lys
275         280         285
Ser Asp Leu Met Leu Gln Cys Lys Met Pro Thr Glu Glu Lys Pro Tyr
290         295         300
Ala Cys Asn Trp Cys Glu Lys Leu Phe Ser Tyr Lys Ser Ser Leu Ile
305         310         315         320
Ile His Gln Arg Ile His Thr Gly Glu Lys Pro Tyr Gly Cys Asn Glu
325         330         335
Cys Gly Lys Thr Phe Arg Arg Lys Ser Phe Leu Thr Leu His Glu Arg
340         345         350
Thr His Thr Gly Asp Lys Pro Tyr Lys Cys Ile Glu Cys Gly Lys Thr
355         360         365

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Phe	His	Cys	Lys	Ser	Leu	Leu	Thr	Leu	His	His	Arg	Thr	His	Ser	Gly	370	375	380
Glu	Lys	Pro	Tyr	Gln	Cys	Ser	Glu	Cys	Gly	Lys	Thr	Phe	Ser	Gln	Lys	385	390	395
Ser	Tyr	Leu	Thr	Ile	His	His	Arg	Thr	His	Thr	Gly	Glu	Lys	Pro	Tyr	405	410	415
Ala	Cys	Asp	His	Cys	Glu	Glu	Ala	Phe	Ser	His	Lys	Ser	Arg	Leu	Thr	420	425	430
Val	His	Gln	Arg	Thr	His	Thr	Gly	Glu	Lys	Pro	Tyr	Glu	Cys	Asn	Glu	435	440	445
Cys	Gly	Lys	Pro	Phe	Ile	Asn	Lys	Ser	Asn	Leu	Arg	Leu	His	Gln	Arg	450	455	460
Thr	His	Thr	Gly	Glu	Lys	Pro	Tyr	Glu	Cys	Asn	Glu	Cys	Gly	Lys	Thr	465	470	475
Phe	His	Arg	Lys	Ser	Phe	Leu	Thr	Ile	His	Gln	Trp	Thr	His	Thr	Gly	485	490	495
Glu	Lys	Pro	Tyr	Glu	Cys	Asn	Glu	Cys	Gly	Lys	Thr	Phe	Arg	Cys	Lys	500	505	510
Ser	Phe	Leu	Thr	Val	His	Gln	Arg	Thr	His	Ala	Gly	Glu	Lys	Pro	Tyr	515	520	525
Ala	Cys	Asn	Glu	Cys	Gly	Lys	Thr	Tyr	Ser	His	Lys	Ser	Tyr	Leu	Thr	530	535	540
Val	His	His	Arg	Thr	His	Thr	Gly	Glu	Lys	Pro	Tyr	Glu	Cys	Asn	Glu	545	550	555
Cys	Gly	Lys	Ser	Phe	His	Cys	Lys	Ser	Phe	Leu	Thr	Ile	His	Gln	Arg	565	570	575
Thr	His	Ala	Gly	Lys	Lys	Pro	Tyr	Glu	Cys	Asn	Glu	Cys	Glu	Lys	Thr	580	585	590
Phe	Ile	Asn	Lys	Leu	Asn	Leu	Gly	Ile	His	Lys	Ile	Thr	His	Thr	Gly	595	600	605
Glu	Arg	Pro	Tyr	Glu	Cys	Asn	Glu	Cys	Gly	Lys	Thr	Phe	Arg	Gln	Lys	610	615	620
Ser	Asn	Leu	Ser	Thr	His	Gln	Gly	Thr	His	Thr	Gly	Glu	Lys	Pro	Tyr	625	630	635
Val	Cys	Gly	Lys	Thr	Phe	His	Arg	Lys	Ser	Phe	Leu	Thr	Ile	His	Gln	645	650	655
Arg	Thr	His	Thr	Gly	Lys	Asn	Arg	Met	Asp	Val	Met	Asn	Val	Glu	Lys	660	665	670
Leu	Phe	Val	Arg	Asn	His	Thr	Leu	Leu	Tyr	Ile	Arg	Glu	Leu	Thr	Pro	675	680	685
Gly	Lys	Ser	Pro	Met	Asn	Val	Met	Asn	Val	Glu	Asn	Pro	Phe	Ile	Arg	690	695	700
Arg	Gln	Ile	Phe	Arg	Ser	Ile	Lys	Val	Phe	Thr	Arg	Gly	Arg	Asn	Pro	705	710	715
Met	Asn	Val	Ala	Asn	Val	Glu	Lys	Pro	Cys	Gln	Lys	Ser	Val	Leu	Thr	725	730	735
Val	His	His	Arg	Thr	His	Thr	Gly	Glu	Lys	Pro	Tyr	Glu	Cys	Asn	Glu	740	745	750
Cys	Gly	Lys	Thr	Phe	Cys	His	Lys	Ser	Asn	Leu	Ser	Thr	His	Gln	Gly	755	760	765
Thr	His	Ser	Gly	Glu	Lys	Pro	Tyr	Glu	Cys	Asp	Glu	Cys	Arg	Lys	Thr			

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770					775					780					
Phe	Tyr	Asp	Lys	Thr	Val	Leu	Thr	Ile	His	Gln	Arg	Thr	His	Thr	Gly
785					790					795					800
Glu	Lys	Pro	Phe	Glu	Cys	Lys	Glu	Cys	Arg	Lys	Thr	Phe	Ser	Gln	Lys
				805					810					815	
Ser	Lys	Leu	Phe	Val	His	His	Arg	Thr	His	Thr	Gly	Glu	Lys	Pro	Phe
			820					825					830		
Arg	Cys	Asn	Glu	Cys	Arg	Lys	Thr	Phe	Ser	Gln	Lys	Ser	Gly	Leu	Ser
		835					840					845			
Ile	His	Gln	Arg	Thr	His	Thr	Gly	Glu	Lys	Pro	Tyr	Glu	Cys	Lys	Glu
	850					855					860				
Cys	Gly	Lys	Thr	Phe	Cys	Gln	Lys	Ser	His	Leu	Ser	Arg	His	Gln	Gln
865						870					875				880
Thr	His	Ile	Gly	Glu	Lys	Ser	Asp	Val	Ala	Glu	Ala	Gly	Tyr	Val	Phe
			885					890						895	
Pro	Gln	Asn	His	Ser	Phe	Phe	Pro								
			900												

<210> SEQ ID NO 12

<211> LENGTH: 930

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 12

Met	Leu	Leu	Gly	Trp	Ala	Ser	Leu	Leu	Leu	Cys	Ala	Phe	Arg	Leu	Pro
1					5					10				15	
Leu	Ala	Ala	Val	Gly	Pro	Ala	Ala	Thr	Pro	Ala	Gln	Asp	Lys	Ala	Gly
			20					25					30		
Gln	Pro	Pro	Thr	Ala	Ala	Ala	Ala	Ala	Gln	Pro	Arg	Arg	Arg	Gln	Gly
		35					40					45			
Glu	Glu	Val	Gln	Glu	Arg	Ala	Glu	Pro	Pro	Gly	His	Pro	His	Pro	Leu
	50					55					60				
Ala	Gln	Arg	Arg	Arg	Ser	Lys	Gly	Leu	Val	Gln	Asn	Ile	Asp	Gln	Leu
65					70						75				80
Tyr	Ser	Gly	Gly	Gly	Lys	Val	Gly	Tyr	Leu	Val	Tyr	Ala	Gly	Gly	Arg
			85						90					95	
Arg	Phe	Leu	Leu	Asp	Leu	Glu	Arg	Asp	Gly	Ser	Val	Gly	Ile	Ala	Gly
		100						105					110		
Phe	Val	Pro	Ala	Gly	Gly	Gly	Thr	Ser	Ala	Pro	Trp	Arg	His	Arg	Ser
		115					120					125			
His	Cys	Phe	Tyr	Arg	Gly	Thr	Val	Asp	Gly	Ser	Pro	Arg	Ser	Leu	Ala
	130					135					140				
Val	Phe	Asp	Leu	Cys	Gly	Gly	Leu	Asp	Gly	Phe	Phe	Ala	Val	Lys	His
145				150							155				160
Ala	Arg	Tyr	Thr	Leu	Lys	Pro	Leu	Leu	Arg	Gly	Pro	Trp	Ala	Glu	Glu
			165						170					175	
Glu	Lys	Gly	Arg	Val	Tyr	Gly	Asp	Gly	Ser	Ala	Arg	Ile	Leu	His	Val
		180					185						190		
Tyr	Thr	Arg	Glu	Gly	Phe	Ser	Phe	Glu	Ala	Leu	Pro	Pro	Arg	Ala	Ser
		195					200					205			
Cys	Glu	Thr	Pro	Ala	Ser	Thr	Pro	Glu	Ala	His	Glu	His	Ala	Pro	Ala
	210					215					220				

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His	Ser	Asn	Pro	Ser	Gly	Arg	Ala	Ala	Leu	Ala	Ser	Gln	Leu	Leu	Asp	225	230	235	240
Gln	Ser	Ala	Leu	Ser	Pro	Ala	Gly	Gly	Ser	Gly	Pro	Gln	Thr	Trp	Trp	245	250	255	
Arg	Arg	Arg	Arg	Arg	Ser	Ile	Ser	Arg	Ala	Arg	Gln	Val	Glu	Leu	Leu	260	265	270	
Leu	Val	Ala	Asp	Ala	Ser	Met	Ala	Arg	Leu	Tyr	Gly	Arg	Gly	Leu	Gln	275	280	285	
His	Tyr	Leu	Leu	Thr	Leu	Ala	Ser	Ile	Ala	Asn	Arg	Leu	Tyr	Ser	His	290	295	300	
Ala	Ser	Ile	Glu	Asn	His	Ile	Arg	Leu	Ala	Val	Val	Lys	Val	Val	Val	305	310	315	320
Leu	Gly	Asp	Lys	Asp	Lys	Ser	Leu	Glu	Val	Ser	Lys	Asn	Ala	Ala	Thr	325	330	335	
Thr	Leu	Lys	Asn	Phe	Cys	Lys	Trp	Gln	His	Gln	His	Asn	Gln	Leu	Gly	340	345	350	
Asp	Asp	His	Glu	Glu	His	Tyr	Asp	Ala	Ala	Ile	Leu	Phe	Thr	Arg	Glu	355	360	365	
Asp	Leu	Cys	Gly	His	His	Ser	Cys	Asp	Thr	Leu	Gly	Met	Ala	Asp	Val	370	375	380	
Gly	Thr	Ile	Cys	Ser	Pro	Glu	Arg	Ser	Cys	Ala	Val	Ile	Glu	Asp	Asp	385	390	395	400
Gly	Leu	His	Ala	Ala	Phe	Thr	Val	Ala	His	Glu	Ile	Gly	His	Leu	Leu	405	410	415	
Gly	Leu	Ser	His	Asp	Asp	Ser	Lys	Phe	Cys	Glu	Glu	Thr	Phe	Gly	Ser	420	425	430	
Thr	Glu	Asp	Lys	Arg	Leu	Met	Ser	Ser	Ile	Leu	Thr	Ser	Ile	Asp	Ala	435	440	445	
Ser	Lys	Pro	Trp	Ser	Lys	Cys	Thr	Ser	Ala	Thr	Ile	Thr	Glu	Phe	Leu	450	455	460	
Asp	Asp	Gly	His	Gly	Asn	Cys	Leu	Leu	Asp	Leu	Pro	Arg	Lys	Gln	Ile	465	470	475	480
Leu	Gly	Pro	Glu	Glu	Leu	Pro	Gly	Gln	Thr	Tyr	Asp	Ala	Thr	Gln	Gln	485	490	495	
Cys	Asn	Leu	Thr	Phe	Gly	Pro	Glu	Tyr	Ser	Val	Cys	Pro	Gly	Met	Asp	500	505	510	
Val	Cys	Ala	Arg	Leu	Trp	Cys	Ala	Val	Val	Arg	Gln	Gly	Gln	Met	Val	515	520	525	
Cys	Leu	Thr	Lys	Lys	Leu	Pro	Ala	Val	Glu	Gly	Thr	Pro	Cys	Gly	Lys	530	535	540	
Gly	Arg	Ile	Cys	Leu	Gln	Gly	Lys	Cys	Val	Asp	Lys	Thr	Lys	Lys	Lys	545	550	555	560
Tyr	Tyr	Ser	Thr	Ser	Ser	His	Gly	Asn	Trp	Gly	Ser	Trp	Gly	Ser	Trp	565	570	575	
Gly	Gln	Cys	Ser	Arg	Ser	Cys	Gly	Gly	Gly	Val	Gln	Phe	Ala	Tyr	Arg	580	585	590	
His	Cys	Asn	Asn	Pro	Ala	Pro	Arg	Asn	Asn	Gly	Arg	Tyr	Cys	Thr	Gly	595	600	605	
Lys	Arg	Ala	Ile	Tyr	Arg	Ser	Cys	Ser	Leu	Met	Pro	Cys	Pro	Pro	Asn	610	615	620	
Gly	Lys	Ser	Phe	Arg	His	Glu	Gln	Cys	Glu	Ala	Lys	Asn	Gly	Tyr	Gln				

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625	630	635	640
Ser Asp Ala Lys Gly Val Lys Thr Phe Val Glu Trp Val Pro Lys Tyr			
	645	650	655
Ala Gly Val Leu Pro Ala Asp Val Cys Lys Leu Thr Cys Arg Ala Lys			
	660	665	670
Gly Thr Gly Tyr Tyr Val Val Phe Ser Pro Lys Val Thr Asp Gly Thr			
	675	680	685
Glu Cys Arg Leu Tyr Ser Asn Ser Val Cys Val Arg Gly Lys Cys Val			
	690	695	700
Arg Thr Gly Cys Asp Gly Ile Ile Gly Ser Lys Leu Gln Tyr Asp Lys			
	705	710	715
Cys Gly Val Cys Gly Gly Asp Asn Ser Ser Cys Thr Lys Ile Val Gly			
	725	730	735
Thr Phe Asn Lys Lys Ser Lys Gly Tyr Thr Asp Val Val Arg Ile Pro			
	740	745	750
Glu Gly Ala Thr His Ile Lys Val Arg Gln Phe Lys Ala Lys Asp Gln			
	755	760	765
Thr Arg Phe Thr Ala Tyr Leu Ala Leu Lys Lys Lys Asn Gly Glu Tyr			
	770	775	780
Leu Ile Asn Gly Lys Tyr Met Ile Ser Thr Ser Glu Thr Ile Ile Asp			
	785	790	795
Ile Asn Gly Thr Val Met Asn Tyr Ser Gly Trp Ser His Arg Asp Asp			
	805	810	815
Phe Leu His Gly Met Gly Tyr Ser Ala Thr Lys Glu Ile Leu Ile Val			
	820	825	830
Gln Ile Leu Ala Thr Asp Pro Thr Lys Pro Leu Asp Val Arg Tyr Ser			
	835	840	845
Phe Phe Val Pro Lys Lys Ser Thr Pro Lys Val Asn Ser Val Thr Ser			
	850	855	860
His Gly Ser Asn Lys Val Gly Ser His Thr Ser Gln Pro Gln Trp Val			
	865	870	875
Thr Gly Pro Trp Leu Ala Cys Ser Arg Thr Cys Asp Thr Gly Trp His			
	885	890	895
Thr Arg Thr Val Gln Cys Gln Asp Gly Asn Arg Lys Leu Ala Lys Gly			
	900	905	910
Cys Pro Leu Ser Gln Arg Pro Ser Ala Phe Lys Gln Cys Leu Leu Lys			
	915	920	925
Lys Cys			
930			

<210> SEQ ID NO 13

<211> LENGTH: 2025

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 13

agagcgagaa gatgatgcat gtgagccctg cccttgggaa gcttccaggt tgggaaatga	60
ggaatgagcc tgacaccag ggcccagaga gaccaggagc agaggcaggt caggaggcag	120
acacgcgctg ctgggtaatg acgacagcac cagtaatcac ggctactcct tgttaagtac	180
ttactaagtg caagactcca agctaagcaa ttaatatagacc ttttcttggt taatccttac	240
cacaattcca taggggtgag taggaagtcc ttcttgaagt ctaatctcaa gaattcatgc	300

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catggattgg gccaatgttc cacctttatt ggagtctggg gacttaggtg gaaaaggcag 360
aactggctgg ttggaggagc cacctccctg caaggggcca ggaggagat tacggaggcg 420
ccaagcccag agctccaaca tgcacctgcc acagctccag gggagatcgg gggcctgcc 480
atttaccocg ccccatgata tcatggctgt gtgcgccagg cactggctca gggagcagca 540
tctcacagag aagatacttg gatgggccac aggcctaaac tgggctaaaa ccctggagag 600
ggggacctgg ctcaaggcca caccacattt ccatgtcatt ttccagtggc accaactgag 660
ctggacagat gttctcacag agctacacat gccacagtca tcaactagtt cgagagtctc 720
cagtgttcat taagcctatt ttgctggaag ttagtccct ggaggattag tccctgcct 780
taaggagtgc caaagaagac ttggaatct aagtcacatc tccctgctg caattttttc 840
ctcttgataa tcaaggaatc tcaactgataa aactctcaat gaggagcagc tctcccatg 900
agcaagttct tgccttcatt gccagggagg cccagctaa ggttatactg ggaaaggaat 960
ctctggggag tacctggaag aggtcagctt ctccctcaaag gaagcctctc cagctggtgt 1020
actgcaaagc cttccaagac caagtccct ctctctggcc tctggaacca gctgtgtgc 1080
cagcgtctgt tccccagtaa cacacatgag tctctccctc agaagccaca gaaggatgca 1140
gagagagacc aaccaggat tgattctcag cacttactaa ttgtgtggac ataactgctc 1200
tggacctcag ctttctatc tataaatga gcacctttt tatagatatg aacactttaa 1260
gagagtcata aagattgaag ttgatttgtg ctgcgctgg cacagaggtc tccctcagta 1320
aatggcagcc actgctattg ggtgccccca acaccccttg cccctgcccc ccattagctt 1380
tcaacttaggc ccacactgag ggtgtggctg ttgtgttggg ggaaggaaaa aaaccatgcc 1440
tgggttgtag gggccccacc accaactggt ctgcttttgt gggaagcatt ttctgatgcc 1500
ctcgccctac cgtgggctg gttgactaag tgtccagcat catgaggaaa agagcagggt 1560
tcaggggacag cttggctcag cctgaaact catctgggcc taggtccaga tgagaggcaa 1620
gcctgggagg ccttccttg taccctgcc taccctgcag caccctctag cctgcaggcc 1680
gctctgggt ggcatagcac cttggatgtc aggtgtgggc cttcccaggc acatggcatg 1740
agggggcact tctgtgggtt gttgggggca ggaagggatg tgtctccagc tggacttggg 1800
ctctcgcat ttggggccca gccatgccag gacagcacac atgggcgctt agtgcgatc 1860
tcatgatgga gcagaggagg agcaaagcaa aaccagggag tccccaggcc cctgcttctg 1920
ccccccag agacagtgga gcaggtgtct ccagcttctt aacctcaacg cacagtaaga 1980
aatacathtt acagcaacca atagacacat acatttaca aacga 2025

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<210> SEQ ID NO 14

<211> LENGTH: 534

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 14

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Met Arg Arg His Ser Glu Thr Asp Val Glu Glu Gln Thr Gln Glu Leu
1           5           10           15

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Lys Thr Ile Thr Gln Leu Gln Glu Gln Cys Arg Ala Leu Gln Ile Gln
20           25           30

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Gly Val Lys Glu Asn Met Asp Gln Asn Lys Ala Thr Leu Ala Leu Leu
35           40           45

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Arg	Ser	Asn	Ile	Arg	Arg	Gly	Ala	Gln	Asp	Trp	Ala	Leu	Ala	Lys	Lys
50						55					60				
Tyr	Asp	Gln	Trp	Thr	Ile	Ser	Lys	Ala	Cys	Gly	Lys	Asn	Leu	Pro	Leu
65					70					75					80
Arg	Leu	Ala	His	Cys	Arg	Ser	Thr	Met	Glu	Val	Val	Arg	Glu	Lys	Leu
			85						90					95	
Arg	Lys	Tyr	Val	Phe	Asp	Arg	Val	Asn	Met	His	Asn	Leu	Leu	Ile	His
			100					105					110		
Leu	Val	Arg	Arg	Arg	Gly	Gln	Lys	Leu	Glu	Ser	Met	Gln	Leu	Glu	Leu
		115				120						125			
Asp	Ser	Leu	Arg	Ser	Gln	Pro	Asp	Ala	Ser	Lys	Glu	Glu	Leu	Arg	Leu
	130					135					140				
Leu	Gln	Ile	Ile	Arg	Gln	Leu	Glu	Asn	Asn	Ile	Glu	Lys	Thr	Met	Ile
145					150					155					160
Lys	Ile	Ile	Thr	Ser	Gln	Asn	Ile	His	Leu	Leu	Tyr	Leu	Asp	Leu	Leu
			165						170					175	
Asp	Tyr	Leu	Lys	Thr	Val	Leu	Ala	Gly	Tyr	Pro	Ile	Glu	Leu	Asp	Lys
			180					185					190		
Leu	Gln	Asn	Leu	Val	Val	Asn	Tyr	Cys	Ser	Glu	Leu	Ser	Asp	Met	Lys
		195					200					205			
Ile	Met	Ser	Gln	Asp	Ala	Met	Met	Ile	Thr	Asp	Glu	Val	Lys	Arg	Asn
	210					215					220				
Met	Arg	Gln	Arg	Glu	Ala	Ser	Phe	Ile	Glu	Glu	Arg	Arg	Ala	Arg	Glu
225					230					235					240
Asn	Arg	Leu	Asn	Gln	Gln	Lys	Lys	Leu	Ile	Asp	Lys	Ile	His	Thr	Lys
			245						250					255	
Glu	Thr	Ser	Glu	Lys	Tyr	Arg	Arg	Gly	Gln	Met	Asp	Leu	Asp	Phe	Pro
			260					265					270		
Ser	Asn	Leu	Met	Ser	Thr	Glu	Thr	Leu	Lys	Leu	Arg	Arg	Lys	Glu	Thr
		275					280					285			
Ser	Thr	Ala	Glu	Met	Glu	Tyr	Gln	Ser	Gly	Val	Thr	Ala	Val	Val	Glu
		290				295					300				
Lys	Val	Lys	Ser	Ala	Val	Arg	Cys	Ser	His	Val	Trp	Asp	Ile	Thr	Ser
305					310					315					320
Arg	Phe	Leu	Ala	Gln	Arg	Asn	Thr	Glu	Glu	Asn	Leu	Glu	Leu	Gln	Met
			325						330					335	
Glu	Asp	Cys	Glu	Glu	Trp	Arg	Val	Gln	Leu	Lys	Ala	Leu	Val	Lys	Gln
			340					345					350		
Leu	Glu	Leu	Glu	Glu	Ala	Val	Leu	Lys	Phe	Arg	Gln	Lys	Pro	Ser	Ser
		355					360					365			
Ile	Ser	Phe	Lys	Ser	Val	Glu	Lys	Lys	Met	Thr	Asp	Met	Leu	Lys	Glu
		370				375					380				
Glu	Glu	Glu	Arg	Leu	Gln	Leu	Ala	His	Ser	Asn	Met	Thr	Lys	Gly	Gln
385					390					395					400
Glu	Leu	Leu	Leu	Thr	Ile	Gln	Met	Gly	Ile	Asp	Asn	Leu	Tyr	Val	Arg
			405						410					415	
Leu	Met	Gly	Ile	Asn	Leu	Pro	Ala	Thr	Gln	Arg	Glu	Val	Val	Leu	Ser
			420					425					430		
Asn	Thr	Leu	Asp	Leu	Asn	Ser	Lys	Leu	Ala	Tyr	Cys	Glu	Gly	Lys	Leu
		435					440					445			
Thr	Tyr	Leu	Ala	Asp	Arg	Val	Gln	Met	Val	Ser	Arg	Thr	Glu	Glu	Gly

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450	455	460
Asp Thr Lys Val Arg	Asp Thr Leu Glu Ser	Ser Thr Leu Met Glu Lys
465	470	475 480
Tyr Asn Thr Arg Ile Ser Phe Glu Asn Arg Glu Glu Asp Met Ile Asp		
	485	490 495
Thr Phe Gln Phe Pro Asp Met Asp His Ser Tyr Val Pro Ser Arg Ala		
	500	505 510
Glu Ile Lys Arg Gln Ala Gln Arg Leu Ile Glu Gly Lys Leu Lys Ala		
	515	520 525
Ala Lys Lys Lys Lys Lys		
530		

<210> SEQ ID NO 15
 <211> LENGTH: 399
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 15

Met Glu Arg Lys Asn Pro Ser Arg Glu Ser Pro Arg Arg Leu Ser Ala
1 5 10 15
Lys Val Gly Lys Gly Thr Glu Met Lys Lys Val Ala Arg Gln Leu Gly
20 25 30
Met Ala Ala Ala Glu Ser Asp Lys Asp Ser Gly Phe Ser Asp Gly Ser
35 40 45
Ser Glu Cys Leu Ser Ser Ala Glu Gln Met Glu Ser Glu Asp Met Leu
50 55 60
Ser Ala Leu Gly Trp Ser Arg Glu Asp Arg Pro Arg Gln Asn Ser Lys
65 70 75 80
Thr Ala Lys Asn Ala Phe Pro Thr Leu Ser Pro Met Val Val Met Lys
85 90 95
Asn Val Leu Val Lys Gln Gly Ser Ser Ser Ser Gln Leu Gln Ser Trp
100 105 110
Thr Val Gln Pro Ser Phe Glu Val Ile Ser Ala Gln Pro Gln Leu Leu
115 120 125
Phe Leu His Pro Pro Val Pro Ser Pro Val Ser Pro Cys His Thr Gly
130 135 140
Glu Lys Lys Ser Asp Ser Arg Asn Tyr Leu Pro Ile Leu Asn Ser Tyr
145 150 155 160
Thr Lys Ile Ala Pro His Pro Gly Lys Arg Gly Leu Ser Leu Gly Pro
165 170 175
Glu Glu Lys Gly Thr Ser Gly Val Gln Lys Lys Ile Cys Thr Glu Arg
180 185 190
Leu Gly Pro Ser Leu Ser Ser Ser Glu Pro Thr Lys Ala Gly Ala Val
195 200 205
Pro Ser Ser Pro Ser Thr Pro Ala Pro Pro Ser Ala Lys Leu Ala Glu
210 215 220
Asp Ser Ala Leu Gln Gly Val Pro Ser Leu Val Ala Gly Gly Ser Pro
225 230 235 240
Gln Thr Leu Gln Pro Val Ser Ser Ser His Val Ala Lys Ala Pro Ser
245 250 255
Leu Thr Phe Ala Ser Pro Ala Ser Pro Val Cys Ala Ser Asp Ser Thr
260 265 270

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Leu His Gly Leu Glu Ser Asn Ser Pro Leu Ser Pro Leu Ser Ala Asn
 275 280 285
 Tyr Ser Ser Pro Leu Trp Ala Ala Glu His Leu Cys Arg Ser Pro Asp
 290 295 300
 Ile Phe Ser Glu Gln Arg Gln Ser Lys His Arg Arg Phe Gln Asn Thr
 305 310 315 320
 Leu Val Val Leu His Lys Ser Gly Leu Leu Glu Ile Thr Leu Lys Thr
 325 330 335
 Lys Glu Leu Ile Arg Gln Asn Gln Ala Thr Gln Val Glu Leu Asp Gln
 340 345 350
 Leu Lys Glu Gln Thr Gln Leu Phe Ile Glu Ala Thr Lys Ser Arg Ala
 355 360 365
 Pro Gln Ala Trp Ala Lys Leu Gln Ala Ser Leu Thr Pro Gly Ser Ser
 370 375 380
 Asn Thr Gly Ser Asp Leu Glu Ala Phe Ser Asp His Pro Ala Ile
 385 390 395

 <210> SEQ ID NO 16
 <211> LENGTH: 768
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

 <400> SEQUENCE: 16

 Met Ser Asn Leu Ser Lys Gly Thr Gly Ser Arg Lys Asp Thr Lys Met
 1 5 10 15
 Arg Ile Arg Ala Phe Pro Met Thr Met Asp Glu Lys Tyr Val Asn Ser
 20 25 30
 Ile Trp Asp Leu Leu Lys Asn Ala Ile Gln Glu Ile Gln Arg Lys Asn
 35 40 45
 Asn Ser Gly Leu Ser Phe Glu Glu Leu Tyr Arg Asn Ala Tyr Thr Met
 50 55 60
 Val Leu His Lys His Gly Glu Lys Leu Tyr Thr Gly Leu Arg Glu Val
 65 70 75 80
 Val Thr Glu His Leu Ile Asn Lys Val Arg Glu Asp Val Leu Asn Ser
 85 90 95
 Leu Asn Asn Asn Phe Leu Gln Thr Leu Asn Gln Ala Trp Asn Asp His
 100 105 110
 Gln Thr Ala Met Val Met Ile Arg Asp Ile Leu Met Tyr Met Asp Arg
 115 120 125
 Val Tyr Val Gln Gln Asn Asn Val Glu Asn Val Tyr Asn Leu Gly Leu
 130 135 140
 Ile Ile Phe Arg Asp Gln Val Val Arg Tyr Gly Cys Ile Arg Asp His
 145 150 155 160
 Leu Arg Gln Thr Leu Leu Asp Met Ile Ala Arg Glu Arg Lys Gly Glu
 165 170 175
 Val Val Asp Arg Gly Ala Ile Arg Asn Ala Cys Gln Met Leu Met Ile
 180 185 190
 Leu Gly Leu Glu Gly Arg Ser Val Tyr Glu Glu Asp Phe Glu Ala Pro
 195 200 205
 Phe Leu Glu Met Ser Ala Glu Phe Phe Gln Met Glu Ser Gln Lys Phe
 210 215 220
 Leu Ala Glu Asn Ser Ala Ser Val Tyr Ile Lys Lys Val Glu Ala Arg
 225 230 235 240

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Ile	Asn	Glu	Glu	Ile	Glu	Arg	Val	Met	His	Cys	Leu	Asp	Lys	Ser	Thr	245	250	255
Glu	Glu	Pro	Ile	Val	Lys	Val	Val	Glu	Arg	Glu	Leu	Ile	Ser	Lys	His	260	265	270
Met	Lys	Thr	Ile	Val	Glu	Met	Glu	Asn	Ser	Gly	Leu	Val	His	Met	Leu	275	280	285
Lys	Asn	Gly	Lys	Thr	Glu	Asp	Leu	Gly	Cys	Met	Tyr	Lys	Leu	Phe	Ser	290	295	300
Arg	Val	Pro	Asn	Gly	Leu	Lys	Thr	Met	Cys	Glu	Cys	Met	Ser	Ser	Tyr	305	310	315
Leu	Arg	Glu	Gln	Gly	Lys	Ala	Leu	Val	Ser	Glu	Glu	Gly	Glu	Gly	Lys	325	330	335
Asn	Pro	Val	Asp	Tyr	Ile	Gln	Gly	Leu	Leu	Asp	Leu	Lys	Ser	Arg	Phe	340	345	350
Asp	Arg	Phe	Leu	Leu	Glu	Ser	Phe	Asn	Asn	Asp	Arg	Leu	Phe	Lys	Gln	355	360	365
Thr	Ile	Ala	Gly	Asp	Phe	Glu	Tyr	Phe	Leu	Asn	Leu	Asn	Ser	Arg	Ser	370	375	380
Pro	Glu	Tyr	Leu	Ser	Leu	Phe	Ile	Asp	Asp	Lys	Leu	Lys	Lys	Gly	Val	385	390	395
Lys	Gly	Leu	Thr	Glu	Gln	Glu	Val	Glu	Thr	Ile	Leu	Asp	Lys	Ala	Met	405	410	415
Val	Leu	Phe	Arg	Phe	Met	Gln	Glu	Lys	Asp	Val	Phe	Glu	Arg	Tyr	Tyr	420	425	430
Lys	Gln	His	Leu	Ala	Arg	Arg	Leu	Leu	Thr	Asn	Lys	Ser	Val	Ser	Asp	435	440	445
Asp	Ser	Glu	Lys	Asn	Met	Ile	Ser	Lys	Leu	Lys	Thr	Glu	Cys	Gly	Cys	450	455	460
Gln	Phe	Thr	Ser	Lys	Leu	Glu	Gly	Met	Phe	Arg	Asp	Met	Ser	Ile	Ser	465	470	475
Asn	Thr	Thr	Met	Asp	Glu	Phe	Arg	Gln	His	Leu	Gln	Ala	Thr	Gly	Val	485	490	495
Ser	Leu	Gly	Gly	Val	Asp	Leu	Thr	Val	Arg	Val	Leu	Thr	Thr	Gly	Tyr	500	505	510
Trp	Pro	Thr	Gln	Ser	Ala	Thr	Pro	Lys	Cys	Asn	Ile	Pro	Pro	Ala	Pro	515	520	525
Arg	His	Ala	Phe	Glu	Ile	Phe	Arg	Arg	Phe	Tyr	Leu	Ala	Lys	His	Ser	530	535	540
Gly	Arg	Gln	Leu	Thr	Leu	Gln	His	His	Met	Gly	Ser	Ala	Asp	Leu	Asn	545	550	555
Ala	Thr	Phe	Tyr	Gly	Pro	Val	Lys	Lys	Glu	Asp	Gly	Ser	Glu	Val	Gly	565	570	575
Val	Gly	Gly	Ala	Gln	Val	Thr	Gly	Ser	Asn	Thr	Arg	Lys	His	Ile	Leu	580	585	590
Gln	Val	Ser	Thr	Phe	Gln	Met	Thr	Ile	Leu	Met	Leu	Phe	Asn	Asn	Arg	595	600	605
Glu	Lys	Tyr	Thr	Phe	Glu	Glu	Ile	Gln	Gln	Glu	Thr	Asp	Ile	Pro	Glu	610	615	620
Arg	Glu	Leu	Val	Arg	Ala	Leu	Gln	Ser	Leu	Ala	Cys	Gly	Lys	Pro	Thr	625	630	635

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Gln Arg Val Leu Thr Lys Glu Pro Lys Ser Lys Glu Ile Glu Asn Gly
 645 650 655
 His Ile Phe Thr Val Asn Asp Gln Phe Thr Ser Lys Leu His Arg Val
 660 665 670
 Lys Ile Gln Thr Val Ala Ala Lys Gln Gly Glu Ser Asp Pro Glu Arg
 675 680 685
 Lys Glu Thr Arg Gln Lys Val Asp Asp Asp Arg Lys His Glu Ile Glu
 690 695 700
 Ala Ala Ile Val Arg Ile Met Lys Ser Arg Lys Lys Met Gln His Asn
 705 710 715 720
 Val Leu Val Ala Glu Val Thr Gln Gln Leu Lys Ala Arg Phe Leu Pro
 725 730 735
 Ser Pro Val Val Ile Lys Lys Arg Ile Glu Gly Leu Ile Glu Arg Glu
 740 745 750
 Tyr Leu Ala Arg Thr Pro Glu Asp Arg Lys Val Tyr Thr Tyr Val Ala
 755 760 765

 <210> SEQ ID NO 17
 <211> LENGTH: 327
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

 <400> SEQUENCE: 17

 Met Arg Ala Leu Pro Gly Leu Leu Glu Ala Arg Ala Arg Thr Pro Arg
 1 5 10 15
 Leu Leu Leu Leu Gln Cys Leu Leu Ala Ala Ala Arg Pro Ser Ser Ala
 20 25 30
 Asp Gly Ser Ala Pro Asp Ser Pro Phe Thr Ser Pro Pro Leu Arg Glu
 35 40 45
 Glu Ile Met Ala Asn Asn Phe Ser Leu Glu Ser His Asn Ile Ser Leu
 50 55 60
 Thr Glu His Ser Ser Met Pro Val Glu Lys Asn Ile Thr Leu Glu Arg
 65 70 75 80
 Pro Ser Asn Val Asn Leu Thr Cys Gln Phe Thr Thr Ser Gly Asp Leu
 85 90 95
 Asn Ala Val Asn Val Thr Trp Lys Lys Asp Gly Glu Gln Leu Glu Asn
 100 105 110
 Asn Tyr Leu Val Ser Ala Thr Gly Ser Thr Leu Tyr Thr Gln Tyr Arg
 115 120 125
 Phe Thr Ile Ile Asn Ser Lys Gln Met Gly Ser Tyr Ser Cys Phe Phe
 130 135 140
 Arg Glu Glu Lys Glu Gln Arg Gly Thr Phe Asn Phe Lys Val Pro Glu
 145 150 155 160
 Leu His Gly Lys Asn Lys Pro Leu Ile Ser Tyr Val Gly Asp Ser Thr
 165 170 175
 Val Leu Thr Cys Lys Cys Gln Asn Cys Phe Pro Leu Asn Trp Thr Trp
 180 185 190
 Tyr Ser Ser Asn Gly Ser Val Lys Val Pro Val Gly Val Gln Met Asn
 195 200 205
 Lys Tyr Val Ile Asn Gly Thr Tyr Ala Asn Glu Thr Lys Leu Lys Ile
 210 215 220
 Thr Gln Leu Leu Glu Glu Asp Gly Glu Ser Tyr Trp Cys Arg Ala Leu
 225 230 235 240

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<210> SEQ ID NO 18
<211> LENGTH: 681
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 18
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Met 1	Thr	Ala	Ser 5	Pro	Asp	Tyr	Leu	Val 10	Val	Leu	Phe	Gly	Ile 15	Thr	Ala
Gly	Ala	Thr	Gly 20	Ala	Lys	Leu	Gly	Ser 25	Asp	Glu	Lys	Glu	Leu 30	Ile	Leu
Leu	Phe	Trp 35	Lys	Val	Val	Asp	Leu 40	Ala	Asn	Lys	Lys	Val 45	Gly	Gln	Leu
His 50	Glu	Val	Leu	Val	Arg	Pro 55	Asp	Gln	Leu	Glu	Leu	Thr 60	Glu	Asp	Cys
Lys 65	Glu	Glu	Thr	Lys	Ile 70	Asp	Val	Glu	Ser	Leu 75	Ser	Ser	Ala	Ser	Gln 80
Leu	Asp	Gln	Ala	Leu 85	Arg	Gln	Phe	Asn 90	Gln	Ser	Val	Ser	Asn 95	Glu	Leu
Asn	Ile	Gly 100	Val	Gly	Thr	Ser	Phe	Cys 105	Leu	Cys	Thr	Asp 110	Gly	Gln	Leu
His 115	Val	Arg	Gln	Ile	Leu	His	Pro 120	Glu	Ala	Ser	Lys	Lys 125	Asn	Val	Leu
Leu	Pro 130	Glu	Cys	Phe	Tyr	Ser 135	Phe	Phe	Asp	Leu	Arg 140	Lys	Glu	Phe	Lys
Lys 145	Cys	Cys	Pro	Gly	Ser 150	Pro	Asp	Ile	Asp 155	Lys	Leu	Asp	Val	Ala	Thr 160
Met	Thr	Glu	Tyr 165	Leu	Asn	Phe	Glu	Lys	Ser 170	Ser	Ser	Val	Ser	Arg 175	Tyr
Gly	Ala	Ser	Gln 180	Val	Glu	Asp	Met	Gly 185	Asn	Ile	Ile	Leu 190	Ala	Met	Ile
Ser	Glu 195	Pro	Tyr	Asn	His	Arg	Phe 200	Ser	Asp	Pro	Glu	Arg 205	Val	Asn	Tyr
Lys 210	Phe	Glu	Ser	Gly	Thr	Cys 215	Ser	Lys	Met	Glu	Leu 220	Ile	Asp	Asp	Asn
Thr 225	Val	Val	Arg	Ala	Arg 230	Gly	Leu	Pro	Trp	Gln 235	Ser	Ser	Asp	Gln	Asp 240
Ile	Ala	Arg	Phe 245	Phe	Lys	Gly	Leu	Asn 250	Ile	Ala	Lys	Gly	Gly	Ala	Ala 255
Leu	Cys	Leu	Asn	Ala	Gln	Gly	Arg	Arg	Asn	Gly	Glu	Ala	Leu	Val	Arg

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260							265					270				
Phe	Val	Ser	Glu	Glu	His	Arg	Asp	Leu	Ala	Leu	Gln	Arg	His	Lys	His	
		275					280					285				
His	Met	Gly	Thr	Arg	Tyr	Ile	Glu	Val	Tyr	Lys	Ala	Thr	Gly	Glu	Asp	
	290					295					300					
Phe	Leu	Lys	Ile	Ala	Gly	Gly	Thr	Ser	Asn	Glu	Val	Ala	Gln	Phe	Leu	
305					310					315					320	
Ser	Lys	Glu	Asn	Gln	Val	Ile	Val	Arg	Met	Arg	Gly	Leu	Pro	Phe	Thr	
			325						330					335		
Ala	Thr	Ala	Glu	Glu	Val	Val	Ala	Phe	Phe	Gly	Gln	His	Cys	Pro	Ile	
		340						345					350			
Thr	Gly	Gly	Lys	Glu	Gly	Ile	Leu	Phe	Val	Thr	Tyr	Pro	Asp	Gly	Arg	
	355						360					365				
Pro	Thr	Gly	Asp	Ala	Phe	Val	Leu	Phe	Ala	Cys	Glu	Glu	Tyr	Ala	Gln	
	370					375					380					
Asn	Ala	Leu	Arg	Lys	His	Lys	Asp	Leu	Leu	Gly	Lys	Arg	Tyr	Ile	Glu	
385					390					395					400	
Leu	Phe	Arg	Ser	Thr	Ala	Ala	Glu	Val	Gln	Gln	Val	Leu	Asn	Arg	Phe	
			405						410					415		
Ser	Ser	Ala	Pro	Leu	Ile	Pro	Leu	Pro	Thr	Pro	Pro	Ile	Ile	Pro	Val	
		420						425					430			
Leu	Pro	Gln	Gln	Phe	Val	Pro	Pro	Thr	Asn	Val	Arg	Asp	Cys	Ile	Arg	
	435						440					445				
Leu	Arg	Gly	Leu	Pro	Tyr	Ala	Ala	Thr	Ile	Glu	Asp	Ile	Leu	Asp	Phe	
	450					455					460					
Leu	Gly	Glu	Phe	Ala	Thr	Asp	Ile	Arg	Thr	His	Gly	Val	His	Met	Val	
465					470					475					480	
Leu	Asn	His	Gln	Gly	Arg	Pro	Ser	Gly	Asp	Ala	Phe	Ile	Gln	Met	Lys	
			485						490					495		
Ser	Ala	Asp	Arg	Ala	Phe	Met	Ala	Ala	Gln	Lys	Cys	His	Lys	Lys	Asn	
		500						505					510			
Met	Lys	Asp	Arg	Tyr	Val	Glu	Val	Phe	Gln	Cys	Ser	Ala	Glu	Glu	Met	
	515						520					525				
Asn	Phe	Val	Leu	Met	Gly	Gly	Thr	Leu	Asn	Arg	Asn	Gly	Leu	Ser	Pro	
	530				535						540					
Pro	Pro	Cys	Lys	Leu	Pro	Cys	Leu	Ser	Pro	Pro	Ser	Tyr	Thr	Phe	Pro	
545				550						555					560	
Ala	Pro	Ala	Ala	Val	Ile	Pro	Thr	Glu	Ala	Ala	Ile	Tyr	Gln	Pro	Ser	
			565						570					575		
Val	Ile	Leu	Asn	Pro	Arg	Ala	Leu	Gln	Pro	Ser	Thr	Ala	Tyr	Tyr	Pro	
		580						585					590			
Ala	Gly	Thr	Gln	Leu	Phe	Met	Asn	Tyr	Thr	Ala	Tyr	Tyr	Pro	Ser	Pro	
	595						600					605				
Pro	Gly	Ser	Pro	Asn	Ser	Leu	Gly	Tyr	Phe	Pro	Thr	Ala	Ala	Asn	Leu	
	610					615					620					
Ser	Gly	Val	Pro	Pro	Gln	Pro	Gly	Thr	Val	Val	Arg	Met	Gln	Gly	Leu	
625					630					635					640	
Ala	Tyr	Asn	Thr	Gly	Val	Lys	Glu	Ile	Leu	Asn	Phe	Phe	Gln	Gly	Tyr	
			645						650					655		
Gln	Tyr	Ala	Thr	Glu	Asp	Gly	Leu	Ile	His	Thr	Asn	Asp	Gln	Ala	Arg	
		660						665					670			

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Thr Leu Pro Lys Glu Trp Val Cys Ile
675 680

<210> SEQ ID NO 19

<211> LENGTH: 735

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 19

Met Thr Pro Glu Leu Met Ile Lys Ala Cys Ser Phe Tyr Thr Gly His
1 5 10 15

Leu Val Lys Thr His Phe Cys Thr Trp Arg Asp Ile Ala Arg Thr Asn
20 25 30

Glu Asn Val Val Leu Ala Glu Lys Met Asn Arg Ala Val Thr Cys Tyr
35 40 45

Asn Phe Arg Leu Gln Lys Ser Val Phe His His Trp His Ser Tyr Met
50 55 60

Glu Asp Gln Lys Glu Lys Leu Lys Asn Ile Leu Leu Arg Ile Gln Gln
65 70 75 80

Ile Ile Tyr Cys His Lys Leu Thr Ile Ile Leu Thr Lys Trp Arg Asn
85 90 95

Thr Ala Arg His Lys Ser Lys Lys Lys Glu Asp Glu Leu Ile Leu Lys
100 105 110

His Glu Leu Gln Leu Lys Lys Trp Lys Asn Arg Leu Ile Leu Lys Arg
115 120 125

Ala Ala Ala Glu Glu Ser Asn Phe Pro Glu Arg Ser Ser Ser Glu Val
130 135 140

Phe Leu Val Asp Glu Thr Leu Lys Cys Asp Ile Ser Leu Leu Pro Glu
145 150 155 160

Arg Ala Ile Leu Gln Ile Phe Phe Tyr Leu Ser Leu Lys Asp Val Ile
165 170 175

Ile Cys Gly Gln Val Asn His Ala Trp Met Leu Met Thr Gln Leu Asn
180 185 190

Ser Leu Trp Asn Ala Ile Asp Phe Ser Ser Val Lys Asn Val Ile Pro
195 200 205

Asp Lys Tyr Ile Val Ser Thr Leu Gln Arg Trp Arg Leu Asn Val Leu
210 215 220

Arg Leu Asn Phe Arg Gly Cys Leu Leu Arg Pro Lys Thr Phe Arg Ser
225 230 235 240

Val Ser His Cys Arg Asn Leu Gln Glu Leu Asn Val Ser Asp Cys Pro
245 250 255

Thr Phe Thr Asp Glu Ser Met Arg His Ile Ser Glu Gly Cys Pro Gly
260 265 270

Val Leu Cys Leu Asn Leu Ser Asn Thr Thr Ile Thr Asn Arg Thr Met
275 280 285

Arg Leu Leu Pro Arg His Phe His Asn Leu Gln Asn Leu Ser Leu Ala
290 295 300

Tyr Cys Arg Arg Phe Thr Asp Lys Gly Leu Gln Tyr Leu Asn Leu Gly
305 310 315 320

Asn Gly Cys His Lys Leu Ile Tyr Leu Asp Leu Ser Gly Cys Thr Gln
325 330 335

Ile Ser Val Gln Gly Phe Arg Tyr Ile Ala Asn Ser Cys Thr Gly Ile

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340					345					350					
Met	His	Leu	Thr	Ile	Asn	Asp	Met	Pro	Thr	Leu	Thr	Asp	Asn	Cys	Val
		355					360					365			
Lys	Ala	Leu	Val	Glu	Lys	Cys	Ser	Arg	Ile	Thr	Ser	Leu	Val	Phe	Thr
		370					375					380			
Gly	Ala	Pro	His	Ile	Ser	Asp	Cys	Thr	Phe	Arg	Ala	Leu	Ser	Ala	Cys
		385					390					395			400
Lys	Leu	Arg	Lys	Ile	Arg	Phe	Glu	Gly	Asn	Lys	Arg	Val	Thr	Asp	Ala
			405						410					415	
Ser	Phe	Lys	Phe	Ile	Asp	Lys	Asn	Tyr	Pro	Asn	Leu	Ser	His	Ile	Tyr
			420					425					430		
Met	Ala	Asp	Cys	Lys	Gly	Ile	Thr	Asp	Ser	Ser	Leu	Arg	Ser	Leu	Ser
		435						440				445			
Pro	Leu	Lys	Gln	Leu	Thr	Val	Leu	Asn	Leu	Ala	Asn	Cys	Val	Arg	Ile
		450					455					460			
Gly	Asp	Met	Gly	Leu	Lys	Gln	Phe	Leu	Asp	Gly	Pro	Ala	Ser	Met	Arg
		465					470					475			480
Ile	Arg	Glu	Leu	Asn	Leu	Ser	Asn	Cys	Val	Arg	Leu	Ser	Asp	Ala	Ser
			485						490					495	
Val	Met	Lys	Leu	Ser	Glu	Arg	Cys	Pro	Asn	Leu	Asn	Tyr	Leu	Ser	Leu
			500					505					510		
Arg	Asn	Cys	Glu	His	Leu	Thr	Ala	Gln	Gly	Ile	Gly	Tyr	Ile	Val	Asn
		515					520					525			
Ile	Phe	Ser	Leu	Val	Ser	Ile	Asp	Leu	Ser	Gly	Thr	Asp	Ile	Ser	Asn
		530					535					540			
Glu	Gly	Leu	Asn	Val	Leu	Ser	Arg	His	Lys	Lys	Leu	Lys	Glu	Leu	Ser
		545					550					555			560
Val	Ser	Glu	Cys	Tyr	Arg	Ile	Thr	Asp	Asp	Gly	Ile	Gln	Ala	Phe	Cys
			565						570					575	
Lys	Ser	Ser	Leu	Ile	Leu	Glu	His	Leu	Asp	Val	Ser	Tyr	Cys	Ser	Gln
			580					585					590		
Leu	Ser	Asp	Met	Ile	Ile	Lys	Ala	Leu	Ala	Ile	Tyr	Cys	Ile	Asn	Leu
		595					600					605			
Thr	Ser	Leu	Ser	Ile	Ala	Gly	Cys	Pro	Lys	Ile	Thr	Asp	Ser	Ala	Met
		610					615					620			
Glu	Met	Leu	Ser	Ala	Lys	Cys	His	Tyr	Leu	His	Ile	Leu	Asp	Ile	Ser
		625					630					635			640
Gly	Cys	Val	Leu	Leu	Thr	Asp	Gln	Ile	Leu	Glu	Asp	Leu	Gln	Ile	Gly
			645						650					655	
Cys	Lys	Gln	Leu	Arg	Ile	Leu	Lys	Met	Gln	Tyr	Cys	Thr	Asn	Ile	Ser
		660						665					670		
Lys	Lys	Ala	Ala	Gln	Arg	Met	Ser	Ser	Lys	Val	Gln	Gln	Gln	Glu	Tyr
		675					680						685		
Asn	Thr	Asn	Asp	Pro	Pro	Arg	Trp	Phe	Gly	Tyr	Asp	Arg	Glu	Gly	Asn
		690					695					700			
Pro	Val	Thr	Glu	Leu	Asp	Asn	Ile	Thr	Ser	Ser	Lys	Gly	Ala	Leu	Glu
		705					710					715			720
Leu	Thr	Val	Lys	Lys	Ser	Thr	Tyr	Ser	Ser	Glu	Asp	Gln	Ala	Ala	
			725						730					735	

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<211> LENGTH: 875

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 20

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Met Ala Ser Leu Asp Leu Pro Tyr Arg Cys Pro Arg Cys Gly Glu His
1          5          10          15

Lys Arg Phe Arg Ser Leu Ser Ser Leu Arg Ala His Leu Glu Tyr Ser
          20          25          30

His Thr Tyr Glu Thr Leu Tyr Ile Leu Ser Lys Thr Asn Ser Ile Cys
          35          40          45

Asp Gly Ala Ala Ala Ala Ala Ala Ala Ala Ala Ala Ala Ser Gly Phe
50          55          60

Pro Leu Ala Pro Glu Pro Ala Ala Leu Leu Ala Val Pro Gly Ala Arg
65          70          75          80

Arg Glu Val Phe Glu Ser Thr Ser Phe Gln Gly Lys Glu Gln Ala Ala
          85          90          95

Gly Pro Ser Pro Ala Ala Pro His Leu Leu His His His His His His
100          105          110

Ala Pro Leu Ala His Phe Pro Gly Asp Leu Val Pro Ala Ser Leu Pro
115          120          125

Cys Glu Glu Leu Ala Glu Pro Gly Leu Val Pro Ala Ala Ala Ala Arg
130          135          140

Tyr Ala Leu Arg Glu Ile Glu Ile Pro Leu Gly Glu Leu Phe Ala Arg
145          150          155          160

Lys Ser Val Ala Ser Ser Ala Cys Ser Thr Pro Pro Pro Gly Pro Gly
165          170          175

Pro Gly Pro Cys Pro Gly Pro Ala Ser Ala Ser Pro Ala Ser Pro Ser
180          185          190

Pro Ala Asp Val Ala Tyr Glu Glu Gly Leu Ala Arg Leu Lys Ile Arg
195          200          205

Ala Leu Glu Lys Leu Glu Val Asp Arg Arg Leu Glu Arg Leu Ser Glu
210          215          220

Glu Val Glu Gln Lys Ile Ala Gly Gln Val Gly Arg Leu Gln Ala Glu
225          230          235          240

Leu Glu Arg Lys Ala Ala Glu Leu Glu Thr Ala Arg Gln Glu Ser Ala
245          250          255

Arg Leu Gly Arg Glu Lys Glu Glu Leu Glu Glu Arg Ala Ser Glu Leu
260          265          270

Ser Arg Gln Val Asp Val Ser Val Glu Leu Leu Ala Ser Leu Lys Gln
275          280          285

Asp Leu Val His Lys Glu Gln Glu Leu Ser Arg Lys Gln Gln Glu Val
290          295          300

Val Gln Ile Asp Gln Phe Leu Lys Glu Thr Ala Ala Arg Glu Ala Ser
305          310          315          320

Ala Lys Leu Arg Leu Gln Gln Phe Ile Glu Glu Leu Leu Glu Arg Ala
325          330          335

Asp Arg Ala Glu Arg Gln Leu Gln Val Ile Ser Ser Ser Cys Gly Ser
340          345          350

Thr Pro Ser Ala Ser Leu Gly Arg Gly Gly Gly Gly Gly Gly Ala Gly
355          360          365

Pro Asn Ala Arg Gly Pro Gly Arg Met Arg Glu His His Val Gly Pro

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370					375					380					
Ala 385	Val	Pro	Asn	Thr	Tyr	Ala 390	Val	Ser	Arg	His	Gly	Ser	Ser	Pro	Ser 400
Thr	Gly	Ala	Ser	Ser 405	Arg	Val	Pro	Ala	Ala 410	Ser	Gln	Ser	Ser	Gly	Cys 415
Tyr	Asp	Ser	Asp	Ser	Leu	Glu	Leu	Pro 425	Arg	Pro	Glu	Glu	Gly	Ala	Pro
Glu	Asp	Ser	Gly	Pro	Gly	Gly	Leu 440	Gly	Thr	Arg	Ala	Gln	Ala	Ala	Asn
Gly	Gly	Ser	Glu	Arg	Ser	Gln	Pro 455	Pro	Arg	Ser	Ser	Gly	Leu	Arg	Arg
Gln 465	Ala	Ile	Gln	Asn	Trp	Gln	Arg	Arg	Pro	Arg 475	Arg	His	Ser	Thr	Glu 480
Gly	Glu	Glu	Gly	Asp	Val	Ser	Asp	Val	Gly	Ser	Arg	Thr	Thr	Glu	Ser 495
Glu	Ala	Glu	Gly	Pro	Leu	Asp	Ala	Pro 505	Arg	Pro	Gly	Pro	Ala	Met	Ala
Gly	Pro	Leu	Ser	Ser	Cys	Arg	Leu 520	Ser	Ala	Arg	Pro	Glu	Gly	Gly	Ser
Gly	Arg	Gly	Arg	Arg	Ala	Glu	Arg 535	Val	Ser	Pro	Ser	Arg	Ser	Asn	Glu
Val 545	Ile	Ser	Pro	Glu	Ile	Leu	Lys	Met	Arg	Ala 555	Ala	Leu	Phe	Cys	Ile 560
Phe	Thr	Tyr	Leu	Asp	Thr	Arg	Thr	Leu	Leu 570	His	Ala	Ala	Glu	Val	Cys 575
Arg	Asp	Trp	Arg	Phe	Val	Ala	Arg	His	Pro	Ala 585	Val	Trp	Thr	Arg	Val
Leu	Leu	Glu	Asn	Ala	Arg	Val	Cys 600	Ser	Lys	Phe	Leu	Ala	Met	Leu	Ala
Gln	Trp	Cys	Thr	Gln	Ala	His	Ser	Leu	Thr	Leu	Gln	Asn	Leu	Lys	Pro
Arg 625	Gln	Arg	Gly	Lys	Lys	Glu	Ser	Lys	Glu	Glu	Tyr	Ala	Arg	Ser	Thr 640
Arg	Gly	Cys	Leu	Glu	Ala	Gly	Leu	Glu	Ser	Leu	Leu	Lys	Ala	Ala	Gly 655
Gly	Asn	Leu	Leu	Ile	Leu	Arg	Ile	Ser	His	Cys	Pro	Asn	Ile	Leu	Thr
Asp	Arg	Ser	Leu	Trp	Leu	Ala	Ser	Cys	Tyr	Cys	Arg	Ala	Leu	Gln	Ala
Val	Thr	Tyr	Arg	Ser	Ala	Thr	Asp	Pro	Val	Gly	His	Glu	Val	Ile	Trp
Ala 705	Leu	Gly	Ala	Gly	Cys	Arg	Glu	Ile	Val	Ser	Leu	Gln	Val	Ala	Pro 720
Leu	His	Pro	Cys	Gln	Gln	Pro	Thr	Arg	Phe	Ser	Asn	Arg	Cys	Leu	Gln
Met	Ile	Gly	Arg	Cys	Trp	Pro	His	Leu	Arg	Ala	Leu	Gly	Val	Gly	Gly
Ala	Gly	Cys	Gly	Val	Gln	Gly	Leu	Ala	Ser	Leu	Ala	Arg	Asn	Cys	Met
Arg	Leu	Gln	Val	Leu	Glu	Leu	Asp	His	Val	Ser	Glu	Ile	Thr	Gln	Glu

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Val Ala Ala Glu Val Cys Arg Glu Gly Leu Lys Gly Leu Glu Met Leu
 785 790 795 800

Val Leu Thr Ala Thr Pro Val Thr Pro Lys Ala Leu Leu His Phe Asn
 805 810 815

Ser Ile Cys Arg Asn Leu Lys Ser Ile Val Val Gln Ile Gly Ile Ala
 820 825 830

Asp Tyr Phe Lys Glu Pro Ser Ser Pro Glu Ala Gln Lys Leu Phe Glu
 835 840 845

Asp Met Val Thr Lys Leu Gln Ala Leu Arg Arg Arg Pro Gly Phe Ser
 850 855 860

Lys Ile Leu His Ile Lys Val Glu Gly Gly Cys
 865 870 875

<210> SEQ ID NO 21

<211> LENGTH: 271

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 21

Met Phe Arg Asp Phe Gly Glu Pro Gly Pro Ser Ser Gly Asn Gly Gly
 1 5 10 15

Gly Tyr Gly Gly Pro Ala Gln Pro Pro Ala Ala Ala Gln Ala Ala Gln
 20 25 30

Gln Lys Phe His Leu Val Pro Ser Ile Asn Thr Met Ser Gly Ser Gln
 35 40 45

Glu Leu Gln Trp Met Val Gln Pro His Phe Leu Gly Pro Ser Ser Tyr
 50 55 60

Pro Arg Pro Leu Thr Tyr Pro Gln Tyr Ser Pro Pro Gln Pro Arg Pro
 65 70 75 80

Gly Val Ile Arg Ala Leu Gly Pro Pro Pro Gly Val Arg Arg Arg Pro
 85 90 95

Cys Glu Gln Ile Ser Pro Glu Glu Glu Glu Arg Arg Arg Val Arg Arg
 100 105 110

Glu Arg Asn Lys Leu Ala Ala Ala Lys Cys Arg Asn Arg Arg Lys Glu
 115 120 125

Leu Thr Asp Phe Leu Gln Ala Glu Thr Asp Lys Leu Glu Asp Glu Lys
 130 135 140

Ser Gly Leu Gln Arg Glu Ile Glu Glu Leu Gln Lys Gln Lys Glu Arg
 145 150 155 160

Leu Glu Leu Val Leu Glu Ala His Arg Pro Ile Cys Lys Ile Pro Glu
 165 170 175

Gly Ala Lys Glu Gly Asp Thr Gly Ser Thr Ser Gly Thr Ser Ser Pro
 180 185 190

Pro Ala Pro Cys Arg Pro Val Pro Cys Ile Ser Leu Ser Pro Gly Pro
 195 200 205

Val Leu Glu Pro Glu Ala Leu His Thr Pro Thr Leu Met Thr Thr Pro
 210 215 220

Ser Leu Thr Pro Phe Thr Pro Ser Leu Val Phe Thr Tyr Pro Ser Thr
 225 230 235 240

Pro Glu Pro Cys Ala Ser Ala His Arg Lys Ser Ser Ser Ser Ser Gly
 245 250 255

Asp Pro Ser Ser Asp Pro Leu Gly Ser Pro Thr Leu Leu Ala Leu

<210> SEQ ID NO 22																
<211> LENGTH: 505																
<212> TYPE: PRT																
<213> ORGANISM: Homo sapiens																
<400> SEQUENCE: 22																
Met	Asp	Pro	Gly	Asn	Glu	Asn	Ser	Ala	Thr	Glu	Ala	Ala	Ala	Ile	Ile	
1				5					10					15		
Asp	Leu	Asp	Pro	Asp	Phe	Glu	Pro	Gln	Ser	Arg	Pro	Arg	Ser	Cys	Thr	
			20					25					30			
Trp	Pro	Leu	Pro	Arg	Pro	Glu	Ile	Ala	Asn	Gln	Pro	Ser	Glu	Pro	Pro	
		35					40					45				
Glu	Val	Glu	Pro	Asp	Leu	Gly	Glu	Lys	Val	His	Thr	Glu	Gly	Arg	Ser	
		50				55					60					
Glu	Pro	Ile	Leu	Leu	Pro	Ser	Arg	Leu	Pro	Glu	Pro	Ala	Gly	Gly	Pro	
65					70					75					80	
Gln	Pro	Gly	Ile	Leu	Gly	Ala	Val	Thr	Gly	Pro	Arg	Lys	Gly	Gly	Ser	
			85						90					95		
Arg	Arg	Asn	Ala	Trp	Gly	Asn	Gln	Ser	Tyr	Ala	Glu	Leu	Ile	Ser	Gln	
			100					105					110			
Ala	Ile	Glu	Ser	Ala	Pro	Glu	Lys	Arg	Leu	Thr	Leu	Ala	Gln	Ile	Tyr	
			115				120					125				
Glu	Trp	Met	Val	Arg	Thr	Val	Pro	Tyr	Phe	Lys	Asp	Lys	Gly	Asp	Ser	
			130			135					140					
Asn	Ser	Ser	Ala	Gly	Trp	Lys	Asn	Ser	Ile	Arg	His	Asn	Leu	Ser	Leu	
145					150					155					160	
His	Ser	Lys	Phe	Ile	Lys	Val	His	Asn	Glu	Ala	Thr	Gly	Lys	Ser	Ser	
			165						170					175		
Trp	Trp	Met	Leu	Asn	Pro	Glu	Gly	Gly	Lys	Ser	Gly	Lys	Ala	Pro	Arg	
			180					185					190			
Arg	Arg	Ala	Ala	Ser	Met	Asp	Ser	Ser	Ser	Lys	Leu	Leu	Arg	Gly	Arg	
			195				200					205				
Ser	Lys	Ala	Pro	Lys	Lys	Lys	Pro	Ser	Val	Leu	Pro	Ala	Pro	Pro	Glu	
						215					220					
Gly	Ala	Thr	Pro	Thr	Ser	Pro	Val	Gly	His	Phe	Ala	Lys	Trp	Ser	Gly	
225					230					235					240	
Ser	Pro	Cys	Ser	Arg	Asn	Arg	Glu	Glu	Ala	Asp	Met	Trp	Thr	Thr	Phe	
				245					250					255		
Arg	Pro	Arg	Ser	Ser	Ser	Asn	Ala	Ser	Ser	Val	Ser	Thr	Arg	Leu	Ser	
			260					265					270			
Pro	Leu	Arg	Pro	Glu	Ser	Glu	Val	Leu	Ala	Glu	Glu	Ile	Pro	Ala	Ser	
		275					280					285				
Val	Ser	Ser	Tyr	Ala	Gly	Gly	Val	Pro	Pro	Thr	Leu	Asn	Glu	Gly	Leu	
			290			295						300				
Glu	Leu	Leu	Asp	Gly	Leu	Asn	Leu	Thr	Ser	Ser	His	Ser	Leu	Leu	Ser	
305					310					315					320	
Arg	Ser	Gly	Leu	Ser	Gly	Phe	Ser	Leu	Gln	His	Pro	Gly	Val	Thr	Gly	
				3												

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Leu Ser Ala Gly Glu Gly Cys Phe Ser Ser Ser Gln Ala Leu Glu Ala
 355 360 365
 Leu Leu Thr Ser Asp Thr Pro Pro Pro Pro Ala Asp Val Leu Met Thr
 370 375 380
 Gln Val Asp Pro Ile Leu Ser Gln Ala Pro Thr Leu Leu Leu Leu Gly
 385 390 395 400
 Gly Leu Pro Ser Ser Ser Lys Leu Ala Thr Gly Val Gly Leu Cys Pro
 405 410 415
 Lys Pro Leu Glu Ala Pro Gly Pro Ser Ser Leu Val Pro Thr Leu Ser
 420 425 430
 Met Ile Ala Pro Pro Pro Val Met Ala Ser Ala Pro Ile Pro Lys Ala
 435 440 445
 Leu Gly Thr Pro Val Leu Thr Pro Pro Thr Glu Ala Ala Ser Gln Asp
 450 455 460
 Arg Met Pro Gln Asp Leu Asp Leu Asp Met Tyr Met Glu Asn Leu Glu
 465 470 475 480
 Cys Asp Met Asp Asn Ile Ile Ser Asp Leu Met Asp Glu Gly Glu Gly
 485 490 495
 Leu Asp Phe Asn Phe Glu Pro Asp Pro
 500 505

<210> SEQ ID NO 23

<211> LENGTH: 418

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 23

Met Gly Glu Glu Gly Thr Gly Gly Thr Val His Leu Leu Cys Leu Ala
 1 5 10 15
 Ala Ser Ser Gly Val Pro Leu Phe Cys Arg Ser Ser Arg Gly Gly Ala
 20 25 30
 Pro Ala Arg Gln Gln Leu Pro Phe Ser Val Ile Gly Ser Leu Asn Gly
 35 40 45
 Val His Met Phe Gly Gln Asn Leu Glu Val Gln Leu Ser Ser Ala Arg
 50 55 60
 Thr Glu Asn Thr Thr Val Val Trp Lys Ser Phe His Asp Ser Ile Thr
 65 70 75 80
 Leu Ile Val Leu Ser Ser Glu Val Gly Ile Ser Glu Leu Arg Leu Glu
 85 90 95
 Arg Leu Leu Gln Met Val Phe Gly Ala Met Val Leu Leu Val Gly Leu
 100 105 110
 Glu Glu Leu Thr Asn Ile Arg Asn Val Glu Arg Leu Lys Lys Asp Leu
 115 120 125
 Arg Ala Ser Tyr Cys Leu Ile Asp Ser Phe Leu Gly Asp Ser Glu Leu
 130 135 140
 Ile Gly Asp Leu Thr Gln Cys Val Asp Cys Val Ile Pro Pro Glu Gly
 145 150 155 160
 Ser Leu Leu Gln Glu Ala Leu Ser Gly Phe Ala Glu Ala Ala Gly Thr
 165 170 175
 Thr Phe Val Ser Leu Val Val Ser Gly Arg Val Val Ala Ala Thr Glu
 180 185 190
 Gly Trp Trp Arg Leu Gly Thr Pro Glu Ala Val Leu Leu Pro Trp Leu
 195 200 205

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Val	Gly	Ser	Leu	Pro	Pro	Gln	Thr	Ala	Arg	Asp	Tyr	Pro	Val	Tyr	Leu
210						215					220				
Pro	His	Gly	Ser	Pro	Thr	Val	Pro	His	Arg	Leu	Leu	Thr	Leu	Thr	Leu
225					230					235					240
Leu	Pro	Ser	Leu	Glu	Leu	Cys	Leu	Leu	Cys	Gly	Pro	Ser	Pro	Pro	Leu
			245						250					255	
Ser	Gln	Leu	Tyr	Pro	Gln	Leu	Leu	Glu	Arg	Trp	Trp	Gln	Pro	Leu	Leu
		260						265					270		
Asp	Pro	Leu	Arg	Ala	Cys	Leu	Pro	Leu	Gly	Pro	Arg	Ala	Leu	Pro	Ser
		275						280				285			
Gly	Phe	Pro	Leu	His	Thr	Asp	Ile	Leu	Gly	Leu	Leu	Leu	Leu	His	Leu
	290					295					300				
Glu	Leu	Lys	Arg	Cys	Leu	Phe	Thr	Val	Glu	Pro	Leu	Gly	Asp	Lys	Glu
305					310					315					320
Pro	Ser	Pro	Glu	Gln	Arg	Arg	Arg	Leu	Leu	Arg	Asn	Phe	Tyr	Thr	Leu
			325						330					335	
Val	Thr	Ser	Thr	His	Phe	Pro	Pro	Glu	Pro	Gly	Pro	Pro	Glu	Lys	Thr
			340					345					350		
Glu	Asp	Glu	Val	Tyr	Gln	Ala	Gln	Leu	Pro	Arg	Ala	Cys	Tyr	Leu	Val
		355					360					365			
Leu	Gly	Thr	Glu	Glu	Pro	Gly	Thr	Gly	Val	Arg	Leu	Val	Ala	Leu	Gln
	370					375					380				
Leu	Gly	Leu	Arg	Arg	Leu	Leu	Leu	Leu	Leu	Ser	Pro	Gln	Ser	Pro	Thr
385					390					395					400
His	Gly	Leu	Arg	Ser	Leu	Ala	Thr	His	Thr	Leu	His	Ala	Leu	Thr	Pro
			405						410					415	

Leu Leu

<210> SEQ ID NO 24
 <211> LENGTH: 464
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 24

Met	Val	Arg	Leu	Ala	Ala	Glu	Leu	Leu	Leu	Leu	Gly	Leu	Leu	Leu	
1			5					10				15			
Leu	Thr	Leu	His	Ile	Thr	Val	Leu	Arg	Gly	Ser	Gly	Ala	Ala	Asp	Gly
	20						25					30			
Pro	Asp	Ala	Ala	Ala	Gly	Asn	Ala	Ser	Gln	Ala	Gln	Leu	Gln	Asn	Asn
	35					40					45				
Leu	Asn	Val	Gly	Ser	Asp	Thr	Thr	Ser	Glu	Thr	Ser	Phe	Ser	Leu	Ser
	50				55					60					
Lys	Glu	Ala	Pro	Arg	Glu	His	Leu	Asp	His	Gln	Ala	Ala	His	Gln	Pro
65					70				75					80	
Phe	Pro	Arg	Pro	Arg	Phe	Arg	Gln	Glu	Thr	Gly	His	Pro	Ser	Leu	Gln
		85					90							95	
Arg	Asp	Phe	Pro	Arg	Ser	Phe	Leu	Leu	Asp	Leu	Pro	Asn	Phe	Pro	Asp
		100					105					110			
Leu	Ser	Lys	Ala	Asp	Ile	Asn	Gly	Gln	Asn	Pro	Asn	Ile	Gln	Val	Thr
	115					120					125				
Ile	Glu	Val	Val	Asp	Gly	Pro	Asp	Ser	Glu	Ala	Asp	Lys	Asp	Gln	His
	130					135					140				

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Pro Glu Asn Lys Pro Ser Trp Ser Val Pro Ser Pro Asp Trp Arg Ala
145                      150                      155                      160

Trp Trp Gln Arg Ser Leu Ser Leu Ala Arg Ala Asn Ser Gly Asp Gln
                      165                      170                      175

Asp Tyr Lys Tyr Asp Ser Thr Ser Asp Asp Ser Asn Phe Leu Asn Pro
180                      185                      190

Pro Arg Gly Trp Asp His Thr Ala Pro Gly His Arg Thr Phe Glu Thr
195                      200                      205

Lys Asp Gln Pro Glu Tyr Asp Ser Thr Asp Gly Glu Gly Asp Trp Ser
210                      215                      220

Leu Trp Ser Val Cys Ser Val Thr Cys Gly Asn Gly Asn Gln Lys Arg
225                      230                      235                      240

Thr Arg Ser Cys Gly Tyr Ala Cys Thr Ala Thr Glu Ser Arg Thr Cys
245                      250                      255

Asp Arg Pro Asn Cys Pro Gly Ile Glu Asp Thr Phe Arg Thr Ala Ala
260                      265                      270

Thr Glu Val Ser Leu Leu Ala Gly Ser Glu Glu Phe Asn Ala Thr Lys
275                      280                      285

Leu Phe Glu Val Asp Thr Asp Ser Cys Glu Arg Trp Met Ser Cys Lys
290                      295                      300

Ser Glu Phe Leu Lys Lys Tyr Met His Lys Val Met Asn Asp Leu Pro
305                      310                      315                      320

Ser Cys Pro Cys Ser Tyr Pro Thr Glu Val Ala Tyr Ser Thr Ala Asp
325                      330                      335

Ile Phe Asp Arg Ile Lys Arg Lys Asp Phe Arg Trp Lys Asp Ala Ser
340                      345                      350

Gly Pro Lys Glu Lys Leu Glu Ile Tyr Lys Pro Thr Ala Arg Tyr Cys
355                      360                      365

Ile Arg Ser Met Leu Ser Leu Glu Ser Thr Thr Leu Ala Ala Gln His
370                      375                      380

Cys Cys Tyr Gly Asp Asn Met Gln Leu Ile Thr Arg Gly Lys Gly Ala
385                      390                      395                      400

Gly Thr Pro Asn Leu Ile Ser Thr Glu Phe Ser Ala Glu Leu His Tyr
405                      410                      415

Lys Val Asp Val Leu Pro Trp Ile Ile Cys Lys Gly Asp Trp Ser Arg
420                      425                      430

Tyr Asn Glu Ala Arg Pro Pro Asn Asn Gly Gln Lys Cys Thr Glu Ser
435                      440                      445

Pro Ser Asp Glu Asp Tyr Ile Lys Gln Phe Gln Glu Ala Arg Glu Tyr
450                      455                      460

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<210> SEQ ID NO 25

<211> LENGTH: 709

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 25

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Met Trp Trp Gly Gly Arg Gly Gln Ser Phe Asn Ile Ala Pro Gln Lys
1                      5                      10                      15

Glu Glu Pro Glu Met Gly Ser Val Gln Glu Asn Arg Met Pro Glu Pro
20                      25                      30

Arg Ser Arg Gln Pro Ser Ser Cys Leu Ala Ser Arg Cys Leu Pro Gly

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35					40					45					
Glu	Gln	Ile	Leu	Ala	Trp	Ala	Pro	Gly	Val	Arg	Lys	Gly	Leu	Glu	Pro
50						55					60				
Glu	Leu	Ser	Gly	Thr	Leu	Ile	Cys	Thr	Asn	Phe	Arg	Val	Thr	Phe	Gln
65					70					75					80
Pro	Cys	Gly	Trp	Gln	Trp	Asn	Gln	Asp	Thr	Pro	Leu	Asn	Ser	Glu	Tyr
				85					90					95	
Asp	Phe	Ala	Leu	Val	Asn	Ile	Gly	Arg	Leu	Glu	Ala	Val	Ser	Gly	Leu
			100					105					110		
Ser	Arg	Val	Gln	Leu	Leu	Arg	Pro	Gly	Ser	Leu	His	Lys	Phe	Ile	Pro
		115					120					125			
Glu	Glu	Ile	Leu	Ile	His	Gly	Arg	Asp	Phe	Arg	Leu	Leu	Arg	Val	Gly
	130					135					140				
Phe	Glu	Ala	Gly	Gly	Leu	Glu	Pro	Gln	Ala	Phe	Gln	Val	Thr	Met	Ala
145					150					155					160
Ile	Val	Gln	Ala	Arg	Ala	Gln	Ser	Asn	Gln	Ala	Gln	Gln	Tyr	Ser	Gly
				165					170					175	
Ile	Thr	Leu	Ser	Lys	Ala	Gly	Gln	Gly	Ser	Gly	Ser	Arg	Lys	Pro	Pro
		180						185					190		
Ile	Pro	Leu	Met	Glu	Thr	Ala	Glu	Asp	Trp	Glu	Thr	Glu	Arg	Lys	Lys
		195					200					205			
Gln	Ala	Ala	Arg	Gly	Trp	Arg	Val	Ser	Thr	Val	Asn	Glu	Arg	Phe	Asp
	210					215					220				
Val	Ala	Thr	Ser	Leu	Pro	Arg	Tyr	Phe	Trp	Val	Pro	Asn	Arg	Ile	Leu
225					230					235					240
Asp	Ser	Glu	Val	Arg	Arg	Ala	Phe	Gly	His	Phe	His	Gln	Gly	Arg	Gly
				245					250					255	
Pro	Arg	Leu	Ser	Trp	His	His	Pro	Gly	Gly	Ser	Asp	Leu	Leu	Arg	Cys
		260						265					270		
Gly	Gly	Phe	Tyr	Thr	Ala	Ser	Asp	Pro	Asn	Lys	Glu	Asp	Ile	Arg	Ala
		275					280					285			
Val	Glu	Leu	Met	Leu	Gln	Ala	Gly	His	Ser	Asp	Val	Val	Leu	Val	Asp
	290					295					300				
Thr	Met	Asp	Glu	Leu	Pro	Ser	Leu	Ala	Asp	Val	Gln	Leu	Ala	His	Leu
305					310					315					320
Arg	Leu	Arg	Ala	Leu	Cys	Leu	Pro	Asp	Ser	Ser	Val	Ala	Glu	Asp	Lys
				325					330					335	
Trp	Leu	Ser	Ala	Leu	Glu	Gly	Thr	Arg	Trp	Leu	Asp	Tyr	Val	Arg	Ala
			340					345					350		
Cys	Leu	Arg	Lys	Ala	Ser	Asp	Ile	Ser	Val	Leu	Val	Thr	Ser	Arg	Val
	355						360					365			
Arg	Ser	Val	Ile	Leu	Gln	Glu	Arg	Gly	Asp	Arg	Asp	Leu	Asn	Gly	Leu
	370					375					380				
Leu	Ser	Ser	Leu	Val	Gln	Leu	Leu	Ser	Ala	Pro	Glu	Ala	Arg	Thr	Leu
385					390					395					400
Phe	Gly	Phe	Gln	Ser	Leu	Val	Gln	Arg	Glu	Trp	Val	Ala	Ala	Gly	His
			405						410					415	
Pro	Phe	Leu	Thr	Arg	Leu	Gly	Gly	Thr	Gly	Ala	Ser	Glu	Glu	Ala	Pro
			420					425					430		
Val	Phe	Leu	Leu	Phe	Leu	Asp	Cys	Val	Trp	Gln	Leu	Leu	Gln	Gln	Phe
	435						440					445			

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Pro Ala Asp Phe Glu Phe Ser Glu Phe Phe Leu Leu Ala Leu His Asp
  450                      455                      460
Ser Val Arg Val Pro Asp Thr Leu Thr Phe Leu Arg Asn Thr Pro Trp
  465                      470                      475                      480
Glu Arg Gly Lys Gln Ser Gly Gln Leu Asn Ser Tyr Thr Gln Val Tyr
                      485                      490                      495
Thr Pro Gly Tyr Ser Gln Pro Pro Ala Gly Asn Ser Phe Asn Leu Gln
                      500                      505                      510
Leu Ser Val Trp Asp Trp Asp Leu Arg Tyr Ser Asn Ala Gln Ile Leu
                      515                      520                      525
Gln Phe Gln Asn Pro Gly Tyr Asp Pro Glu His Cys Pro Asp Ser Trp
                      530                      535                      540
Leu Pro Arg Pro Gln Pro Ser Phe Met Val Pro Gly Pro Pro Ser Ser
  545                      550                      555                      560
Val Trp Leu Phe Ser Arg Gly Ala Leu Thr Pro Leu Asn Gln Leu Cys
                      565                      570                      575
Pro Trp Arg Asp Ser Pro Ser Leu Leu Ala Val Ser Ser Arg Trp Leu
                      580                      585                      590
Pro Arg Pro Ala Ile Ser Ser Glu Ser Leu Ala Asp Gln Glu Trp Gly
                      595                      600                      605
Leu Pro Ser His Trp Gly Ala Cys Pro Leu Pro Pro Gly Leu Leu Leu
                      610                      615                      620
Pro Gly Tyr Leu Gly Pro Gln Ile Arg Leu Trp Arg Arg Cys Tyr Leu
  625                      630                      635                      640
Arg Gly Arg Pro Glu Val Gln Met Gly Leu Ser Ala Pro Thr Ile Ser
                      645                      650                      655
Gly Leu Gln Asp Glu Leu Ser His Leu Gln Glu Leu Leu Arg Lys Trp
                      660                      665                      670
Thr Pro Arg Ile Ser Pro Glu Asp His Ser Lys Lys Arg Asp Pro His
                      675                      680                      685
Thr Ile Leu Asn Pro Thr Glu Ile Ala Gly Ile Leu Lys Gly Arg Ala
                      690                      695                      700
Glu Gly Asp Leu Gly
705

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<210> SEQ ID NO 26

<211> LENGTH: 375

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 26

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Met Asp Glu Leu Gln Asp Val Gln Leu Thr Glu Ile Lys Pro Leu Leu
  1                      5                      10                      15
Asn Asp Lys Asn Gly Thr Arg Asn Phe Gln Asp Phe Asp Cys Gln Glu
                      20                      25                      30
His Asp Ile Glu Thr Thr His Gly Val Val His Val Thr Ile Arg Gly
                      35                      40                      45
Leu Pro Lys Gly Asn Arg Pro Val Ile Leu Thr Tyr His Asp Ile Gly
                      50                      55                      60
Leu Asn His Lys Ser Cys Phe Asn Ala Phe Phe Asn Phe Glu Asp Met
                      65                      70                      75                      80
Gln Glu Ile Thr Gln His Phe Ala Val Cys His Val Asp Ala Pro Gly

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85							90				95				
Gln	Gln	Glu	Gly	Ala	Pro	Ser	Phe	Pro	Thr	Gly	Tyr	Gln	Tyr	Pro	Thr
100							105				110				
Met	Asp	Glu	Leu	Ala	Glu	Met	Leu	Pro	Pro	Val	Leu	Thr	His	Leu	Ser
115							120				125				
Leu	Lys	Ser	Ile	Ile	Gly	Ile	Gly	Val	Gly	Ala	Gly	Ala	Tyr	Ile	Leu
130							135				140				
Ser	Arg	Phe	Ala	Leu	Asn	His	Pro	Glu	Leu	Val	Glu	Gly	Leu	Val	Leu
145							150				155				
Ile	Asn	Val	Asp	Pro	Cys	Ala	Lys	Gly	Trp	Ile	Asp	Trp	Ala	Ala	Ser
165							170				175				
Lys	Leu	Ser	Gly	Leu	Thr	Thr	Asn	Val	Val	Asp	Ile	Ile	Leu	Ala	His
180							185				190				
His	Phe	Gly	Gln	Glu	Glu	Leu	Gln	Ala	Asn	Leu	Asp	Leu	Ile	Gln	Thr
195							200				205				
Tyr	Arg	Met	His	Ile	Ala	Gln	Asp	Ile	Asn	Gln	Asp	Asn	Leu	Gln	Leu
210							215				220				
Phe	Leu	Asn	Ser	Tyr	Asn	Gly	Arg	Arg	Asp	Leu	Glu	Ile	Glu	Arg	Pro
225							230				235				
Ile	Leu	Gly	Gln	Asn	Asp	Asn	Lys	Ser	Lys	Thr	Leu	Lys	Cys	Ser	Thr
245							250				255				
Leu	Leu	Val	Val	Gly	Asp	Asn	Ser	Pro	Ala	Val	Glu	Ala	Val	Val	Glu
260							265				270				
Cys	Asn	Ser	Arg	Leu	Asn	Pro	Ile	Asn	Thr	Thr	Leu	Leu	Lys	Met	Ala
275							280				285				
Asp	Cys	Gly	Gly	Leu	Pro	Gln	Val	Val	Gln	Pro	Gly	Lys	Leu	Thr	Glu
290							295				300				
Ala	Phe	Lys	Tyr	Phe	Leu	Gln	Gly	Met	Gly	Tyr	Ile	Pro	Ser	Ala	Ser
305							310				315				
Met	Thr	Arg	Leu	Ala	Arg	Ser	Arg	Thr	His	Ser	Thr	Ser	Ser	Ser	Leu
325							330				335				
Gly	Ser	Gly	Glu	Ser	Pro	Phe	Ser	Arg	Ser	Val	Thr	Ser	Asn	Gln	Ser
340							345				350				
Asp	Gly	Thr	Gln	Glu	Ser	Cys	Glu	Ser	Pro	Asp	Val	Leu	Asp	Arg	His
355							360				365				
Gln	Thr	Met	Glu	Val	Ser	Cys									
370							375								

<210> SEQ ID NO 27

<211> LENGTH: 608

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 27

Met	Ala	Glu	Ser	Ile	Ile	Ile	Arg	Val	Gln	Ser	Pro	Asp	Gly	Val	Lys
1				5					10					15	
Arg	Ile	Thr	Ala	Thr	Lys	Arg	Glu	Thr	Ala	Ala	Thr	Phe	Leu	Lys	Lys
		20					25						30		
Val	Ala	Lys	Glu	Phe	Gly	Phe	Gln	Asn	Asn	Gly	Phe	Ser	Val	Tyr	Ile
		35					40					45			
Asn	Arg	Asn	Lys	Thr	Gly	Glu	Ile	Thr	Ala	Ser	Ser	Asn	Lys	Ser	Leu
	50					55					60				

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Asn	Leu	Leu	Lys	Ile	Lys	His	Gly	Asp	Leu	Leu	Phe	Leu	Phe	Pro	Ser	65	70	75	80
Ser	Leu	Ala	Gly	Pro	Ser	Ser	Glu	Met	Glu	Thr	Ser	Val	Pro	Pro	Gly	85	90	95	
Phe	Lys	Val	Phe	Gly	Ala	Pro	Asn	Val	Val	Glu	Asp	Glu	Ile	Asp	Gln	100	105	110	
Tyr	Leu	Ser	Lys	Gln	Asp	Gly	Lys	Ile	Tyr	Arg	Ser	Arg	Asp	Pro	Gln	115	120	125	
Leu	Cys	Arg	His	Gly	Pro	Leu	Gly	Lys	Cys	Val	His	Cys	Val	Pro	Leu	130	135	140	
Glu	Pro	Phe	Asp	Glu	Asp	Tyr	Leu	Asn	His	Leu	Glu	Pro	Pro	Val	Lys	145	150	155	160
His	Met	Ser	Phe	His	Ala	Tyr	Ile	Arg	Lys	Leu	Thr	Gly	Gly	Ala	Asp	165	170	175	
Lys	Gly	Lys	Phe	Val	Ala	Leu	Glu	Asn	Ile	Ser	Cys	Lys	Ile	Lys	Ser	180	185	190	
Gly	Cys	Glu	Gly	His	Leu	Pro	Trp	Pro	Asn	Gly	Ile	Cys	Thr	Lys	Cys	195	200	205	
Gln	Pro	Ser	Ala	Ile	Thr	Leu	Asn	Arg	Gln	Lys	Tyr	Arg	His	Val	Asp	210	215	220	
Asn	Ile	Met	Phe	Glu	Asn	His	Thr	Val	Ala	Asp	Arg	Phe	Leu	Asp	Phe	225	230	235	240
Trp	Arg	Lys	Thr	Gly	Asn	Gln	His	Phe	Gly	Tyr	Leu	Tyr	Gly	Arg	Tyr	245	250	255	
Thr	Glu	His	Lys	Asp	Ile	Pro	Leu	Gly	Ile	Arg	Ala	Glu	Val	Ala	Ala	260	265	270	
Ile	Tyr	Glu	Pro	Pro	Gln	Ile	Gly	Thr	Gln	Asn	Ser	Leu	Glu	Leu	Leu	275	280	285	
Glu	Asp	Pro	Lys	Ala	Glu	Val	Val	Asp	Glu	Ile	Ala	Ala	Lys	Leu	Gly	290	295	300	
Leu	Arg	Lys	Val	Gly	Trp	Ile	Phe	Thr	Asp	Leu	Val	Ser	Glu	Asp	Thr	305	310	315	320
Arg	Lys	Gly	Thr	Val	Arg	Tyr	Ser	Arg	Asn	Lys	Asp	Thr	Tyr	Phe	Leu	325	330	335	
Ser	Ser	Glu	Glu	Cys	Ile	Thr	Ala	Gly	Asp	Phe	Gln	Asn	Lys	His	Pro	340	345	350	
Asn	Met	Cys	Arg	Leu	Ser	Pro	Asp	Gly	His	Phe	Gly	Ser	Lys	Phe	Val	355	360	365	
Thr	Ala	Val	Ala	Thr	Gly	Gly	Pro	Asp	Asn	Gln	Val	His	Phe	Glu	Gly	370	375	380	
Tyr	Gln	Val	Ser	Asn	Gln	Cys	Met	Ala	Leu	Val	Arg	Asp	Glu	Cys	Leu	385	390	395	400
Leu	Pro	Cys	Lys	Asp	Ala	Pro	Glu	Leu	Gly	Tyr	Ala	Lys	Glu	Ser	Ser	405	410	415	
Ser	Glu	Gln	Tyr	Val	Pro	Asp	Val	Phe	Tyr	Lys	Asp	Val	Asp	Lys	Phe	420	425	430	
Gly	Asn	Glu	Ile	Thr	Gln	Leu	Ala	Arg	Pro	Leu	Pro	Val	Glu	Tyr	Leu	435	440	445	
Ile	Ile	Asp	Ile	Thr	Thr	Thr	Phe	Pro	Lys	Asp	Pro	Val	Tyr	Thr	Phe	450	455	460	
Ser	Ile	Ser	Gln	Asn	Pro	Phe	Pro	Ile	Glu	Asn	Arg	Asp	Val	Leu	Gly				

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465	470	475	480
Glu Thr Gln Asp Phe His Ser Leu Ala Thr Tyr Leu Ser Gln Asn Thr			
	485	490	495
Ser Ser Val Phe Leu Asp Thr Ile Ser Asp Phe His Leu Leu Leu Phe			
	500	505	510
Leu Val Thr Asn Glu Val Met Pro Leu Gln Asp Ser Ile Ser Leu Leu			
	515	520	525
Leu Glu Ala Val Arg Thr Arg Asn Glu Glu Leu Ala Gln Thr Trp Lys			
	530	535	540
Arg Ser Glu Gln Trp Ala Thr Ile Glu Gln Leu Cys Ser Thr Val Gly			
	545	550	555
Gly Gln Leu Pro Gly Leu His Glu Tyr Gly Ala Val Gly Gly Ser Thr			
	565	570	575
His Thr Ala Thr Ala Ala Met Trp Ala Cys Gln His Cys Thr Phe Met			
	580	585	590
Asn Gln Pro Gly Thr Gly His Cys Glu Met Cys Ser Leu Pro Arg Thr			
	595	600	605

<210> SEQ ID NO 28
 <211> LENGTH: 596
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 28

Met Gly Arg Gly Ala Arg Ala Ala Ala Ala Gln Ser Arg Trp Gly Arg			
1	5	10	15
Ala Ser Arg Ala Ser Val Ser Pro Gly Arg Thr Ile Arg Ser Ala Pro			
	20	25	30
Ala Val Gly Glu Ala Gln Glu Thr Glu Ala Ala Pro Glu Lys Glu Asn			
	35	40	45
Arg Val Asp Val Gly Ala Glu Glu Arg Ala Ala Ala Thr Arg Pro Arg			
	50	55	60
Gln Lys Ser Trp Leu Val Arg His Phe Ser Leu Leu Leu Arg Arg Asp			
	65	70	75
Arg Gln Ala Gln Lys Ala Gly Gln Leu Phe Ser Gly Leu Leu Ala Leu			
	85	90	95
Asn Val Val Phe Leu Gly Gly Ala Phe Ile Cys Ser Met Ile Phe Asn			
	100	105	110
Lys Val Ala Val Thr Leu Gly Asp Val Trp Ile Leu Leu Ala Thr Leu			
	115	120	125
Lys Val Leu Ser Leu Leu Trp Leu Leu Tyr Tyr Val Ala Ser Thr Thr			
	130	135	140
Arg Arg Pro His Ala Val Leu Tyr Gln Asp Pro His Ala Gly Pro Leu			
	145	150	155
Trp Val Arg Gly Ser Leu Val Leu Phe Gly Ser Cys Thr Phe Cys Leu			
	165	170	175
Asn Ile Phe Arg Val Gly Tyr Asp Val Ser His Ile Arg Cys Lys Ser			
	180	185	190
Gln Leu Asp Leu Val Phe Ser Val Ile Glu Met Val Phe Ile Gly Val			
	195	200	205
Gln Thr Trp Val Leu Trp Lys His Cys Lys Asp Cys Val Arg Val Gln			
	210	215	220

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Thr	Asn	Phe	Thr	Arg	Cys	Gly	Leu	Met	Leu	Thr	Leu	Ala	Thr	Asn	Leu
225					230					235					240
Leu	Leu	Trp	Val	Leu	Ala	Val	Thr	Asn	Asp	Ser	Met	His	Arg	Glu	Ile
			245					250						255	
Glu	Ala	Glu	Leu	Gly	Ile	Leu	Met	Glu	Lys	Ser	Thr	Gly	Asn	Glu	Thr
			260					265					270		
Asn	Thr	Cys	Leu	Cys	Leu	Asn	Ala	Thr	Ala	Cys	Glu	Ala	Phe	Arg	Arg
		275					280					285			
Gly	Phe	Leu	Met	Leu	Tyr	Pro	Phe	Ser	Thr	Glu	Tyr	Cys	Leu	Ile	Cys
	290					295					300				
Cys	Ala	Val	Leu	Phe	Val	Met	Trp	Lys	Asn	Val	Gly	Arg	His	Val	Ala
305					310					315					320
Pro	His	Met	Gly	Ala	His	Pro	Ala	Thr	Ala	Pro	Phe	His	Leu	His	Gly
			325					330						335	
Ala	Ile	Phe	Gly	Pro	Leu	Leu	Gly	Leu	Leu	Val	Leu	Leu	Ala	Gly	Val
			340					345					350		
Cys	Val	Phe	Val	Leu	Phe	Gln	Ile	Glu	Ala	Ser	Gly	Pro	Ala	Ile	Ala
		355					360					365			
Cys	Gln	Tyr	Phe	Thr	Leu	Tyr	Tyr	Ala	Phe	Tyr	Val	Ala	Val	Leu	Pro
	370					375					380				
Thr	Met	Ser	Leu	Ala	Cys	Leu	Ala	Gly	Thr	Ala	Ile	His	Gly	Leu	Glu
385					390					395					400
Glu	Arg	Glu	Leu	Asp	Thr	Val	Lys	Asn	Pro	Thr	Arg	Ser	Leu	Asp	Val
				405					410					415	
Val	Leu	Leu	Met	Gly	Ala	Ala	Leu	Gly	Gln	Met	Gly	Ile	Ala	Tyr	Phe
			420					425					430		
Ser	Ile	Val	Ala	Ile	Val	Ala	Lys	Arg	Pro	His	Glu	Leu	Leu	Asn	Arg
		435					440					445			
Leu	Ile	Leu	Ala	Tyr	Ser	Leu	Leu	Leu	Ile	Leu	Gln	His	Ile	Ala	Gln
	450					455					460				
Asn	Leu	Phe	Ile	Ile	Glu	Gly	Leu	His	Arg	Arg	Pro	Leu	Trp	Glu	Thr
465					470					475					480
Val	Pro	Glu	Gly	Leu	Ala	Gly	Lys	Gln	Glu	Ala	Glu	Pro	Pro	Arg	Arg
				485					490					495	
Gly	Ser	Leu	Leu	Glu	Leu	Gly	Gln	Gly	Leu	Gln	Arg	Ala	Ser	Leu	Ala
			500					505					510		
Tyr	Ile	His	Ser	Tyr	Ser	His	Leu	Asn	Trp	Lys	Arg	Arg	Ala	Leu	Lys
		515					520					525			
Glu	Ile	Ser	Leu	Phe	Leu	Ile	Leu	Cys	Asn	Ile	Thr	Leu	Trp	Met	Met
	530					535					540				
Pro	Ala	Phe	Gly	Ile	His	Pro	Glu	Phe	Glu	Asn	Gly	Leu	Glu	Lys	Asp
545					550					555					560
Phe	Tyr	Gly	Tyr	Gln	Ile	Trp	Phe	Ala	Ile	Val	Asn	Phe	Gly	Leu	Pro
				565					570					575	
Leu	Gly	Val	Phe	Tyr	Arg	Met	His	Ser	Val	Gly	Gly	Leu	Val	Glu	Val
			580					585					590		
Tyr	Leu	Gly	Ala												
			595												

<210> SEQ ID NO 29

<211> LENGTH: 926

<212> TYPE: PRT

-continued

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 29

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Met Val Ser Ser Val Leu Pro Asn Pro Thr Ser Ala Glu Cys Trp Ala
 1           5           10           15

Ala Leu Leu His Asp Pro Met Thr Leu Asp Met Asp Ala Val Leu Ser
 20           25           30

Asp Phe Val Arg Ser Thr Gly Ala Glu Pro Gly Leu Ala Arg Asp Leu
 35           40           45

Leu Glu Gly Lys Asn Trp Asp Leu Thr Ala Ala Leu Ser Asp Tyr Glu
 50           55           60

Gln Leu Arg Gln Val His Thr Ala Asn Leu Pro His Val Phe Asn Glu
 65           70           75           80

Gly Arg Gly Pro Lys Gln Pro Glu Arg Glu Pro Gln Pro Gly His Lys
 85           90           95

Val Glu Arg Pro Cys Leu Gln Arg Gln Asp Asp Ile Ala Gln Glu Lys
100           105           110

Arg Leu Ser Arg Gly Ile Ser His Ala Ser Ser Ala Ile Val Ser Leu
115           120           125

Ala Arg Ser His Val Ala Ser Glu Cys Asn Asn Glu Gln Phe Pro Leu
130           135           140

Glu Met Pro Ile Tyr Thr Phe Gln Leu Pro Asp Leu Ser Val Tyr Ser
145           150           155           160

Glu Asp Phe Arg Ser Phe Ile Glu Arg Asp Leu Ile Glu Gln Ala Thr
165           170           175

Met Val Ala Leu Glu Gln Ala Gly Arg Leu Asn Trp Trp Ser Thr Val
180           185           190

Cys Thr Ser Cys Lys Arg Leu Leu Pro Leu Ala Thr Thr Gly Asp Gly
195           200           205

Asn Cys Leu Leu His Ala Ala Ser Leu Gly Met Trp Gly Phe His Asp
210           215           220

Arg Asp Leu Val Leu Arg Lys Ala Leu Tyr Thr Met Met Arg Thr Gly
225           230           235           240

Ala Glu Arg Glu Ala Leu Lys Arg Arg Trp Arg Trp Gln Gln Thr Gln
245           250           255

Gln Asn Lys Glu Glu Glu Trp Glu Arg Glu Trp Thr Glu Leu Leu Lys
260           265           270

Leu Ala Ser Ser Glu Pro Arg Thr His Phe Ser Lys Asn Gly Gly Thr
275           280           285

Gly Gly Gly Val Asp Asn Ser Glu Asp Pro Val Tyr Glu Ser Leu Glu
290           295           300

Glu Phe His Val Phe Val Leu Ala His Ile Leu Arg Arg Pro Ile Val
305           310           315           320

Val Val Ala Asp Thr Met Leu Arg Asp Ser Gly Gly Glu Ala Phe Ala
325           330           335

Pro Ile Pro Phe Gly Gly Ile Tyr Leu Pro Leu Glu Val Pro Pro Asn
340           345           350

Arg Cys His Cys Ser Pro Leu Val Leu Ala Tyr Asp Gln Ala His Phe
355           360           365

Ser Ala Leu Val Ser Met Glu Gln Arg Asp Gln Gln Arg Glu Gln Ala
370           375           380

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Val	Ile	Pro	Leu	Thr	Asp	Ser	Glu	His	Lys	Leu	Leu	Pro	Leu	His	Phe	385	390	395	400
Ala	Val	Asp	Pro	Gly	Lys	Asp	Trp	Glu	Trp	Gly	Lys	Asp	Asp	Asn	Asp	405	410	415	
Asn	Ala	Arg	Leu	Ala	His	Leu	Ile	Leu	Ser	Leu	Glu	Ala	Lys	Leu	Asn	420	425	430	
Leu	Leu	His	Ser	Tyr	Met	Asn	Val	Thr	Trp	Ile	Arg	Ile	Pro	Ser	Glu	435	440	445	
Thr	Arg	Ala	Pro	Leu	Ala	Gln	Pro	Glu	Ser	Pro	Thr	Ala	Ser	Ala	Gly	450	455	460	
Glu	Asp	Val	Gln	Ser	Leu	Ala	Asp	Ser	Leu	Asp	Ser	Asp	Arg	Asp	Ser	465	470	475	480
Val	Cys	Ser	Asn	Ser	Asn	Ser	Asn	Asn	Gly	Lys	Asn	Gly	Lys	Asp	Lys	485	490	495	
Glu	Lys	Glu	Lys	Gln	Arg	Lys	Glu	Lys	Asp	Lys	Thr	Arg	Ala	Asp	Ser	500	505	510	
Val	Ala	Asn	Lys	Leu	Gly	Ser	Phe	Ser	Lys	Thr	Leu	Gly	Ile	Lys	Leu	515	520	525	
Lys	Lys	Asn	Met	Gly	Gly	Leu	Gly	Gly	Leu	Val	His	Gly	Lys	Met	Gly	530	535	540	
Arg	Ala	Asn	Ser	Ala	Asn	Gly	Lys	Asn	Gly	Asp	Ser	Ala	Glu	Arg	Gly	545	550	555	560
Lys	Glu	Lys	Lys	Ala	Lys	Ser	Arg	Lys	Gly	Ser	Lys	Glu	Glu	Ser	Gly	565	570	575	
Ala	Ser	Ala	Ser	Thr	Ser	Pro	Ser	Glu	Lys	Thr	Thr	Pro	Ser	Pro	Thr	580	585	590	
Asp	Lys	Ala	Ala	Gly	Ala	Ser	Pro	Ala	Glu	Lys	Gly	Gly	Gly	Pro	Arg	595	600	605	
Gly	Asp	Ala	Trp	Lys	Tyr	Ser	Thr	Asp	Val	Lys	Leu	Ser	Leu	Asn	Ile	610	615	620	
Leu	Arg	Ala	Ala	Met	Gln	Gly	Glu	Arg	Lys	Phe	Ile	Phe	Ala	Gly	Leu	625	630	635	640
Leu	Leu	Thr	Ser	His	Arg	His	Gln	Phe	His	Glu	Glu	Met	Ile	Gly	Tyr	645	650	655	
Tyr	Leu	Thr	Ser	Ala	Gln	Glu	Arg	Phe	Ser	Ala	Glu	Gln	Glu	Gln	Arg	660	665	670	
Arg	Arg	Asp	Ala	Ala	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	675	680	685	
Ala	Thr	Ala	Lys	Arg	Pro	Pro	Arg	Arg	Pro	Glu	Thr	Glu	Gly	Val	Pro	690	695	700	
Val	Pro	Glu	Arg	Ala	Ser	Pro	Gly	Pro	Pro	Thr	Gln	Leu	Val	Leu	Lys	705	710	715	720
Leu	Lys	Glu	Arg	Pro	Ser	Pro	Gly	Pro	Ala	Ala	Gly	Arg	Ala	Ala	Arg	725	730	735	
Ala	Ala	Ala	Gly	Gly	Thr	Ala	Ser	Pro	Gly	Gly	Gly	Ala	Arg	Arg	Ala	740	745	750	
Ser	Ala	Ser	Gly	Pro	Val	Pro	Gly	Arg	Ser	Pro	Pro	Ala	Pro	Ala	Arg	755	760	765	
Gln	Ser	Val	Ile	His	Val	Gln	Ala	Ser	Gly	Ala	Arg	Asp	Glu	Ala	Cys	770	775	780	
Ala	Pro	Ala	Val	Gly	Ala	Leu	Arg	Pro	Cys	Ala	Thr	Tyr	Pro	Gln	Gln				

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785	790	795	800
Asn Arg Ser Leu Ser Ser Gln Ser Tyr Ser Pro Ala Arg Ala Ala Ala			
	805	810	815
Leu Arg Thr Val Asn Thr Val Glu Ser Leu Ala Arg Ala Val Pro Gly			
	820	825	830
Ala Leu Pro Gly Ala Ala Gly Thr Ala Gly Ala Ala Glu His Lys Ser			
	835	840	845
Gln Thr Tyr Thr Asn Gly Phe Gly Ala Leu Arg Asp Gly Leu Glu Phe			
	850	855	860
Ala Asp Ala Asp Ala Pro Thr Ala Arg Ser Asn Gly Glu Cys Gly Arg			
	865	870	875
Gly Gly Pro Gly Pro Val Gln Arg Arg Cys Gln Arg Glu Asn Cys Ala			
	885	890	895
Phe Tyr Gly Arg Ala Glu Thr Glu His Tyr Cys Ser Tyr Cys Tyr Arg			
	900	905	910
Glu Glu Leu Arg Arg Arg Arg Glu Ala Arg Gly Ala Arg Pro			
	915	920	925

<210> SEQ ID NO 30

<211> LENGTH: 1141

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 30

Met Ala Val Pro Gly Asp Ala Ala Arg Val Arg Asp Lys Pro Val His			
1	5	10	15
Ser Gly Val Ser Gln Ala Pro Thr Ala Gly Arg Asp Cys His His Arg			
	20	25	30
Ala Asp Pro Ala Ser Pro Arg Asp Ser Gly Cys Arg Gly Cys Trp Gly			
	35	40	45
Asp Leu Val Leu Gln Pro Leu Arg Ser Ser Arg Lys Leu Ser Ser Ala			
	50	55	60
Leu Cys Ala Gly Ser Leu Ser Phe Leu Leu Ala Leu Leu Val Arg Leu			
	65	70	75
Val Arg Gly Glu Val Gly Cys Asp Leu Glu Gln Cys Lys Glu Ala Ala			
	85	90	95
Ala Ala Glu Glu Glu Glu Ala Ala Pro Gly Ala Glu Gly Gly Val Phe			
	100	105	110
Pro Gly Pro Arg Gly Gly Ala Pro Gly Gly Gly Ala Arg Leu Ser Pro			
	115	120	125
Trp Leu Gln Pro Ser Ala Leu Leu Phe Ser Leu Leu Cys Ala Phe Phe			
	130	135	140
Trp Met Gly Leu Tyr Leu Leu Arg Ala Gly Val Arg Leu Pro Leu Ala			
	145	150	155
Val Ala Leu Leu Ala Ala Cys Cys Gly Gly Glu Ala Leu Val Gln Ile			
	165	170	175
Gly Leu Gly Val Gly Glu Asp His Leu Leu Ser Leu Pro Ala Ala Gly			
	180	185	190
Val Val Leu Ser Cys Leu Ala Ala Ala Thr Trp Leu Val Leu Arg Leu			
	195	200	205
Arg Leu Gly Val Leu Met Ile Ala Leu Thr Ser Ala Val Arg Thr Val			
	210	215	220

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Ser	Leu	Ile	Ser	Leu	Glu	Arg	Phe	Lys	Val	Ala	Trp	Arg	Pro	Tyr	Leu	225	230	235	240
Ala	Tyr	Leu	Ala	Gly	Val	Leu	Gly	Ile	Leu	Leu	Ala	Arg	Tyr	Val	Glu	245	250	255	
Gln	Ile	Leu	Pro	Gln	Ser	Ala	Glu	Ala	Ala	Pro	Arg	Glu	His	Leu	Gly	260	265	270	
Ser	Gln	Leu	Ile	Ala	Gly	Thr	Lys	Glu	Asp	Ile	Pro	Val	Phe	Lys	Arg	275	280	285	
Arg	Arg	Arg	Ser	Ser	Ser	Val	Val	Ser	Ala	Glu	Met	Ser	Gly	Cys	Ser	290	295	300	
Ser	Lys	Ser	His	Arg	Arg	Thr	Ser	Leu	Pro	Cys	Ile	Pro	Arg	Glu	Gln	305	310	315	320
Leu	Met	Gly	His	Ser	Glu	Trp	Asp	His	Lys	Arg	Gly	Pro	Arg	Gly	Ser	325	330	335	
Gln	Ser	Ser	Gly	Thr	Ser	Ile	Thr	Val	Asp	Ile	Ala	Val	Met	Gly	Glu	340	345	350	
Ala	His	Gly	Leu	Ile	Thr	Asp	Leu	Leu	Ala	Asp	Pro	Ser	Leu	Pro	Pro	355	360	365	
Asn	Val	Cys	Thr	Ser	Leu	Arg	Ala	Val	Ser	Asn	Leu	Leu	Ser	Thr	Gln	370	375	380	
Leu	Thr	Phe	Gln	Ala	Ile	His	Lys	Pro	Arg	Val	Asn	Pro	Val	Thr	Ser	385	390	395	400
Leu	Ser	Glu	Asn	Tyr	Thr	Cys	Ser	Asp	Ser	Glu	Glu	Ser	Ser	Glu	Lys	405	410	415	
Asp	Lys	Leu	Ala	Ile	Pro	Lys	Arg	Leu	Arg	Arg	Ser	Leu	Pro	Pro	Gly	420	425	430	
Leu	Leu	Arg	Arg	Val	Ser	Ser	Thr	Trp	Thr	Thr	Thr	Thr	Ser	Ala	Thr	435	440	445	
Gly	Leu	Pro	Thr	Leu	Glu	Pro	Ala	Pro	Val	Arg	Arg	Asp	Arg	Ser	Thr	450	455	460	
Ser	Ile	Lys	Leu	Gln	Glu	Ala	Pro	Ser	Ser	Ser	Pro	Asp	Ser	Trp	Asn	465	470	475	480
Asn	Pro	Val	Met	Met	Thr	Leu	Thr	Lys	Ser	Arg	Ser	Phe	Thr	Ser	Ser	485	490	495	
Tyr	Ala	Ile	Ser	Ala	Ala	Asn	His	Val	Lys	Ala	Lys	Lys	Gln	Ser	Arg	500	505	510	
Pro	Gly	Ala	Leu	Ala	Lys	Ile	Ser	Pro	Leu	Ser	Ser	Pro	Cys	Ser	Ser	515	520	525	
Pro	Leu	Gln	Gly	Thr	Pro	Ala	Ser	Ser	Leu	Val	Ser	Lys	Ile	Ser	Ala	530	535	540	
Val	Gln	Phe	Pro	Glu	Ser	Ala	Asp	Thr	Thr	Ala	Lys	Gln	Ser	Leu	Gly	545	550	555	560
Ser	His	Arg	Ala	Leu	Thr	Tyr	Thr	Gln	Ser	Ala	Pro	Asp	Leu	Ser	Pro	565	570	575	
Gln	Ile	Leu	Thr	Pro	Pro	Val	Ile	Cys	Ser	Ser	Cys	Gly	Arg	Pro	Tyr	580	585	590	
Ser	Gln	Gly	Asn	Pro	Ala	Asp	Glu	Pro	Leu	Glu	Arg	Ser	Gly	Val	Ala	595	600	605	
Thr	Arg	Thr	Pro	Ser	Arg	Thr	Asp	Asp	Thr	Ala	Gln	Val	Thr	Ser	Asp	610	615	620	
Tyr	Glu	Thr	Asn	Asn	Asn	Ser	Asp	Ser	Ser	Asp	Ile	Val	Gln	Asn	Glu				

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625					630						635					640
Asp	Glu	Thr	Glu	Cys	Leu	Arg	Glu	Pro	Leu	Arg	Lys	Ala	Ser	Ala	Cys	
				645					650						655	
Ser	Thr	Tyr	Ala	Pro	Glu	Thr	Met	Met	Phe	Leu	Asp	Lys	Pro	Ile	Leu	
			660					665					670			
Ala	Pro	Glu	Pro	Leu	Val	Met	Asp	Asn	Leu	Asp	Ser	Ile	Met	Glu	Gln	
		675					680					685				
Leu	Asn	Thr	Trp	Asn	Phe	Pro	Ile	Phe	Asp	Leu	Val	Glu	Asn	Ile	Gly	
	690				695						700					
Arg	Lys	Cys	Gly	Arg	Ile	Leu	Ser	Gln	Val	Ser	Tyr	Arg	Leu	Phe	Glu	
	705				710					715					720	
Asp	Met	Gly	Leu	Phe	Glu	Ala	Phe	Lys	Ile	Pro	Ile	Arg	Glu	Phe	Met	
			725					730						735		
Asn	Tyr	Phe	His	Ala	Leu	Glu	Ile	Gly	Tyr	Arg	Asp	Ile	Pro	Tyr	His	
		740						745					750			
Asn	Arg	Ile	His	Ala	Thr	Asp	Val	Leu	His	Ala	Val	Trp	Tyr	Leu	Thr	
	755						760					765				
Thr	Gln	Pro	Ile	Pro	Gly	Leu	Ser	Thr	Val	Ile	Asn	Asp	His	Gly	Ser	
	770					775					780					
Thr	Ser	Asp	Ser	Asp	Ser	Asp	Ser	Gly	Phe	Thr	His	Gly	His	Met	Gly	
	785				790					795					800	
Tyr	Val	Phe	Ser	Lys	Thr	Tyr	Asn	Val	Thr	Asp	Asp	Lys	Tyr	Gly	Cys	
			805						810					815		
Leu	Ser	Gly	Asn	Ile	Pro	Ala	Leu	Glu	Leu	Met	Ala	Leu	Tyr	Val	Ala	
			820					825					830			
Ala	Ala	Met	His	Asp	Tyr	Asp	His	Pro	Gly	Arg	Thr	Asn	Ala	Phe	Leu	
		835					840					845				
Val	Ala	Thr	Ser	Ala	Pro	Gln	Ala	Val	Leu	Tyr	Asn	Asp	Arg	Ser	Val	
	850				855						860					
Leu	Glu	Asn	His	His	Ala	Ala	Ala	Ala	Trp	Asn	Leu	Phe	Met	Ser	Arg	
	865				870					875					880	
Pro	Glu	Tyr	Asn	Phe	Leu	Ile	Asn	Leu	Asp	His	Val	Glu	Phe	Lys	His	
			885						890					895		
Phe	Arg	Phe	Leu	Val	Ile	Glu	Ala	Ile	Leu	Ala	Thr	Asp	Leu	Lys	Lys	
		900						905						910		
His	Phe	Asp	Phe	Val	Ala	Lys	Phe	Asn	Gly	Lys	Val	Asn	Asp	Asp	Val	
	915						920					925				
Gly	Ile	Asp	Trp	Thr	Asn	Glu	Asn	Asp	Arg	Leu	Leu	Val	Cys	Gln	Met	
	930				935						940					
Cys	Ile	Lys	Leu	Ala	Asp	Ile	Asn	Gly	Pro	Ala	Lys	Cys	Lys	Glu	Leu	
	945				950					955					960	
His	Leu	Gln	Trp	Thr	Asp	Gly	Ile	Val	Asn	Glu	Phe	Tyr	Glu	Gln	Gly	
			965					970						975		
Asp	Glu	Glu	Ala	Ser	Leu	Gly	Leu	Pro	Ile	Ser	Pro	Phe	Met	Asp	Arg	
			980					985					990			
Ser	Ala	Pro	Gln	Leu	Ala	Asn	Leu	Gln	Glu	Ser	Phe	Ile	Ser	His	Ile	
		995					1000					1005				
Val	Gly	Pro	Leu	Cys	Asn	Ser	Tyr	Asp	Ser	Ala	Gly	Leu	Met	Pro	Gly	
	1010					1015					1020					
Lys	Trp	Val	Glu	Asp	Ser	Asp	Glu	Ser	Gly	Asp	Thr	Asp	Asp	Pro	Glu	
	1025					1030				1035					1040	

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Glu Glu Glu Glu Glu Ala Pro Ala Pro Asn Glu Glu Glu Thr Cys Glu
 1045 1050 1055
 Asn Asn Glu Ser Pro Lys Lys Lys Thr Phe Lys Arg Arg Lys Ile Tyr
 1060 1065 1070
 Cys Gln Ile Thr Gln His Leu Leu Gln Asn His Lys Met Trp Lys Lys
 1075 1080 1085
 Val Ile Glu Glu Glu Gln Arg Leu Ala Gly Ile Glu Asn Gln Ser Leu
 1090 1095 1100
 Asp Gln Thr Pro Gln Ser His Ser Ser Glu Gln Ile Gln Ala Ile Lys
 1105 1110 1115 1120
 Glu Glu Glu Glu Glu Lys Gly Lys Pro Arg Gly Glu Glu Ile Pro Thr
 1125 1130 1135
 Gln Lys Pro Asp Gln
 1140

<210> SEQ ID NO 31

<211> LENGTH: 870

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 31

Met Asp Ser Thr Lys Glu Lys Cys Asp Ser Tyr Lys Asp Asp Leu Leu
 1 5 10 15
 Leu Arg Met Gly Leu Asn Asp Asn Lys Ala Gly Met Glu Gly Leu Asp
 20 25 30
 Lys Glu Lys Ile Asn Lys Ile Ile Met Glu Ala Thr Lys Gly Ser Arg
 35 40 45
 Phe Tyr Gly Asn Glu Leu Lys Lys Glu Lys Gln Val Asn Gln Arg Ile
 50 55 60
 Glu Asn Met Met Gln Gln Lys Ala Gln Ile Thr Ser Gln Gln Leu Arg
 65 70 75 80
 Lys Ala Gln Leu Gln Val Asp Arg Phe Ala Met Glu Leu Glu Gln Ser
 85 90 95
 Arg Asn Leu Ser Asn Thr Ile Val His Ile Asp Met Asp Ala Phe Tyr
 100 105 110
 Ala Ala Val Glu Met Arg Asp Asn Pro Glu Leu Lys Asp Lys Pro Ile
 115 120 125
 Ala Val Gly Ser Met Ser Met Leu Ser Thr Ser Asn Tyr His Ala Arg
 130 135 140
 Arg Phe Gly Val Arg Ala Ala Met Pro Gly Phe Ile Ala Lys Arg Leu
 145 150 155 160
 Cys Pro Gln Leu Ile Ile Val Pro Pro Asn Phe Asp Lys Tyr Arg Ala
 165 170 175
 Val Ser Lys Glu Val Lys Glu Ile Leu Ala Asp Tyr Asp Pro Asn Phe
 180 185 190
 Met Ala Met Ser Leu Asp Glu Ala Tyr Leu Asn Ile Thr Lys His Leu
 195 200 205
 Glu Glu Arg Gln Asn Trp Pro Glu Asp Lys Arg Arg Tyr Phe Ile Lys
 210 215 220
 Met Gly Ser Ser Val Glu Asn Asp Asn Pro Gly Lys Glu Val Asn Lys
 225 230 235 240
 Leu Ser Glu His Glu Arg Ser Ile Ser Pro Leu Leu Phe Glu Glu Ser

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245																250								255							
Pro	Ser	Asp	Val	Gln	Pro	Pro	Gly	Asp	Pro	Phe	Gln	Val	Asn	Phe	Glu																
260																265								270							
Glu	Gln	Asn	Asn	Pro	Gln	Ile	Leu	Gln	Asn	Ser	Val	Val	Phe	Gly	Thr																
275																280								285							
Ser	Ala	Gln	Glu	Val	Val	Lys	Glu	Ile	Arg	Phe	Arg	Ile	Glu	Gln	Lys																
290																295								300							
Thr	Thr	Leu	Thr	Ala	Ser	Ala	Gly	Ile	Ala	Pro	Asn	Thr	Met	Leu	Ala																
305																310								315							
Lys	Val	Cys	Ser	Asp	Lys	Asn	Lys	Pro	Asn	Gly	Gln	Tyr	Gln	Ile	Leu																
325																330								335							
Pro	Asn	Arg	Gln	Ala	Val	Met	Asp	Phe	Ile	Lys	Asp	Leu	Pro	Ile	Arg																
340																345								350							
Lys	Val	Ser	Gly	Ile	Gly	Lys	Val	Thr	Glu	Lys	Met	Leu	Lys	Ala	Leu																
355																360								365							
Gly	Ile	Ile	Thr	Cys	Thr	Glu	Leu	Tyr	Gln	Gln	Arg	Ala	Leu	Leu	Ser																
370																375								380							
Leu	Leu	Phe	Ser	Glu	Thr	Ser	Trp	His	Tyr	Phe	Leu	His	Ile	Ser	Leu																
385																390								395							
Gly	Leu	Gly	Ser	Thr	His	Leu	Thr	Arg	Asp	Gly	Glu	Arg	Lys	Ser	Met																
405																410								415							
Ser	Val	Glu	Arg	Thr	Phe	Ser	Glu	Ile	Asn	Lys	Ala	Glu	Glu	Gln	Tyr																
420																425								430							
Ser	Leu	Cys	Gln	Glu	Leu	Cys	Ser	Glu	Leu	Ala	Gln	Asp	Leu	Gln	Lys																
435																440								445							
Glu	Arg	Leu	Lys	Gly	Arg	Thr	Val	Thr	Ile	Lys	Leu	Lys	Asn	Val	Asn																
450																455								460							
Phe	Glu	Val	Lys	Thr	Arg	Ala	Ser	Thr	Val	Ser	Ser	Val	Val	Ser	Thr																
465																470								475							
Ala	Glu	Glu	Ile	Phe	Ala	Ile	Ala	Lys	Glu	Leu	Leu	Lys	Thr	Glu	Ile																
485																490								495							
Asp	Ala	Asp	Phe	Pro	His	Pro	Leu	Arg	Leu	Arg	Leu	Met	Gly	Val	Arg																
500																505								510							
Ile	Ser	Ser	Phe	Pro	Asn	Glu	Glu	Asp	Arg	Lys	His	Gln	Gln	Arg	Ser																
515																520								525							
Ile	Ile	Gly	Phe	Leu	Gln	Ala	Gly	Asn	Gln	Ala	Leu	Ser	Ala	Thr	Glu																
530																535								540							
Cys	Thr	Leu	Glu	Lys	Thr	Asp	Lys	Asp	Lys	Phe	Val	Lys	Pro	Leu	Glu																
545																550								555							
Met	Ser	His	Lys	Lys	Ser	Phe	Phe	Asp	Lys	Lys	Arg	Ser	Glu	Arg	Lys																
565																570								575							
Trp	Ser	His	Gln	Asp	Thr	Phe	Lys	Cys	Glu	Ala	Val	Asn	Lys	Gln	Ser																
580																585								590							
Phe	Gln	Thr	Ser	Gln	Pro	Phe	Gln	Val	Leu	Lys	Lys	Lys	Met	Asn	Glu																
595																600								605							
Asn	Leu	Glu	Ile	Ser	Glu	Asn	Ser	Asp	Asp	Cys	Gln	Ile	Leu	Thr	Cys																
610																615								620							
Pro	Val	Cys	Phe	Arg	Ala	Gln	Gly	Cys	Ile	Ser	Leu	Glu	Ala	Leu	Asn																
625																630								635							
Lys	His	Val	Asp	Glu	Cys	Leu	Asp	Gly	Pro	Ser	Ile	Ser	Glu	Asn	Phe																
645																650								655							

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Lys Met Phe Ser Cys Ser His Val Ser Ala Thr Lys Val Asn Lys Lys
 660 665 670
 Glu Asn Val Pro Ala Ser Ser Leu Cys Glu Lys Gln Asp Tyr Glu Ala
 675 680 685
 His Pro Lys Ile Lys Glu Ile Ser Ser Val Asp Cys Ile Ala Leu Val
 690 695 700
 Asp Thr Ile Asp Asn Ser Ser Lys Ala Glu Ser Ile Asp Ala Leu Ser
 705 710 715 720
 Asn Lys His Ser Lys Glu Glu Cys Ser Ser Leu Pro Ser Lys Ser Phe
 725 730 735
 Asn Ile Glu His Cys His Gln Asn Ser Ser Ser Thr Val Ser Leu Glu
 740 745 750
 Asn Glu Asp Val Gly Ser Phe Arg Gln Glu Tyr Arg Gln Pro Tyr Leu
 755 760 765
 Cys Glu Val Lys Thr Gly Gln Ala Leu Val Cys Pro Val Cys Asn Val
 770 775 780
 Glu Gln Lys Thr Ser Asp Leu Thr Leu Phe Asn Val His Val Asp Val
 785 790 795 800
 Cys Leu Asn Lys Ser Phe Ile Gln Glu Leu Arg Lys Asp Lys Phe Asn
 805 810 815
 Pro Val Asn Gln Pro Lys Glu Ser Ser Arg Ser Thr Gly Ser Ser Ser
 820 825 830
 Gly Val Gln Lys Ala Val Thr Arg Thr Lys Arg Pro Gly Leu Met Thr
 835 840 845
 Lys Tyr Ser Thr Ser Lys Lys Ile Lys Pro Asn Asn Pro Lys His Thr
 850 855 860
 Leu Asp Ile Phe Phe Lys
 865 870

<210> SEQ ID NO 32
 <211> LENGTH: 57
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 32

Met Leu Cys Ala His Phe Ser Asp Gln Gly Pro Ala His Leu Thr Thr
 1 5 10 15
 Ser Lys Ser Ala Phe Leu Ser Asn Lys Lys Thr Ser Thr Leu Lys His
 20 25 30
 Leu Leu Gly Glu Thr Arg Ser Asp Gly Ser Ala Cys Asn Ser Gly Ile
 35 40 45
 Ser Gly Gly Arg Gly Arg Lys Ile Pro
 50 55

<210> SEQ ID NO 33
 <211> LENGTH: 418
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 33

Met Gln Ala Leu Val Leu Leu Leu Cys Ile Gly Ala Leu Leu Gly His
 1 5 10 15
 Ser Ser Cys Gln Asn Pro Ala Ser Pro Pro Glu Glu Gly Ser Pro Asp
 20 25 30

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Pro	Asp	Ser	Thr	Gly	Ala	Leu	Val	Glu	Glu	Glu	Asp	Pro	Phe	Phe	Lys
	35					40					45				
Val	Pro	Val	Asn	Lys	Leu	Ala	Ala	Ala	Val	Ser	Asn	Phe	Gly	Tyr	Asp
	50				55						60				
Leu	Tyr	Arg	Val	Arg	Ser	Ser	Thr	Ser	Pro	Thr	Thr	Asn	Val	Leu	Leu
65				70					75					80	
Ser	Pro	Leu	Ser	Val	Ala	Thr	Ala	Leu	Ser	Ala	Leu	Ser	Leu	Gly	Ala
		85						90					95		
Glu	Gln	Arg	Thr	Glu	Ser	Ile	Ile	His	Arg	Ala	Leu	Tyr	Tyr	Asp	Leu
		100						105					110		
Ile	Ser	Ser	Pro	Asp	Ile	His	Gly	Thr	Tyr	Lys	Glu	Leu	Leu	Asp	Thr
	115					120					125				
Val	Thr	Ala	Pro	Gln	Lys	Asn	Leu	Lys	Ser	Ala	Ser	Arg	Ile	Val	Phe
	130					135				140					
Glu	Lys	Lys	Leu	Arg	Ile	Lys	Ser	Ser	Phe	Val	Ala	Pro	Leu	Glu	Lys
145				150					155					160	
Ser	Tyr	Gly	Thr	Arg	Pro	Arg	Val	Leu	Thr	Gly	Asn	Pro	Arg	Leu	Asp
		165					170						175		
Leu	Gln	Glu	Ile	Asn	Asn	Trp	Val	Gln	Ala	Gln	Met	Lys	Gly	Lys	Leu
	180					185						190			
Ala	Arg	Ser	Thr	Lys	Glu	Ile	Pro	Asp	Glu	Ile	Ser	Ile	Leu	Leu	Leu
	195					200					205				
Gly	Val	Ala	His	Phe	Lys	Gly	Gln	Trp	Val	Thr	Lys	Phe	Asp	Ser	Arg
	210				215					220					
Lys	Thr	Ser	Leu	Glu	Asp	Phe	Tyr	Leu	Asp	Glu	Glu	Arg	Thr	Val	Arg
225				230					235					240	
Val	Pro	Met	Met	Ser	Asp	Pro	Lys	Ala	Val	Leu	Arg	Tyr	Gly	Leu	Asp
		245					250						255		
Ser	Asp	Leu	Ser	Cys	Lys	Ile	Ala	Gln	Leu	Pro	Leu	Thr	Gly	Ser	Met
	260						265					270			
Ser	Ile	Ile	Phe	Phe	Leu	Pro	Leu	Lys	Val	Thr	Gln	Asn	Leu	Thr	Leu
	275				280						285				
Ile	Glu	Glu	Ser	Leu	Thr	Ser	Glu	Phe	Ile	His	Asp	Ile	Asp	Arg	Glu
	290				295					300					
Leu	Lys	Thr	Val	Gln	Ala	Val	Leu	Thr	Val	Pro	Lys	Leu	Lys	Leu	Ser
305				310					315					320	
Tyr	Glu	Gly	Glu	Val	Thr	Lys	Ser	Leu	Gln	Glu	Met	Lys	Leu	Gln	Ser
		325						330					335		
Leu	Phe	Asp	Ser	Pro	Asp	Phe	Ser	Lys	Ile	Thr	Gly	Lys	Pro	Ile	Lys
	340						345					350			
Leu	Thr	Gln	Val	Glu	His	Arg	Ala	Gly	Phe	Glu	Trp	Asn	Glu	Asp	Gly
	355					360					365				
Ala	Gly	Thr	Thr	Pro	Ser	Pro	Gly	Leu	Gln	Pro	Ala	His	Leu	Thr	Phe
	370				375					380					
Pro	Leu	Asp	Tyr	His	Leu	Asn	Gln	Pro	Phe	Ile	Phe	Val	Leu	Arg	Asp
385				390					395					400	
Thr	Asp	Thr	Gly	Ala	Leu	Leu	Phe	Ile	Gly	Lys	Ile	Leu	Asp	Pro	Arg
		405					410						415		

Gly Pro

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<210> SEQ ID NO 34
<211> LENGTH: 353
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 34
Met Pro Ser Pro Val Gly Leu Leu Arg Ala Leu Pro Leu Pro Trp Pro
1          5          10          15
Gln Phe Leu Ala Cys Thr Leu Arg Arg Leu Ala Gly Pro Arg Glu Ser
20          25          30
Thr Gly Pro Ser Gln Lys Pro Pro Pro Leu Cys Ser Val Pro Cys Arg
35          40          45
Val Pro Ala Met Thr Glu Glu Val Ala Arg Glu Ala Leu Leu Ser Phe
50          55          60
Val Asp Ser Lys Cys Cys Tyr Ser Ser Thr Val Ala Gly Asp Leu Val
65          70          75          80
Ile Gln Glu Leu Lys Arg Gln Thr Leu Cys Arg Tyr Arg Leu Glu Thr
85          90          95
Phe Ser Glu Ser Arg Ile Ser Glu Trp Thr Phe Gln Pro Phe Thr Asn
100         105         110
His Ser Val Asp Gly Pro Gln Arg Gly Ala Ser Pro Arg Leu Trp Asp
115         120         125
Ile Lys Val Gln Gly Pro Pro Met Phe Gln Glu Asp Thr Arg Lys Phe
130         135         140
Gln Val Pro His Ser Ser Leu Val Lys Glu Cys His Lys Cys His Gly
145         150         155         160
Arg Gly Arg Tyr Lys Cys Ser Gly Cys His Gly Ala Gly Thr Val Arg
165         170         175
Cys Pro Ser Cys Cys Gly Ala Lys Arg Lys Ala Lys Gln Ser Arg Arg
180         185         190
Cys Gln Leu Cys Ala Gly Ser Gly Arg Arg Arg Cys Ser Thr Cys Ser
195         200         205
Gly Arg Gly Asn Lys Thr Cys Ala Thr Cys Lys Gly Glu Lys Lys Leu
210         215         220
Leu His Phe Ile Gln Leu Val Ile Met Trp Lys Asn Ser Leu Phe Glu
225         230         235         240
Phe Val Ser Glu His Arg Leu Asn Cys Pro Arg Glu Leu Leu Ala Lys
245         250         255
Ala Lys Gly Glu Asn Leu Phe Lys Asp Glu Asn Ser Val Val Tyr Pro
260         265         270
Ile Val Asp Phe Pro Leu Arg Asp Ile Ser Leu Ala Ser Gln Arg Gly
275         280         285
Ile Ala Glu His Ser Ala Ala Leu Ala Ser Arg Ala Arg Val Leu Gln
290         295         300
Gln Arg Gln Thr Ile Glu Leu Ile Pro Leu Thr Glu Val His Tyr Trp
305         310         315         320
Tyr Gln Gly Lys Thr Tyr Val Tyr Tyr Ile Tyr Gly Thr Asp His Gln
325         330         335
Val Tyr Ala Val Asp Tyr Pro Glu Arg Tyr Cys Cys Gly Cys Thr Ile
340         345         350
Val

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-continued

<210> SEQ ID NO 35
<211> LENGTH: 202
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 35

Met Cys Thr Gly Gly Cys Ala Arg Cys Leu Gly Gly Thr Leu Ile Pro
1 5 10 15

Leu Ala Phe Phe Gly Phe Leu Ala Asn Ile Leu Leu Phe Phe Pro Gly
20 25 30

Gly Lys Val Ile Asp Asp Asn Asp His Leu Ser Gln Glu Ile Trp Phe
35 40 45

Phe Gly Gly Ile Leu Gly Ser Gly Val Leu Met Ile Phe Pro Ala Leu
50 55 60

Val Phe Leu Gly Leu Lys Asn Asn Asp Cys Cys Gly Cys Cys Gly Asn
65 70 75 80

Glu Gly Cys Gly Lys Arg Phe Ala Met Phe Thr Ser Thr Ile Phe Ala
85 90 95

Val Val Gly Phe Leu Gly Ala Gly Tyr Ser Phe Ile Ile Ser Ala Ile
100 105 110

Ser Ile Asn Lys Gly Pro Lys Cys Leu Met Ala Asn Ser Thr Trp Gly
115 120 125

Tyr Pro Phe His Asp Gly Asp Tyr Leu Asn Asp Glu Ala Leu Trp Asn
130 135 140

Lys Cys Arg Glu Pro Leu Asn Val Val Pro Trp Asn Leu Thr Leu Phe
145 150 155 160

Ser Ile Leu Leu Val Val Gly Gly Ile Gln Met Val Leu Cys Ala Ile
165 170 175

Gln Val Val Asn Gly Leu Leu Gly Thr Leu Cys Gly Asp Cys Gln Cys
180 185 190

Cys Gly Cys Cys Gly Gly Asp Gly Pro Val
195 200

<210> SEQ ID NO 36
<211> LENGTH: 1499
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 36

Met Glu Arg Glu Pro Ala Gly Thr Glu Glu Pro Gly Pro Pro Gly Arg
1 5 10 15

Arg Arg Arg Arg Glu Gly Arg Thr Arg Thr Val Arg Ser Asn Leu Leu
20 25 30

Pro Pro Pro Gly Ala Glu Asp Pro Ala Ala Gly Ala Ala Lys Gly Glu
35 40 45

Arg Arg Arg Arg Arg Gly Cys Ala Gln His Leu Ala Asp Asn Arg Leu
50 55 60

Lys Thr Thr Lys Tyr Thr Leu Leu Ser Phe Leu Pro Lys Asn Leu Phe
65 70 75 80

Glu Gln Phe His Arg Pro Ala Asn Val Tyr Phe Val Phe Ile Ala Leu
85 90 95

Leu Asn Phe Val Pro Ala Val Asn Ala Phe Gln Pro Gly Leu Ala Leu
100 105 110

Ala Pro Val Leu Phe Ile Leu Ala Ile Thr Ala Phe Arg Asp Leu Trp

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115	120	125
Glu Asp Tyr Ser Arg His Arg Ser Asp His Lys Ile Asn His Leu Gly		
130	135	140
Cys Leu Val Phe Ser Arg Glu Glu Lys Lys Tyr Val Asn Arg Phe Trp		
145	150	155
Lys Glu Ile His Val Gly Asp Phe Val Arg Leu Arg Cys Asn Glu Ile		
165	170	175
Phe Pro Ala Asp Ile Leu Leu Leu Ser Ser Ser Asp Pro Asp Gly Leu		
180	185	190
Cys His Ile Glu Thr Ala Asn Leu Asp Gly Glu Thr Asn Leu Lys Arg		
195	200	205
Arg Gln Val Val Arg Gly Phe Ser Glu Leu Val Ser Glu Phe Asn Pro		
210	215	220
Leu Thr Phe Thr Ser Val Ile Glu Cys Glu Lys Pro Asn Asn Asp Leu		
225	230	235
Ser Arg Phe Arg Gly Cys Ile Ile His Asp Asn Gly Lys Lys Ala Gly		
245	250	255
Leu Tyr Lys Glu Asn Leu Leu Leu Arg Gly Cys Thr Leu Arg Asn Thr		
260	265	270
Asp Ala Val Val Gly Ile Val Ile Tyr Ala Gly His Glu Thr Lys Ala		
275	280	285
Leu Leu Asn Asn Ser Gly Pro Arg Tyr Lys Arg Ser Lys Leu Glu Arg		
290	295	300
Gln Met Asn Cys Asp Val Leu Trp Cys Val Leu Leu Val Cys Met		
305	310	315
Ser Leu Phe Ser Ala Val Gly His Gly Leu Trp Ile Trp Arg Tyr Gln		
325	330	335
Glu Lys Lys Ser Leu Phe Tyr Val Pro Lys Ser Asp Gly Ser Ser Leu		
340	345	350
Ser Pro Val Thr Ala Ala Val Tyr Ser Phe Leu Thr Met Ile Ile Val		
355	360	365
Leu Gln Val Leu Ile Pro Ile Ser Leu Tyr Val Ser Ile Glu Ile Val		
370	375	380
Lys Ala Cys Gln Val Tyr Phe Ile Asn Gln Asp Met Gln Leu Tyr Asp		
385	390	395
Glu Glu Thr Asp Ser Gln Leu Gln Cys Arg Ala Leu Asn Ile Thr Glu		
405	410	415
Asp Leu Gly Gln Ile Gln Tyr Ile Phe Ser Asp Lys Thr Gly Thr Leu		
420	425	430
Thr Glu Asn Lys Met Val Phe Arg Arg Cys Thr Val Ser Gly Val Glu		
435	440	445
Tyr Ser His Asp Ala Asn Ala Gln Arg Leu Ala Arg Tyr Gln Glu Ala		
450	455	460
Asp Ser Glu Glu Glu Glu Val Val Pro Arg Gly Gly Ser Val Ser Gln		
465	470	475
Arg Gly Ser Ile Gly Ser His Gln Ser Val Arg Val Val His Arg Thr		
485	490	495
Gln Ser Thr Lys Ser His Arg Arg Thr Gly Ser Arg Ala Glu Ala Lys		
500	505	510
Arg Ala Ser Met Leu Ser Lys His Thr Ala Phe Ser Ser Pro Met Glu		
515	520	525

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Lys	Asp	Ile	Thr	Pro	Asp	Pro	Lys	Leu	Leu	Glu	Lys	Val	Ser	Glu	Cys
530						535					540				
Asp	Lys	Ser	Leu	Ala	Val	Ala	Arg	His	Gln	Glu	His	Leu	Leu	Ala	His
545					550					555					560
Leu	Ser	Pro	Glu	Leu	Ser	Asp	Val	Phe	Asp	Phe	Phe	Ile	Ala	Leu	Thr
			565						570						575
Ile	Cys	Asn	Thr	Val	Val	Val	Thr	Ser	Pro	Asp	Gln	Pro	Arg	Thr	Lys
			580					585					590		
Val	Arg	Val	Arg	Phe	Glu	Leu	Lys	Ser	Pro	Val	Lys	Thr	Ile	Glu	Asp
		595					600					605			
Phe	Leu	Arg	Arg	Phe	Thr	Pro	Ser	Cys	Leu	Thr	Ser	Gly	Cys	Ser	Ser
610						615					620				
Ile	Gly	Ser	Leu	Ala	Ala	Asn	Lys	Ser	Ser	His	Lys	Leu	Gly	Ser	Ser
625					630					635					640
Phe	Pro	Ser	Thr	Pro	Ser	Ser	Asp	Gly	Met	Leu	Leu	Arg	Leu	Glu	Glu
			645					650							655
Arg	Leu	Gly	Gln	Pro	Thr	Ser	Ala	Ile	Ala	Ser	Asn	Gly	Tyr	Ser	Ser
			660				665						670		
Gln	Ala	Asp	Asn	Trp	Ala	Ser	Glu	Leu	Ala	Gln	Glu	Gln	Glu	Ser	Glu
		675					680					685			
Arg	Glu	Leu	Arg	Tyr	Glu	Ala	Glu	Ser	Pro	Asp	Glu	Ala	Ala	Leu	Val
	690					695					700				
Tyr	Ala	Ala	Arg	Ala	Tyr	Asn	Cys	Val	Leu	Val	Glu	Arg	Leu	His	Asp
705					710					715					720
Gln	Val	Ser	Val	Glu	Leu	Pro	His	Leu	Gly	Arg	Leu	Thr	Phe	Glu	Leu
			725					730						735	
Leu	His	Thr	Leu	Gly	Phe	Asp	Ser	Val	Arg	Lys	Arg	Met	Ser	Val	Val
		740					745						750		
Ile	Arg	His	Pro	Leu	Thr	Asp	Glu	Ile	Asn	Val	Tyr	Thr	Lys	Gly	Ala
		755					760					765			
Asp	Ser	Val	Val	Met	Asp	Leu	Leu	Gln	Pro	Cys	Ser	Ser	Val	Asp	Ala
	770					775					780				
Arg	Gly	Arg	His	Gln	Lys	Lys	Ile	Arg	Ser	Lys	Thr	Gln	Asn	Tyr	Leu
785					790					795					800
Asn	Val	Tyr	Ala	Ala	Glu	Gly	Leu	Arg	Thr	Leu	Cys	Ile	Ala	Lys	Arg
			805					810						815	
Val	Leu	Ser	Lys	Glu	Glu	Tyr	Ala	Cys	Trp	Leu	Gln	Ser	His	Leu	Glu
			820					825					830		
Ala	Glu	Ser	Ser	Leu	Glu	Asn	Ser	Glu	Glu	Leu	Leu	Phe	Gln	Ser	Ala
		835					840					845			
Ile	Arg	Leu	Glu	Thr	Asn	Leu	His	Leu	Leu	Gly	Ala	Thr	Gly	Ile	Glu
	850					855					860				
Asp	Arg	Leu	Gln	Asp	Gly	Val	Pro	Glu	Thr	Ile	Ser	Lys	Leu	Arg	Gln
865					870					875					880
Ala	Gly	Leu	Gln	Ile	Trp	Val	Leu	Thr	Gly	Asp	Lys	Gln	Glu	Thr	Ala
			885						890						895
Val	Asn	Ile	Ala	Tyr	Ala	Cys	Lys	Leu	Leu	Asp	His	Asp	Glu	Glu	Val
		900						905					910		
Ile	Thr	Leu	Asn	Ala	Thr	Ser	Gln	Glu	Ala	Cys	Ala	Ala	Leu	Leu	Asp
		915					920						925		

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Gln	Cys	Leu	Cys	Tyr	Val	Gln	Ser	Arg	Gly	Leu	Gln	Arg	Ala	Pro	Glu
930						935					940				
Lys	Thr	Lys	Gly	Lys	Val	Ser	Met	Arg	Phe	Ser	Ser	Leu	Cys	Pro	Pro
945					950					955					960
Ser	Thr	Ser	Thr	Ala	Ser	Gly	Arg	Arg	Pro	Ser	Leu	Val	Ile	Asp	Gly
				965					970					975	
Arg	Ser	Leu	Ala	Tyr	Ala	Leu	Glu	Lys	Asn	Leu	Glu	Asp	Lys	Phe	Leu
			980					985					990		
Phe	Leu	Ala	Lys	Gln	Cys	Arg	Ser	Val	Leu	Cys	Cys	Arg	Ser	Thr	Pro
		995					1000					1005			
Leu	Gln	Lys	Ser	Met	Val	Val	Lys	Leu	Val	Arg	Ser	Lys	Leu	Lys	Ala
1010						1015					1020				
Met	Thr	Leu	Ala	Ile	Gly	Asp	Gly	Ala	Asn	Asp	Val	Ser	Met	Ile	Gln
1025					1030					1035					1040
Val	Ala	Asp	Val	Gly	Val	Gly	Ile	Ser	Gly	Gln	Glu	Gly	Met	Gln	Ala
				1045					1050					1055	
Val	Met	Ala	Ser	Asp	Phe	Ala	Val	Pro	Lys	Phe	Arg	Tyr	Leu	Glu	Arg
			1060					1065					1070		
Leu	Leu	Ile	Leu	His	Gly	His	Trp	Cys	Tyr	Ser	Arg	Leu	Ala	Asn	Met
		1075					1080					1085			
Val	Leu	Tyr	Phe	Phe	Tyr	Lys	Asn	Thr	Met	Phe	Val	Gly	Leu	Leu	Phe
		1090				1095						1100			
Trp	Phe	Gln	Phe	Phe	Cys	Gly	Phe	Ser	Ala	Ser	Thr	Met	Ile	Asp	Gln
1105					1110						1115				1120
Trp	Tyr	Leu	Ile	Phe	Phe	Asn	Leu	Leu	Phe	Ser	Ser	Leu	Pro	Pro	Leu
				1125					1130					1135	
Val	Thr	Gly	Val	Leu	Asp	Arg	Asp	Val	Pro	Ala	Asn	Val	Leu	Leu	Thr
			1140					1145					1150		
Asn	Pro	Gln	Leu	Tyr	Lys	Ser	Gly	Gln	Asn	Met	Glu	Glu	Tyr	Arg	Pro
		1155					1160					1165			
Arg	Thr	Phe	Trp	Phe	Asn	Met	Ala	Asp	Ala	Ala	Phe	Gln	Ser	Leu	Val
		1170				1175					1180				
Cys	Phe	Ser	Ile	Pro	Tyr	Leu	Ala	Tyr	Tyr	Asp	Ser	Asn	Val	Asp	Leu
1185					1190					1195					1200
Phe	Thr	Trp	Gly	Thr	Pro	Ile	Val	Thr	Ile	Ala	Leu	Leu	Thr	Phe	Leu
				1205					1210					1215	
Leu	His	Leu	Gly	Ile	Glu	Thr	Lys	Thr	Trp	Thr	Trp	Leu	Asn	Trp	Ile
			1220					1225					1230		
Thr	Cys	Gly	Phe	Ser	Val	Leu	Leu	Phe	Phe	Thr	Val	Ala	Leu	Ile	Tyr
		1235					1240					1245			
Asn	Ala	Ser	Cys	Ala	Thr	Cys	Tyr	Pro	Pro	Ser	Asn	Pro	Tyr	Trp	Thr
		1250				1255					1260				
Met	Gln	Ala	Leu	Leu	Gly	Asp	Pro	Val	Phe	Tyr	Leu	Thr	Cys	Leu	Met
1265					1270					1275					1280
Thr	Pro	Val	Ala	Ala	Leu	Leu	Pro	Arg	Leu	Phe	Phe	Arg	Ser	Leu	Gln
			1285						1290					1295	
Gly	Arg	Val	Phe	Pro	Thr	Gln	Leu	Gln	Leu	Ala	Arg	Gln	Leu	Thr	Arg
			1300					1305					1310		
Lys	Ser	Pro	Arg	Arg	Cys	Ser	Ala	Pro	Lys	Glu	Thr	Phe	Ala	Gln	Gly
		1315					1320					1325			
Arg	Leu	Pro	Lys	Asp	Ser	Gly	Thr	Glu	His	Ser	Ser	Gly	Arg	Thr	Val

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1330	1335	1340
Lys Thr Ser Val Pro Leu Ser Gln Pro Ser Trp His Thr Gln Gln Pro		
1345	1350	1355 1360
Val Cys Ser Leu Glu Ala Ser Gly Glu Pro Ser Thr Val Asp Met Ser		
	1365	1370 1375
Met Pro Val Arg Glu His Thr Leu Leu Glu Gly Leu Ser Ala Pro Ala		
	1380	1385 1390
Pro Met Ser Ser Ala Pro Gly Glu Ala Val Leu Arg Ser Pro Gly Gly		
	1395	1400 1405
Cys Pro Glu Glu Ser Lys Val Arg Ala Ala Ser Thr Gly Arg Val Thr		
	1410	1415 1420
Pro Leu Ser Ser Leu Phe Ser Leu Pro Thr Phe Ser Leu Leu Asn Trp		
1425	1430	1435 1440
Ile Ser Ser Trp Ser Leu Val Ser Arg Leu Gly Ser Val Leu Gln Phe		
	1445	1450 1455
Ser Arg Thr Glu Gln Leu Ala Asp Gly Gln Ala Gly Arg Gly Leu Pro		
	1460	1465 1470
Val Gln Pro His Ser Gly Arg Ser Gly Leu Gln Gly Pro Asp His Arg		
	1475	1480 1485
Leu Leu Ile Gly Ala Ser Ser Arg Arg Ser Gln		
1490	1495	

<210> SEQ ID NO 37

<211> LENGTH: 80

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 37

Met Ser Gln Gln Lys Gln Gln Ser Trp Lys Pro Pro Asn Val Pro Lys
1 5 10 15
Cys Ser Pro Pro Gln Arg Ser Asn Pro Cys Leu Ala Pro Tyr Ser Thr
20 25 30
Pro Cys Gly Ala Pro His Ser Glu Gly Cys His Ser Ser Ser Gln Arg
35 40 45
Pro Glu Val Gln Lys Pro Arg Arg Ala Arg Gln Lys Leu Arg Cys Leu
50 55 60
Ser Arg Gly Thr Thr Tyr His Cys Lys Glu Glu Glu Cys Glu Gly Asp
65 70 75 80

<210> SEQ ID NO 38

<211> LENGTH: 227

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 38

Met Ala Pro Tyr His Ile Arg Lys Tyr Gln Glu Ser Asp Arg Lys Ser
1 5 10 15
Val Val Gly Leu Leu Ser Gly Gly Met Ala Glu His Ala Pro Ala Thr
20 25 30
Phe Arg Arg Leu Leu Lys Leu Pro Arg Thr Leu Ile Leu Leu Gly
35 40 45
Gly Ala Leu Ala Leu Leu Val Ser Gly Ser Trp Ile Leu Ala Leu
50 55 60
Val Phe Ser Leu Ser Leu Leu Pro Ala Leu Trp Phe Leu Ala Lys Lys

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65		70		75		80									
Pro	Trp	Thr	Arg	Tyr	Val	Asp	Ile	Ala	Leu	Arg	Thr	Asp	Met	Ser	Asp
			85						90					95	
Ile	Thr	Lys	Ser	Tyr	Leu	Ser	Glu	Cys	Gly	Ser	Cys	Phe	Trp	Val	Ala
		100						105					110		
Glu	Ser	Glu	Glu	Lys	Val	Val	Gly	Thr	Val	Gly	Ala	Leu	Pro	Val	Asp
		115					120					125			
Asp	Pro	Thr	Leu	Arg	Glu	Lys	Arg	Leu	Gln	Leu	Phe	His	Leu	Ser	Val
	130					135					140				
Asp	Asn	Glu	His	Arg	Gly	Gln	Gly	Ile	Ala	Lys	Ala	Leu	Val	Arg	Thr
145				150					155					160	
Val	Leu	Gln	Phe	Ala	Arg	Asp	Gln	Gly	Tyr	Ser	Glu	Val	Val	Leu	Asp
		165						170						175	
Thr	Ser	Asn	Ile	Gln	Leu	Ser	Ala	Met	Gly	Leu	Tyr	Gln	Ser	Leu	Gly
		180						185					190		
Phe	Lys	Lys	Thr	Gly	Gln	Ser	Phe	Phe	His	Val	Trp	Ala	Arg	Leu	Val
	195						200					205			
Asp	Leu	His	Thr	Val	His	Phe	Ile	Tyr	His	Leu	Pro	Ser	Ala	Gln	Ala
	210					215					220				
Gly	Arg	Leu													
225															

<210> SEQ ID NO 39

<211> LENGTH: 150

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 39

Met	Ser	Gly	Glu	Leu	Ser	Asn	Arg	Phe	Gln	Gly	Gly	Lys	Ala	Phe	Gly
1				5					10					15	
Leu	Leu	Lys	Ala	Arg	Gln	Glu	Arg	Arg	Leu	Ala	Glu	Ile	Asn	Arg	Glu
		20						25					30		
Phe	Leu	Cys	Asp	Gln	Lys	Tyr	Ser	Asp	Glu	Glu	Asn	Leu	Pro	Glu	Lys
		35					40					45			
Leu	Thr	Ala	Phe	Lys	Glu	Lys	Tyr	Met	Glu	Phe	Asp	Leu	Asn	Asn	Glu
	50					55					60				
Gly	Glu	Ile	Asp	Leu	Met	Ser	Leu	Lys	Arg	Met	Met	Glu	Lys	Leu	Gly
65				70					75					80	
Val	Pro	Lys	Thr	His	Leu	Glu	Met	Lys	Lys	Met	Ile	Ser	Glu	Val	Thr
			85						90					95	
Gly	Gly	Val	Ser	Asp	Thr	Ile	Ser	Tyr	Arg	Asp	Phe	Val	Asn	Met	Met
		100						105					110		
Leu	Gly	Lys	Arg	Ser	Ala	Val	Leu	Lys	Leu	Val	Met	Met	Phe	Glu	Gly
		115					120					125			
Lys	Ala	Asn	Glu	Ser	Ser	Pro	Lys	Pro	Val	Gly	Pro	Pro	Pro	Glu	Arg
	130					135					140				
Asp	Ile	Ala	Ser	Leu	Pro										
145					150										

<210> SEQ ID NO 40

<211> LENGTH: 201

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 40

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Met Thr Glu Lys Ala Pro Glu Pro His Val Glu Glu Asp Asp Asp Asp
1      5              10              15
Glu Leu Asp Ser Lys Leu Asn Tyr Lys Pro Pro Pro Gln Lys Ser Leu
20      25              30
Lys Glu Leu Gln Glu Met Asp Lys Asp Asp Glu Ser Leu Ile Lys Tyr
35      40              45
Lys Lys Thr Leu Leu Gly Asp Gly Pro Val Val Thr Asp Pro Lys Ala
50      55              60
Pro Asn Val Val Val Thr Arg Leu Thr Leu Val Cys Glu Ser Ala Pro
65      70              75              80
Gly Pro Ile Thr Met Asp Leu Thr Gly Asp Leu Glu Ala Leu Lys Lys
85      90              95
Glu Thr Ile Val Leu Lys Glu Gly Ser Glu Tyr Arg Val Lys Ile His
100     105            110
Phe Lys Val Asn Arg Asp Ile Val Ser Gly Leu Lys Tyr Val Gln His
115     120            125
Thr Tyr Arg Thr Gly Val Lys Val Asp Lys Ala Thr Phe Met Val Gly
130     135            140
Ser Tyr Gly Pro Arg Pro Glu Glu Tyr Glu Phe Leu Thr Pro Val Glu
145     150            155            160
Glu Ala Pro Lys Gly Met Leu Ala Arg Gly Thr Tyr His Asn Lys Ser
165     170            175
Phe Phe Thr Asp Asp Asp Lys Gln Asp His Leu Ser Trp Glu Trp Asn
180     185            190
Leu Ser Ile Lys Lys Glu Trp Thr Glu
195     200

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<210> SEQ ID NO 41

<211> LENGTH: 1417

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 41

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Met Ala Ala Val Pro Gln Asn Asn Leu Gln Glu Gln Leu Glu Arg His
1      5              10              15
Ser Ala Arg Thr Leu Asn Asn Lys Leu Ser Leu Ser Lys Pro Lys Phe
20      25              30
Ser Gly Phe Thr Phe Lys Lys Lys Thr Ser Ser Asp Asn Asn Val Ser
35      40              45
Val Thr Asn Val Ser Val Ala Lys Thr Pro Val Leu Arg Asn Lys Asp
50      55              60
Val Asn Val Thr Glu Asp Phe Ser Phe Ser Glu Pro Leu Pro Asn Thr
65      70              75              80
Thr Asn Gln Gln Arg Val Lys Asp Phe Phe Lys Asn Ala Pro Ala Gly
85      90              95
Gln Glu Thr Gln Arg Gly Gly Ser Lys Ser Leu Leu Pro Asp Phe Leu
100     105            110
Gln Thr Pro Lys Glu Val Val Cys Thr Thr Gln Asn Thr Pro Thr Val
115     120            125
Lys Lys Ser Arg Asp Thr Ala Leu Lys Lys Leu Glu Phe Ser Ser Ser
130     135            140

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Pro	Asp	Ser	Leu	Ser	Thr	Ile	Asn	Asp	Trp	Asp	Asp	Met	Asp	Asp	Phe	145	150	155	160
Asp	Thr	Ser	Glu	Thr	Ser	Lys	Ser	Phe	Val	Thr	Pro	Pro	Gln	Ser	His	165	170	175	
Phe	Val	Arg	Val	Ser	Thr	Ala	Gln	Lys	Ser	Lys	Lys	Gly	Lys	Arg	Asn	180	185	190	
Phe	Phe	Lys	Ala	Gln	Leu	Tyr	Thr	Thr	Asn	Thr	Val	Lys	Thr	Asp	Leu	195	200	205	
Pro	Pro	Pro	Ser	Ser	Glu	Ser	Glu	Gln	Ile	Asp	Leu	Thr	Glu	Glu	Gln	210	215	220	
Lys	Asp	Asp	Ser	Glu	Trp	Leu	Ser	Ser	Asp	Val	Ile	Cys	Ile	Asp	Asp	225	230	235	240
Gly	Pro	Ile	Ala	Glu	Val	His	Ile	Asn	Glu	Asp	Ala	Gln	Glu	Ser	Asp	245	250	255	
Ser	Leu	Lys	Thr	His	Leu	Glu	Asp	Glu	Arg	Asp	Asn	Ser	Glu	Lys	Lys	260	265	270	
Lys	Asn	Leu	Glu	Glu	Ala	Glu	Leu	His	Ser	Thr	Glu	Lys	Val	Pro	Cys	275	280	285	
Ile	Glu	Phe	Asp	Asp	Asp	Asp	Tyr	Asp	Thr	Asp	Phe	Val	Pro	Pro	Ser	290	295	300	
Pro	Glu	Glu	Ile	Ile	Ser	Ala	Ser	Ser	Ser	Ser	Ser	Lys	Cys	Leu	Ser	305	310	315	320
Thr	Leu	Lys	Asp	Leu	Asp	Thr	Ser	Asp	Arg	Lys	Glu	Asp	Val	Leu	Ser	325	330	335	
Thr	Ser	Lys	Asp	Leu	Leu	Ser	Lys	Pro	Glu	Lys	Met	Ser	Met	Gln	Glu	340	345	350	
Leu	Asn	Pro	Glu	Thr	Ser	Thr	Asp	Cys	Asp	Ala	Arg	Gln	Ile	Ser	Leu	355	360	365	
Gln	Gln	Gln	Leu	Ile	His	Val	Met	Glu	His	Ile	Cys	Lys	Leu	Ile	Asp	370	375	380	
Thr	Ile	Pro	Asp	Asp	Lys	Leu	Lys	Leu	Leu	Asp	Cys	Gly	Asn	Glu	Leu	385	390	395	400
Leu	Gln	Gln	Arg	Asn	Ile	Arg	Arg	Lys	Leu	Leu	Thr	Glu	Val	Asp	Phe	405	410	415	
Asn	Lys	Ser	Asp	Ala	Ser	Leu	Leu	Gly	Ser	Leu	Trp	Arg	Tyr	Arg	Pro	420	425	430	
Asp	Ser	Leu	Asp	Gly	Pro	Met	Glu	Gly	Asp	Ser	Cys	Pro	Thr	Gly	Asn	435	440	445	
Ser	Met	Lys	Glu	Leu	Asn	Phe	Ser	His	Leu	Pro	Ser	Asn	Ser	Val	Ser	450	455	460	
Pro	Gly	Asp	Cys	Leu	Leu	Thr	Thr	Thr	Leu	Gly	Lys	Thr	Gly	Phe	Ser	465	470	475	480
Ala	Thr	Arg	Lys	Asn	Leu	Phe	Glu	Arg	Pro	Leu	Phe	Asn	Thr	His	Leu	485	490	495	
Gln	Lys	Ser	Phe	Val	Ser	Ser	Asn	Trp	Ala	Glu	Thr	Pro	Arg	Leu	Gly	500	505	510	
Lys	Lys	Asn	Glu	Ser	Ser	Tyr	Phe	Pro	Gly	Asn	Val	Leu	Thr	Ser	Thr	515	520	525	
Ala	Val	Lys	Asp	Gln	Asn	Lys	His	Thr	Ala	Ser	Ile	Asn	Asp	Leu	Glu	530	535	540	
Arg	Glu	Thr	Gln	Pro	Ser	Tyr	Asp	Ile	Asp	Asn	Phe	Asp	Ile	Asp	Asp				

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545	550							555							560		
Phe	Asp	Asp	Asp	Asp	565	Trp	Glu	Asp	Ile	Met	His	Asn	Leu	Ala	Ala		
									570					575			
Ser	Lys	Ser	Ser	Thr	Ala	Ala	Tyr	Gln	Pro	Ile	Lys	Glu	Gly	Arg	Pro		
								585					590				
Ile	Lys	Ser	Val	Ser	Glu	Arg	Leu	Ser	Ser	Ala	Lys	Thr	Asp	Cys	Leu		
			595				600					605					
Pro	Val	Ser	Ser	Thr	Ala	Gln	Asn	Ile	Asn	Phe	Ser	Glu	Ser	Ile	Gln		
	610					615					620						
Asn	Tyr	Thr	Asp	Lys	Ser	Ala	Gln	Asn	Leu	Ala	Ser	Arg	Asn	Leu	Lys		
625					630					635					640		
His	Glu	Arg	Phe	Gln	Ser	Leu	Ser	Phe	Pro	His	Thr	Lys	Glu	Met	Met		
				645					650					655			
Lys	Ile	Phe	His	Lys	Lys	Phe	Gly	Leu	His	Asn	Phe	Arg	Thr	Asn	Gln		
			660					665					670				
Leu	Glu	Ala	Ile	Asn	Ala	Ala	Leu	Leu	Gly	Glu	Asp	Cys	Phe	Ile	Leu		
		675					680					685					
Met	Pro	Thr	Gly	Gly	Gly	Lys	Ser	Leu	Cys	Tyr	Gln	Leu	Pro	Ala	Cys		
	690					695					700						
Val	Ser	Pro	Gly	Val	Thr	Val	Val	Ile	Ser	Pro	Leu	Arg	Ser	Leu	Ile		
705					710					715					720		
Val	Asp	Gln	Val	Gln	Lys	Leu	Thr	Ser	Leu	Asp	Ile	Pro	Ala	Thr	Tyr		
			725						730					735			
Leu	Thr	Gly	Asp	Lys	Thr	Asp	Ser	Glu	Ala	Thr	Asn	Ile	Tyr	Leu	Gln		
			740					745					750				
Leu	Ser	Lys	Lys	Asp	Pro	Ile	Ile	Lys	Leu	Leu	Tyr	Val	Thr	Pro	Glu		
		755					760					765					
Lys	Ile	Cys	Ala	Ser	Asn	Arg	Leu	Ile	Ser	Thr	Leu	Glu	Asn	Leu	Tyr		
	770					775					780						
Glu	Arg	Lys	Leu	Leu	Ala	Arg	Phe	Val	Ile	Asp	Glu	Ala	His	Cys	Val		
785					790					795					800		
Ser	Gln	Trp	Gly	His	Asp	Phe	Arg	Gln	Asp	Tyr	Lys	Arg	Met	Asn	Met		
				805					810					815			
Leu	Arg	Gln	Lys	Phe	Pro	Ser	Val	Pro	Val	Met	Ala	Leu	Thr	Ala	Thr		
			820					825					830				
Ala	Asn	Pro	Arg	Val	Gln	Lys	Asp	Ile	Leu	Thr	Gln	Leu	Lys	Ile	Leu		
		835					840					845					
Arg	Pro	Gln	Val	Phe	Ser	Met	Ser	Phe	Asn	Arg	His	Asn	Leu	Lys	Tyr		
	850					855					860						
Tyr	Val	Leu	Pro	Lys	Lys	Pro	Lys	Lys	Val	Ala	Phe	Asp	Cys	Leu	Glu		
865				870						875					880		
Trp	Ile	Arg	Lys	His	His	Pro	Tyr	Asp	Ser	Gly	Ile	Ile	Tyr	Cys	Leu		
				885				890						895			
Ser	Arg	Arg	Glu	Cys	Asp	Thr	Met	Ala	Asp	Thr	Leu	Gln	Arg	Asp	Gly		
			900					905					910				
Leu	Ala	Ala	Leu	Ala	Tyr	His	Ala	Gly	Leu	Ser	Asp	Ser	Ala	Arg	Asp		
		915					920					925					
Glu	Val	Gln	Gln	Lys	Trp	Ile	Asn	Gln	Asp	Gly	Cys	Gln	Val	Ile	Cys		
	930					935					940						
Ala	Thr	Ile	Ala	Phe	Gly	Met	Gly	Ile	Asp	Lys	Pro	Asp	Val	Arg	Phe		
945					950					955					960		

Val	Ile	His	Ala	Ser	Leu	Pro	Lys	Ser	Val	Glu	Gly	Tyr	Tyr	Gln	Glu	
				965					970					975		
Ser	Gly	Arg	Ala	Gly	Arg	Asp	Gly	Glu	Ile	Ser	His	Cys	Leu	Leu	Phe	
				980					985					990		
Tyr	Thr	Tyr	His	Asp	Val	Thr	Arg	Leu	Lys	Arg	Leu	Ile	Met	Met	Glu	
				995					1000					1005		
Lys	Asp	Gly	Asn	His	His	Thr	Arg	Glu	Thr	His	Phe	Asn	Asn	Leu	Tyr	
				1010					1015					1020		
Ser	Met	Val	His	Tyr	Cys	Glu	Asn	Ile	Thr	Glu	Cys	Arg	Arg	Ile	Gln	
				1025					1030					1035		
Leu	Leu	Ala	Tyr	Phe	Gly	Glu	Asn	Gly	Phe	Asn	Pro	Asp	Phe	Cys	Lys	
				1045					1050					1055		
Lys	His	Pro	Asp	Val	Ser	Cys	Asp	Asn	Cys	Cys	Lys	Thr	Lys	Asp	Tyr	
				1060					1065					1070		
Lys	Thr	Arg	Asp	Val	Thr	Asp	Asp	Val	Lys	Ser	Ile	Val	Arg	Phe	Val	
				1075					1080					1085		
Gln	Glu	His	Ser	Ser	Ser	Gln	Gly	Met	Arg	Asn	Ile	Lys	His	Val	Gly	
				1090					1095					1100		
Pro	Ser	Gly	Arg	Phe	Thr	Met	Asn	Met	Leu	Val	Asp	Ile	Phe	Leu	Gly	
				1105					1110					1115		
Ser	Lys	Ser	Ala	Lys	Ile	Gln	Ser	Gly	Ile	Phe	Gly	Lys	Gly	Ser	Ala	
				1125					1130					1135		
Tyr	Ser	Arg	His	Asn	Ala	Glu	Arg	Leu	Phe	Lys	Lys	Leu	Ile	Leu	Asp	
				1140					1145					1150		
Lys	Ile	Leu	Asp	Glu	Asp	Leu	Tyr	Ile	Asn	Ala	Asn	Asp	Gln	Ala	Ile	
				1155					1160					1165		
Ala	Tyr	Val	Met	Leu	Gly	Asn	Lys	Ala	Gln	Thr	Val	Leu	Asn	Gly	Asn	
				1170					1175					1180		
Leu	Lys	Val	Asp	Phe	Met	Glu	Thr	Glu	Asn	Ser	Ser	Ser	Val	Lys	Lys	
				1185					1190					1195		
Gln	Lys	Ala	Leu	Val	Ala	Lys	Val	Ser	Gln	Arg	Glu	Glu	Met	Val	Lys	
				1205					1210					1215		
Lys	Cys	Leu	Gly	Glu	Leu	Thr	Glu	Val	Cys	Lys	Ser	Leu	Gly	Lys	Val	
				1220					1225					1230		
Phe	Gly	Val	His	Tyr	Phe	Asn	Ile	Phe	Asn	Thr	Val	Thr	Leu	Lys	Lys	
				1235					1240					1245		
Leu	Ala	Glu	Ser	Leu	Ser	Ser	Asp	Pro	Glu	Val	Leu	Leu	Gln	Ile	Asp	
				1250					1255					1260		
Gly	Val	Thr	Glu	Asp	Lys	Leu	Glu	Lys	Tyr	Gly	Ala	Glu	Val	Ile	Ser	
				1265					1270					1275		
Val	Leu	Gln	Lys	Tyr	Ser	Glu	Trp	Thr	Ser	Pro	Ala	Glu	Asp	Ser	Ser	
				1285					1290					1295		
Pro	Gly	Ile	Ser	Leu	Ser	Ser	Ser	Arg	Gly	Pro	Gly	Arg	Ser	Ala	Ala	
				1300					1305					1310		
Glu	Glu	Leu	Asp	Glu	Glu	Ile	Pro	Val	Ser	Ser	His	Tyr	Phe	Ala	Ser	
				1315					1320					1325		
Lys	Thr	Arg	Asn	Glu	Arg	Lys	Arg	Lys	Lys	Met	Pro	Ala	Ser	Gln	Arg	
				1330					1335					1340		
Ser	Lys	Arg	Arg	Lys	Thr	Ala	Ser	Ser	Gly	Ser	Lys	Ala	Lys	Gly	Gly	
				1345					1350					1355		

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Ser Ala Thr Cys Arg Lys Ile Ser Ser Lys Thr Lys Ser Ser Ser Ile
      1365                      1370                      1375

Ile Gly Ser Ser Ser Ala Ser His Thr Ser Gln Ala Thr Ser Gly Ala
      1380                      1385                      1390

Asn Ser Lys Leu Gly Ile Met Ala Pro Pro Lys Pro Ile Asn Arg Pro
      1395                      1400                      1405

Phe Leu Lys Pro Ser Tyr Ala Phe Ser
      1410                      1415

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<210> SEQ ID NO 42
<211> LENGTH: 648
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 42

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Met Glu Ala Lys Phe Leu Ala Ser Ala Leu Ser Phe Leu Ser Ile Phe
1      5      10      15

Leu Ala Ile Tyr Ala Gln Ser Leu Ala Asn Asp Leu Ser Lys Glu Ser
20     25     30

Thr Glu Phe Glu Glu Ser Pro Thr Ile Tyr Tyr Gly Asp Pro Val Val
35     40     45

Asn Leu Gly Gln Pro Phe Ser Ile Thr Cys Ile Ile Pro Ile Thr Asp
50     55     60

Gln Ile His Trp Leu Lys Asn Gly Glu Pro Ile Thr Arg His Asn Leu
65     70     75     80

Arg His Gly Arg Asp Asp His Ala Tyr Val Leu Ser Glu Ser Ala Ile
85     90     95

Glu Gly Glu Lys His Lys Ile Glu Ala His Leu Ser Val Arg His Ala
100    105    110

Leu Lys Val His Glu Gly Arg Tyr Gln Cys Asn Arg Arg Arg Gly Ser
115    120    125

Tyr Ile Leu His Val Arg Asp Pro Lys Gly Val Gly Ala Gly Ala Gly
130    135    140

Glu Pro Thr Glu Ser Lys Tyr Gln Thr Ile Asp Glu Leu Thr Pro Asn
145    150    155    160

Ser Ala Asp Asp Phe Phe Thr Arg Ala Trp Leu Glu Gln Gln Gln
165    170    175

Gln Gln Gln Leu Pro His Gln Ser His Lys Leu His Lys Ser His Leu
180    185    190

Gly Tyr Gly Asn Ala Ser Leu Ser Gly Ser Gln Pro Trp His Pro Ser
195    200    205

Ala Gly Gly Gly Gly Ile His Arg Val Tyr Ser Ala Thr Pro Pro Asp
210    215    220

Phe Pro Pro Pro Arg Leu Asn Leu Leu Glu Gln Thr Val Ala Pro Pro
225    230    235    240

Glu Pro Pro Thr Ile Leu Tyr Asn Pro Asn Pro Thr His Pro Thr Ala
245    250    255

Ser Ala Thr Ala Thr Glu Thr Ser Val Leu Leu Thr Thr Ala His His
260    265    270

His Ala His His Gln Gln Gln Leu Gln Gln Gln Ser Gln His Thr Leu
275    280    285

Asn Ala Phe Gln Leu Pro Leu Pro Pro Arg Pro Asn Pro Gly Gln Asn
290    295    300

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Glu Arg Tyr Gln Thr Tyr Ala Pro His Tyr Val Pro Pro Val Val Val
 305 310 315 320
 Ser Gly Ala Gly Ala Gly Ala Gly Ala Asp Pro Gly Ala Gly Ala Ser
 325 330 335
 Gly Glu Gln Thr Thr Ile Ser Ala Ala Thr Ser Thr Arg Ala Met Met
 340 345 350
 Gly Gly Gly Gly Gly Val Ala Gly Ala Gly Phe Ser Ala Gly Ala Ser
 355 360 365
 Gly Pro Met Leu Gly Ala Gly Gly His Met Leu Met Gly Gly Gln Gly
 370 375 380
 His Gln Val His Leu Gln His Gln Thr Leu Leu Pro Val Lys Met Asp
 385 390 395 400
 Lys Leu Val Pro Asn Tyr Asp Asn Ala Glu His Gln Met Lys Phe Tyr
 405 410 415
 Asp Ile Arg Ser Pro Leu Val Leu Ser Cys Asn Val Lys Asp Gly Thr
 420 425 430
 Pro Gly Gly Val Leu Ile Trp Lys Lys Asn Gly Thr Ala Val Thr Asp
 435 440 445
 Val Pro Ser Leu Arg Gly Arg Phe Lys Leu Ile Ala Asp Glu Asn Lys
 450 455 460
 Phe Ile Ile Asp Lys Thr Asp Thr Asn Asp Asp Gly Lys Tyr Ser Cys
 465 470 475 480
 Glu Phe Asp Gly Val Ser Lys Glu Ile Glu Val Ile Ala Arg Val Val
 485 490 495
 Val Arg Val Pro Ser Asn Thr Ala Val Val Glu Gly Glu Lys Met Ser
 500 505 510
 Val Thr Cys Ser Val Val Gly Thr Lys Pro Glu Leu Thr Trp Thr Phe
 515 520 525
 Ala Asn Val Thr Leu Thr Asn Ala Thr Asp Arg Phe Ile Leu Lys Pro
 530 535 540
 Asp Asp Asn Gly Val Pro Asn Ala Ile Leu Thr Leu Asp Asn Val Thr
 545 550 555 560
 Leu Asp Asp Arg Gly Glu Tyr Lys Cys Ile Gly Arg Asn Ala Ala Asn
 565 570 575
 Val Tyr Gly Gly Asn Thr Thr Thr Pro Ala Ser Asp Val Thr Thr Val
 580 585 590
 Arg Val Lys Gly Lys Phe Ala Ala Leu Trp Pro Phe Leu Gly Ile Cys
 595 600 605
 Ala Glu Val Leu Ile Leu Cys Ile Ile Ile Leu Ile Tyr Glu Lys Arg
 610 615 620
 Arg Asn Lys Ser Glu Leu Glu Glu Ser Asp Thr Asp Pro Gln Glu Gln
 625 630 635 640
 Lys Lys Lys Arg Arg Asn Tyr Asp
 645

<210> SEQ ID NO 43

<211> LENGTH: 525

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 43

Met Ala Ala Gly Gly Ser Gly Gly Arg Ala Ser Cys Pro Pro Gly Val

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1	5	10	15
Gly Val Gly	Pro Gly Thr	Gly Gly Ser	Pro Gly Pro Ser Ala Asn Ala
	20	25	30
Ala Ala Thr	Pro Ala Pro	Gly Asn Ala Ala Ala	Ala Ala Ala Ala
	35	40	45
Ala Ala Ala	Ala Ala Ala	Pro Gly Pro Thr	Pro Pro Ala Pro Pro Gly
	50	55	60
Pro Gly Thr	Asp Ala Gln	Ala Ala Gly Ala	Glu Arg Ala Glu Glu Ala
	65	70	75
Ala Gly Pro	Gly Ala Ala Ala	Leu Gln Arg	Glu Ala Ala Tyr Asn Trp
	85	90	95
Gln Ala Ser	Lys Pro Thr	Val Gln Glu Arg	Phe Ala Phe Leu Phe Asn
	100	105	110
Asn Glu Val	Leu Cys Asp	Val His Phe Leu	Val Gly Lys Gly Leu Ser
	115	120	125
Ser Gln Arg	Ile Pro Ala	His Arg Phe Val	Leu Ala Val Gly Ser Ala
	130	135	140
Val Phe Asp	Ala Met Phe	Asn Gly Gly Met	Ala Thr Thr Ser Thr Glu
	145	150	155
Ile Glu Leu	Pro Asp Val	Glu Pro Ala Ala	Phe Leu Ala Leu Leu Lys
	165	170	175
Phe Leu Tyr	Ser Asp Glu	Val Gln Ile Gly	Pro Glu Thr Val Met Thr
	180	185	190
Thr Leu Tyr	Thr Ala Lys	Lys Tyr Ala Val	Pro Ala Leu Glu Ala His
	195	200	205
Cys Val Glu	Phe Leu Lys	Lys Asn Leu Arg	Ala Asp Asn Ala Phe Met
	210	215	220
Leu Leu Thr	Gln Ala Arg	Leu Phe Asp Glu	Pro Gln Leu Ala Ser Leu
	225	230	235
Cys Leu Glu	Asn Ile Asp	Lys Asn Thr	Ala Asp Ala Ile Thr Ala Glu
	245	250	255
Gly Phe Thr	Asp Ile Asp	Leu Asp Thr	Leu Val Ala Val Leu Glu Arg
	260	265	270
Asp Thr Leu	Gly Ile Arg	Glu Val Arg Leu	Phe Asn Ala Val Val Arg
	275	280	285
Trp Ser Glu	Ala Glu Cys	Gln Arg Gln Gln	Leu Gln Val Thr Pro Glu
	290	295	300
Asn Arg Arg	Lys Val Leu	Gly Lys Ala Leu	Gly Leu Ile Arg Phe Pro
	305	310	315
Leu Met Thr	Ile Glu Glu	Phe Ala Ala Gly	Pro Ala Gln Ser Gly Ile
	325	330	335
Leu Val Asp	Arg Glu Val	Val Ser Leu Phe	Leu His Phe Thr Val Asn
	340	345	350
Pro Lys Pro	Arg Val Glu	Phe Ile Asp Arg	Pro Arg Cys Cys Leu Arg
	355	360	365
Gly Lys Glu	Cys Ser Ile	Asn Arg Phe Gln	Gln Val Glu Ser Arg Trp
	370	375	380
Gly Tyr Ser	Gly Thr Ser	Asp Arg Ile Arg	Phe Ser Val Asn Lys Arg
	385	390	395
Ile Phe Val	Val Gly Phe	Gly Leu Tyr Gly	Ser Ile His Gly Pro Thr
	405	410	415

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Asp Tyr Gln Val Asn Ile Gln Ile Ile His Thr Asp Ser Asn Thr Val
 420 425 430
 Leu Gly Gln Asn Asp Thr Gly Phe Ser Cys Asp Gly Ser Ala Ser Thr
 435 440 445
 Phe Arg Val Met Phe Lys Glu Pro Val Glu Val Leu Pro Asn Val Asn
 450 455 460
 Tyr Thr Ala Cys Ala Thr Leu Lys Gly Pro Asp Ser His Tyr Gly Thr
 465 470 475 480
 Lys Gly Leu Arg Lys Val Thr His Glu Ser Pro Thr Thr Gly Ala Lys
 485 490 495
 Thr Cys Phe Thr Phe Cys Tyr Ala Ala Gly Asn Asn Asn Gly Thr Ser
 500 505 510
 Val Glu Asp Gly Gln Ile Pro Glu Val Ile Phe Tyr Thr
 515 520 525

<210> SEQ ID NO 44

<211> LENGTH: 329

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 44

Met Glu Leu His Ile Leu Glu His Arg Leu Gln Val Ala Ser Val Ala
 1 5 10 15
 Lys Glu Ser Ile Pro Leu Phe Thr Tyr Gly Leu Ile Lys Leu Ala Phe
 20 25 30
 Leu Ser Ser Lys Thr Arg Cys Lys Phe Phe Ser Leu Thr Glu Thr Pro
 35 40 45
 Glu Asp Tyr Thr Ile Ile Val Asp Glu Glu Gly Phe Leu Glu Leu Pro
 50 55 60
 Ser Ser Glu His Leu Ser Val Ala Asp Ala Thr Trp Leu Ala Leu Asn
 65 70 75 80
 Val Val Ser Gly Gly Gly Ser Phe Ser Ser Ser Gln Pro Ile Gly Val
 85 90 95
 Thr Lys Ile Ala Lys Ser Val Ile Ala Pro Leu Ala Asp Gln Asn Ile
 100 105 110
 Ser Val Phe Met Leu Ser Thr Tyr Gln Thr Asp Phe Ile Leu Val Arg
 115 120 125
 Glu Arg Asp Leu Pro Phe Val Thr His Thr Leu Ser Ser Glu Phe Thr
 130 135 140
 Ile Leu Arg Val Val Asn Gly Glu Thr Val Ala Ala Glu Asn Leu Gly
 145 150 155 160
 Ile Thr Asn Gly Phe Val Lys Pro Lys Leu Val Gln Arg Pro Val Ile
 165 170 175
 His Pro Leu Ser Ser Pro Ser Asn Arg Phe Cys Val Thr Ser Leu Asp
 180 185 190
 Pro Asp Thr Leu Pro Ala Val Ala Thr Leu Leu Met Asp Val Met Phe
 195 200 205
 Tyr Ser Asn Gly Val Lys Asp Pro Met Ala Thr Gly Asp Asp Cys Gly
 210 215 220
 His Ile Arg Phe Phe Ser Phe Ser Leu Ile Glu Gly Tyr Ile Ser Leu
 225 230 235 240
 Val Met Asp Val Gln Thr Gln Gln Arg Phe Pro Ser Asn Leu Leu Phe

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Thr	Ser	Ala	Ser	Gly	Glu	Leu	Trp	Lys	Met	Val	Arg	Ile	Gly	Gly	Gln
			260					265					270		
Pro	Leu	Gly	Phe	Asp	Glu	Cys	Gly	Ile	Val	Ala	Gln	Ile	Ser	Glu	Pro
		275					280					285			
Leu	Ala	Ala	Ala	Asp	Ile	Pro	Ala	Tyr	Tyr	Ile	Ser	Thr	Phe	Lys	Phe
	290					295					300				
Asp	His	Ala	Leu	Val	Pro	Glu	Glu	Asn	Ile	Asn	Gly	Val	Ile	Ser	Ala
305					310					315					320
Leu	Lys	Val	Ser	Gln	Ala	Glu	Lys	His							
			325												

<210> SEQ ID NO 45

<211> LENGTH: 900

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 45

Met	Gly	Asp	Ser	Gly	Ser	Arg	Arg	Ser	Thr	Leu	Val	Ser	Arg	Leu	Pro
1				5					10					15	
Ile	Phe	Arg	Arg	Ser	Ile	Asn	Arg	Arg	His	Asp	Ser	Leu	Pro	Ser	Ser
		20					25					30			
Pro	Ser	Ser	Ser	Asn	Thr	Val	Gly	Val	His	Ser	Ser	Ser	Pro	Ser	Ser
		35				40						45			
Thr	Asn	Ser	Ser	Ser	Gly	Ser	Thr	Gly	Lys	Arg	Arg	Ser	Ile	Phe	Arg
	50					55				60					
Thr	Pro	Ser	Ile	Ser	Phe	His	His	Lys	Lys	Gly	Ser	Glu	Pro	Lys	Gln
65					70					75				80	
Glu	Pro	Thr	Asn	Gln	Asn	Leu	Ser	Ile	Ser	Asn	Gly	Ala	Gln	Pro	Gly
			85					90						95	
His	Ser	Asn	Met	Gln	Lys	Leu	Ser	Leu	Glu	Glu	His	Ile	Lys	Thr	Arg
		100						105					110		
Gly	Arg	His	Ser	Val	Gly	Phe	Ser	Ser	Ser	Arg	Asn	Lys	Lys	Ile	Thr
		115				120						125			
Arg	Ser	Leu	Thr	Glu	Asp	Phe	Glu	Arg	Glu	Lys	Glu	His	Ser	Thr	Asn
	130					135					140				
Lys	Asn	Val	Phe	Ile	Asn	Cys	Leu	Ser	Ser	Gly	Lys	Ser	Glu	Gly	Asp
145					150					155				160	
Asp	Ser	Gly	Phe	Thr	Glu	Asp	Gln	Thr	Arg	Arg	Ser	Val	Lys	Gln	Ser
			165					170					175		
Thr	Arg	Lys	Leu	Leu	Pro	Lys	Ser	Phe	Ser	Ser	His	Tyr	Lys	Phe	Ser
		180						185					190		
Lys	Pro	Val	Leu	Gln	Ser	Gln	Ser	Ile	Ser	Leu	Val	Gln	Gln	Ser	Glu
		195				200						205			
Phe	Ser	Leu	Glu	Val	Thr	Gln	Tyr	Gln	Glu	Arg	Glu	Pro	Val	Leu	Val
	210					215					220				
Arg	Ala	Ser	Pro	Ser	Cys	Ser	Val	Asp	Val	Thr	Glu	Arg	Ala	Gly	Ser
225					230					235				240	
Ser	Leu	Gln	Ser	Pro	Leu	Leu	Ser	Ala	Asp	Leu	Thr	Thr	Ala	Gln	Thr
			245					250					255		
Pro	Ser	Glu	Phe	Leu	Ala	Leu	Thr	Glu	Asp	Ser	Val	Ser	Glu	Met	Asp
		260						265					270		

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Ala	Phe	Ser	Lys	Ser	Gly	Ser	Met	Ala	Ser	His	Cys	Asp	Asn	Phe	Gly
	275						280					285			
His	Asn	Asp	Ser	Thr	Ser	Gln	Met	Ser	Leu	Asn	Ser	Ala	Ala	Val	Thr
	290					295				300					
Lys	Thr	Thr	Thr	Glu	Leu	Thr	Gly	Thr	Val	Pro	Cys	Ala	Ile	Met	Ser
305					310					315					320
Pro	Gly	Lys	Tyr	Arg	Leu	Glu	Gly	Gln	Cys	Ser	Thr	Glu	Ser	Asn	Ser
				325					330					335	
Leu	Pro	Glu	Thr	Ser	Ala	Ala	Asn	Gln	Lys	Glu	Val	Leu	Leu	Gln	Ile
			340					345					350		
Ala	Glu	Leu	Pro	Ala	Thr	Ser	Val	Ser	His	Ser	Glu	Ser	Asn	Leu	Pro
	355						360					365			
Ala	Asp	Ser	Glu	Arg	Glu	Glu	Asn	Ile	Gly	Leu	Gln	Asn	Gly	Glu	Thr
	370					375					380				
Met	Leu	Gly	Thr	Asn	Ser	Pro	Arg	Lys	Leu	Gly	Phe	Tyr	Glu	Gln	His
385					390					395					400
Lys	Ala	Ile	Ala	Glu	His	Val	Lys	Gly	Ile	His	Pro	Ile	Ser	Asp	Ser
			405						410					415	
Lys	Ile	Ile	Pro	Thr	Ser	Gly	Asp	His	His	Ile	Phe	Asn	Lys	Thr	Ser
			420					425					430		
His	Gly	Tyr	Glu	Ala	Asn	Pro	Ala	Lys	Val	Leu	Ala	Ser	Ser	Leu	Ser
	435						440					445			
Pro	Phe	Arg	Glu	Gly	Arg	Phe	Ile	Glu	Arg	Arg	Leu	Arg	Ser	Ser	Ser
	450					455					460				
Glu	Gly	Thr	Ala	Gly	Ser	Ser	Arg	Met	Ile	Leu	Lys	Pro	Lys	Asp	Gly
465					470					475					480
Asn	Ile	Glu	Glu	Val	Asn	Ser	Leu	Arg	Lys	Gln	Arg	Ala	Gly	Ser	Ser
			485						490					495	
Ser	Ser	Lys	Met	Asn	Ser	Leu	Asp	Val	Leu	Asn	Asn	Leu	Gly	Ser	Cys
			500					505					510		
Glu	Leu	Asp	Glu	Asp	Asp	Leu	Met	Leu	Asp	Leu	Glu	Phe	Leu	Glu	Glu
	515						520					525			
Gln	Ser	Leu	His	Pro	Ser	Val	Cys	Arg	Glu	Asp	Ser	Tyr	His	Ser	Val
	530					535					540				
Val	Ser	Cys	Ala	Ala	Val	Val	Leu	Thr	Pro	Met	Glu	Pro	Met	Ile	Glu
545					550					555					560
Met	Lys	Lys	Arg	Glu	Glu	Pro	Glu	Phe	Pro	Glu	Pro	Ser	Lys	Gln	Asn
			565					570						575	
Leu	Ser	Leu	Lys	Leu	Thr	Lys	Asp	Val	Asp	Gln	Glu	Ala	Arg	Cys	Ser
			580					585					590		
His	Ile	Ser	Arg	Met	Pro	Asn	Ser	Pro	Ser	Ala	Asp	Trp	Pro	Leu	Gln
	595						600					605			
Gly	Val	Glu	Glu	Asn	Gly	Gly	Ile	Asp	Ser	Leu	Pro	Phe	Arg	Leu	Met
	610				615						620				
Leu	Gln	Asp	Cys	Thr	Ala	Val	Lys	Thr	Leu	Leu	Leu	Lys	Met	Lys	Arg
625					630					635					640
Val	Leu	Gln	Glu	Ser	Ala	Asp	Met	Ser	Pro	Ala	Ser	Ser	Thr	Thr	Ser
			645					650						655	
Leu	Pro	Val	Ser	Pro	Leu	Thr	Glu	Glu	Pro	Val	Pro	Phe	Lys	Asp	Ile
		660					665					670			
Met	Lys	Asp	Glu	Cys	Ser	Met	Leu	Lys	Leu	Gln	Leu	Lys	Glu	Lys	Asp

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675	680	685
Glu Leu Ile Ser Gln Leu Gln Glu Glu Leu Gly Lys Val Arg His Leu		
690	695	700
Gln Lys Ala Phe Ala Ser Arg Val Asp Lys Ser Thr Gln Thr Glu Leu		
705	710	715
Leu Cys Tyr Asp Gly Leu Asn Leu Lys Arg Leu Glu Thr Val Gln Gly		
725	730	735
Gly Arg Glu Ala Thr Tyr Arg Asn Arg Ile Val Ser Gln Asn Leu Ser		
740	745	750
Thr Arg Asp Arg Lys Ala Ile His Thr Pro Thr Glu Asp Arg Phe Arg		
755	760	765
Tyr Ser Ala Ala Asp Gln Thr Ser Pro Tyr Lys Asn Lys Thr Cys Gln		
770	775	780
Leu Pro Ser Leu Cys Leu Ser Asn Phe Leu Lys Asp Lys Glu Leu Ala		
785	790	795
Glu Val Ile Lys His Ser Arg Gly Thr Tyr Glu Thr Leu Thr Ser Asp		
805	810	815
Val Thr Gln Asn Leu Arg Ala Thr Val Gly Gln Ser Ser Leu Lys Pro		
820	825	830
Thr Ala Lys Thr Glu Gly Leu Ser Thr Phe Leu Glu Lys Pro Lys Asp		
835	840	845
Gln Val Ala Thr Ala Arg Gln His Ser Thr Phe Thr Gly Arg Phe Gly		
850	855	860
Gln Pro Pro Arg Gly Pro Ile Ser Leu His Met Tyr Ser Arg Lys Asn		
865	870	875
Val Phe Leu His His Asn Leu His Ser Thr Glu Leu Gln Thr Leu Gly		
885	890	895
Gln Gln Asp Gly		
900		

<210> SEQ ID NO 46
 <211> LENGTH: 898
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 46

Met Ala Ala Ala Ala Ala Glu Glu Gly Met Glu Pro Arg Ala Leu Gln
1 5 10 15
Tyr Glu Gln Thr Leu Met Tyr Gly Arg Tyr Thr Gln Asp Leu Gly Ala
20 25 30
Phe Ala Lys Glu Glu Ala Ala Arg Ile Arg Leu Gly Gly Pro Glu Pro
35 40 45
Trp Lys Gly Pro Pro Ser Ser Arg Ala Ala Pro Glu Leu Leu Glu Tyr
50 55 60
Gly Arg Ser Arg Cys Ala Arg Cys Arg Val Cys Ser Val Arg Cys His
65 70 75 80
Lys Phe Leu Val Ser Arg Val Gly Glu Asp Trp Ile Phe Leu Val Leu
85 90 95
Leu Gly Leu Leu Met Ala Leu Val Ser Trp Val Met Asp Tyr Ala Ile
100 105 110
Ala Ala Cys Leu Gln Ala Gln Gln Trp Met Ser Arg Gly Leu Asn Thr
115 120 125

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Ser	Ile	Leu	Leu	Gln	Tyr	Leu	Ala	Trp	Val	Thr	Tyr	Pro	Val	Val	Leu
130						135					140				
Ile	Thr	Phe	Ser	Ala	Gly	Phe	Thr	Gln	Ile	Leu	Ala	Pro	Gln	Ala	Val
145					150					155					160
Gly	Ser	Gly	Ile	Pro	Glu	Met	Lys	Thr	Ile	Leu	Arg	Gly	Val	Val	Leu
				165					170					175	
Lys	Glu	Tyr	Leu	Thr	Leu	Lys	Thr	Phe	Ile	Ala	Lys	Val	Ile	Gly	Leu
			180					185					190		
Thr	Cys	Ala	Leu	Gly	Ser	Gly	Met	Pro	Leu	Gly	Lys	Glu	Gly	Pro	Phe
		195					200					205			
Val	His	Ile	Ala	Ser	Met	Cys	Ala	Ala	Leu	Leu	Ser	Lys	Phe	Leu	Ser
	210					215					220				
Leu	Phe	Gly	Gly	Ile	Tyr	Glu	Asn	Glu	Ser	Arg	Asn	Thr	Glu	Met	Leu
225					230					235					240
Ala	Ala	Ala	Cys	Ala	Val	Gly	Val	Gly	Cys	Cys	Phe	Ala	Ala	Pro	Ile
			245						250					255	
Gly	Gly	Val	Leu	Phe	Ser	Ile	Glu	Val	Thr	Ser	Thr	Phe	Phe	Ala	Val
			260					265					270		
Arg	Asn	Tyr	Trp	Arg	Gly	Phe	Phe	Ala	Ala	Thr	Phe	Ser	Ala	Phe	Ile
		275					280					285			
Phe	Arg	Val	Leu	Ala	Val	Trp	Asn	Arg	Asp	Glu	Glu	Thr	Ile	Thr	Ala
		290				295					300				
Leu	Phe	Lys	Thr	Arg	Phe	Arg	Leu	Asp	Phe	Pro	Phe	Asp	Leu	Gln	Glu
305					310					315					320
Leu	Pro	Ala	Phe	Ala	Val	Ile	Gly	Ile	Ala	Ser	Gly	Phe	Gly	Gly	Ala
			325						330					335	
Leu	Phe	Val	Tyr	Leu	Asn	Arg	Lys	Ile	Val	Gln	Val	Met	Arg	Lys	Gln
			340					345					350		
Lys	Thr	Ile	Asn	Arg	Phe	Leu	Met	Arg	Lys	Arg	Leu	Leu	Phe	Pro	Ala
		355					360					365			
Leu	Val	Thr	Leu	Leu	Ile	Ser	Thr	Leu	Thr	Phe	Pro	Pro	Gly	Phe	Gly
		370				375					380				
Gln	Phe	Met	Ala	Gly	Gln	Leu	Ser	Gln	Lys	Glu	Thr	Leu	Val	Thr	Leu
385					390					395					400
Phe	Asp	Asn	Arg	Thr	Trp	Val	Arg	Gln	Gly	Leu	Val	Glu	Glu	Leu	Glu
			405						410					415	
Pro	Pro	Ser	Thr	Ser	Gln	Ala	Trp	Asn	Pro	Pro	Arg	Ala	Asn	Val	Phe
			420					425					430		
Leu	Thr	Leu	Val	Ile	Phe	Ile	Leu	Met	Lys	Phe	Trp	Met	Ser	Ala	Leu
		435					440					445			
Ala	Thr	Thr	Ile	Pro	Val	Pro	Cys	Gly	Ala	Phe	Met	Pro	Val	Phe	Val
		450				455					460				
Ile	Gly	Ala	Ala	Phe	Gly	Arg	Leu	Val	Gly	Glu	Ser	Met	Ala	Ala	Trp
465					470					475					480
Phe	Pro	Asp	Gly	Ile	His	Thr	Asp	Ser	Ser	Thr	Tyr	Arg	Ile	Val	Pro
			485					490					495		
Gly	Gly	Tyr	Ala	Val	Val	Gly	Ala	Ala	Ala	Leu	Ala	Gly	Ala	Val	Thr
			500				505					510			
His	Thr	Val	Ser	Thr	Ala	Val	Ile	Val	Phe	Glu	Leu	Thr	Gly	Gln	Ile
		515					520					525			
Ala	His	Ile	Leu	Pro	Val	Met	Ile	Ala	Val	Ile	Leu	Ala	Asn	Ala	Val

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530				535				540							
Ala	Gln	Ser	Leu	Gln	Pro	Ser	Leu	Tyr	Asp	Ser	Ile	Ile	Arg	Ile	Lys
545				550					555						560
Lys	Leu	Pro	Tyr	Leu	Pro	Glu	Leu	Gly	Trp	Gly	Arg	His	Gln	Gln	Tyr
			565						570						575
Arg	Val	Arg	Val	Glu	Asp	Ile	Met	Val	Arg	Asp	Val	Pro	His	Val	Ala
			580						585						590
Leu	Ser	Cys	Thr	Phe	Arg	Asp	Leu	Arg	Leu	Ala	Leu	His	Arg	Thr	Lys
		595					600						605		
Gly	Arg	Met	Leu	Ala	Leu	Val	Glu	Ser	Pro	Glu	Ser	Met	Ile	Leu	Leu
	610					615					620				
Gly	Ser	Ile	Glu	Arg	Ser	Gln	Val	Val	Ala	Leu	Leu	Gly	Ala	Gln	Leu
625					630					635					640
Ser	Pro	Ala	Arg	Arg	Arg	Gln	His	Met	Gln	Glu	Arg	Arg	Ala	Thr	Gln
			645						650						655
Thr	Ser	Pro	Leu	Ser	Asp	Gln	Glu	Gly	Pro	Pro	Thr	Pro	Glu	Ala	Ser
			660						665						670
Val	Cys	Phe	Gln	Val	Asn	Thr	Glu	Asp	Ser	Ala	Phe	Pro	Ala	Ala	Arg
		675					680						685		
Gly	Glu	Thr	His	Lys	Pro	Leu	Lys	Pro	Ala	Leu	Lys	Arg	Gly	Pro	Ser
	690					695					700				
Val	Thr	Arg	Asn	Leu	Gly	Glu	Ser	Pro	Thr	Gly	Ser	Ala	Glu	Ser	Ala
705					710					715					720
Gly	Ile	Ala	Leu	Arg	Ser	Leu	Phe	Cys	Gly	Ser	Pro	Pro	Pro	Glu	Ala
			725						730						735
Ala	Ser	Glu	Lys	Leu	Glu	Ser	Cys	Glu	Lys	Arg	Lys	Leu	Lys	Arg	Val
		740							745						750
Arg	Ile	Ser	Leu	Ala	Ser	Asp	Ala	Asp	Leu	Glu	Gly	Glu	Met	Ser	Pro
		755					760						765		
Glu	Glu	Ile	Leu	Glu	Trp	Glu	Glu	Gln	Gln	Leu	Asp	Glu	Pro	Val	Asn
	770					775					780				
Phe	Ser	Asp	Cys	Lys	Ile	Asp	Pro	Ala	Pro	Phe	Gln	Leu	Val	Glu	Arg
785					790					795					800
Thr	Ser	Leu	His	Lys	Thr	His	Thr	Ile	Phe	Ser	Leu	Leu	Gly	Val	Asp
			805						810						815
His	Ala	Tyr	Val	Thr	Ser	Ile	Gly	Arg	Leu	Ile	Gly	Ile	Val	Thr	Leu
			820						825						830
Lys	Glu	Leu	Arg	Lys	Ala	Ile	Glu	Gly	Ser	Val	Thr	Ala	Gln	Gly	Val
	835						840						845		
Lys	Val	Arg	Pro	Pro	Leu	Ala	Ser	Phe	Arg	Asp	Ser	Ala	Thr	Ser	Ser
	850					855					860				
Ser	Asp	Thr	Glu	Thr	Glu	Val	His	Ala	Leu	Trp	Gly	Pro	His	Ser	
865					870					875					880
Arg	His	Gly	Leu	Pro	Arg	Glu	Gly	Ser	Pro	Ser	Asp	Ser	Asp	Asp	Lys
			885						890						895

Cys Gln

<210> SEQ ID NO 47

<211> LENGTH: 575

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

-continued

<400> SEQUENCE: 47

Met	Phe	Gly	Ile	Gln	Asp	Thr	Leu	Gly	Arg	Gly	Pro	Thr	Leu	Lys	Glu
1			5					10					15		
Lys	Ser	Leu	Gly	Ala	Glu	Met	Asp	Ser	Val	Arg	Ser	Trp	Val	Arg	Asn
		20					25					30			
Val	Gly	Val	Val	Asp	Ala	Asn	Val	Ala	Ala	Gln	Ser	Gly	Val	Ala	Leu
	35					40					45				
Ser	Arg	Ala	His	Phe	Glu	Lys	Gln	Pro	Pro	Ser	Asn	Leu	Arg	Lys	Ser
	50				55					60					
Asn	Phe	Phe	His	Phe	Val	Leu	Ala	Leu	Tyr	Asp	Arg	Gln	Gly	Gln	Pro
65			70						75					80	
Val	Glu	Ile	Glu	Arg	Thr	Ala	Phe	Val	Asp	Phe	Val	Glu	Asn	Asp	Lys
			85					90					95		
Glu	Gln	Gly	Asn	Glu	Lys	Thr	Asn	Asn	Gly	Thr	His	Tyr	Lys	Leu	Gln
	100						105					110			
Leu	Leu	Tyr	Ser	Asn	Gly	Val	Arg	Thr	Glu	Gln	Asp	Leu	Tyr	Val	Arg
	115					120					125				
Leu	Ile	Asp	Ser	Val	Thr	Lys	Gln	Pro	Ile	Ala	Tyr	Glu	Gly	Gln	Asn
	130				135					140					
Lys	Asn	Pro	Glu	Met	Cys	Arg	Val	Leu	Leu	Thr	His	Glu	Val	Met	Cys
145				150					155					160	
Ser	Arg	Cys	Cys	Glu	Lys	Lys	Ser	Cys	Gly	Asn	Arg	Asn	Glu	Thr	Pro
		165						170					175		
Ser	Asp	Pro	Val	Ile	Ile	Asp	Arg	Phe	Phe	Leu	Lys	Phe	Phe	Leu	Lys
	180					185						190			
Cys	Asn	Gln	Asn	Cys	Leu	Lys	Thr	Ala	Gly	Asn	Pro	Arg	Asp	Met	Arg
	195					200					205				
Arg	Phe	Gln	Val	Val	Leu	Ser	Thr	Thr	Val	Asn	Val	Asp	Gly	His	Val
	210				215						220				
Leu	Ala	Val	Ser	Asp	Asn	Met	Phe	Val	His	Asn	Asn	Ser	Lys	His	Gly
225				230					235					240	
Arg	Arg	Ala	Arg	Arg	Leu	Asp	Pro	Ser	Glu	Ala	Thr	Pro	Cys	Ile	Lys
		245						250					255		
Ala	Ile	Ser	Pro	Ser	Glu	Gly	Trp	Thr	Thr	Gly	Gly	Ala	Met	Val	Ile
	260						265						270		
Ile	Ile	Gly	Asp	Asn	Phe	Phe	Asp	Gly	Leu	Gln	Val	Val	Phe	Gly	Thr
	275					280					285				
Met	Leu	Val	Trp	Ser	Glu	Leu	Ile	Thr	Pro	His	Ala	Ile	Arg	Val	Gln
	290				295					300					
Thr	Pro	Pro	Arg	His	Ile	Pro	Gly	Val	Val	Glu	Val	Thr	Leu	Ser	Tyr
305				310					315					320	
Lys	Ser	Lys	Gln	Phe	Cys	Lys	Gly	Ala	Pro	Gly	Arg	Phe	Ile	Tyr	Thr
		325						330					335		
Ala	Leu	Asn	Glu	Pro	Thr	Ile	Asp	Tyr	Gly	Phe	Gln	Arg	Leu	Gln	Lys
		340					345					350			
Val	Ile	Pro	Arg	His	Pro	Gly	Asp	Pro	Glu	Arg	Leu	Ala	Lys	Glu	Met
	355					360					365				
Leu	Leu	Lys	Arg	Ala	Ala	Asp	Leu	Val	Glu	Ala	Leu	Tyr	Gly	Thr	Pro
	370				375						380				
His	Asn	Asn	Gln	Asp	Ile	Ile	Leu	Lys	Arg	Ala	Ala	Asp	Ile	Ala	Glu
385					390				395					400	

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Ala Leu Tyr Ser Val Pro Arg Asn Pro Ser Gln Leu Pro Ala Leu Ser
405 410 415

Ser Ser Pro Ala His Ser Gly Met Met Gly Ile Asn Ser Tyr Gly Ser
420 425 430

Gln Leu Gly Val Ser Ile Ser Glu Ser Thr Gln Gly Asn Asn Gln Gly
435 440 445

Tyr Ile Arg Asn Thr Ser Ser Ile Ser Pro Arg Gly Tyr Ser Ser Ser
450 455 460

Ser Thr Pro Gln Gln Ser Asn Tyr Ser Thr Ser Ser Asn Ser Met Asn
465 470 475 480

Gly Tyr Ser Asn Val Pro Met Ala Asn Leu Gly Val Pro Gly Ser Pro
485 490 495

Gly Phe Leu Asn Gly Ser Pro Thr Gly Ser Pro Tyr Gly Ile Met Ser
500 505 510

Ser Ser Pro Thr Val Gly Ser Ser Ser Thr Ser Ser Ile Leu Pro Phe
515 520 525

Ser Ser Ser Val Phe Pro Ala Val Lys Gln Lys Ser Ala Phe Ala Pro
530 535 540

Val Ile Arg Pro Gln Gly Ser Pro Ser Pro Ala Cys Ser Ser Gly Asn
545 550 555 560

Gly Asn Gly Phe Arg Ala Met Thr Gly Leu Val Val Pro Pro Met
565 570 575

<210> SEQ ID NO 48

<211> LENGTH: 434

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 48

Met Ser Arg Ser Arg His Leu Gly Lys Ile Arg Lys Arg Leu Glu Asp
1 5 10 15

Val Lys Ser Gln Trp Val Arg Pro Ala Arg Ala Asp Phe Ser Asp Asn
20 25 30

Glu Ser Ala Arg Leu Ala Thr Asp Ala Leu Leu Asp Gly Gly Ser Glu
35 40 45

Ala Tyr Trp Arg Val Leu Ser Gln Glu Gly Glu Val Asp Phe Leu Ser
50 55 60

Ser Val Glu Ala Gln Tyr Ile Gln Ala Gln Ala Arg Glu Pro Pro Cys
65 70 75 80

Pro Pro Asp Thr Leu Gly Gly Ala Glu Ala Gly Pro Lys Gly Leu Asp
85 90 95

Ser Ser Ser Leu Gln Ser Gly Thr Tyr Phe Pro Val Ala Ser Glu Gly
100 105 110

Ser Glu Pro Ala Leu Leu His Ser Trp Ala Ser Ala Glu Lys Pro Tyr
115 120 125

Leu Lys Glu Lys Ser Ser Ala Thr Val Tyr Phe Gln Thr Val Lys His
130 135 140

Asn Asn Ile Arg Asp Leu Val Arg Arg Cys Ile Thr Arg Thr Ser Gln
145 150 155 160

Val Leu Val Ile Leu Met Asp Val Phe Thr Asp Val Glu Ile Phe Cys
165 170 175

Asp Ile Leu Glu Ala Ala Asn Lys Arg Gly Val Phe Val Cys Val Leu

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180					185					190					
Leu	Asp	Gln	Gly	Gly	Val	Lys	Leu	Phe	Gln	Glu	Met	Cys	Asp	Lys	Val
	195						200					205			
Gln	Ile	Ser	Asp	Ser	His	Leu	Lys	Asn	Ile	Ser	Ile	Arg	Ser	Val	Glu
	210					215				220					
Gly	Glu	Ile	Tyr	Cys	Ala	Lys	Ser	Gly	Arg	Lys	Phe	Ala	Gly	Gln	Ile
225				230					235					240	
Arg	Glu	Lys	Phe	Ile	Ile	Ser	Asp	Trp	Arg	Phe	Val	Leu	Ser	Gly	Ser
			245						250					255	
Tyr	Ser	Phe	Thr	Trp	Leu	Cys	Gly	His	Val	His	Arg	Asn	Ile	Leu	Ser
			260					265					270		
Lys	Phe	Thr	Gly	Gln	Ala	Val	Glu	Leu	Phe	Asp	Glu	Glu	Phe	Arg	His
		275					280					285			
Leu	Tyr	Ala	Ser	Ser	Lys	Pro	Val	Met	Gly	Leu	Lys	Ser	Pro	Arg	Leu
	290				295						300				
Val	Ala	Pro	Val	Pro	Pro	Gly	Ala	Ala	Pro	Ala	Asn	Gly	Arg	Leu	Ser
305					310					315				320	
Ser	Ser	Ser	Gly	Ser	Ala	Ser	Asp	Arg	Thr	Ser	Ser	Asn	Pro	Phe	Ser
			325						330					335	
Gly	Arg	Ser	Ala	Gly	Ser	His	Pro	Gly	Thr	Arg	Ser	Val	Ser	Ala	Ser
		340						345					350		
Ser	Gly	Pro	Cys	Ser	Pro	Ala	Ala	Pro	His	Pro	Pro	Pro	Pro	Pro	Arg
		355					360					365			
Phe	Gln	Pro	His	Gln	Gly	Pro	Trp	Gly	Ala	Pro	Ser	Pro	Gln	Ala	His
	370				375						380				
Leu	Ser	Pro	Arg	Pro	His	Asp	Gly	Pro	Pro	Ala	Ala	Val	Tyr	Ser	Asn
385					390					395				400	
Leu	Gly	Ala	Tyr	Arg	Pro	Thr	Arg	Leu	Gln	Leu	Glu	Gln	Leu	Gly	Leu
			405					410						415	
Val	Pro	Arg	Leu	Thr	Pro	Thr	Trp	Arg	Pro	Phe	Leu	Gln	Ala	Ser	Pro
			420					425					430		

His Phe

<210> SEQ ID NO 49

<211> LENGTH: 379

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 49

Met	Ser	Ser	Ala	Pro	Glu	Lys	Gln	Gln	Pro	Pro	His	Gly	Gly	Gly	Gly
1			5						10					15	
Gly	Gly	Gly	Gly	Gly	Gly	Gly	Ala	Ala	Met	Asp	Pro	Ala	Ser	Ser	Gly
		20					25					30			
Pro	Ser	Lys	Ala	Lys	Lys	Thr	Asn	Ala	Gly	Ile	Arg	Arg	Pro	Glu	Lys
		35				40					45				
Pro	Pro	Tyr	Ser	Tyr	Ile	Ala	Leu	Ile	Val	Met	Ala	Ile	Gln	Ser	Ser
		50			55					60					
Pro	Thr	Lys	Arg	Leu	Thr	Leu	Ser	Glu	Ile	Tyr	Gln	Phe	Leu	Gln	Ser
65				70					75					80	
Arg	Phe	Pro	Phe	Phe	Arg	Gly	Ser	Tyr	Gln	Gly	Trp	Lys	Asn	Ser	Val
			85					90					95		
Arg	His	Asn	Leu	Ser	Leu	Asn	Glu	Cys	Phe	Ile	Lys	Leu	Pro	Lys	Gly

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100				105				110							
Leu	Gly	Arg	Pro	Gly	Lys	Gly	His	Tyr	Trp	Thr	Ile	Asp	Pro	Ala	Ser
	115						120					125			
Glu	Phe	Met	Phe	Glu	Glu	Gly	Ser	Phe	Arg	Arg	Arg	Pro	Arg	Gly	Phe
	130					135					140				
Arg	Arg	Lys	Cys	Gln	Ala	Leu	Lys	Pro	Met	Tyr	Ser	Met	Met	Asn	Gly
	145				150					155					160
Leu	Gly	Phe	Asn	His	Leu	Pro	Asp	Thr	Tyr	Gly	Phe	Gln	Gly	Ser	Ala
			165						170					175	
Gly	Gly	Leu	Ser	Cys	Pro	Pro	Asn	Ser	Leu	Ala	Leu	Glu	Gly	Gly	Leu
		180						185				190			
Gly	Met	Met	Asn	Gly	His	Leu	Pro	Gly	Asn	Val	Asp	Gly	Met	Ala	Leu
	195						200					205			
Pro	Ser	His	Ser	Val	Pro	His	Leu	Pro	Ser	Asn	Gly	Gly	His	Ser	Tyr
	210					215					220				
Met	Gly	Gly	Cys	Gly	Gly	Ala	Ala	Ala	Gly	Glu	Tyr	Pro	His	His	Asp
	225				230					235					240
Ser	Ser	Val	Pro	Ala	Ser	Pro	Leu	Leu	Pro	Thr	Gly	Ala	Gly	Gly	Val
			245						250					255	
Met	Glu	Pro	His	Ala	Val	Tyr	Ser	Gly	Ser	Ala	Ala	Ala	Trp	Pro	Pro
		260						265					270		
Ser	Ala	Ser	Ala	Ala	Leu	Asn	Ser	Gly	Ala	Ser	Tyr	Ile	Lys	Gln	Gln
		275				280						285			
Pro	Leu	Ser	Pro	Cys	Asn	Pro	Ala	Ala	Asn	Pro	Leu	Ser	Gly	Ser	Leu
	290				295						300				
Ser	Thr	His	Ser	Leu	Glu	Gln	Pro	Tyr	Leu	His	Gln	Asn	Ser	His	Asn
	305				310					315					320
Ala	Pro	Ala	Glu	Leu	Gln	Gly	Ile	Pro	Arg	Tyr	His	Ser	Gln	Ser	Pro
			325						330					335	
Ser	Met	Cys	Asp	Arg	Lys	Glu	Phe	Val	Phe	Ser	Phe	Asn	Ala	Met	Ala
		340					345						350		
Ser	Ser	Ser	Met	His	Ser	Ala	Gly	Gly	Gly	Ser	Tyr	Tyr	His	Gln	Gln
		355				360						365			
Val	Thr	Tyr	Gln	Asp	Ile	Lys	Pro	Cys	Val	Met					
	370					375									

<210> SEQ ID NO 50

<211> LENGTH: 420

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 50

Met	Ala	Leu	Tyr	Cys	Gly	Asp	Asn	Phe	Gly	Val	Tyr	Ser	Gln	Pro	Gly
1				5					10					15	
Leu	Pro	Pro	Pro	Ala	Ala	Thr	Ala	Ala	Ala	Pro	Gly	Ala	Pro	Pro	Ala
			20					25					30		
Ala	Arg	Ala	Pro	Tyr	Gly	Leu	Ala	Asp	Tyr	Ala	Ala	Pro	Pro	Ala	Ala
		35					40					45			
Ala	Ala	Asn	Pro	Tyr	Leu	Trp	Leu	Asn	Gly	Pro	Gly	Val	Gly	Gly	Pro
	50					55					60				
Pro	Ser	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Tyr	Leu	Gly	Ala	Pro	Pro
	65				70					75				80	

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Pro	Pro	Pro	Pro	Pro	Gly	Ala	Ala	Ala	Gly	Pro	Phe	Leu	Gln	Pro	Pro	85	90	95
Pro	Ala	Ala	Gly	Thr	Phe	Gly	Cys	Ser	Gln	Arg	Pro	Phe	Ala	Gln	Pro	100	105	110
Ala	Pro	Ala	Ala	Pro	Ala	Ser	Pro	Ala	Ala	Pro	Ala	Gly	Pro	Gly	Glu	115	120	125
Leu	Gly	Trp	Leu	Ser	Met	Ala	Ser	Arg	Glu	Asp	Leu	Met	Lys	Met	Val	130	135	140
Arg	Pro	Pro	Tyr	Ser	Tyr	Ser	Ala	Leu	Ile	Ala	Met	Ala	Ile	Gln	Ser	145	150	155
Ala	Pro	Glu	Arg	Lys	Leu	Thr	Leu	Ser	His	Ile	Tyr	Gln	Phe	Val	Ala	165	170	175
Asp	Ser	Phe	Pro	Phe	Tyr	Gln	Arg	Ser	Lys	Ala	Gly	Trp	Gln	Asn	Ser	180	185	190
Ile	Arg	His	Asn	Leu	Ser	Leu	Asn	Asp	Cys	Phe	Lys	Lys	Val	Pro	Arg	195	200	205
Asp	Glu	Asp	Asp	Pro	Gly	Lys	Gly	Asn	Tyr	Trp	Thr	Leu	Asp	Pro	Asn	210	215	220
Cys	Glu	Lys	Met	Phe	Asp	Asn	Gly	Asn	Phe	Arg	Arg	Lys	Arg	Lys	Arg	225	230	235
Arg	Ser	Glu	Ala	Ser	Asn	Gly	Ser	Thr	Val	Ala	Ala	Gly	Thr	Ser	Lys	245	250	255
Ser	Glu	Glu	Gly	Leu	Ser	Ser	Gly	Leu	Gly	Ser	Gly	Val	Gly	Gly	Lys	260	265	270
Pro	Glu	Glu	Glu	Ser	Pro	Ser	Thr	Leu	Leu	Arg	Pro	Ser	His	Ser	Pro	275	280	285
Glu	Pro	Pro	Glu	Gly	Thr	Lys	Ser	Thr	Ala	Ser	Ser	Pro	Gly	Gly	Pro	290	295	300
Met	Leu	Thr	Ser	Thr	Pro	Cys	Leu	Asn	Thr	Phe	Phe	Ser	Ser	Leu	Ser	305	310	315
Ser	Leu	Ser	Val	Ser	Ser	Ser	Val	Ser	Thr	Gln	Arg	Ala	Leu	Pro	Gly	325	330	335
Ser	Arg	His	Leu	Gly	Ile	Gln	Gly	Ala	Gln	Leu	Pro	Ser	Ser	Gly	Val	340	345	350
Phe	Ser	Pro	Thr	Ser	Ile	Ser	Glu	Ala	Ser	Ala	Asp	Thr	Leu	Gln	Leu	355	360	365
Ser	Asn	Ser	Thr	Ser	Asn	Ser	Thr	Gly	Gln	Arg	Ser	Ser	Tyr	Tyr	Ser	370	375	380
Pro	Phe	Pro	Ala	Ser	Thr	Ser	Gly	Gly	Gln	Ser	Ser	Pro	Phe	Ser	Ser	385	390	395
Pro	Phe	His	Asn	Phe	Ser	Met	Val	Asn	Ser	Leu	Ile	Tyr	Pro	Arg	Glu	405	410	415
Gly	Ser	Glu	Val													420		

<210> SEQ ID NO 51

<211> LENGTH: 175

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 51

Met	Thr	Arg	Thr	Pro	Val	Gly	Ser	Ala	Arg	Thr	Arg	Pro	Lys	Pro	Arg	1	5	10	15
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	---	---	----	----

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Lys Leu Gly Pro Gln Arg Gly Lys Ala Leu Gln Ala Ser Ser Arg Leu
 20 25 30
 Ser Glu Ser Pro Ala Leu Val Lys Lys Arg Met Pro Asp Ala Cys Thr
 35 40 45
 Leu Gly Arg Ala Gly Ile Gly Leu Pro Lys Met Cys Leu His Met Ala
 50 55 60
 Val Arg His Ser Lys Ala Gln Lys Thr Gly Pro Gly Ile Leu Gln Gln
 65 70 75 80
 Arg Gln Lys Pro Pro Ala Pro Arg Ala Ser Gly Gly Pro Ala Leu Leu
 85 90 95
 Gly Lys Arg Arg Gly Cys Ser Glu Ala Gly Ser Ala Ser Leu Glu Pro
 100 105 110
 Leu Ser Ser Ser Arg Ala Ala Ala Gly Cys Leu Asn Gln Val Pro Leu
 115 120 125
 Ser Pro Phe Leu Ala Gly Pro Arg Asn Thr Arg Arg Leu Pro Ala Pro
 130 135 140
 Glu Arg Glu Arg Ile Glu Leu Ala Ala Thr Leu Cys Leu Glu Gly Trp
 145 150 155 160
 Pro Leu Arg Cys Leu Ala Ser Lys Gly Lys Leu His Cys Val Tyr
 165 170 175

<210> SEQ ID NO 52

<211> LENGTH: 299

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 52

Met Glu Glu Leu Leu Pro Asp Gly Gln Ile Trp Ala Asn Met Asp Pro
 1 5 10 15
 Glu Glu Arg Met Leu Ala Ala Ala Thr Ala Phe Thr His Ile Cys Ala
 20 25 30
 Gly Gln Gly Glu Gly Asp Val Arg Arg Glu Ala Gln Ser Ile Gln Tyr
 35 40 45
 Asp Pro Tyr Ser Lys Ala Ser Val Ala Pro Gly Lys Arg Pro Ala Leu
 50 55 60
 Pro Val Gln Leu Gln Tyr Pro His Val Glu Ser Asn Val Pro Ser Glu
 65 70 75 80
 Thr Val Ser Glu Ala Ser Gln Arg Leu Arg Lys Pro Val Met Lys Arg
 85 90 95
 Lys Val Leu Arg Arg Lys Pro Asp Gly Glu Val Leu Val Thr Asp Glu
 100 105 110
 Ser Ile Ile Ser Glu Ser Glu Ser Gly Thr Glu Asn Asp Gln Asp Leu
 115 120 125
 Trp Asp Leu Arg Gln Arg Leu Met Asn Val Gln Phe Gln Glu Asp Lys
 130 135 140
 Glu Ser Ser Phe Asp Val Ser Gln Lys Phe Asn Leu Pro His Glu Tyr
 145 150 155 160
 Gln Gly Ile Ser Gln Asp Gln Leu Ile Cys Ser Leu Gln Arg Glu Gly
 165 170 175
 Met Gly Ser Pro Ala Tyr Glu Gln Asp Leu Ile Val Ala Ser Arg Pro
 180 185 190
 Lys Ser Phe Ile Leu Pro Lys Leu Asp Gln Leu Ser Arg Asn Arg Gly

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195					200					205					
Lys	Thr	Asp	Arg	Val	Ala	Arg	Tyr	Phe	Glu	Tyr	Lys	Arg	Asp	Trp	Asp
210					215					220					
Ser	Ile	Arg	Leu	Pro	Gly	Glu	Asp	His	Arg	Lys	Glu	Leu	Arg	Trp	Gly
225					230					235					240
Val	Arg	Glu	Gln	Met	Leu	Cys	Arg	Ala	Glu	Pro	Gln	Ser	Lys	Pro	Gln
				245					250					255	
His	Ile	Tyr	Val	Pro	Asn	Asn	Tyr	Leu	Val	Pro	Thr	Glu	Lys	Lys	Arg
			260					265					270		
Ser	Ala	Leu	Arg	Trp	Gly	Val	Arg	Cys	Asp	Leu	Ala	Asn	Gly	Val	Ile
		275					280					285			
Pro	Arg	Lys	Leu	Pro	Phe	Pro	Leu	Ser	Pro	Ser					
	290					295									

<210> SEQ ID NO 53
 <211> LENGTH: 1786
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 53

Met	Gly	Leu	Leu	Gln	Leu	Leu	Ala	Phe	Ser	Phe	Leu	Ala	Leu	Cys	Arg
1				5					10					15	
Ala	Arg	Val	Arg	Ala	Gln	Glu	Pro	Glu	Phe	Ser	Tyr	Gly	Cys	Ala	Glu
		20						25					30		
Gly	Ser	Cys	Tyr	Pro	Ala	Thr	Gly	Asp	Leu	Leu	Ile	Gly	Arg	Ala	Gln
		35					40					45			
Lys	Leu	Ser	Val	Thr	Ser	Thr	Cys	Gly	Leu	His	Lys	Pro	Glu	Pro	Tyr
	50					55					60				
Cys	Ile	Val	Ser	His	Leu	Gln	Glu	Asp	Lys	Lys	Cys	Phe	Ile	Cys	Asn
65				70						75					80
Ser	Gln	Asp	Pro	Tyr	His	Glu	Thr	Leu	Asn	Pro	Asp	Ser	His	Leu	Ile
			85						90					95	
Glu	Asn	Val	Val	Thr	Thr	Phe	Ala	Pro	Asn	Arg	Leu	Lys	Ile	Trp	Trp
		100						105					110		
Gln	Ser	Glu	Asn	Gly	Val	Glu	Asn	Val	Thr	Ile	Gln	Leu	Asp	Leu	Glu
		115					120					125			
Ala	Glu	Phe	His	Phe	Thr	His	Leu	Ile	Met	Thr	Phe	Lys	Thr	Phe	Arg
	130					135					140				
Pro	Ala	Ala	Met	Leu	Ile	Glu	Arg	Ser	Ser	Asp	Phe	Gly	Lys	Thr	Trp
145				150						155					160
Gly	Val	Tyr	Arg	Tyr	Phe	Ala	Tyr	Asp	Cys	Glu	Ala	Ser	Phe	Pro	Gly
			165					170						175	
Ile	Ser	Thr	Gly	Pro	Met	Lys	Lys	Val	Asp	Asp	Ile	Ile	Cys	Asp	Ser
			180					185					190		
Arg	Tyr	Ser	Asp	Ile	Glu	Pro	Ser	Thr	Glu	Gly	Glu	Val	Ile	Phe	Arg
		195					200					205			
Ala	Leu	Asp	Pro	Ala	Phe	Lys	Ile	Glu	Asp	Pro	Tyr	Ser	Pro	Arg	Ile
	210					215					220				
Gln	Asn	Leu	Leu	Lys	Ile	Thr	Asn	Leu	Arg	Ile	Lys	Phe	Val	Lys	Leu
225				230						235					240
His	Thr	Leu	Gly	Asp	Asn	Leu	Leu	Asp	Ser	Arg	Met	Glu	Ile	Arg	Glu
				245					250					255	

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Lys	Tyr	Tyr	Tyr	Ala	Val	Tyr	Asp	Met	Val	Val	Arg	Gly	Asn	Cys	Phe	260	265	270	
Cys	Tyr	Gly	His	Ala	Ser	Glu	Cys	Ala	Pro	Val	Asp	Gly	Phe	Asn	Glu	275	280	285	
Glu	Val	Glu	Gly	Met	Val	His	Gly	His	Cys	Met	Cys	Arg	His	Asn	Thr	290	295	300	
Lys	Gly	Leu	Asn	Cys	Glu	Leu	Cys	Met	Asp	Phe	Tyr	His	Asp	Leu	Pro	305	310	315	320
Trp	Arg	Pro	Ala	Glu	Gly	Arg	Asn	Ser	Asn	Ala	Cys	Lys	Lys	Cys	Asn	325	330	335	
Cys	Asn	Glu	His	Ser	Ile	Ser	Cys	His	Phe	Asp	Met	Ala	Val	Tyr	Leu	340	345	350	
Ala	Thr	Gly	Asn	Val	Ser	Gly	Gly	Val	Cys	Asp	Asp	Cys	Gln	His	Asn	355	360	365	
Thr	Met	Gly	Arg	Asn	Cys	Glu	Gln	Cys	Lys	Pro	Phe	Tyr	Tyr	Gln	His	370	375	380	
Pro	Glu	Arg	Asp	Ile	Arg	Asp	Pro	Asn	Phe	Cys	Glu	Arg	Cys	Thr	Cys	385	390	395	400
Asp	Pro	Ala	Gly	Ser	Gln	Asn	Glu	Gly	Ile	Cys	Asp	Ser	Tyr	Thr	Asp	405	410	415	
Phe	Ser	Thr	Gly	Leu	Ile	Ala	Gly	Gln	Cys	Arg	Cys	Lys	Leu	Asn	Val	420	425	430	
Glu	Gly	Glu	His	Cys	Asp	Val	Cys	Lys	Glu	Gly	Phe	Tyr	Asp	Leu	Ser	435	440	445	
Ser	Glu	Asp	Pro	Phe	Gly	Cys	Lys	Ser	Cys	Ala	Cys	Asn	Pro	Leu	Gly	450	455	460	
Thr	Ile	Pro	Gly	Gly	Asn	Pro	Cys	Asp	Ser	Glu	Thr	Gly	His	Cys	Tyr	465	470	475	480
Cys	Lys	Arg	Leu	Val	Thr	Gly	Gln	His	Cys	Asp	Gln	Cys	Leu	Pro	Glu	485	490	495	
His	Trp	Gly	Leu	Ser	Asn	Asp	Leu	Asp	Gly	Cys	Arg	Pro	Cys	Asp	Cys	500	505	510	
Asp	Leu	Gly	Gly	Ala	Leu	Asn	Asn	Ser	Cys	Phe	Ala	Glu	Ser	Gly	Gln	515	520	525	
Cys	Ser	Cys	Arg	Pro	His	Met	Ile	Gly	Arg	Gln	Cys	Asn	Glu	Val	Glu	530	535	540	
Pro	Gly	Tyr	Tyr	Phe	Ala	Thr	Leu	Asp	His	Tyr	Leu	Tyr	Glu	Ala	Glu	545	550	555	560
Glu	Ala	Asn	Leu	Gly	Pro	Gly	Val	Ser	Ile	Val	Glu	Arg	Gln	Tyr	Ile	565	570	575	
Gln	Asp	Arg	Ile	Pro	Ser	Trp	Thr	Gly	Ala	Gly	Phe	Val	Arg	Val	Pro	580	585	590	
Glu	Gly	Ala	Tyr	Leu	Glu	Phe	Phe	Ile	Asp	Asn	Ile	Pro	Tyr	Ser	Met	595	600	605	
Glu	Tyr	Asp	Ile	Leu	Ile	Arg	Tyr	Glu	Pro	Gln	Leu	Pro	Asp	His	Trp	610	615	620	
Glu	Lys	Ala	Val	Ile	Thr	Val	Gln	Arg	Pro	Gly	Arg	Ile	Pro	Thr	Ser	625	630	635	640
Ser	Arg	Cys	Gly	Asn	Thr	Ile	Pro	Asp	Asp	Asp	Asn	Gln	Val	Val	Ser	645	650	655	
Leu	Ser	Pro	Gly	Ser	Arg	Tyr	Val	Val	Leu	Pro	Arg	Pro	Val	Cys	Phe				

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660						665						670					
Glu	Lys	Gly	Thr	Asn	Tyr	Thr	Val	Arg	Leu	Glu	Leu	Pro	Gln	Tyr	Thr		
		675					680					685					
Ser	Ser	Asp	Ser	Asp	Val	Glu	Ser	Pro	Tyr	Thr	Leu	Ile	Asp	Ser	Leu		
		690				695					700						
Val	Leu	Met	Pro	Tyr	Cys	Lys	Ser	Leu	Asp	Ile	Phe	Thr	Val	Gly	Gly		
705					710					715					720		
Ser	Gly	Asp	Gly	Val	Val	Thr	Asn	Ser	Ala	Trp	Glu	Thr	Phe	Gln	Arg		
				725					730					735			
Tyr	Arg	Cys	Leu	Glu	Asn	Ser	Arg	Ser	Val	Val	Lys	Thr	Pro	Met	Thr		
			740					745					750				
Asp	Val	Cys	Arg	Asn	Ile	Ile	Phe	Ser	Ile	Ser	Ala	Leu	Leu	His	Gln		
		755					760					765					
Thr	Gly	Leu	Ala	Cys	Glu	Cys	Asp	Pro	Gln	Gly	Ser	Leu	Ser	Ser	Val		
		770				775					780						
Cys	Asp	Pro	Asn	Gly	Gly	Gln	Cys	Gln	Cys	Arg	Pro	Asn	Val	Val	Gly		
785					790					795					800		
Arg	Thr	Cys	Asn	Arg	Cys	Ala	Pro	Gly	Thr	Phe	Gly	Phe	Gly	Pro	Ser		
			805						810					815			
Gly	Cys	Lys	Pro	Cys	Glu	Cys	His	Leu	Gln	Gly	Ser	Val	Asn	Ala	Phe		
			820					825					830				
Cys	Asn	Pro	Val	Thr	Gly	Gln	Cys	His	Cys	Phe	Gln	Gly	Val	Tyr	Ala		
		835					840					845					
Arg	Gln	Cys	Asp	Arg	Cys	Leu	Pro	Gly	His	Trp	Gly	Phe	Pro	Ser	Cys		
		850				855					860						
Gln	Pro	Cys	Gln	Cys	Asn	Gly	His	Ala	Asp	Asp	Cys	Asp	Pro	Val	Thr		
865					870					875					880		
Gly	Glu	Cys	Leu	Asn	Cys	Gln	Asp	Tyr	Thr	Met	Gly	His	Asn	Cys	Glu		
				885					890					895			
Arg	Cys	Leu	Ala	Gly	Tyr	Tyr	Gly	Asp	Pro	Ile	Ile	Gly	Ser	Gly	Asp		
			900					905					910				
His	Cys	Arg	Pro	Cys	Pro	Cys	Pro	Asp	Gly	Pro	Asp	Ser	Gly	Arg	Gln		
		915					920					925					
Phe	Ala	Arg	Ser	Cys	Tyr	Gln	Asp	Pro	Val	Thr	Leu	Gln	Leu	Ala	Cys		
		930				935					940						
Val	Cys	Asp	Pro	Gly	Tyr	Ile	Gly	Ser	Arg	Cys	Asp	Asp	Cys	Ala	Ser		
945					950					955					960		
Gly	Tyr	Phe	Gly	Asn	Pro	Ser	Glu	Val	Gly	Gly	Ser	Cys	Gln	Pro	Cys		
				965					970					975			
Gln	Cys	His	Asn	Asn	Ile	Asp	Thr	Thr	Asp	Pro	Glu	Ala	Cys	Asp	Lys		
			980					985					990				
Glu	Thr	Gly	Arg	Cys	Leu	Lys	Cys	Leu	Tyr	His	Thr	Glu	Gly	Glu	His		
		995					1000					1005					
Cys	Gln	Phe	Cys	Arg	Phe	Gly	Tyr	Tyr	Gly	Asp	Ala	Leu	Gln	Gln	Asp		
		1010				1015					1020						
Cys	Arg	Lys	Cys	Val	Cys	Asn	Tyr	Leu	Gly	Thr	Val	Gln	Glu	His	Cys		
1025					1030						1035				1040		
Asn	Gly	Ser	Asp	Cys	Gln	Cys	Asp	Lys	Ala	Thr	Gly	Gln	Cys	Leu	Cys		
				1045					1050					1055			
Leu	Pro	Asn	Val	Ile	Gly	Gln	Asn	Cys	Asp	Arg	Cys	Ala	Pro	Asn	Thr		
		1060						1065					1070				

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Trp	Gln	Leu	Ala	Ser	Gly	Thr	Gly	Cys	Asp	Pro	Cys	Asn	Cys	Asn	Ala	1075	1080	1085
Ala	His	Ser	Phe	Gly	Pro	Ser	Cys	Asn	Glu	Phe	Thr	Gly	Gln	Cys	Gln	1090	1095	1100
Cys	Met	Pro	Gly	Phe	Gly	Gly	Arg	Thr	Cys	Ser	Glu	Cys	Gln	Glu	Leu	1105	1110	1115
Phe	Trp	Gly	Asp	Pro	Asp	Val	Glu	Cys	Arg	Ala	Cys	Asp	Cys	Asp	Pro	1125	1130	1135
Arg	Gly	Ile	Glu	Thr	Pro	Gln	Cys	Asp	Gln	Ser	Thr	Gly	Gln	Cys	Val	1140	1145	1150
Cys	Val	Glu	Gly	Val	Glu	Gly	Pro	Arg	Cys	Asp	Lys	Cys	Thr	Arg	Gly	1155	1160	1165
Tyr	Ser	Gly	Val	Phe	Pro	Asp	Cys	Thr	Pro	Cys	His	Gln	Cys	Phe	Ala	1170	1175	1180
Leu	Trp	Asp	Val	Ile	Ile	Ala	Glu	Leu	Thr	Asn	Arg	Thr	His	Arg	Phe	1185	1190	1195
Leu	Glu	Lys	Ala	Lys	Ala	Leu	Lys	Ile	Ser	Gly	Val	Ile	Gly	Pro	Tyr	1205	1210	1215
Arg	Glu	Thr	Val	Asp	Ser	Val	Glu	Arg	Lys	Val	Ser	Glu	Ile	Lys	Asp	1220	1225	1230
Ile	Leu	Ala	Gln	Ser	Pro	Ala	Ala	Glu	Pro	Leu	Lys	Asn	Ile	Gly	Asn	1235	1240	1245
Leu	Phe	Glu	Glu	Ala	Glu	Lys	Leu	Ile	Lys	Asp	Val	Thr	Glu	Met	Met	1250	1255	1260
Ala	Gln	Val	Glu	Val	Lys	Leu	Ser	Asp	Thr	Thr	Ser	Gln	Ser	Asn	Ser	1265	1270	1275
Thr	Ala	Lys	Glu	Leu	Asp	Ser	Leu	Gln	Thr	Glu	Ala	Glu	Ser	Leu	Asp	1285	1290	1295
Asn	Thr	Val	Lys	Glu	Leu	Ala	Glu	Gln	Leu	Glu	Phe	Ile	Lys	Asn	Ser	1300	1305	1310
Asp	Ile	Arg	Gly	Ala	Leu	Asp	Ser	Ile	Thr	Lys	Tyr	Phe	Gln	Met	Ser	1315	1320	1325
Leu	Glu	Ala	Glu	Glu	Arg	Val	Asn	Ala	Ser	Thr	Thr	Glu	Pro	Asn	Ser	1330	1335	1340
Thr	Val	Glu	Gln	Ser	Ala	Leu	Met	Arg	Asp	Arg	Val	Glu	Asp	Val	Met	1345	1350	1355
Met	Glu	Arg	Glu	Ser	Gln	Phe	Lys	Glu	Lys	Gln	Glu	Glu	Gln	Ala	Arg	1365	1370	1375
Leu	Leu	Asp	Glu	Leu	Ala	Gly	Lys	Leu	Gln	Ser	Leu	Asp	Leu	Ser	Ala	1380	1385	1390
Ala	Ala	Glu	Met	Thr	Cys	Gly	Thr	Pro	Pro	Gly	Ala	Ser	Cys	Ser	Glu	1395	1400	1405
Thr	Glu	Cys	Gly	Gly	Pro	Asn	Cys	Arg	Thr	Asp	Glu	Gly	Glu	Arg	Lys	1410	1415	1420
Cys	Gly	Gly	Pro	Gly	Cys	Gly	Gly	Leu	Val	Thr	Val	Ala	His	Asn	Ala	1425	1430	1435
Trp	Gln	Lys	Ala	Met	Asp	Leu	Asp	Gln	Asp	Val	Leu	Ser	Ala	Leu	Ala	1445	1450	1455
Glu	Val	Glu	Gln	Leu	Ser	Lys	Met	Val	Ser	Glu	Ala	Lys	Leu	Arg	Ala	1460	1465	1470

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Asp Glu Ala Lys Gln Ser Ala Glu Asp Ile Leu Leu Lys Thr Asn Ala
 1475 1480 1485
 Thr Lys Glu Lys Met Asp Lys Ser Asn Glu Glu Leu Arg Asn Leu Ile
 1490 1495 1500
 Lys Gln Ile Arg Asn Phe Leu Thr Gln Asp Ser Ala Asp Leu Asp Ser
 1505 1510 1515 1520
 Ile Glu Ala Val Ala Asn Glu Val Leu Lys Met Glu Met Pro Ser Thr
 1525 1530 1535
 Pro Gln Gln Leu Gln Asn Leu Thr Glu Asp Ile Arg Glu Arg Val Glu
 1540 1545 1550
 Ser Leu Ser Gln Val Glu Val Ile Leu Gln His Ser Ala Ala Asp Ile
 1555 1560 1565
 Ala Arg Ala Glu Met Leu Leu Glu Glu Ala Lys Arg Ala Ser Lys Ser
 1570 1575 1580
 Ala Thr Asp Val Lys Val Thr Ala Asp Met Val Lys Glu Ala Leu Glu
 1585 1590 1595 1600
 Glu Ala Glu Lys Ala Gln Val Ala Ala Glu Lys Ala Ile Lys Gln Ala
 1605 1610 1615
 Asp Glu Asp Ile Gln Gly Thr Gln Asn Leu Leu Thr Ser Ile Glu Ser
 1620 1625 1630
 Glu Thr Ala Ala Ser Glu Glu Thr Leu Phe Asn Ala Ser Gln Arg Ile
 1635 1640 1645
 Ser Glu Leu Glu Arg Asn Val Glu Glu Leu Lys Arg Lys Ala Ala Gln
 1650 1655 1660
 Asn Ser Gly Glu Ala Glu Tyr Ile Glu Lys Val Val Tyr Thr Val Lys
 1665 1670 1675 1680
 Gln Ser Ala Glu Asp Val Lys Lys Thr Leu Asp Gly Glu Leu Asp Glu
 1685 1690 1695
 Lys Tyr Lys Lys Val Glu Asn Leu Ile Ala Lys Lys Thr Glu Glu Ser
 1700 1705 1710
 Ala Asp Ala Arg Arg Lys Ala Glu Met Leu Gln Asn Glu Ala Lys Thr
 1715 1720 1725
 Leu Leu Ala Gln Ala Asn Ser Lys Leu Gln Leu Leu Lys Asp Leu Glu
 1730 1735 1740
 Arg Lys Tyr Glu Asp Asn Gln Arg Tyr Leu Glu Asp Lys Ala Gln Glu
 1745 1750 1755 1760
 Leu Ala Arg Leu Glu Gly Glu Val Arg Ser Leu Leu Lys Asp Ile Ser
 1765 1770 1775
 Gln Lys Val Ala Val Tyr Ser Thr Cys Leu
 1780 1785

<210> SEQ ID NO 54

<211> LENGTH: 61

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 54

Met Gln Pro Arg Thr Gln Pro Leu Ala Gln Thr Leu Pro Phe Phe Leu
 1 5 10 15
 Gly Gly Ala Pro Arg Asp Thr Gly Leu Arg Val Pro Val Ile Lys Met
 20 25 30
 Gly Thr Gly Trp Glu Gly Phe Gln Arg Thr Leu Lys Glu Val Ala Tyr
 35 40 45

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Ile Leu Leu Cys Cys Trp Cys Ile Lys Glu Leu Leu Asp
 50                      55                      60

<210> SEQ ID NO 55
<211> LENGTH: 713
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 55

Met Ala Arg Ala Arg Gly Ser Pro Cys Pro Pro Leu Pro Pro Gly Arg
 1                      5                      10                      15

Met Ser Trp Pro His Gly Ala Leu Leu Phe Leu Trp Leu Phe Ser Pro
 20                      25                      30

Pro Leu Gly Ala Gly Gly Gly Val Ala Val Thr Ser Ala Ala Gly
 35                      40                      45

Gly Gly Ser Pro Pro Ala Thr Ser Cys Pro Val Ala Cys Ser Cys Ser
 50                      55                      60

Asn Gln Ala Ser Arg Val Ile Cys Thr Arg Arg Asp Leu Ala Glu Val
 65                      70                      75                      80

Pro Ala Ser Ile Pro Val Asn Thr Arg Tyr Leu Asn Leu Gln Glu Asn
 85                      90                      95

Gly Ile Gln Val Ile Arg Thr Asp Thr Phe Lys His Leu Arg His Leu
 100                     105                     110

Glu Ile Leu Gln Leu Ser Lys Asn Leu Val Arg Lys Ile Glu Val Gly
 115                     120                     125

Ala Phe Asn Gly Leu Pro Ser Leu Asn Thr Leu Glu Leu Phe Asp Asn
 130                     135                     140

Arg Leu Thr Thr Val Pro Thr Gln Ala Phe Glu Tyr Leu Ser Lys Leu
 145                     150                     155                     160

Arg Glu Leu Trp Leu Arg Asn Asn Pro Ile Glu Ser Ile Pro Ser Tyr
 165                     170                     175

Ala Phe Asn Arg Val Pro Ser Leu Arg Arg Leu Asp Leu Gly Glu Leu
 180                     185                     190

Lys Arg Leu Glu Tyr Ile Ser Glu Ala Ala Phe Glu Gly Leu Val Asn
 195                     200                     205

Leu Arg Tyr Leu Asn Leu Gly Met Cys Asn Leu Lys Asp Ile Pro Asn
 210                     215                     220

Leu Thr Ala Leu Val Arg Leu Glu Glu Leu Glu Leu Ser Gly Asn Arg
 225                     230                     235                     240

Leu Asp Leu Ile Arg Pro Gly Ser Phe Gln Gly Leu Thr Ser Leu Arg
 245                     250                     255

Lys Leu Trp Leu Met His Ala Gln Val Ala Thr Ile Glu Arg Asn Ala
 260                     265                     270

Phe Asp Asp Leu Lys Ser Leu Glu Glu Leu Asn Leu Ser His Asn Asn
 275                     280                     285

Leu Met Ser Leu Pro His Asp Leu Phe Thr Pro Leu His Arg Leu Glu
 290                     295                     300

Arg Val His Leu Asn His Asn Pro Trp His Cys Asn Cys Asp Val Leu
 305                     310                     315                     320

Trp Leu Ser Trp Trp Leu Lys Glu Thr Val Pro Ser Asn Thr Thr Cys
 325                     330                     335

Cys Ala Arg Cys His Ala Pro Ala Gly Leu Lys Gly Arg Tyr Ile Gly

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340				345				350							
Glu	Leu	Asp	Gln	Ser	His	Phe	Thr	Cys	Tyr	Ala	Pro	Val	Ile	Val	Glu
	355						360					365			
Pro	Pro	Thr	Asp	Leu	Asn	Val	Thr	Glu	Gly	Met	Ala	Ala	Glu	Leu	Lys
	370				375						380				
Cys	Arg	Thr	Gly	Thr	Ser	Met	Thr	Ser	Val	Asn	Trp	Leu	Thr	Pro	Asn
385					390					395				400	
Gly	Thr	Leu	Met	Thr	His	Gly	Ser	Tyr	Arg	Val	Arg	Ile	Ser	Val	Leu
			405						410					415	
His	Asp	Gly	Thr	Leu	Asn	Phe	Thr	Asn	Val	Thr	Val	Gln	Asp	Thr	Gly
	420							425					430		
Gln	Tyr	Thr	Cys	Met	Val	Thr	Asn	Ser	Ala	Gly	Asn	Thr	Thr	Ala	Ser
	435						440					445			
Ala	Thr	Leu	Asn	Val	Ser	Ala	Val	Asp	Pro	Val	Ala	Ala	Gly	Gly	Thr
	450				455					460					
Gly	Ser	Gly	Gly	Gly	Gly	Pro	Gly	Gly	Ser	Gly	Gly	Val	Gly	Gly	Gly
465					470					475					480
Ser	Gly	Gly	Tyr	Thr	Tyr	Phe	Thr	Thr	Val	Thr	Val	Glu	Thr	Leu	Glu
			485						490					495	
Thr	Gln	Pro	Gly	Glu	Glu	Ala	Leu	Gln	Pro	Arg	Gly	Thr	Glu	Lys	Glu
			500					505					510		
Pro	Pro	Gly	Pro	Thr	Thr	Asp	Gly	Val	Trp	Gly	Gly	Gly	Arg	Pro	Gly
	515						520					525			
Asp	Ala	Ala	Gly	Pro	Ala	Ser	Ser	Ser	Thr	Thr	Ala	Pro	Ala	Pro	Arg
	530				535						540				
Ser	Ser	Arg	Pro	Thr	Glu	Lys	Ala	Phe	Thr	Val	Pro	Ile	Thr	Asp	Val
545					550					555				560	
Thr	Glu	Asn	Ala	Leu	Lys	Asp	Leu	Asp	Asp	Val	Met	Lys	Thr	Thr	Lys
			565						570					575	
Ile	Ile	Ile	Gly	Cys	Phe	Val	Ala	Ile	Thr	Phe	Met	Ala	Ala	Val	Met
			580					585					590		
Leu	Val	Ala	Phe	Tyr	Lys	Leu	Arg	Lys	Gln	His	Gln	Leu	His	Lys	His
	595						600					605			
His	Gly	Pro	Thr	Arg	Thr	Val	Glu	Ile	Ile	Asn	Val	Glu	Asp	Glu	Leu
	610				615					620					
Pro	Ala	Ala	Ser	Ala	Val	Ser	Val	Ala	Ala	Ala	Ala	Ala	Val	Ala	Ser
625					630					635				640	
Gly	Gly	Gly	Val	Gly	Gly	Asp	Ser	His	Leu	Ala	Leu	Pro	Ala	Leu	Glu
			645						650					655	
Arg	Asp	His	Leu	Asn	His	His	His	Tyr	Val	Ala	Ala	Ala	Phe	Lys	Ala
	660							665					670		
His	Tyr	Ser	Ser	Asn	Pro	Ser	Gly	Gly	Gly	Cys	Gly	Gly	Lys	Gly	Pro
	675						680					685			
Pro	Gly	Leu	Asn	Ser	Ile	His	Glu	Pro	Leu	Leu	Phe	Lys	Ser	Gly	Ser
	690					695					700				
Lys	Glu	Asn	Val	Gln	Glu	Thr	Gln	Ile							
705					710										

<210> SEQ ID NO 56

<211> LENGTH: 359

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

-continued

<400> SEQUENCE: 56

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Met Ile Ala Ala Gln Ala Lys Leu Val Tyr Gln Leu Asn Lys Tyr Tyr
 1             5             10             15

Thr Glu Arg Cys Gln Ala Arg Lys Ala Ala Ile Ala Lys Thr Ile Arg
 20             25             30

Glu Val Cys Lys Val Val Ser Asp Val Leu Lys Glu Val Glu Val Gln
 35             40             45

Glu Pro Arg Phe Ile Ser Ser Leu Ser Glu Ile Asp Ala Arg Tyr Glu
 50             55             60

Gly Leu Glu Val Ile Ser Pro Thr Glu Phe Glu Val Val Leu Tyr Leu
 65             70             75             80

Asn Gln Met Gly Val Phe Asn Phe Val Asp Asp Gly Ser Leu Pro Gly
 85             90             95

Cys Ala Val Leu Lys Leu Ser Asp Gly Arg Lys Arg Ser Met Ser Leu
100             105             110

Trp Val Glu Phe Ile Thr Ala Ser Gly Tyr Leu Ser Ala Arg Lys Ile
115             120             125

Arg Ser Arg Phe Gln Thr Leu Val Ala Gln Ala Val Asp Lys Cys Ser
130             135             140

Tyr Arg Asp Val Val Lys Met Ile Ala Asp Thr Ser Glu Val Lys Leu
145             150             155             160

Arg Ile Arg Glu Arg Tyr Val Val Gln Ile Thr Pro Ala Phe Lys Cys
165             170             175

Thr Gly Ile Trp Pro Arg Ser Ala Ala Gln Trp Pro Met Pro His Ile
180             185             190

Pro Trp Pro Gly Pro Asn Arg Val Ala Glu Val Lys Ala Glu Gly Phe
195             200             205

Asn Leu Leu Ser Lys Glu Cys Tyr Ser Leu Thr Gly Lys Gln Ser Ser
210             215             220

Ala Glu Ser Asp Ala Trp Val Leu Gln Phe Gly Glu Ala Glu Asn Arg
225             230             235             240

Leu Leu Met Gly Gly Cys Arg Asn Lys Cys Leu Ser Val Leu Lys Thr
245             250             255

Leu Arg Asp Arg His Leu Glu Leu Pro Gly Gln Pro Leu Asn Asn Tyr
260             265             270

His Met Lys Thr Leu Leu Leu Tyr Glu Cys Glu Lys His Pro Arg Glu
275             280             285

Thr Asp Trp Asp Glu Ser Cys Leu Gly Asp Arg Leu Asn Gly Ile Leu
290             295             300

Leu Gln Leu Ile Ser Cys Leu Gln Cys Arg Arg Cys Pro His Tyr Phe
305             310             315             320

Leu Pro Asn Leu Asp Leu Phe Gln Gly Lys Pro His Ser Ala Leu Glu
325             330             335

Ser Ala Ala Lys Gln Thr Trp Arg Leu Ala Arg Glu Ile Leu Thr Asn
340             345             350

Pro Lys Ser Leu Asp Lys Leu
355

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<210> SEQ ID NO 57

<211> LENGTH: 497

<212> TYPE: PRT

-continued

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 57

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Met Ala Ser Pro Ala Pro Pro Glu His Ala Glu Glu Gly Cys Pro Ala
 1          5          10          15

Pro Ala Ala Glu Glu Gln Ala Pro Pro Ser Pro Pro Pro Pro Gln Ala
 20          25          30

Ser Pro Ala Glu Arg Gln Gln Gln Glu Glu Glu Ala Gln Glu Ala Gly
 35          40          45

Ala Ala Glu Gly Ala Gly Leu Gln Val Glu Glu Ala Ala Gly Arg Ala
 50          55          60

Ala Ala Ala Val Thr Trp Leu Leu Gly Glu Pro Val Leu Trp Leu Gly
 65          70          75          80

Cys Arg Ala Asp Glu Leu Leu Ser Trp Lys Arg Pro Leu Arg Ser Leu
 85          90          95

Leu Gly Phe Val Ala Ala Asn Leu Leu Phe Trp Phe Leu Ala Leu Thr
100          105          110

Pro Trp Arg Val Tyr His Leu Ile Ser Val Met Ile Leu Gly Arg Val
115          120          125

Ile Met Gln Ile Ile Lys Asp Met Val Leu Ser Arg Thr Arg Gly Ala
130          135          140

Gln Leu Trp Arg Ser Leu Ser Glu Ser Trp Glu Val Ile Asn Ser Lys
145          150          155          160

Pro Asp Glu Arg Pro Arg Leu Ser His Cys Ile Ala Glu Ser Trp Met
165          170          175

Asn Phe Ser Ile Phe Leu Gln Glu Met Ser Leu Phe Lys Gln Gln Ser
180          185          190

Pro Gly Lys Phe Cys Leu Leu Val Cys Ser Val Cys Thr Phe Phe Thr
195          200          205

Ile Leu Gly Ser Tyr Ile Pro Gly Val Ile Leu Ser Tyr Leu Leu Leu
210          215          220

Leu Cys Ala Phe Leu Cys Pro Leu Phe Lys Cys Asn Asp Ile Gly Gln
225          230          235          240

Lys Ile Tyr Ser Lys Ile Lys Ser Val Leu Leu Lys Leu Asp Phe Gly
245          250          255

Ile Gly Glu Tyr Ile Asn Gln Lys Lys Arg Glu Arg Ser Glu Ala Asp
260          265          270

Lys Glu Lys Ser His Lys Asp Asp Ser Glu Leu Asp Phe Ser Ala Leu
275          280          285

Cys Pro Lys Ile Ser Leu Thr Val Ala Ala Lys Glu Leu Ser Val Ser
290          295          300

Asp Thr Asp Val Ser Glu Val Ser Trp Thr Asp Asn Gly Thr Phe Asn
305          310          315          320

Leu Ser Glu Gly Tyr Thr Pro Gln Thr Asp Thr Ser Asp Asp Leu Asp
325          330          335

Arg Pro Ser Glu Glu Val Phe Ser Arg Asp Leu Ser Asp Phe Pro Ser
340          345          350

Leu Glu Asn Gly Met Gly Thr Asn Asp Glu Asp Glu Leu Ser Leu Gly
355          360          365

Leu Pro Thr Glu Leu Lys Arg Lys Lys Glu Gln Leu Asp Ser Gly His
370          375          380

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Arg	Pro	Ser	Lys	Glu	Thr	Gln	Ser	Ala	Ala	Gly	Leu	Thr	Leu	Pro	Leu
385					390					395					400
Asn	Ser	Asp	Gln	Thr	Phe	His	Leu	Met	Ser	Asn	Leu	Ala	Gly	Asp	Val
			405					410					415		
Ile	Thr	Ala	Ala	Val	Thr	Ala	Ala	Ile	Lys	Asp	Gln	Leu	Glu	Gly	Val
		420					425					430			
Gln	Gln	Ala	Leu	Ser	Gln	Ala	Ala	Pro	Ile	Pro	Glu	Glu	Asp	Thr	Asp
		435				440					445				
Thr	Glu	Glu	Gly	Asp	Asp	Phe	Glu	Leu	Leu	Asp	Gln	Ser	Glu	Leu	Asp
	450				455					460					
Gln	Ile	Glu	Ser	Glu	Leu	Gly	Leu	Thr	Gln	Asp	Gln	Glu	Ala	Glu	Ala
465				470					475					480	
Gln	Gln	Asn	Lys	Lys	Ser	Ser	Gly	Phe	Leu	Ser	Asn	Leu	Leu	Gly	Gly
			485					490						495	

His

<210> SEQ ID NO 58

<211> LENGTH: 467

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 58

Met	His	Ser	Thr	Thr	Pro	Ile	Ser	Ser	Leu	Phe	Ser	Phe	Thr	Ser	Pro
1				5					10					15	
Ala	Val	Lys	Arg	Leu	Leu	Gly	Trp	Lys	Gln	Gly	Asp	Glu	Glu	Glu	Lys
		20					25					30			
Trp	Ala	Glu	Lys	Ala	Val	Asp	Ser	Leu	Val	Lys	Lys	Leu	Lys	Lys	Lys
	35				40					45					
Lys	Gly	Ala	Met	Asp	Glu	Leu	Glu	Arg	Ala	Leu	Ser	Cys	Pro	Gly	Gln
50					55				60						
Pro	Ser	Lys	Cys	Val	Thr	Ile	Pro	Arg	Ser	Leu	Asp	Gly	Arg	Leu	Gln
65				70				75						80	
Val	Ser	His	Arg	Lys	Gly	Leu	Pro	His	Val	Ile	Tyr	Cys	Arg	Val	Trp
		85					90						95		
Arg	Trp	Pro	Asp	Leu	Gln	Ser	His	His	Glu	Leu	Lys	Pro	Leu	Glu	Cys
		100					105					110			
Cys	Glu	Phe	Pro	Phe	Gly	Ser	Lys	Gln	Lys	Glu	Val	Cys	Ile	Asn	Pro
	115				120					125					
Tyr	His	Tyr	Arg	Arg	Val	Glu	Thr	Pro	Val	Leu	Pro	Pro	Val	Leu	Val
130				135					140						
Pro	Arg	His	Ser	Glu	Tyr	Asn	Pro	Gln	Leu	Ser	Leu	Leu	Ala	Lys	Phe
145				150				155						160	
Arg	Ser	Ala	Ser	Leu	His	Ser	Glu	Pro	Leu	Met	Pro	His	Asn	Ala	Thr
		165					170						175		
Tyr	Pro	Asp	Ser	Phe	Gln	Gln	Pro	Pro	Cys	Ser	Ala	Leu	Pro	Pro	Ser
	180						185					190			
Pro	Ser	His	Ala	Phe	Ser	Gln	Ser	Pro	Cys	Thr	Ala	Ser	Tyr	Pro	His
	195					200					205				
Ser	Pro	Gly	Ser	Pro	Ser	Glu	Pro	Glu	Ser	Pro	Tyr	Gln	His	Ser	Val
210					215					220					
Asp	Thr	Pro	Pro	Leu	Pro	Tyr	His	Ala	Thr	Glu	Ala	Ser	Glu	Thr	Gln
225				230				235						240	

-continued

Ser	Gly	Gln	Pro	Val	Asp	Ala	Thr	Ala	Asp	Arg	His	Val	Val	Leu	Ser
				245					250					255	
Ile	Pro	Asn	Gly	Asp	Phe	Arg	Pro	Val	Cys	Tyr	Glu	Glu	Pro	Gln	His
			260					265					270		
Trp	Cys	Ser	Val	Ala	Tyr	Tyr	Glu	Leu	Asn	Asn	Arg	Val	Gly	Glu	Thr
		275					280					285			
Phe	Gln	Ala	Ser	Ser	Arg	Ser	Val	Leu	Ile	Asp	Gly	Phe	Thr	Asp	Pro
	290					295					300				
Ser	Asn	Asn	Arg	Asn	Arg	Phe	Cys	Leu	Gly	Leu	Leu	Ser	Asn	Val	Asn
305				310						315					320
Arg	Asn	Ser	Thr	Ile	Glu	Asn	Thr	Arg	Arg	His	Ile	Gly	Lys	Gly	Val
			325					330						335	
His	Leu	Tyr	Tyr	Val	Gly	Gly	Glu	Val	Tyr	Ala	Glu	Cys	Val	Ser	Asp
			340					345					350		
Ser	Ser	Ile	Phe	Val	Gln	Ser	Arg	Asn	Cys	Asn	Tyr	Gln	His	Gly	Phe
		355					360					365			
His	Pro	Ala	Thr	Val	Cys	Lys	Ile	Pro	Ser	Gly	Cys	Ser	Leu	Lys	Val
	370					375					380				
Phe	Asn	Asn	Gln	Leu	Phe	Ala	Gln	Leu	Leu	Ala	Gln	Ser	Val	His	His
385					390					395					400
Gly	Phe	Glu	Val	Val	Tyr	Glu	Leu	Thr	Lys	Met	Cys	Thr	Ile	Arg	Met
			405					410						415	
Ser	Phe	Val	Lys	Gly	Trp	Gly	Ala	Glu	Tyr	His	Arg	Gln	Asp	Val	Thr
			420					425					430		
Ser	Thr	Pro	Cys	Trp	Ile	Glu	Ile	His	Leu	His	Gly	Pro	Leu	Gln	Trp
		435					440					445			
Leu	Asp	Lys	Val	Leu	Thr	Gln	Met	Gly	Ser	Pro	His	Asn	Pro	Ile	Ser
	450					455					460				
Ser	Val	Ser													
465															

<210> SEQ ID NO 59

<211> LENGTH: 1347

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 59

Met	Glu	Asn	Leu	Pro	Phe	Pro	Leu	Lys	Leu	Leu	Ser	Ala	Ser	Ser	Leu
1				5					10					15	
Asn	Ala	Pro	Ser	Ser	Thr	Pro	Trp	Val	Leu	Asp	Ile	Phe	Leu	Thr	Leu
		20					25						30		
Val	Phe	Ala	Leu	Gly	Phe	Phe	Phe	Leu	Leu	Leu	Pro	Tyr	Leu	Ser	Tyr
	35					40						45			
Phe	Arg	Cys	Asp	Asp	Pro	Pro	Ser	Pro	Ser	Pro	Gly	Lys	Arg	Lys	Cys
	50				55						60				
Pro	Val	Gly	Arg	Arg	Arg	Arg	Pro	Arg	Gly	Arg	Met	Lys	Asn	His	Ser
65					70					75				80	
Leu	Arg	Ala	Gly	Arg	Glu	Cys	Pro	Arg	Gly	Leu	Gln	Glu	Thr	Ser	Asp
			85						90					95	
Leu	Leu	Ser	Gln	Leu	Gln	Ser	Leu	Leu	Gly	Pro	His	Leu	Asp	Lys	Gly
			100						105					110	
Asp	Phe	Gly	Gln	Leu	Ser	Gly	Pro	Asp	Pro	Pro	Gly	Glu	Val	Gly	Glu
		115					120						125		

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Arg	Ala	Pro	Asp	Gly	Ala	Ser	Gln	Ser	Ser	His	Glu	Pro	Met	Glu	Asp
130						135					140				
Ala	Ala	Pro	Ile	Leu	Ser	Pro	Leu	Ala	Ser	Pro	Asp	Pro	Gln	Ala	Lys
145					150					155					160
His	Pro	Gln	Asp	Leu	Ala	Ser	Thr	Pro	Ser	Pro	Gly	Pro	Met	Thr	Thr
				165					170					175	
Ser	Val	Ser	Ser	Leu	Ser	Ala	Ser	Gln	Pro	Pro	Glu	Pro	Ser	Leu	Pro
				180				185					190		
Leu	Glu	His	Pro	Ser	Pro	Glu	Pro	Pro	Ala	Leu	Phe	Pro	His	Pro	Pro
		195					200					205			
His	Thr	Pro	Asp	Pro	Leu	Ala	Cys	Ser	Pro	Pro	Pro	Pro	Lys	Gly	Phe
	210					215						220			
Thr	Ala	Pro	Pro	Leu	Arg	Asp	Ser	Thr	Leu	Ile	Thr	Pro	Ser	His	Cys
225					230					235					240
Asp	Ser	Val	Ala	Leu	Pro	Leu	Gly	Thr	Val	Pro	Gln	Ser	Leu	Ser	Pro
				245					250					255	
His	Glu	Asp	Leu	Val	Ala	Ser	Val	Pro	Ala	Ile	Ser	Gly	Leu	Gly	Gly
			260					265					270		
Ser	Asn	Ser	His	Val	Ser	Ala	Ser	Ser	Arg	Trp	Gln	Glu	Thr	Ala	Arg
		275					280					285			
Thr	Ser	Cys	Ala	Phe	Asn	Ser	Ser	Val	Gln	Gln	Asp	His	Leu	Ser	Arg
	290					295					300				
His	Pro	Pro	Glu	Thr	Tyr	Gln	Met	Glu	Ala	Gly	Ser	Leu	Phe	Leu	Leu
305					310					315					320
Ser	Ser	Asp	Gly	Gln	Asn	Ala	Val	Gly	Ile	Gln	Val	Thr	Glu	Thr	Ala
				325					330					335	
Lys	Val	Asn	Ile	Trp	Glu	Glu	Lys	Glu	Asn	Val	Gly	Ser	Phe	Thr	Asp
			340					345					350		
Arg	Met	Thr	Pro	Glu	Lys	His	Leu	Asn	Ser	Leu	Arg	Asn	Leu	Ala	Lys
		355					360					365			
Ser	Leu	Asp	Ala	Glu	Gln	Asp	Thr	Thr	Asn	Pro	Lys	Pro	Phe	Trp	Asn
	370					375					380				
Met	Gly	Glu	Asn	Ser	Lys	Gln	Leu	Pro	Gly	Pro	Gln	Lys	Leu	Ser	Asp
385					390					395					400
Pro	Arg	Leu	Trp	Gln	Glu	Ser	Phe	Trp	Lys	Asn	Tyr	Ser	Gln	Leu	Phe
				405					410					415	
Trp	Gly	Leu	Pro	Ser	Leu	His	Ser	Glu	Ser	Leu	Val	Ala	Asn	Ala	Trp
			420					425					430		
Val	Thr	Asp	Arg	Ser	Tyr	Thr	Leu	Gln	Ser	Pro	Pro	Phe	Leu	Phe	Asn
		435					440					445			
Glu	Met	Ser	Asn	Val	Cys	Pro	Ile	Gln	Arg	Glu	Thr	Thr	Met	Ser	Pro
	450					455					460				
Leu	Leu	Phe	Gln	Ala	Gln	Pro	Pro	Ser	His	Leu	Gly	Pro	Glu	Cys	Gln
465					470					475					480
Pro	Phe	Ile	Ser	Ser	Thr	Pro	Gln	Phe	Arg	Pro	Thr	Pro	Met	Ala	Gln
				485					490					495	
Ala	Glu	Ala	Gln	Ala	His	Leu	Gln	Ser	Ser	Phe	Pro	Val	Leu	Ser	Pro
			500					505					510		
Ala	Phe	Pro	Ser	Leu	Ile	Lys	Asn	Thr	Gly	Val	Ala	Cys	Pro	Ala	Ser
	515						520					525			

Gln 530	Asn	Lys	Val	Gln	Ala	Leu 535	Ser	Leu	Pro	Glu	Thr 540	Gln	His	Pro	Glu
Trp 545	Pro	Leu	Leu	Arg	Arg 550	Gln	Leu	Glu	Gly	Arg 555	Leu	Ala	Leu	Pro	Ser 560
Arg	Val	Gln	Lys	Ser 565	Gln	Asp	Val	Phe	Ser 570	Val	Ser	Thr	Pro	Asn 575	Leu
Pro	Gln	Glu	Ser 580	Leu	Thr	Ser	Ile	Leu 585	Pro	Glu	Asn	Phe	Pro 590	Val	Ser
Pro	Glu	Leu 595	Arg	Arg	Gln	Leu	Glu 600	Gln	His	Ile	Lys	Lys 605	Trp	Ile	Ile
Gln 610	His	Trp	Gly	Asn	Leu	Gly 615	Arg	Ile	Gln	Glu	Ser 620	Leu	Asp	Leu	Met
Gln 625	Leu	Arg	Asp	Glu 630	Ser	Pro	Gly	Thr	Ser	Gln 635	Ala	Lys	Gly	Lys	Pro 640
Ser	Pro	Trp	Gln	Ser 645	Ser	Met	Ser	Thr	Gly 650	Glu	Ser	Ser	Lys	Glu 655	Ala
Gln	Lys	Val	Lys 660	Phe	Gln	Leu	Glu	Arg 665	Asp	Pro	Cys	Pro	His 670	Leu	Gly
Gln 675	Ile	Leu	Gly	Glu	Thr	Pro	Gln 680	Asn	Leu	Ser	Arg	Asp 685	Met	Lys	Ser
Phe 690	Pro	Arg	Lys	Val	Leu	Gly 695	Val	Thr	Ser	Glu	Glu 700	Ser	Glu	Arg	Asn
Leu 705	Arg	Lys	Pro	Leu 710	Arg	Ser	Asp	Ser	Gly 715	Ser	Asp	Leu	Leu	Arg	Cys 720
Thr	Glu	Arg	Thr	His 725	Ile	Glu	Asn	Ile	Leu 730	Lys	Ala	His	Met	Gly 735	Arg
Asn	Leu	Gly	Gln 740	Thr	Asn	Glu	Gly	Leu 745	Ile	Pro	Val	Arg 750	Val	Arg	Arg
Ser	Trp	Leu 755	Ala	Val	Asn	Gln	Ala 760	Leu	Pro	Val	Ser	Asn 765	Thr	His	Val
Lys 770	Thr	Ser	Asn	Leu	Ala	Ala 775	Pro	Lys	Ser	Gly	Lys 780	Ala	Cys	Val	Asn
Thr 785	Ala	Gln	Val	Leu 790	Ser	Phe	Leu	Glu	Pro	Cys 795	Thr	Gln	Gln	Gly	Leu 800
Gly	Ala	His	Ile 805	Val	Arg	Phe	Trp	Ala	Lys 810	His	Arg	Trp	Gly	Leu 815	Pro
Leu	Arg	Val	Leu 820	Lys	Pro	Ile	Gln	Cys 825	Phe	Lys	Leu	Glu 830	Lys	Val	Ser
Ser	Leu 835	Ser	Leu	Thr	Gln	Leu	Ala 840	Gly	Pro	Ser	Ser	Ala 845	Thr	Cys	Glu
Ser 850	Gly	Ala	Gly	Ser	Glu	Val 855	Glu	Val	Asp	Met	Phe 860	Leu	Arg	Lys	Pro
Pro 865	Met	Ala	Ser	Leu 870	Arg	Lys	Gln	Val	Leu	Thr 875	Lys	Ala	Ser	Asp	His 880
Met	Pro	Glu	Ser 885	Leu	Leu	Ala	Ser	Ser	Pro 890	Ala	Trp	Lys	Gln	Phe 895	Gln
Arg	Ala	Pro	Arg 900	Gly	Ile	Pro	Ser	Trp 905	Asn	Asp	His	Gly 910	Pro	Leu	Lys
Pro	Pro	Pro 915	Ala	Gly	Gln	Glu	Gly 920	Arg	Trp	Pro	Ser	Lys 925	Pro	Leu	Thr
Tyr	Ser	Leu	Thr	Gly	Ser	Thr	Gln	Gln	Ser	Arg	Ser	Leu	Gly	Ala	Gln

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930					935					940					
Ser 945	Ser	Lys	Ala	Gly	Glu 950	Thr	Arg	Glu	Ala	Val 955	Pro	Gln	Cys	Arg	Val 960
Pro	Leu	Glu	Thr	Cys 965	Met	Leu	Ala	Asn	Leu 970	Gln	Ala	Thr	Ser	Glu	Asp 975
Met	His	Gly	Phe 980	Glu	Ala	Pro	Gly	Thr 985	Ser	Lys	Ser	Ser	Leu 990	His	Pro
Arg	Val	Ser 995	Val	Ser	Gln	Asp	Pro 1000	Arg	Lys	Leu	Cys	Leu 1005	Met	Glu	Glu
Val 1010	Val	Asn	Glu	Phe	Glu	Pro 1015	Gly	Met	Ala	Thr	Lys 1020	Ser	Glu	Thr	Gln
Pro 1025	Gln	Val	Cys	Ala	Ala 1030	Val	Val	Leu	Leu	Pro 1035	Asp	Gly	Gln	Ala	Ser 1040
Val	Val	Pro	His 1045	Ala	Ser	Glu	Asn	Leu 1050	Val	Ser	Gln	Val	Pro	Gln	Gly 1055
His	Leu	Gln	Ser 1060	Met	Pro	Ala	Gly	Asn 1065	Met	Arg	Ala	Ser	Gln 1070	Glu	Leu
His 1075	Asp	Leu	Met	Ala	Ala	Arg	Arg 1080	Ser	Lys	Leu	Val	His 1085	Glu	Glu	Pro
Arg 1090	Asn	Pro	Asn	Cys	Gln	Gly 1095	Ser	Cys	Lys	Asn	Gln 1100	Arg	Pro	Met	Phe
Pro 1105	Pro	Ile	His	Lys	Ser 1110	Glu	Lys	Ser	Arg	Lys 1115	Pro	Asn	Leu	Glu	Lys 1120
His	Glu	Glu	Arg 1125	Leu	Glu	Gly	Leu	Arg	Thr 1130	Pro	Gln	Leu	Thr	Pro	Val 1135
Arg	Lys	Thr	Glu 1140	Asp	Thr	His	Gln	Asp	Glu 1145	Gly	Val	Gln	Leu	Leu	Pro 1150
Ser	Lys	Lys 1155	Gln	Pro	Pro	Ser	Val 1160	Ser	His	Phe	Gly	Gly 1165	Asn	Ile	Lys
Gln 1170	Phe	Phe	Gln	Trp	Ile	Phe 1175	Ser	Lys	Lys	Lys	Ser 1180	Lys	Pro	Ala	Pro
Val 1185	Thr	Ala	Glu	Ser	Gln 1190	Lys	Thr	Val	Lys	Asn 1195	Arg	Ser	Cys	Val	Tyr 1200
Ser	Ser	Ser	Ala 1205	Glu	Ala	Gln	Gly	Leu	Met 1210	Thr	Ala	Val	Gly	Gln	Met 1215
Leu	Asp	Glu	Lys 1220	Met	Ser	Leu	Cys	His	Ala 1225	Arg	His	Ala	Ser	Lys	Val 1230
Asn	Gln	His 1235	Lys	Gln	Lys	Phe	Gln 1240	Ala	Pro	Val	Cys	Gly 1245	Phe	Pro	Cys
Asn 1250	His	Arg	His	Leu	Phe	Tyr 1255	Ser	Glu	His	Gly	Arg 1260	Ile	Leu	Ser	Tyr
Ala 1265	Ala	Ser	Ser	Gln	Gln 1270	Ala	Thr	Leu	Lys	Ser 1275	Gln	Gly	Cys	Pro	Asn 1280
Arg	Asp	Arg	Gln 1285	Ile	Arg	Asn	Gln	Gln	Pro 1290	Leu	Lys	Ser	Val	Arg	Cys 1295
Asn	Asn	Glu 1300	Gln	Trp	Gly	Leu	Arg	His 1305	Pro	Gln	Ile	Leu	His 1310	Pro	Lys
Lys	Ala	Val 1315	Ser	Pro	Val	Ser	Pro 1320	Leu	Gln	His	Trp	Pro 1325	Lys	Thr	Ser
Gly 1330	Ala	Ser	Ser	His	His 1335	His	His	Cys	Pro	Arg	His 1340	Cys	Leu	Leu	Trp

-continued

Glu Gly Ile
1345

<210> SEQ ID NO 60

<211> LENGTH: 1031

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 60

Met Pro Pro Gly Arg Trp His Ala Ala Tyr Pro Ala Gln Ala Gln Ser
1 5 10 15

Ser Arg Glu Arg Gly Arg Leu Gln Thr Val Lys Lys Glu Glu Glu Asp
20 25 30

Glu Ser Tyr Thr Pro Val Gln Ala Ala Arg Pro Gln Thr Leu Asn Arg
35 40 45

Pro Gly Gln Glu Leu Phe Arg Gln Leu Phe Arg Gln Leu Arg Tyr His
50 55 60

Glu Ser Ser Gly Pro Leu Glu Thr Leu Ser Arg Leu Arg Glu Leu Cys
65 70 75 80

Arg Trp Trp Leu Arg Pro Asp Val Leu Ser Lys Ala Gln Ile Leu Glu
85 90 95

Leu Leu Val Leu Glu Gln Phe Leu Ser Ile Leu Pro Gly Glu Leu Arg
100 105 110

Val Trp Val Gln Leu His Asn Pro Glu Ser Gly Glu Glu Ala Val Ala
115 120 125

Leu Leu Glu Glu Leu Gln Arg Asp Leu Asp Gly Thr Ser Trp Arg Asp
130 135 140

Pro Gly Pro Ala Gln Ser Pro Asp Val His Trp Met Gly Thr Gly Ala
145 150 155 160

Leu Arg Ser Ala Gln Ile Trp Ser Leu Ala Ser Pro Leu Arg Ser Ser
165 170 175

Ser Ala Leu Gly Asp His Leu Glu Pro Pro Tyr Glu Ile Glu Ala Arg
180 185 190

Asp Phe Leu Ala Gly Gln Ser Asp Thr Pro Ala Ala Gln Met Pro Ala
195 200 205

Leu Phe Pro Arg Glu Gly Cys Pro Gly Asp Gln Val Thr Pro Thr Arg
210 215 220

Ser Leu Thr Ala Gln Leu Gln Glu Thr Met Thr Phe Lys Asp Val Glu
225 230 235 240

Val Thr Phe Ser Gln Asp Glu Trp Gly Trp Leu Asp Ser Ala Gln Arg
245 250 255

Asn Leu Tyr Arg Asp Val Met Leu Glu Asn Tyr Arg Asn Met Ala Ser
260 265 270

Leu Val Gly Pro Phe Thr Lys Pro Ala Leu Ile Ser Trp Leu Glu Ala
275 280 285

Arg Glu Pro Trp Gly Leu Asn Met Gln Ala Ala Gln Pro Lys Gly Asn
290 295 300

Pro Val Ala Ala Pro Thr Gly Asp Asp Leu Gln Ser Lys Thr Asn Lys
305 310 315 320

Phe Ile Leu Asn Gln Glu Pro Leu Glu Glu Ala Glu Thr Leu Ala Val
325 330 335

Ser Ser Gly Cys Pro Ala Thr Ser Val Ser Glu Gly Ile Gly Leu Arg

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340							345					350				
Glu	Ser	Phe	Gln	Gln	Lys	Ser	Arg	Gln	Lys	Asp	Gln	Cys	Glu	Asn	Pro	
355						360						365				
Ile	Gln	Val	Arg	Val	Lys	Lys	Glu	Glu	Thr	Asn	Phe	Ser	His	Arg	Thr	
370						375				380						
Gly	Lys	Asp	Ser	Glu	Val	Ser	Gly	Ser	Asn	Ser	Leu	Asp	Leu	Lys	His	
385				390						395				400		
Val	Thr	Tyr	Leu	Arg	Val	Ser	Gly	Arg	Lys	Glu	Ser	Leu	Lys	His	Gly	
			405						410			415				
Cys	Gly	Lys	His	Phe	Arg	Met	Ser	Ser	His	His	Tyr	Asp	Tyr	Lys	Lys	
			420						425				430			
Tyr	Gly	Lys	Gly	Leu	Arg	His	Met	Ile	Gly	Gly	Phe	Ser	Leu	His	Gln	
		435				440						445				
Arg	Ile	His	Ser	Gly	Leu	Lys	Gly	Asn	Lys	Lys	Asp	Val	Cys	Gly	Lys	
450						455						460				
Asp	Phe	Ser	Leu	Ser	Ser	His	His	Gln	Arg	Gly	Gln	Ser	Leu	His	Thr	
465				470						475				480		
Val	Gly	Val	Ser	Phe	Lys	Cys	Ser	Asp	Cys	Gly	Arg	Thr	Phe	Ser	His	
			485						490				495			
Ser	Ser	His	Leu	Ala	Tyr	His	Gln	Arg	Leu	His	Thr	Gln	Glu	Lys	Ala	
			500						505				510			
Phe	Lys	Cys	Arg	Val	Cys	Gly	Lys	Ala	Phe	Arg	Trp	Ser	Ser	Asn	Cys	
		515				520						525				
Ala	Arg	His	Glu	Lys	Ile	His	Thr	Gly	Val	Lys	Pro	Tyr	Lys	Cys	Asp	
530						535						540				
Leu	Cys	Glu	Lys	Ala	Phe	Arg	Arg	Leu	Ser	Ala	Tyr	Arg	Leu	His	Arg	
545				550						555				560		
Glu	Thr	His	Ala	Lys	Lys	Lys	Phe	Leu	Glu	Leu	Asn	Gln	Tyr	Arg	Ala	
			565						570				575			
Ala	Leu	Thr	Tyr	Ser	Ser	Gly	Phe	Asp	His	His	Leu	Gly	Asp	Gln	Ser	
			580					585				590				
Gly	Glu	Lys	Leu	Phe	Asp	Cys	Ser	Gln	Cys	Arg	Lys	Ser	Phe	His	Cys	
		595				600						605				
Lys	Ser	Tyr	Val	Leu	Glu	His	Gln	Arg	Ile	His	Thr	Gln	Glu	Lys	Pro	
610						615						620				
Tyr	Lys	Cys	Thr	Lys	Cys	Arg	Lys	Thr	Phe	Arg	Trp	Arg	Ser	Asn	Phe	
625				630						635				640		
Thr	Arg	His	Met	Arg	Leu	His	Glu	Glu	Glu	Lys	Phe	Tyr	Lys	Gln	Asp	
			645						650				655			
Glu	Cys	Arg	Glu	Gly	Phe	Arg	Gln	Ser	Pro	Asp	Cys	Ser	Gln	Pro	Gln	
			660						665				670			
Gly	Ala	Pro	Ala	Val	Glu	Lys	Thr	Phe	Leu	Cys	Gln	Gln	Cys	Gly	Lys	
		675				680						685				
Thr	Phe	Thr	Arg	Lys	Lys	Thr	Leu	Val	Asp	His	Gln	Arg	Ile	His	Thr	
690						695						700				
Gly	Glu	Lys	Pro	Tyr	Gln	Cys	Ser	Asp	Cys	Gly	Lys	Asp	Phe	Ala	Tyr	
705				710						715				720		
Arg	Ser	Ala	Phe	Ile	Val	His	Lys	Lys	Lys	His	Ala	Met	Lys	Arg	Lys	
			725						730				735			
Pro	Glu	Gly	Gly	Pro	Ser	Phe	Ser	Gln	Asp	Thr	Val	Phe	Gln	Val	Pro	
		740						745				750				

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Gln Ser Ser His Ser Lys Glu Glu Pro Tyr Lys Cys Ser Gln Cys Gly
 755 760 765
 Lys Ala Phe Arg Asn His Ser Phe Leu Leu Ile His Gln Arg Val His
 770 775 780
 Thr Gly Glu Lys Pro Tyr Lys Cys Arg Glu Cys Gly Lys Ala Phe Arg
 785 790 795 800
 Trp Ser Ser Asn Leu Tyr Arg His Gln Arg Ile His Ser Leu Gln Lys
 805 810 815
 Gln Tyr Asp Cys His Glu Ser Glu Lys Thr Pro Asn Val Glu Pro Lys
 820 825 830
 Ile Leu Thr Gly Glu Lys Arg Phe Trp Cys Gln Glu Cys Gly Lys Thr
 835 840 845
 Phe Thr Arg Lys Arg Thr Leu Leu Asp His Lys Gly Ile His Ser Gly
 850 855 860
 Glu Lys Arg Tyr Lys Cys Asn Leu Cys Gly Lys Ser Tyr Asp Arg Asn
 865 870 875 880
 Tyr Arg Leu Val Asn His Gln Arg Ile His Ser Thr Glu Arg Pro Phe
 885 890 895
 Lys Cys Gln Trp Cys Gly Lys Glu Phe Ile Gly Arg His Thr Leu Ser
 900 905 910
 Ser His Gln Arg Lys His Thr Arg Ala Ala Gln Ala Glu Arg Ser Pro
 915 920 925
 Pro Ala Arg Ser Ser Ser Gln Asp Thr Lys Leu Arg Leu Gln Lys Leu
 930 935 940
 Lys Pro Ser Glu Glu Met Pro Leu Glu Asp Cys Lys Glu Ala Cys Ser
 945 950 955 960
 Gln Ser Ser Arg Leu Thr Gly Leu Gln Asp Ile Ser Ile Gly Lys Lys
 965 970 975
 Cys His Lys Cys Ser Ile Cys Gly Lys Thr Phe Asn Lys Ser Ser Gln
 980 985 990
 Leu Ile Ser His Lys Arg Phe His Thr Arg Glu Arg Pro Phe Lys Cys
 995 1000 1005
 Ser Lys Cys Gly Lys Thr Phe Arg Trp Ser Ser Asn Leu Ala Arg His
 1010 1015 1020
 Met Lys Asn His Ile Arg Asp
 1025 1030

<210> SEQ ID NO 61
 <211> LENGTH: 1075
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 61

Met Arg Gly Gly Lys Cys Asn Met Leu Ser Ser Leu Gly Cys Leu Leu
 1 5 10 15
 Leu Cys Gly Ser Ile Thr Leu Ala Leu Gly Asn Ala Gln Lys Leu Pro
 20 25 30
 Lys Gly Lys Arg Pro Asn Leu Lys Val His Ile Asn Thr Thr Ser Asp
 35 40 45
 Ser Ile Leu Leu Lys Phe Leu Arg Pro Ser Pro Asn Val Lys Leu Glu
 50 55 60
 Gly Leu Leu Leu Gly Tyr Gly Ser Asn Val Ser Pro Asn Gln Tyr Phe

Thr	Lys	Pro	Glu	Arg	Thr	Thr	Ser	Ala	Gly	Thr	Ile	Thr	Pro	Lys	Ile	
				485												
Ser	Lys	Ser	Pro	Glu	Pro	Thr	Trp	Thr	Thr	Pro	Ala	Pro	Gly	Lys	Thr	
				500												
Gln	Phe	Ile	Ser	Leu	Lys	Pro	Lys	Ile	Pro	Leu	Ser	Pro	Glu	Val	Thr	
				515												
His	Thr	Lys	Pro	Ala	Pro	Lys	Gln	Thr	Pro	Arg	Ala	Pro	Pro	Lys	Pro	
				530												
Lys	Thr	Ser	Pro	Arg	Pro	Arg	Ile	Pro	Gln	Thr	Gln	Pro	Val	Pro	Lys	
				545												
Val	Pro	Gln	Arg	Val	Thr	Ala	Lys	Pro	Lys	Thr	Ser	Pro	Ser	Pro	Glu	
				565												
Val	Ser	Tyr	Thr	Thr	Pro	Ala	Pro	Lys	Asp	Val	Leu	Leu	Pro	His	Lys	
				580												
Pro	Tyr	Pro	Glu	Val	Ser	Gln	Ser	Glu	Pro	Ala	Pro	Leu	Glu	Thr	Arg	
				595												
Gly	Ile	Pro	Phe	Ile	Pro	Met	Ile	Ser	Pro	Ser	Pro	Ser	Gln	Glu	Glu	
				610												
Leu	Gln	Thr	Thr	Leu	Glu	Glu	Thr	Asp	Gln	Ser	Thr	Gln	Glu	Pro	Phe	
				625												
Thr	Thr	Lys	Ile	Pro	Arg	Thr	Thr	Glu	Leu	Ala	Lys	Thr	Thr	Gln	Ala	
				645												
Pro	His	Arg	Phe	Tyr	Thr	Thr	Val	Arg	Pro	Arg	Thr	Ser	Asp	Lys	Pro	
				660												
His	Ile	Arg	Pro	Gly	Val	Lys	Gln	Ala	Pro	Arg	Pro	Ser	Gly	Ala	Asp	
				675												
Arg	Asn	Val	Ser	Val	Asp	Ser	Thr	His	Pro	Thr	Lys	Lys	Pro	Gly	Thr	
				690												
Arg	Arg	Pro	Pro	Leu	Pro	Pro	Arg	Pro	Thr	His	Pro	Arg	Arg	Lys	Pro	
				705												
Leu	Pro	Pro	Asn	Asn	Val	Thr	Gly	Lys	Pro	Gly	Ser	Ala	Gly	Ile	Ile	
				725												
Ser	Ser	Gly	Pro	Ile	Thr	Thr	Pro	Pro	Leu	Arg	Ser	Thr	Pro	Arg	Pro	
				740												
Thr	Gly	Thr	Pro	Leu	Glu	Arg	Ile	Glu	Thr	Asp	Ile	Lys	Gln	Pro	Thr	
				755												
Val	Pro	Ala	Ser	Gly	Glu	Glu	Leu	Glu	Asn	Ile	Thr	Asp	Phe	Ser	Ser	
				770												
Ser	Pro	Thr	Arg	Glu	Thr	Asp	Pro	Leu	Gly	Lys	Pro	Arg	Phe	Lys	Gly	
				785												
Pro	His	Val	Arg	Tyr	Ile	Gln	Lys	Pro	Asp	Asn	Ser	Pro	Cys	Ser	Ile	
				805												
Thr	Asp	Ser	Val	Lys	Arg	Phe	Pro	Lys	Glu	Glu	Ala	Thr	Glu	Gly	Asn	
				820												
Ala	Thr	Ser	Pro	Pro	Gln	Asn	Pro	Pro	Thr	Asn	Leu	Thr	Val	Val	Thr	
				835												
Val	Glu	Gly	Cys	Pro	Ser	Phe	Val	Ile	Leu	Asp	Trp	Glu	Lys	Pro	Leu	
				850												
Asn	Asp	Thr	Val	Thr	Glu	Tyr	Glu	Val	Ile	Ser	Arg	Glu	Asn	Gly	Ser	
				865												

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Phe	Ser	Gly	Lys	Asn	Lys	Ser	Ile	Gln	Met	Thr	Asn	Gln	Thr	Phe	Ser	885	890	895
Thr	Val	Glu	Asn	Leu	Lys	Pro	Asn	Thr	Ser	Tyr	Glu	Phe	Gln	Val	Lys	900	905	910
Pro	Lys	Asn	Pro	Leu	Gly	Glu	Gly	Pro	Val	Ser	Asn	Thr	Val	Ala	Phe	915	920	925
Ser	Thr	Glu	Ser	Ala	Asp	Pro	Arg	Val	Ser	Glu	Pro	Val	Ser	Ala	Gly	930	935	940
Arg	Asp	Ala	Ile	Trp	Thr	Glu	Arg	Pro	Phe	Asn	Ser	Asp	Ser	Tyr	Ser	945	950	955
Glu	Cys	Lys	Gly	Lys	Gln	Tyr	Val	Lys	Arg	Thr	Trp	Tyr	Lys	Lys	Phe	965	970	975
Val	Gly	Val	Gln	Leu	Cys	Asn	Ser	Leu	Arg	Tyr	Lys	Ile	Tyr	Leu	Ser	980	985	990
Asp	Ser	Leu	Thr	Gly	Lys	Phe	Tyr	Asn	Ile	Gly	Asp	Gln	Arg	Gly	His	995	1000	1005
Gly	Glu	Asp	His	Cys	Gln	Phe	Val	Asp	Ser	Phe	Leu	Asp	Gly	Arg	Thr	1010	1015	1020
Gly	Gln	Gln	Leu	Thr	Ser	Asp	Gln	Leu	Pro	Ile	Lys	Glu	Gly	Tyr	Phe	1025	1030	1035
Arg	Ala	Val	Arg	Gln	Glu	Pro	Val	Gln	Phe	Gly	Glu	Ile	Gly	Gly	His	1045	1050	1055
Thr	Gln	Ile	Asn	Tyr	Val	Gln	Trp	Tyr	Glu	Cys	Gly	Thr	Thr	Ile	Pro	1060	1065	1070
Gly	Lys	Trp														1075		

<210> SEQ ID NO 62
 <211> LENGTH: 272
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 62

Met	Gly	Asn	Thr	Thr	Ser	Asp	Arg	Val	Ser	Gly	Glu	Arg	His	Gly	Ala	1	5	10	15
Lys	Ala	Ala	Arg	Ser	Glu	Gly	Ala	Gly	Gly	His	Ala	Pro	Gly	Lys	Glu	20	25	30	
His	Lys	Ile	Met	Val	Gly	Ser	Thr	Asp	Asp	Pro	Ser	Val	Phe	Ser	Leu	35	40	45	
Pro	Asp	Ser	Lys	Leu	Pro	Gly	Asp	Lys	Glu	Phe	Val	Ser	Trp	Gln	Gln	50	55	60	
Asp	Leu	Glu	Asp	Ser	Val	Lys	Pro	Thr	Gln	Gln	Ala	Arg	Pro	Thr	Val	65	70	75	80
Ile	Arg	Trp	Ser	Glu	Gly	Gly	Lys	Glu	Val	Phe	Ile	Ser	Gly	Ser	Phe	85	90	95	
Asn	Asn	Trp	Ser	Thr	Lys	Ile	Pro	Leu	Ile	Lys	Ser	His	Asn	Asp	Phe	100	105	110	
Val	Ala	Ile	Leu	Asp	Leu	Pro	Glu	Gly	Glu	His	Gln	Tyr	Lys	Phe	Phe	115	120	125	
Val	Asp	Gly	Gln	Trp	Val	His	Asp	Pro	Ser	Glu	Pro	Val	Val	Thr	Ser	130	135	140	
Gln	Leu	Gly	Thr	Ile	Asn	Asn	Leu	Ile	His	Val	Lys	Lys	Ser	Asp	Phe	145	150	155	160

Glu	Val	Phe	Asp	Ala	Leu	Lys	Leu	Asp	Ser	Met	Glu	Ser	Ser	Glu	Thr
			165					170						175	
Ser	Cys	Arg	Asp	Leu	Ser	Ser	Ser	Pro	Pro	Gly	Pro	Tyr	Gly	Gln	Glu
			180					185					190		
Met	Tyr	Ala	Phe	Arg	Ser	Glu	Glu	Arg	Phe	Lys	Ser	Pro	Pro	Ile	Leu
		195					200					205			
Pro	Pro	His	Leu	Leu	Gln	Val	Ile	Leu	Asn	Lys	Asp	Thr	Asn	Ile	Ser
	210					215					220				
Cys	Asp	Pro	Ala	Leu	Leu	Pro	Glu	Pro	Asn	His	Val	Met	Leu	Asn	His
225				230					235					240	
Leu	Tyr	Ala	Leu	Ser	Ile	Lys	Asp	Ser	Val	Met	Val	Leu	Ser	Ala	Thr
			245						250					255	
His	Arg	Tyr	Lys	Lys	Tyr	Val	Thr	Thr	Leu	Leu	Tyr	Lys	Pro	Ile	
	260						265					270			

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<210> SEQ ID NO 63
<211> LENGTH: 295
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
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<400> SEQUENCE: 63

Met	Glu	Leu	Ile	Gln	Asp	Thr	Ser	Arg	Pro	Pro	Leu	Glu	Tyr	Val	Lys
1				5					10					15	
Gly	Val	Pro	Leu	Ile	Lys	Tyr	Phe	Ala	Glu	Ala	Leu	Gly	Pro	Leu	Gln
			20					25					30		
Ser	Phe	Gln	Ala	Arg	Pro	Asp	Asp	Leu	Leu	Ile	Asn	Thr	Tyr	Pro	Lys
		35					40					45			
Ser	Gly	Thr	Thr	Trp	Val	Ser	Gln	Ile	Leu	Asp	Met	Ile	Tyr	Gln	Gly
	50					55					60				
Gly	Asp	Leu	Glu	Lys	Cys	Asn	Arg	Ala	Pro	Ile	Tyr	Val	Arg	Val	Pro
65					70					75					80
Phe	Leu	Glu	Val	Asn	Asp	Pro	Gly	Glu	Pro	Ser	Gly	Leu	Glu	Thr	Leu
				85					90					95	
Lys	Asp	Thr	Pro	Pro	Pro	Arg	Leu	Ile	Lys	Ser	His	Leu	Pro	Leu	Ala
			100					105					110		
Leu	Leu	Pro	Gln	Thr	Leu	Leu	Asp	Gln	Lys	Val	Lys	Val	Val	Tyr	Val
		115					120					125			
Ala	Arg	Asn	Pro	Lys	Asp	Val	Ala	Val	Ser	Tyr	Tyr	His	Phe	His	Arg
		130				135						140			
Met	Glu	Lys	Ala	His	Pro	Glu	Pro	Gly	Thr	Trp	Asp	Ser	Phe	Leu	Glu
145					150					155					160
Lys	Phe	Met	Ala	Gly	Glu	Val	Ser	Tyr	Gly	Ser	Trp	Tyr	Gln	His	Val
				165					170					175	
Gln	Glu	Trp	Trp	Glu	Leu	Ser	Arg	Thr	His	Pro	Val	Leu	Tyr	Leu	Phe
			180					185					190		
Tyr	Glu	Asp	Met	Lys	Glu	Asn	Pro	Lys	Arg	Glu	Ile	Gln	Lys	Ile	Leu
		195					200					205			
Glu	Phe	Val	Gly	Arg	Ser	Leu	Pro	Glu	Glu	Thr	Met	Asp	Phe	Met	Val
		210				215					220				
Gln	His	Thr	Ser	Phe	Lys	Glu	Met	Lys	Lys	Asn	Pro	Met	Thr	Asn	Tyr
225					230					235					240
Thr	Thr	Val	Pro	Gln	Glu	Leu	Met	Asp	His	Ser	Ile	Ser	Pro	Phe	Met

245																250								255			
Arg	Lys	Gly	Met	Ala	Gly	Asp	Trp	Lys	Thr	Thr	Phe	Thr	Val	Ala	Gln												
			260						265						270												
Asn	Glu	Arg	Phe	Asp	Ala	Asp	Tyr	Ala	Glu	Lys	Met	Ala	Gly	Cys	Ser												
			275						280						285												
Leu	Ser	Phe	Arg	Ser	Glu	Leu																					
			290						295																		
<210> SEQ ID NO 64																											
<211> LENGTH: 491																											
<212> TYPE: PRT																											
<213> ORGANISM: Homo sapiens																											
<400> SEQUENCE: 64																											
Met	Ala	Gly	Gln	Gly	Leu	Pro	Leu	His	Val	Ala	Thr	Leu	Leu	Thr	Gly												
1				5						10						15											
Leu	Leu	Glu	Cys	Leu	Gly	Phe	Ala	Gly	Val	Leu	Phe	Gly	Trp	Pro	Ser												
			20						25						30												
Leu	Val	Phe	Val	Phe	Lys	Asn	Glu	Asp	Tyr	Phe	Lys	Asp	Leu	Cys	Gly												
			35						40						45												
Pro	Asp	Ala	Gly	Pro	Ile	Gly	Asn	Ala	Thr	Gly	Gln	Ala	Asp	Cys	Lys												
			50						55						60												
Ala	Gln	Asp	Glu	Arg	Phe	Ser	Leu	Ile	Phe	Thr	Leu	Gly	Ser	Phe	Met												
65				70						75						80											
Asn	Asn	Phe	Met	Thr	Phe	Pro	Thr	Gly	Tyr	Ile	Phe	Asp	Arg	Phe	Lys												
			85						90						95												
Thr	Thr	Val	Ala	Arg	Leu	Ile	Ala	Ile	Phe	Phe	Tyr	Thr	Thr	Ala	Thr												
			100						105						110												
Leu	Ile	Ile	Ala	Phe	Thr	Ser	Ala	Gly	Ser	Ala	Val	Leu	Leu	Phe	Leu												
			115						120						125												
Ala	Met	Pro	Met	Leu	Thr	Ile	Gly	Gly	Ile	Leu	Phe	Leu	Ile	Thr	Asn												
			130						135						140												
Leu	Gln	Ile	Gly	Asn	Leu	Phe	Gly	Gln	His	Arg	Ser	Thr	Ile	Ile	Thr												
145				150						155						160											
Leu	Tyr	Asn	Gly	Ala	Phe	Asp	Ser	Ser	Ser	Ala	Val	Phe	Leu	Ile	Ile												
			165						170						175												
Lys	Leu	Leu	Tyr	Glu	Lys	Gly	Ile	Ser	Leu	Arg	Ala	Ser	Phe	Ile	Phe												
			180						185						190												
Ile	Ser	Val	Cys	Ser	Thr	Trp	His	Val	Ala	Arg	Thr	Phe	Leu	Leu	Met												
			195						200						205												
Pro	Arg	Gly	His	Ile	Pro	Tyr	Pro	Leu	Pro	Pro	Asn	Tyr	Ser	Tyr	Gly												
			210						215						220												
Leu	Cys	Pro	Gly	Asn	Gly	Thr	Thr	Lys	Glu	Glu	Lys	Glu	Thr	Ala	Glu												
225				230						235						240											
His	Glu	Asn	Arg	Glu	Leu	Gln	Ser	Lys	Glu	Phe	Leu	Ser	Ala	Lys	Glu												
			245						250						255												
Glu	Thr	Pro	Gly	Ala	Gly	Gln	Lys	Gln	Glu	Leu	Arg	Ser	Phe	Trp	Ser												
			260						265						270												
Tyr	Ala	Phe	Ser	Arg	Arg	Phe	Ala	Trp	His	Leu	Val	Trp	Leu	Ser	Val												
			275						280						285												
Ile	Gln	Leu	Trp	His	Tyr	Leu	Phe	Ile	Gly	Thr	Leu	Asn	Ser	Leu	Leu												
			290						295						300												

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Thr	Asn	Met	Ala	Gly	Gly	Asp	Met	Ala	Arg	Val	Ser	Thr	Tyr	Thr	Asn
305					310					315					320
Ala	Phe	Ala	Phe	Thr	Gln	Phe	Gly	Val	Leu	Cys	Ala	Pro	Trp	Asn	Gly
				325					330					335	
Leu	Leu	Met	Asp	Arg	Leu	Lys	Gln	Lys	Tyr	Gln	Lys	Glu	Ala	Arg	Lys
			340					345					350		
Thr	Gly	Ser	Ser	Thr	Leu	Ala	Val	Ala	Leu	Cys	Ser	Thr	Val	Pro	Ser
		355					360					365			
Leu	Ala	Leu	Thr	Ser	Leu	Leu	Cys	Leu	Gly	Phe	Ala	Leu	Cys	Ala	Ser
	370					375					380				
Val	Pro	Ile	Leu	Pro	Leu	Gln	Tyr	Leu	Thr	Phe	Ile	Leu	Gln	Val	Ile
385					390					395					400
Ser	Arg	Ser	Phe	Leu	Tyr	Gly	Ser	Asn	Ala	Ala	Phe	Leu	Thr	Leu	Ala
				405					410					415	
Phe	Pro	Ser	Glu	His	Phe	Gly	Lys	Leu	Phe	Gly	Leu	Val	Met	Ala	Leu
			420					425					430		
Ser	Ala	Val	Val	Ser	Leu	Leu	Gln	Phe	Pro	Ile	Phe	Thr	Leu	Ile	Lys
		435					440					445			
Gly	Ser	Leu	Gln	Asn	Asp	Pro	Phe	Tyr	Val	Asn	Val	Met	Phe	Met	Leu
	450					455					460				
Ala	Ile	Leu	Leu	Thr	Phe	Phe	His	Pro	Phe	Leu	Val	Tyr	Arg	Glu	Cys
465					470					475					480
Arg	Thr	Trp	Lys	Glu	Ser	Pro	Ser	Ala	Ile	Ala					
			485						490						

<210> SEQ ID NO 65

<211> LENGTH: 519

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 65

Met	Ser	Tyr	Pro	Gln	Phe	Gly	Tyr	Pro	Tyr	Ser	Ser	Ala	Pro	Gln	Phe
1				5					10					15	
Leu	Met	Ala	Thr	Asn	Ser	Leu	Ser	Thr	Cys	Cys	Glu	Ser	Gly	Gly	Arg
			20					25					30		
Thr	Leu	Ala	Asp	Ser	Gly	Pro	Ala	Ala	Ser	Ala	Gln	Ala	Pro	Val	Tyr
		35				40					45				
Cys	Pro	Val	Tyr	Glu	Ser	Arg	Leu	Leu	Ala	Thr	Ala	Arg	His	Glu	Leu
	50				55					60					
Asn	Ser	Ala	Ala	Ala	Leu	Gly	Val	Tyr	Gly	Gly	Pro	Tyr	Gly	Gly	Ser
65				70					75					80	
Gln	Gly	Tyr	Gly	Asn	Tyr	Val	Thr	Tyr	Gly	Ser	Glu	Ala	Ser	Ala	Phe
			85					90						95	
Tyr	Ser	Leu	Asn	Ser	Phe	Asp	Ser	Lys	Asp	Gly	Ser	Gly	Ser	Ala	His
		100						105					110		
Gly	Gly	Leu	Ala	Pro	Ala	Ala	Ala	Ala	Tyr	Tyr	Pro	Tyr	Glu	Pro	Ala
		115				120						125			
Leu	Gly	Gln	Tyr	Pro	Tyr	Asp	Arg	Tyr	Gly	Thr	Met	Asp	Ser	Gly	Thr
	130					135					140				
Arg	Arg	Lys	Asn	Ala	Thr	Arg	Glu	Thr	Thr	Ser	Thr	Leu	Lys	Ala	Trp
145				150						155					160
Leu	Gln	Glu	His	Arg	Lys	Asn	Pro	Tyr	Pro	Thr	Lys	Gly	Glu	Lys	Ile
			165					170						175	

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Met	Leu	Ala	Ile	Ile	Thr	Lys	Met	Thr	Leu	Thr	Gln	Val	Ser	Thr	Trp	180	185	190	
Phe	Ala	Asn	Ala	Arg	Arg	Arg	Leu	Lys	Lys	Glu	Asn	Lys	Met	Thr	Trp	195	200	205	
Pro	Pro	Arg	Asn	Lys	Cys	Ala	Asp	Glu	Lys	Arg	Pro	Tyr	Ala	Glu	Gly	210	215	220	
Glu	Glu	Glu	Glu	Gly	Gly	Glu	Glu	Glu	Ala	Arg	Glu	Glu	Pro	Leu	Lys	225	230	235	240
Ser	Ser	Lys	Asn	Ala	Glu	Pro	Val	Gly	Lys	Glu	Glu	Lys	Glu	Leu	Glu	245	250	255	
Leu	Ser	Asp	Leu	Asp	Asp	Phe	Asp	Pro	Leu	Glu	Ala	Glu	Pro	Pro	Ala	260	265	270	
Cys	Glu	Leu	Lys	Pro	Pro	Phe	His	Ser	Leu	Asp	Gly	Gly	Leu	Glu	Arg	275	280	285	
Val	Pro	Ala	Ala	Pro	Asp	Gly	Pro	Val	Lys	Glu	Ala	Ser	Gly	Ala	Leu	290	295	300	
Arg	Met	Ser	Leu	Ala	Ala	Gly	Gly	Gly	Ala	Ala	Leu	Asp	Glu	Asp	Leu	305	310	315	320
Glu	Arg	Ala	Arg	Ser	Cys	Leu	Arg	Ser	Ala	Ala	Ala	Gly	Pro	Glu	Pro	325	330	335	
Leu	Pro	Gly	Ala	Glu	Gly	Gly	Pro	Gln	Val	Cys	Glu	Ala	Lys	Leu	Gly	340	345	350	
Phe	Val	Pro	Ala	Gly	Ala	Ser	Ala	Gly	Leu	Glu	Ala	Lys	Pro	Arg	Ile	355	360	365	
Trp	Ser	Leu	Ala	His	Thr	Ala	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Thr	Ser	370	375	380	
Leu	Ser	Gln	Thr	Glu	Phe	Pro	Ser	Cys	Met	Leu	Lys	Arg	Gln	Gly	Pro	385	390	395	400
Ala	Ala	Pro	Ala	Ala	Val	Ser	Ser	Ala	Pro	Ala	Thr	Ser	Pro	Ser	Val	405	410	415	
Ala	Leu	Pro	His	Ser	Gly	Ala	Leu	Asp	Arg	His	Gln	Asp	Ser	Pro	Val	420	425	430	
Thr	Ser	Leu	Arg	Asn	Trp	Val	Asp	Gly	Val	Phe	His	Asp	Pro	Ile	Leu	435	440	445	
Arg	His	Ser	Thr	Leu	Asn	Gln	Ala	Trp	Ala	Thr	Ala	Lys	Gly	Ala	Leu	450	455	460	
Leu	Asp	Pro	Gly	Pro	Leu	Gly	Arg	Ser	Leu	Gly	Ala	Gly	Ala	Asn	Val	465	470	475	480
Leu	Thr	Ala	Pro	Leu	Ala	Arg	Ala	Phe	Pro	Pro	Ala	Val	Pro	Gln	Asp	485	490	495	
Ala	Pro	Ala	Ala	Gly	Ala	Ala	Arg	Glu	Leu	Leu	Ala	Leu	Pro	Lys	Ala	500	505	510	
Gly	Gly	Lys	Pro	Phe	Cys	Ala										515			

<210> SEQ ID NO 66

<211> LENGTH: 9

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 66

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uucaagaga 9

<210> SEQ ID NO 67
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 67

ccaccgctgt ccaccgcctc 20

<210> SEQ ID NO 68
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 68

gaccaaggc cagcgccgc 20

<210> SEQ ID NO 69
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 69

ggagggagga gaaccagcag 20

<210> SEQ ID NO 70
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 70

aaattgcagc cggagcgcg 20

<210> SEQ ID NO 71
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 71

tccggataaa ccgaggcacg 20

<210> SEQ ID NO 72
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 72

gcgaagcatc ttcaccaccc 20

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<210> SEQ ID NO 73
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 73

gacccagcac tgcagcctgg 20

<210> SEQ ID NO 74
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 74

aaacttatgc ggcgtttcct 20

<210> SEQ ID NO 75
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 75

atcactttaa aaccacctct 20

<210> SEQ ID NO 76
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 76

ctgaaaaagg aaggagttga 20

<210> SEQ ID NO 77
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 77

cgcgagcaca gctaaggcca 20

<210> SEQ ID NO 78
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 78

gagtagcgcg agcacagcta 20

<210> SEQ ID NO 79
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 79

ggccgagatg tctcgctccg 20

<210> SEQ ID NO 80
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 80

aacaaagcac cctccccact 20

<210> SEQ ID NO 81
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 81

caaacaggcg tatctgtgtg 20

<210> SEQ ID NO 82
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 82

ctctgcaacc aggagcccag 20

<210> SEQ ID NO 83
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 83

cactgttgct ttatcagacg 20

<210> SEQ ID NO 84
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 84

ctcgtctgat aagacaacag 20

<210> SEQ ID NO 85
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 85

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gtttgttctt agggtaacag 20

<210> SEQ ID NO 86
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 86

actcagccat tttattagaa 20

<210> SEQ ID NO 87
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 87

gagcagcctg tctttatcgg 20

<210> SEQ ID NO 88
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 88

gttctcggg cccctctac 20

<210> SEQ ID NO 89
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 89

cgctcgtgc gcgccgttg 20

<210> SEQ ID NO 90
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 90

gaggccacac tgcagcgagg 20

<210> SEQ ID NO 91
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 91

tgggtctgtc tcatcgtaa 20

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<210> SEQ ID NO 92
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 92

gtggcacagg tgggcataat 20

<210> SEQ ID NO 93
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 93

gggtggagag ttgggacacc 20

<210> SEQ ID NO 94
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 94

gctcggctgg gccagtcgcc 20

<210> SEQ ID NO 95
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 95

tccattcagt cagtctttgg 20

<210> SEQ ID NO 96
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 96

gagagctatc acctaagtgt 20

<210> SEQ ID NO 97
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 97

tatctggccc tgggtgcttg 20

<210> SEQ ID NO 98
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 98

ggtgccgccc ccagcgtagg 20

<210> SEQ ID NO 99
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 99

gacaggctca gctatgaaga 20

<210> SEQ ID NO 100
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 100

gcggcagtcg ggcggatgta 20

<210> SEQ ID NO 101
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 101

tccccctcca gctgcaacgg 20

<210> SEQ ID NO 102
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 102

ggaggactac caagaggggg 20

<210> SEQ ID NO 103
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 103

acagtcgtgc agtgcagctg 20

<210> SEQ ID NO 104
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 104

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ggggctctga gagcctctga 20

<210> SEQ ID NO 105
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 105

gagtcaacgg ggaataccat 20

<210> SEQ ID NO 106
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 106

gaaccattag atcaatgcga 20

<210> SEQ ID NO 107
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 107

ggggagcggc gctaaaggcg 20

<210> SEQ ID NO 108
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 108

gaaaacacgg ggtcacgcgc 20

<210> SEQ ID NO 109
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 109

acctgcaccc gcggccgttg 20

<210> SEQ ID NO 110
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 110

gccgagtcgc gcaggcgggg 20

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<210> SEQ ID NO 111
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 111

ttgggagtgg tcggattggt 20

<210> SEQ ID NO 112
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 112

ggcacaaggc agccagatct 20

<210> SEQ ID NO 113
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 113

aaagggtgtg tggagaaatg 20

<210> SEQ ID NO 114
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 114

gagcgcgcaa ccaggaccac 20

<210> SEQ ID NO 115
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 115

ccaggcctgt gtcgagtggg 20

<210> SEQ ID NO 116
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 116

gaggaggagc catggggcgg 20

<210> SEQ ID NO 117
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 117

ggaagtagaa tgtgcctgga t 21

<210> SEQ ID NO 118
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 118

gagaccacag ttagagaacc ac 22

<210> SEQ ID NO 119
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 119

cgacagttca gccatcactt 20

<210> SEQ ID NO 120
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 120

cactgttttt ccaagacctc a 21

<210> SEQ ID NO 121
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 121

ctccagaggt ttgagttctt ct 22

<210> SEQ ID NO 122
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 122

gcgaagagag ccacttctg 19

<210> SEQ ID NO 123
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 123

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ctggtgttgc ctcttgtgat 20

<210> SEQ ID NO 124
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 124

cagcttgac actggatctc 20

<210> SEQ ID NO 125
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 125

aagcagacgg aaagtgagg 19

<210> SEQ ID NO 126
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 126

gacctcaag tactttcagc ca 22

<210> SEQ ID NO 127
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 127

tgatctcaa gtctgtctgc 20

<210> SEQ ID NO 128
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 128

acaacttcgt gcactcca 18

<210> SEQ ID NO 129
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 129

tctgcgtgaa tcctagattt ctg 23

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<210> SEQ ID NO 130
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 130

gccgaacttc tcacagca 18

<210> SEQ ID NO 131
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 131

gtttttgatc cagaccaga tg 22

<210> SEQ ID NO 132
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 132

caagactgag atgcacaagt g 21

<210> SEQ ID NO 133
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 133

gtaagcagga agagaagcca 20

<210> SEQ ID NO 134
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 134

tcttgctatt gaccgatgct t 21

<210> SEQ ID NO 135
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 135

gcaacaaaaa gaaacgagat gac 23

<210> SEQ ID NO 136
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 136

ttcctgctat gattttctcc ca 22

<210> SEQ ID NO 137
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 137

aaactcacca ggatgctcac 20

<210> SEQ ID NO 138
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 138

gtgtacttgc tgatcaactg c 21

<210> SEQ ID NO 139
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

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agtgtcagtg gtgttggc 18

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gtggctatgg ttggaggtc 19

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<212> TYPE: DNA
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<220> FEATURE:
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<210> SEQ ID NO 144
<211> LENGTH: 18
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 144

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<210> SEQ ID NO 145
<211> LENGTH: 21
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

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<211> LENGTH: 21
<212> TYPE: DNA
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<400> SEQUENCE: 147

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<210> SEQ ID NO 148
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 148

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<210> SEQ ID NO 149
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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ccacagatca gaaacccgat gaaggc 26

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<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A synthetic sequence

<400> SEQUENCE: 150

taaagctgta gcccgttgcc tgc 23

<210> SEQ ID NO 151
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<212> TYPE: DNA
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<220> FEATURE:
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tgtaacaccc cagaccctc gaa 23

<210> SEQ ID NO 155
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<220> FEATURE:
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aggtgtgcgg gctcaggat 19

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<220> FEATURE:
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<400> SEQUENCE: 158

cgaggggtgga ggctgaatt ttga 24

<210> SEQ ID NO 159
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<210> SEQ ID NO 160
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<220> FEATURE:
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<400> SEQUENCE: 160

acgggtgttct gtttctcctg gca 23

<210> SEQ ID NO 161
<211> LENGTH: 27
<212> TYPE: DNA
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<400> SEQUENCE: 161

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agagagcaga ctggaagaaa acagtgg 27

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<400> SEQUENCE: 162

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cagatgtccc agttcctgtg cctt 24

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<210> SEQ ID NO 163
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<400> SEQUENCE: 163

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tgcagggttt caccaggatc cac 23

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<400> SEQUENCE: 164

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<400> SEQUENCE: 165

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gtttaagagc tatg 14

1. A method comprising ex vivo modification of any of the genes listed in Tables 1-7 or FIGS. 1-4 within at least one lymphoid or myeloid cell, or a combination thereof to generate at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells.

2. The method of claim 1, wherein the modification is one or more deletion, substitution or insertion into one or more endogenous genomic sites of any of the genes listed in Tables 1-7 or FIGS. 1-4.

3. The method of claim 1, wherein the modification is reduction of expression or translation of any of the genes listed in Tables 1-7 or FIGS. 1-4.

4. The method of claim 3, wherein the reduction of expression or translation is by an inhibitory nucleic acid.

5. The method of claim 1, wherein the modification is increased expression of any of the genes listed in Tables 1-7 or FIGS. 1-4.

6. The method of claim 5, wherein the increased expression is by modification of one or more promoters of any of the genes listed in Tables 1-7 or FIGS. 1-4.

7. The method of claim 1, wherein the modification is one or more CRISPR-mediated modifications or activations of any of the genes listed in Tables 1-7 or FIGS. 1-4.

8. The method of claim 1, wherein the modification is transformation of at least one lymphoid or myeloid cell, or a combination thereof, with one or more expression cassettes comprising a promoter operably linked to a nucleic acid segment comprising a coding region of any of the genes listed in Tables 1-7 or FIGS. 1-4.

9. The method of claim 1, further comprising administering at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to a subject.

10. The method of claim 1, further comprising incubating the at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to form a population of modified cells.

11. The method of claim **10**, further comprising administering the population of modified cells to a subject.

12. The method of claim **9**, wherein the subject has a disease or condition.

13. The method of claim **12**, wherein the disease or condition is an immune condition or cancer.

14. A method comprising contacting at least one test agent with test cells to provide a test assay mixture, and measuring:

- a. cellular proliferation of the test cells, cytokine release by the test cells, or a combination thereof;
- b. activation of the test cells;
- c. expression or activity of any of the regulators listed in Tables 1-7 or FIGS. 1-4 in the cells; or
- d. a combination thereof.

15. The method of claim **14**, further comprising comparing the measured results to control results.

16. The method of claim **15**, wherein control results are results of the test cells measured without any of the test agents.

17. The method of claim **14**, wherein the test cells comprise lymphoid and/or myeloid cells.

18. The method of claim **14**, wherein the test cells comprise cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, chimeric antigen receptor (CAR) cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune cells, or a combination thereof.

19. The method of claim **13**, wherein the immune condition is an autoimmune disorder, Graves disease, arthritis, psoriasis, Celiac disease, vitiligo, rheumatoid arthritis, lupus, Crohn's disease, multiple sclerosis, type 1 diabetes, alopecia, inflammatory bowel disease (IBD), Guillain-Barre syndrome, chronic inflammatory demyelinating polyneuropathy, or a combination thereof.

20. The method of claim **13**, wherein the cancer is leukemia, lymphoma, Hodgkin's disease, sarcomas of the soft tissue and bone, lung cancer, mesothelioma, esophagus cancer, stomach cancer, pancreatic cancer, hepatobiliary cancer, small intestinal cancer, colon cancer, colorectal cancer, rectum cancer, kidney cancer, urethral cancer, bladder cancer, prostate cancer, testis cancer, cervical cancer, ovarian cancer, breast cancer, endocrine system cancer, skin cancer, central nervous system cancer, melanoma, cancer associated with AIDS, or a combination thereof.

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